

# ADT 7120 / 7130

## Dicing Series



# Maintenance Manual

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# Chapter 1. Introduction

The following chapter introduces the Model 71XX system and includes the following sections:

- ❖ Model Description (section 1.1)
- ❖ Overview (section 1.2)
- ❖ Standard Features (section 1.4)
- ❖ Options (section 1.5)

## 1.1 Model Description

Model 71XX is a semi-automatic dicing system; manufactured by Advanced Dicing Technologies Ltd.



**Figure 1-1. Model 7132 (Large Area 2")**

Model 71XX can dice various materials, such as: LED packages and wafers, Discrete device wafers, Power device wafers, Semiconductor, Glass, Silicon and many others. The workpieces can be of sizes up to 300 mm in diameter. Dicing is performed with either a 2-inch or a 4-inch spindle.

All Model 71XX cutting procedures can be performed either in automatic mode or in manual mode.

## 1.2 Overview

The 71XX Series can be used to cut different types and shapes of workpieces, each of which may require different processing procedures. The process applied to each workpiece is determined by the assigned recipe, which can be used to dice an individual workpiece or used to dice a series of workpieces.

Workpieces are loaded by the user on the cutting chuck. The Model 71XX moves each workpiece through the alignment and cutting procedures, according to the parameters and algorithms defined in the recipe assigned to that individual workpiece. If required, any part of the operation (e.g., alignment) can be performed manually. After the process is complete, the diced workpiece is manually removed by the user from the cutting chuck.



## 1.3 Available Models

The 71XX Series includes four main models:

- **7124** - 4" DC Spindle
- **7134** - 4" DC Spindle; Large Area Travel
- **7122** - 2" DC Spindle
- **7132** - 2" DC Spindle; Large Area Travel

In addition, the following two special configuration models are available:

- **71TS** - Tilted configuration with 2" DC Spindle

This option enables the System to cut wafers at any angle that falls between 0 and 15 degrees by changing the orientation of the Spindle and the Microscope. Systems that include the Tilted Spindle option are delivered pre-configured with all the settings necessary for cutting at both standard and tilted orientation.

- **71MD** - Medical configuration with 2" DC Spindle

## 1.4 Standard Features

The Model 71XX Series includes the following standard features:

- **Single Spindle dicing machine**
- **Single Magnification Vision System**
- **Fully automated capabilities:**
  - Alignment.
  - Kerf-check monitoring and auto Y-offset correction.
  - Cut verification monitoring and auto cut-position correction.
- **Brushless Spindle:**
  - DC, 60,000 RPM, speed controlled Front Mount Air Bearing 2" Spindle (Models 7122 and 7132)
  - DC, 30,000 RPM, speed controlled Air Bearing 4" Spindle (Models 7124 and 7134)
- **Quick-change chucks;** holding workpieces in place by vacuum suction.
- **Load Monitoring;** enables the User to measure the load on the Spindle (measured in Amps) caused by resistance encountered by the blade during operation.
- **Water cleaning jets;** for additional workpiece cleaning procedure.
- Daily database backup.
- **Water Separator;** collects and drains water from the workpiece vacuum space.

## 1.5 Options

The Model 71XX can include one or more of the following options:

- **Multi-language Interface:** enables display of the GUI in either English or a local language.
- **Magnification:** ×30, ×60, ×120 (standard), ×240.
- **Digital Magnification:** enables improving system performance with reference to alignment and cut accuracy. The image can be adjusted to different magnification levels, used during the alignment and cutting procedures, to accommodate specific needs. The feature supports digital magnifications of: ×4, ×2, ×1
- **Dress Station:** enables automatic reshaping and/or dressing the blade to enhance or maintain cut quality without interrupting the cutting process.
- **Broken Blade Detector (BBD):** uses an optical sensor to detect partial or full blade breakage.
- **High Torque Spindle:** a high torque, 2.4 KW, 2" DC-brushless air bearing 60 krpm spindle; optimized for package singulation and hard material applications.
- **Oblique LED Illumination:** illumination that surrounds the microscope; recommended for BGA (Ball Grid Array) or metal workpieces, which have an uneven surface height.
- **Air Purification Kit:** external unit containing a multi-level air filtration that prevents oil/water from entering the machine air inlets and the system itself.
- **Broken Wafer Utility:** For incomplete workpieces, enables defining the precise workpiece dimensions in order to confine the cutting procedure only to the workpiece.
- **Height Measuring Tool (HMT):** enable adjusting the depth (or cut depth) according to the measured height of the workpiece and/or tape.
- **Wash Pipe** - used in addition to the spray bars for washing during the dicing process.

The Model 71XX Series has several configurations, each offering different features and options, as described in following Table 1-1.



**Note:** Not all the configurations can be upgraded on the customer's site.

**Table 1-1. Available Models of the 71XX Dicing Series**

| <b>Models/ Features</b>                 | <b>7122<br/>(Std. 2")</b> | <b>7132<br/>(LA 2")</b> | <b>7124<br/>(Std. 4")</b> | <b>7134<br/>(LA 4")</b> | <b>71TS<br/>(Tilted 2")</b> | <b>71MD<br/>(Medical)</b> |
|-----------------------------------------|---------------------------|-------------------------|---------------------------|-------------------------|-----------------------------|---------------------------|
| 2" DC Spindle                           | Standard                  | Standard                | NA                        | NA                      | Standard                    | Standard                  |
| 4" DC Spindle                           | NA                        | NA                      | Standard                  | Standard                |                             | TBA                       |
| Closed Loop Theta Table                 | Standard                  | Standard                | Standard                  | Standard                | Standard                    | Optional                  |
| Closed Loop Theta Table (high accuracy) | Optional                  | Optional                | Optional                  | Optional                | Optional                    | Standard                  |
| Digital Camera                          | Standard                  | Standard                | Standard                  | Standard                | Standard                    | Standard                  |
| Non-Contact Height Device (NCHD)        | Standard                  | Standard                | Standard                  | Standard                | Standard                    | Standard                  |
| Mechanical Height Button                | Optional                  | Optional                | Optional                  | Optional                | Optional                    | Optional                  |
| Dress Station                           | Optional                  | Optional                | Optional                  | Optional                | Optional                    | Optional                  |
| 2"/ 3" Cooling Block                    | Standard                  | Standard                | NA                        | NA                      | Special                     | Standard                  |
| 4"/ 5" Cooling Block                    | NA                        | NA                      | Standard                  | Standard                | NA                          | TBA                       |
| Three-Flow Cooling Control              | Standard                  | Standard                | Standard                  | Standard                | Standard                    | Standard                  |
| Cleaning Jets                           | Standard                  | Standard                | Standard                  | Standard                | Standard                    | Standard                  |
| Spray Bars                              | Standard                  | Standard                | NA                        | NA                      | N/A                         | TBA                       |
| High Cooling                            | NA                        | NA                      | Optional                  | Optional                | N/A                         | TBA                       |

| Table 1-1. Available Models of the 71XX Dicing Series                                  |                   |                 |                   |                 |                     |                   |
|----------------------------------------------------------------------------------------|-------------------|-----------------|-------------------|-----------------|---------------------|-------------------|
| Models/ Features                                                                       | 7122<br>(Std. 2") | 7132<br>(LA 2") | 7124<br>(Std. 4") | 7134<br>(LA 4") | 71TS<br>(Tilted 2") | 71MD<br>(Medical) |
| Water Separator                                                                        | Standard          | Standard        | Standard          | Standard        | Standard            | Standard          |
| Venturi Vacuum Pump                                                                    | Standard          | Standard        | Standard          | Standard        | Standard            | Standard          |
| Y Axis Linear Encoder                                                                  | Standard          | Standard        | Standard          | Standard        | Standard            | Standard          |
| Z Axis Rotary Encoder                                                                  | Standard          | Standard        | Standard          | Standard        | Standard            | Optional          |
| Z Axis Linear Encoder                                                                  | Optional          | Optional        | Optional          | Optional        | Optional            | Standard          |
| Vacuum Sensor                                                                          | Standard          | Standard        | Standard          | Standard        | Standard            | Standard          |
| High Definition Vision System with x100 Magnification                                  | Standard          | Standard        | Standard          | Standard        | Standard            | Standard          |
| Vision Magnification x30                                                               | Optional          | Optional        | Optional          | Optional        | Optional            | Optional          |
| Vision Magnification x60, x240                                                         | Optional          | Optional        | Optional          | Optional        | Optional            | Optional          |
| Pattern Recognition System (PRS) with Vertical and Oblique LED Illumination            | Standard          | Standard        | Standard          | Standard        | Standard            | Standard          |
| Pattern Recognition System (PRS) with Vertical and Oblique (Ring) Halogen Illumination | Optional          | Optional        | Optional          | Optional        | Optional            | Optional          |

**Table 1-1. Available Models of the 71XX Dicing Series**

| <b>Models/ Features</b>                                                    | <b>7122<br/>(Std. 2")</b> | <b>7132<br/>(LA 2")</b> | <b>7124<br/>(Std. 4")</b> | <b>7134<br/>(LA 4")</b> | <b>71TS<br/>(Tilted 2")</b> | <b>71MD<br/>(Medical)</b> |
|----------------------------------------------------------------------------|---------------------------|-------------------------|---------------------------|-------------------------|-----------------------------|---------------------------|
| Load Monitor                                                               | Standard                  | Standard                | Standard                  | Standard                | Standard                    | Standard                  |
| Chuck Frame Adapter                                                        | Optional                  | Optional                | Optional                  | Optional                | Optional                    | Optional                  |
| Quick-Change Cutting Chuck                                                 | Standard                  | Standard                | Standard                  | Standard                | Standard                    | Standard                  |
| Broken Blade Detector (BBD)                                                | Optional                  | Optional                | Optional                  | Optional                | Optional                    | Optional                  |
| Light Tower                                                                | Standard                  | Standard                | Standard                  | Standard                | Standard                    | Standard                  |
| Light Tower with Buzzer                                                    | Optional                  | Optional                | Optional                  | Optional                | Optional                    | Optional                  |
| Air Purification Kit                                                       | Optional                  | Optional                | Optional                  | Optional                | Optional                    | Optional                  |
| CE Interlocks                                                              | Optional                  | Optional                | Optional                  | Optional                | Optional                    | Optional                  |
| Large Area (for dicing 300 mm circular or 12" x 9" rectangular workpieces) | N/A                       | Standard                | N/A                       | Standard                | N/A                         | N/A                       |
| Tilted Spindle (0 to 15 degrees)                                           | NA                        | N/A                     | NA                        | NA                      | Standard                    | TBA                       |

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## Chapter 2. Getting Started

The Getting Started chapter provides an overview of system aspects that require a degree of familiarity prior to working with the system.

The Getting Started chapter includes the following sections:

- ❖ Before You Begin (section 2.1).
- ❖ Safety Information (section 2.2).
- ❖ System Description (section 2.3).
- ❖ Graphical User Interface (section 2.3.2).
- ❖ Menu Navigation Chart (section 2.4).
- ❖ Modes of Operation (section 2.5).

## 2.1 Before You Begin

Before you begin to work with 71XX Dicing Series, read the following

- The safety instructions.
- This manual: the Model 71XX Dicing System Maintenance Manual provides information and procedures to aid in the performance of preventative maintenance, as well as troubleshooting and repair procedures.
- The Model 71XX Dicing System Operation Manual, which explains how to operate and work with the 71XX dicing machines.
- Conventions: explains the symbols and navigation descriptions used in this manual, simplifying the identification and understanding for performing procedures.
- The Model 71XX glossary: explains the technical terms used in describing the system, and system functions. Use the glossary as a reference tool while working with the manual.

Before beginning a procedure:

- Read the procedure through to the end. Thorough understanding of the entire procedure prevents unnecessary loss of time and error.

### 2.1.1 Reference Documents

For further information, see the following document:

- **ADT Model 71XX Semi-Automatic Dicing System Maintenance Manual.** It provides information and procedures to aid with performing preventative maintenance, as well as troubleshooting and repair procedures.
- **ADT Model 71XX Semi-Automatic Dicing System Schematics and Illustrated Parts Book (IPB).** It provides drawings and schematics, including Bill of Material (BOM), of the Model 71XX main assemblies.
- **ADT Main System Controller (MSC) documentation.** It provides information on handling and configuring the MSC user interface.
- **Software Release Notes.** It provides the most updated information on new options and features.
- **Option Documentation.** It provides details on special options that may be installed on your dicing machine.



## 2.1.2 Conventions

The Conventions section explains basic symbols and navigation concept used in this manual and includes a glossary of abbreviations and acronyms.

### 2.1.2.1 Symbols

The following symbols have been inserted on the left hand side of the operating instructions in order to make it easier for the User to perform procedures:

| Symbol                                                                              | Description                                                                                                                                                                                                 |
|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|    | <b>Note:</b> Information given in a note describes how the System functions or provides a tip on how best to use it.                                                                                        |
|    | <b>Caution:</b> Information given in a message labeled “caution” refers to the safe operation of the System and provides warnings where the possibility for loss of data or damage to the equipment exists. |
|    | <b>Danger:</b> Information given in a message labeled “danger” warns of possible hazard to personnel and extreme hazard to the Model.                                                                       |
|  | A wrench indicates that a procedure deals with hardware and requires physical User intervention.                                                                                                            |

### 2.1.2.2 Navigation

We have adopted the following shortened instruction form for navigating to the various application elements.

| Shortened Instruction                                                                          | Replaces                                                                                                                                                                                                                      |
|------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| In the Main Menu window, select:<br><b>Maintenance &gt; System &gt; Hardware Configuration</b> | In the Main Menu window:<br>1) Press Maintenance; the Maintenance window appears.<br>2) Press System; the System view appears.<br>3) Press Hardware Configuration; the Hardware Configuration options and value list appears. |

### 2.1.2.3 Abbreviations and Acronyms

**Amp**----- Ampere

**BBD**----- Broken Blade Detector

**ESD** ----- Electrostatic Discharge Device

**Gf** ----- Grams of force

**GUI**----- Graphical User Interface

**Hz**----- Hertz (frequency)

**in.** ----- Inch

**in. Hg**----- Inch of Mercury (vacuum pressure unit)

**kgf\*cm** --- Kilograms of force times lever length, in centimeters (torque unit)

**mil**----- 0.001 inch

**mm** ----- Millimeter

**MPU** ----- Main Power Unit

**ms**----- Millisecond

**NCH**----- Non-contact Height device

**psi** ----- Pounds per Square Inch

**scfm**----- Standard Cubic Feet per Minute (gas flow unit).

**UPS** ----- Uninterruptible Power Supply

**UVL** ----- Ultraviolet Light

**V**----- Volts

**VAC** ----- AC Voltage

**VDC**----- DC Voltage

## 2.2 Safety Information

Advanced Dicing Technologies Ltd. believes that the safety of personnel working with and around our Systems is the utmost important consideration. Before operating the system or performing maintenance operations, read and be familiarized with the safety information:

- Obey and follow all warnings and cautions documented in the manual.
- Comply with all approved and established precautions for operating electrical and mechanical equipment.
- Allow only trained and authorized personnel to perform maintenance tasks.
- Verify that all power, air, and water facilities are turned off before beginning maintenance procedures, or before replacing or repairing parts (including inserting or removing connectors or boards).

### 2.2.1 Hazards

The following hazards are documented in the manual.

#### 2.2.1.1 Electrical Shock



**Danger:** High voltage is present at points throughout the system. Contact can result in injury or death. Before opening system panels or performing maintenance tasks ensure that there is no voltage present:

1. Power down the system.
2. Disconnect the AC power cord plug from the wall outlet and wait ten minutes for any remaining current to dissipate.
3. Turn off the Uninterruptible Power Supply (UPS) on systems where a UPS is present.

#### 2.2.1.2 Moving Parts



**Danger:** There are two types of hazards that deal with moving parts:

- Components (e.g. spindle and blade): rotate at high speeds and can cause injury even after the power has been turned off.
  - Do not touch either the spindle or the blade while they are still in motion. Wait until the rotation stops completely before working on or near moving parts.
  - Do not operate the system while any of the covers are partially or completely open.
- Loose clothing, jewelry and other loose or dangling items: can become caught in moving or rotating parts.

### 2.2.1.3 Burns

Do not touch the solenoid valves inside the system. Some solenoid valves may reach high temperatures and direct contact can cause burns.

### 2.2.1.4 Bodily Injury

Do not drop any of the covers while opening or closing. Release the covers after they are completely opened or closed. Failure to do so can cause bodily injury or damage to the covers.

### 2.2.1.5 High Pressure Water and Air

Observe standard operating procedures to avoid contact with high-pressure air and water, which can cause physical harm, especially to eyes.

## 2.2.2 Safety Features

Model 71XX is equipped with the following safety features. Look at them and become familiar with these features.

- **Emergency Stop Button:** press the red Emergency Stop button to stop all system activity and cut off power to the system.
- **Circuit Breaker Switch:** close the circuit breaker switch when it is necessary to immediately cut-off power to the system.
- **Grounding Cables:** grounding cables are attached to the inside of each system panel and must be disconnected when a system panel is removed. Likewise, the grounding cables must be reconnected after a system panel is replaced. An ESD (Electrostatic Discharge Device) should be worn when handling the grounding cables.
- **Interlocks:** lock the covers so that they cannot be opened during dicing.
  - **Spindle Interlock**

The Spindle Interlock prevents injury to the User by:

    - ♦ Preventing the Spindle Cover from opening while the Spindle is rotating
    - ♦ Preventing the Axes from being moved while the Spindle Cover is open
    - ♦ Stopping an Axis if the Spindle Cover is opened while the Axis is moving
  - **Load/Unload Interlock (71XXEUR only)**

The Load/Unload Interlock prevents the Load/Unload Cover from being opened while the Model 71XX is operating. The Interlock is only disengaged when Workpieces are loaded or unloaded from the Cutting Chuck.

### 2.2.2.1 Emergency Stop

Use the Emergency Stop button to immediately stop all system operation and avert potential harm to the user or system.

The Emergency Stop button is located on the system's front panel.

#### **To perform an emergency stop:**

Press the Emergency Stop button.



**Note:** After pressing the Emergency Stop button:

- ◆ The main circuit breaker shuts off, cutting power to the entire system.
- ◆ All system operations come to an immediate halt.

It should be noted that the air tank continues supplying air to the spindle for a short period of time to prevent damage to the spindle.

#### **To regain power after an emergency stop:**

1. Insert the Emergency Stop button key.
2. Turn the key and pull out the Emergency Stop button.
3. Turn the main circuit breaker on.

## 2.3 System Description

The System Description section is composed of the following two subsections:

- Main Subsystems Description (front panel, dicer and vision system).
- Graphic User Interface (Model 71XX Series application description).

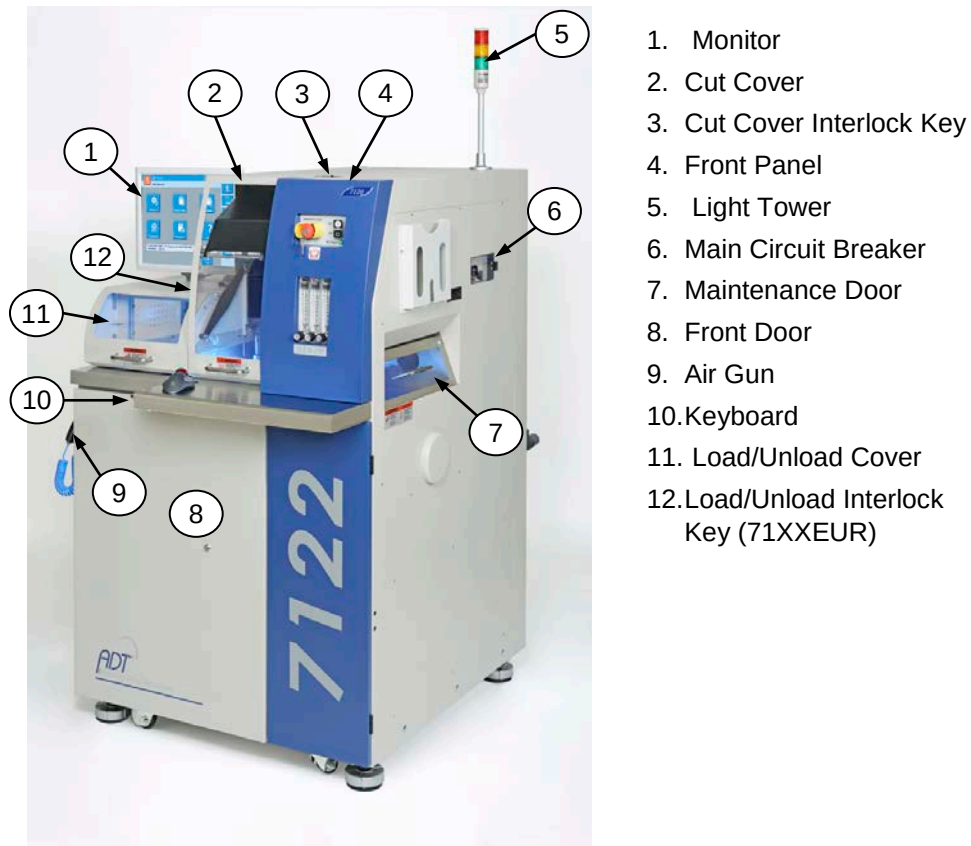


Figure 2-1. Model 71XX External Parts at a Glance

### 2.3.1 Main Subsystems

Main Subsystems are those elements located on the visual surface of the Model 71XX systems.

### 2.3.1.1 Front Panel

The Front Panel is located on the front of the system and includes the following controls and indicators:

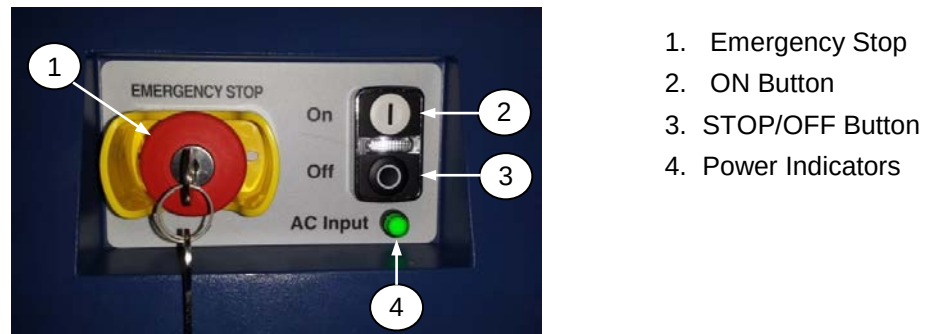


Figure 2-2. Front Panel

#### 2.3.1.1.1 Emergency Stop

The Emergency Stop button is used when a hazardous condition develops in the system or when the system must be stopped immediately.

#### 2.3.1.1.2 ON and OFF Buttons

The ON and STOP > OFF buttons provide the following functionality:



The ON button powers on the components and the system computer, which then begins loading the application.



The STOP > OFF button stops the system operation (if pressed once) and shuts down the System (if pressed twice). To start the system up again, press ON and wait for the User Password window to appear on the monitor.

This button enables the operator to immediately stop any activity in order to prevent self-injury or damaging the material.

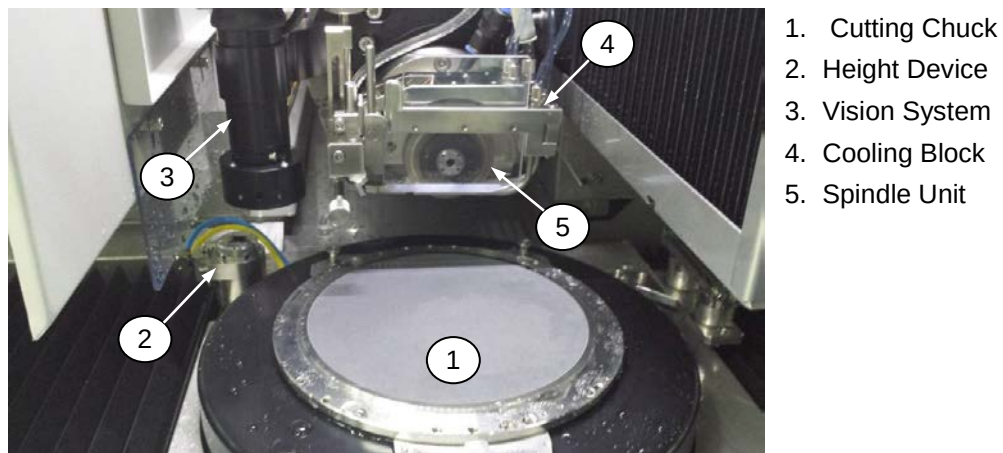
#### 2.3.1.1.3 Power Lights

The following lighting indications signify the power status in the MPU (Main Power Unit):

- First indication: AC power is running in the system.
- Second indication: the system is ON.

### 2.3.1.2 Dicer

The Dicer is where Workpieces are aligned and then cut.



**Figure 2-3. Dicer**

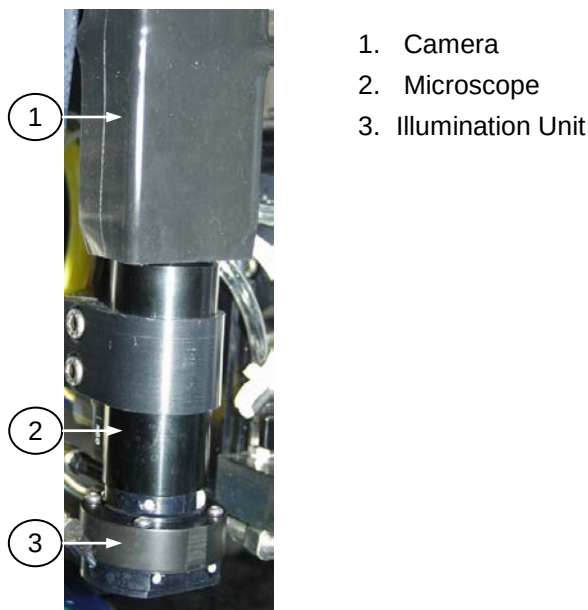
The dicer includes the following main and sub-components:

- Cutting Axes:
  - X-Axis: the axis on which the cutting table moves to the left and right of the operator.
  - Y-Axis: the axis on which the spindle moves away from and towards the system operator.
  - Z-Axis: the axis on which the spindle moves up and down in relation to the cutting table.
  - Theta Axis: the axis on which the cutting table moves in a circular pattern.
  - Rotational Axis (Spindle): the axes on which the spindle turns.
- Spindle Unit (2" or 4").
- Cooling Block (2"/3" or 5").
- Height Device (Non-contact Height device or Mechanical Button).
- Vision System (Microscope, Camera and Illumination).
- Cutting Chuck (quick-change).



### 2.3.1.2.1 Vision System

A high-performance vision system executes precise positioning of workpieces for the purpose of alignment and kerf checking.



**Figure 2-4. Vision System**

The vision system includes the following components:

- Camera.
- Microscope( $\times 30$ ,  $\times 60$ ,  $\times 120$  (standard), or  $\times 240$  magnification).
- LED Illumination Unit (Coaxial and Oblique).
  - **Coaxial:** Vertical illumination passing through the Microscope; generally used on silicon and similar Workpieces, which have reflective and flat surfaces.
  - **Oblique:** Ring/Side illumination around or on the sides of the Microscope; generally used on Ball Grid Array (BGA) or metal Workpieces and substrates which have uneven or non-flat surfaces.

### 2.3.1.3 Light Tower

The Light Tower indicates the system's working status, and includes three lights:

- Green: normal system operation.
- Yellow: need for user intervention.
- Red: an error has occurred.

The Light Tower also includes an integral audio alarm buzzer.



The Light Tower indicates the system's working status, and includes three lights:

- Green: normal system operation.
- Yellow: need for user intervention.
- Red: an error has occurred.

The Light Tower also includes an integral audio alarm buzzer.

**Table 2-1. Light Tower States**

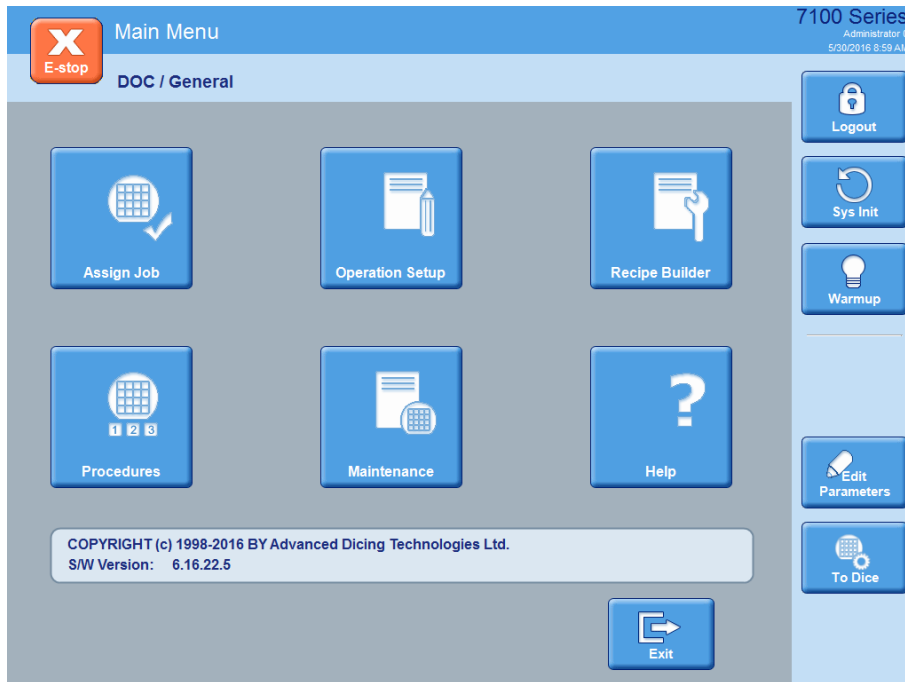
| Red       | Yellow      | Green | System State                                      |
|-----------|-------------|-------|---------------------------------------------------|
| Med Flash | Off         | Off   | Error.                                            |
| Off       | Off         | Off   | The system is not initiated.                      |
| Off       | On          | Off   | The system is idle                                |
| Off       | Med Flash   | Off   | Process paused, operator call.                    |
| Off       | Off         | On    | The system is running is Auto mode                |
| Off       | Off         | On    | Manual Operation                                  |
| Off       | Rapid Flash | Off   | User interference required (operator call)        |
| Off       | Med Flash   | Off   | Wizard is displayed, operator action is required. |



**Note:** The Light Tower indicators can be configured to meet individual customer requirements.

## 2.3.2 Graphical User Interface

When the application is launched, the Main Menu window appears.



**Figure 2-5. Main Menu Window**

The Main Menu window displays the following buttons, which access windows of the same name:

**Table 2-2. Graphical User Interface Description-1**

| Button Name     | Description                                                                                                                                                                                                                                                                                                               |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Assign Job      | Enables selecting a predefined recipe for the job.                                                                                                                                                                                                                                                                        |
| Operation Setup | Enables activating the various elements involving the: <ul style="list-style-type: none"> <li>• Dicer</li> <li>• Head System</li> <li>• Cutting Table</li> <li>• Light Tower</li> <li>• DCE Parameters</li> </ul>                                                                                                         |
| Recipe Builder  | Enables you to create a new recipe, define and modify the settings and parameters of specific recipes, including teaching the indexes, alignment, and the align focus to the system for dicing numerous workpieces using the same recipe.<br>Importing and Exporting recipes is performed from the Recipe Builder screen. |
| Procedures      | General Operator procedures such as Changing Chuck, Change Dress Block, and performing height related operations: Chuck Height, Chuck to NCH/Button Delta and Sample Blade Calibration.                                                                                                                                   |

| Button Name | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Maintenance | <p>Enables accessing the following windows:</p> <p><b><u>Maintenance</u></b></p> <ul style="list-style-type: none"> <li>• Axis Setup: Enables performing setup procedures prior to running the system.</li> <li>• Calibration: Enables calibrating: <ul style="list-style-type: none"> <li>▪ Axes (Error Mapping)</li> <li>▪ Illumination</li> <li>▪ Pixel Size</li> <li>▪ Sensors</li> </ul> </li> <li>• Functions and I/O: Enables performing manual procedures within the system in order to manage and/or repair system-generated problems. Functions and I/O tasks must be performed by authorized personnel only, as these tasks can cause damage to the system or workpiece.</li> <li>• RS232: Enables performing such maintenance procedures as debugging and troubleshooting.</li> </ul> <p><b><u>Administration:</u></b></p> <ul style="list-style-type: none"> <li>• Permissions: for defining screens and parameters available to the various user groups.</li> <li>• System: for configuring users, flags, application, and hardware.</li> <li>• Calculator: Accesses the online calculator.</li> <li>• Virtual keyboard: for loading the virtual keyboard.</li> </ul> <p><b><u>File management:</u></b></p> <ul style="list-style-type: none"> <li>• Backup: Enables defining, creating, storing, and restoring backup files.</li> <li>• Log File: Accesses the Log File Viewer, which displays all of the events that occur in the system, including: <ul style="list-style-type: none"> <li>▪ Event ID (system-generated identification number)</li> <li>▪ Date and time (of the event)</li> <li>▪ Event Name</li> <li>▪ Attribute Name</li> <li>▪ Value (of the event)</li> </ul> <p>From the Log File, you can filter the displayed events, export the information to an external system, and access the statistics window.</p> </li> <li>• Statistics: accesses the statistics window, which enables viewing the statistics of all of the system activities.</li> <li>• Dmp backup: for creating a “dump file” after a crash or for debugging purposes.</li> </ul> |
| Help        | Accesses the system manuals and other documentation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

In addition, the application displays some or all of the following buttons, which enable performing specific tasks:

**Table 2-3. Graphical User Interface Description-2**

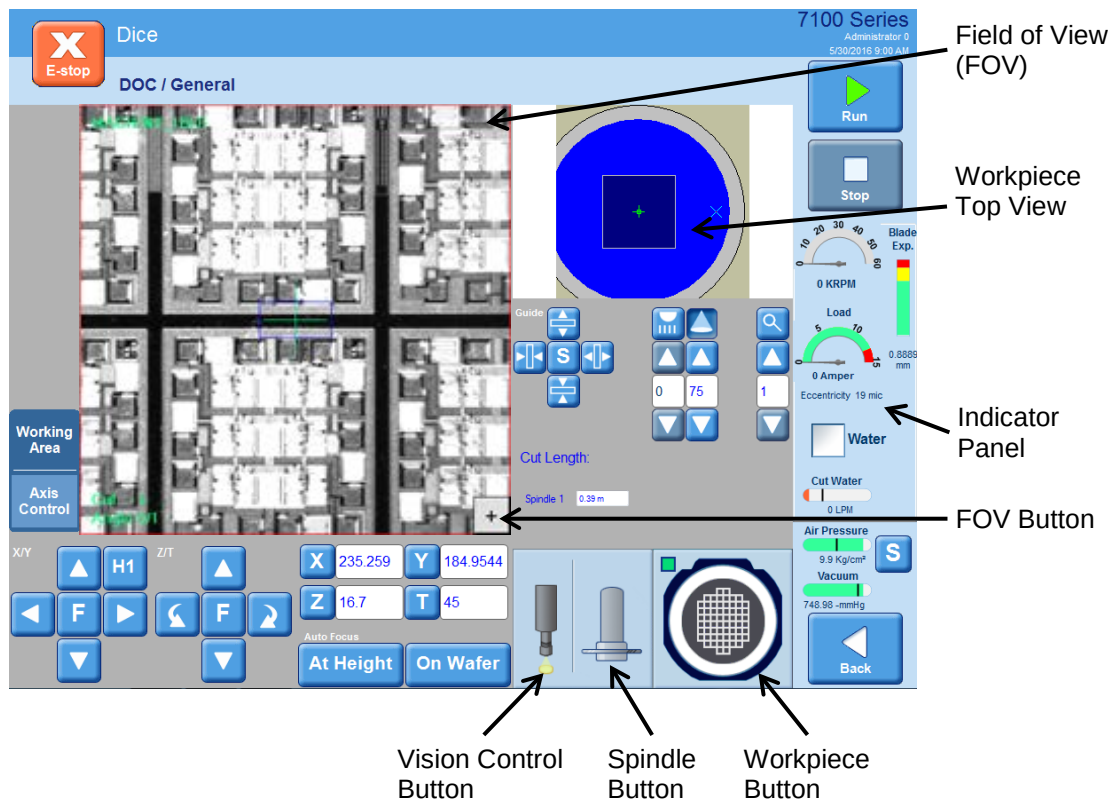
| Button Name     | Description                                                                                                                                                                           |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| X E-stop        | Stops all procedures and cutting. Operation can be resumed from where the system stopped. Abort should only be performed in non-standard situations as it can destabilize the system. |
| Logout          | Enables logging out as the current user and logging back in as a different user.                                                                                                      |
| Sys Init        | Enables manually initializing the system.                                                                                                                                             |
| Warmup          | Initiates a system warm up, whereby the spindle rotates at a predefined velocity and for a predefined amount of time, and the water supply is activated.                              |
| Edit Parameters | Enables editing the assigned recipe.                                                                                                                                                  |
| Exit            | Enables closing the application.                                                                                                                                                      |
| To Dice         | Accesses the Dice window.                                                                                                                                                             |



**Note:** Some of the windows have additional buttons for performing additional tasks.

### 2.3.2.1 Dice Screen Description

Pressing the **To Dice** button opens the Axis Control dice screen.



**Figure 2-6. Dice Window**

Pressing the animation figures opens a list of operations that be performed, and are detailed in following Table 2-4 (some are configuration dependent).

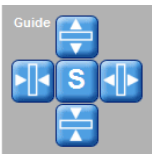
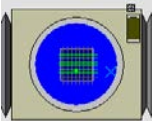
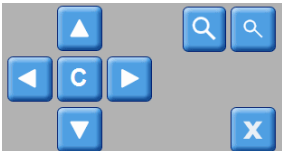
**Table 2-4. Axis Control Buttons-1**

| Button                                                                                                     | Operations                                                                                                      | Details                                                                               |
|------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| <b>Vision Control</b><br> | Align                                                                                                           | Auto, Manual                                                                          |
|                                                                                                            | Cut Verify                                                                                                      | Auto, Manual for Head 1                                                               |
|                                                                                                            | Kerf Check                                                                                                      | Auto, Manual for Head 1, Update Focus and Illumination Front/Rear, Kerf Check Results |
|                                                                                                            | Init Vision                                                                                                     |                                                                                       |
|                                                                                                            | Camera Parameters                                                                                               | (see details in section 9.2)                                                          |
| <b>Spindle</b><br>        | Blade Info                                                                                                      | Operation on the Spindle and Blade                                                    |
|                                                                                                            | Change Blade                                                                                                    |                                                                                       |
|                                                                                                            | Operate Spindle                                                                                                 |                                                                                       |
|                                                                                                            | Height                                                                                                          |                                                                                       |
|                                                                                                            | Dressing                                                                                                        |                                                                                       |
|                                                                                                            | Dressing Block (option):<br>Dress, Dress Both, Change                                                           |                                                                                       |
|                                                                                                            | Override                                                                                                        |                                                                                       |
|                                                                                                            | Open Water                                                                                                      |                                                                                       |
|                                                                                                            | Y Offset                                                                                                        |                                                                                       |
|                                                                                                            | Depth compensation on tape                                                                                      |                                                                                       |
| <b>Workpiece</b><br>    | Load Workpiece                                                                                                  | Move axes to load station                                                             |
|                                                                                                            | Unload Workpiece                                                                                                | Move axes to unload station                                                           |
|                                                                                                            | Partial Wafer Cut                                                                                               | For cutting a partially cut workpiece                                                 |
|                                                                                                            | Broken Wafer                                                                                                    | For broken wafer option                                                               |
|                                                                                                            | View Alarm                                                                                                      | Opens an existing error window                                                        |
|                                                                                                            | Wash Air Pipe                                                                                                   | Manually operate the wash pipe                                                        |
| <b>FOV</b><br>          | Enlarge Video to displays a large view of the loaded workpiece. (press minimize to return to the previous view) | Minimize                                                                              |
|                                                                                                            |                                                                                                                 | Refresh                                                                               |
|                                                                                                            |                                                                                                                 | Measure: select and press in the video window.                                        |
|                                                                                                            |                                                                                                                 | Save Image (browse to location)                                                       |
|                                                                                                            |                                                                                                                 | Load Image (browse to location)                                                       |
|                                                                                                            |                                                                                                                 | Tech Vision: Inspection Illumination, Focus, Tape Focus and Illumination              |
|                                                                                                            |                                                                                                                 | Teach Kerf-Check                                                                      |
|                                                                                                            |                                                                                                                 | Preprocess                                                                            |
|                                                                                                            |                                                                                                                 | Teach Tools: Line Profile, Histogram Graph                                            |

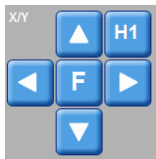
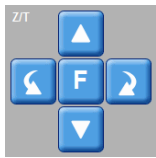
| Button | Operations                                                        | Details                                                                                                                                                                                                                        |
|--------|-------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|        | Enlarge Animation (press minimize to return to the previous view) | Displays a large view of the dicer animation, with buttons to:<br>Enlarging and reducing the displayed<br>Move the animation<br>"C" moves the chuck animation to the center.<br>"X" returns to previous (main) animation view. |
|        | Refresh                                                           | Refresh the vision window                                                                                                                                                                                                      |
|        | Save Image                                                        | Browse to location                                                                                                                                                                                                             |
|        | Load Image                                                        | Browse to location                                                                                                                                                                                                             |

Pressing the buttons performs various operations, as detailed in Table 2-5.

**Table 2-5. Axis Control Buttons-2**

| Button                                                                                                          | Operations                                                                                           | Details                                                                                                                                  |
|-----------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| Illumination and Magnification                                                                                  | Allow changing the illumination levels                                                               | 0–255                                                                                                                                    |
|                                                                                                                 | Allow changing the magnification                                                                     | From ×1 to ×2                                                                                                                            |
| Guides<br>                   | Buttons to change the size of the boxes.                                                             |                                                                                                                                          |
|                                                                                                                 | Center button can change between:                                                                    | <b>S:</b> to modify the Search Box size<br><b>T:</b> to modify the Teach Box size<br><b>R:</b> to modify the Reticle (cross) size        |
| Alternative Illumination<br> | see details in section 9.2                                                                           |                                                                                                                                          |
| Dicer Animation<br>          | Displays the dicer components: chuck, workpiece, Height device, Dress block (if configured), Spindle |                                                                                                                                          |
|                                                                                                                 | Right clicking with a mouse allows showing the spindle.                                              |                                                                                                                                          |
|                                                                                                                 | Clicking on the background opens the following available buttons                                     | Enlarging and reducing what is displayed                                                                                                 |
|                                                                                                                 |                   | Moving the animation<br>Pressing "C" moves the chuck animation to the center.<br>Pressing "X" returns to previous (main) animation view. |



| Button                                                                                                | Operations                                                        | Details                                                                                                                                                                                                                                                                                 |
|-------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>X/Y Axis</b><br>  | Arrows (Up/Down/Right/Left) for moving the X or Y axes            |                                                                                                                                                                                                                                                                                         |
|                                                                                                       | H1                                                                | Moves to the last cut position (active only after dicing starts)                                                                                                                                                                                                                        |
|                                                                                                       | Center button can change between:                                 | <b>F</b> : axis movement fast velocity<br><b>S</b> : axis movement in slow velocity<br><b>P</b> : axis movement in pixel increments<br><b>M</b> : axis movement in micron increments<br><b>I</b> : Y-axis movement by Index<br><b>T</b> : move to Tape height out of the workpiece area |
| <b>Z/T Axis</b><br>  | Arrows (Up/Down Rotate Right/Left) for moving the Z or Theta axes | <b>F</b> : axis movement fast velocity<br><b>S</b> : axis movement in slow velocity<br><b>A</b> : axis (T) by angle<br><b>P</b> axis movement in pixel increments<br><b>M</b> : axis movement in micron increments<br><b>T</b> : move to Tape height out of the workpiece area          |
|                                                                                                       |                                                                   |                                                                                                                                                                                                                                                                                         |
| <b>Sensors</b><br> | Opens the Sensors screen (see Figure 2-7).                        | Spindle Water<br>Air/Vacuum<br>Accessories (NCH, Spindle, BBD, Cut cover)                                                                                                                                                                                                               |

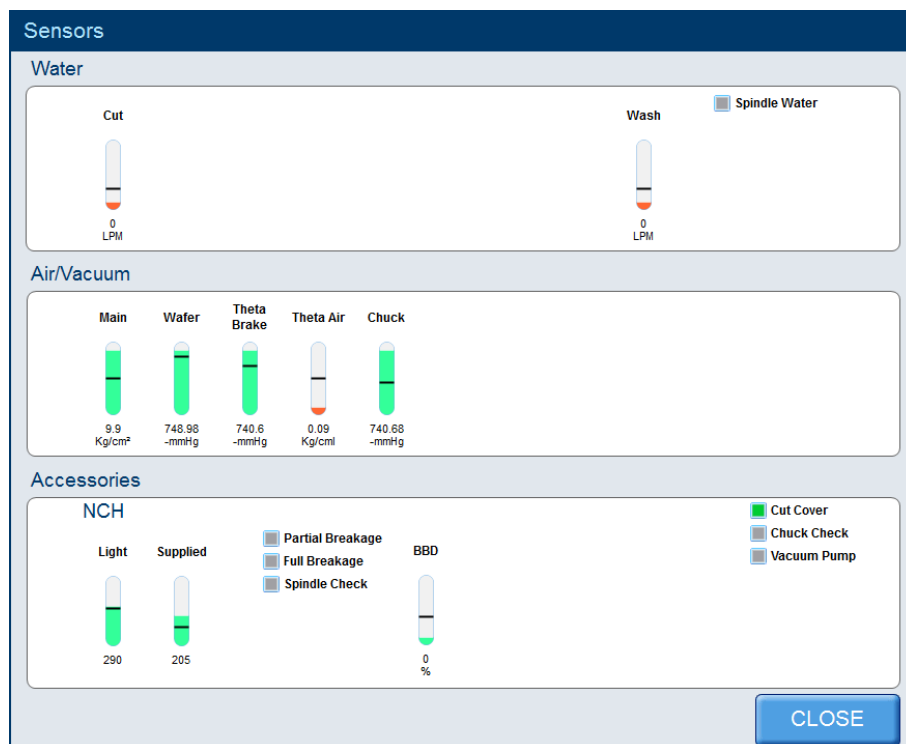


Figure 2-7. Sensors screen

### 2.3.2.1.1 Working Area

Pressing the Working Area button changes the Working Area dice screen.

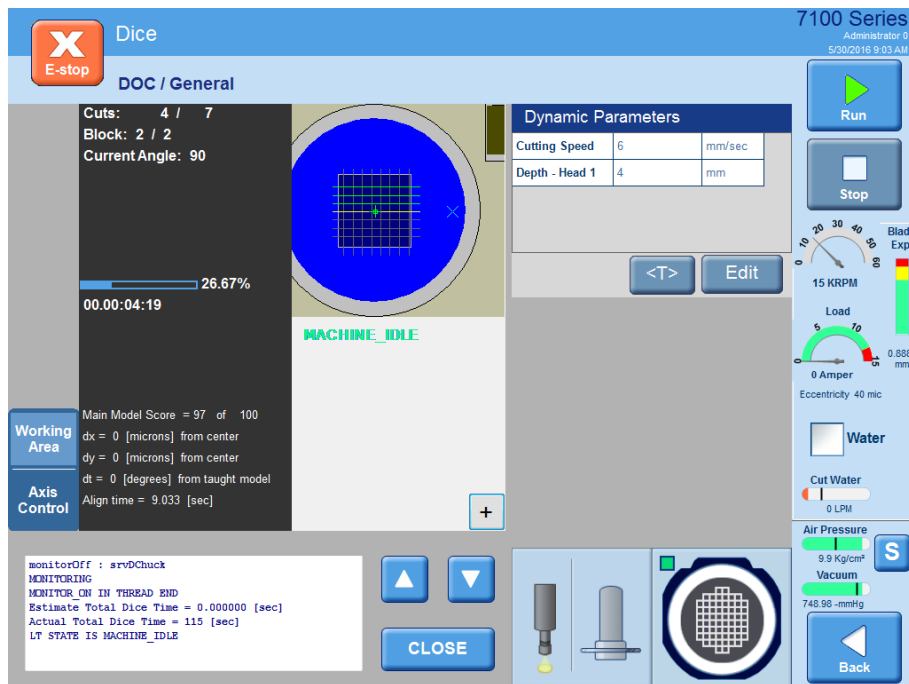
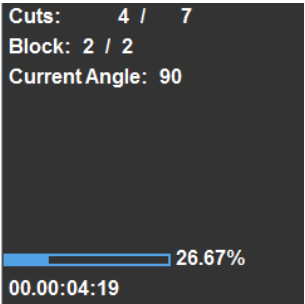
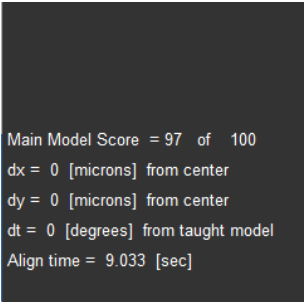
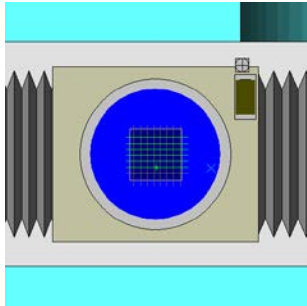
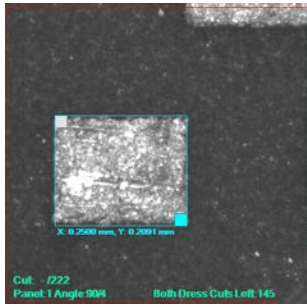
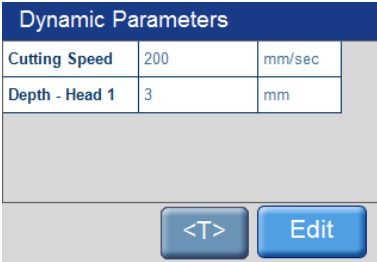
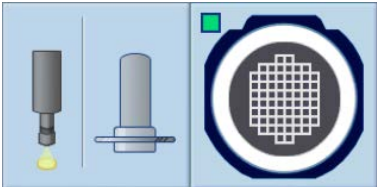


Figure 2-8. Dice Window (Working Area)

The Working Area has five sections, each for viewing or performing different actions.

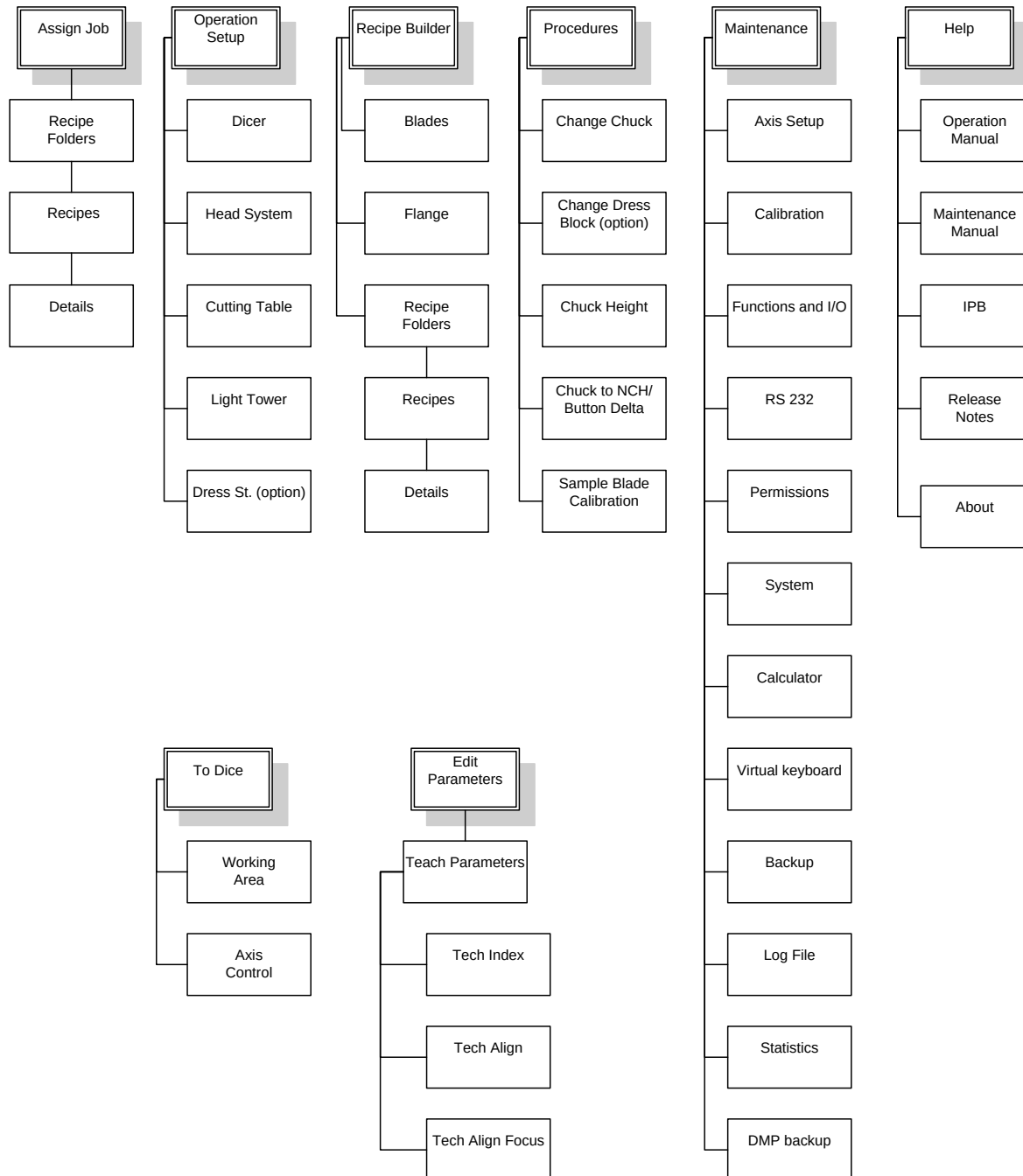
Table 2-6. Working Area

| Section                                         | Description                                                                                                       | Details                    |
|-------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|----------------------------|
| Process display<br>(divided into four sections) | Dicing process information<br> | Cuts, Blocks, Angles, Time |
|                                                 | Alignment information<br>      | Model score. Delta, Time   |

| Section                                                                                                   | Description                                                                                          | Details                                                                                                                 |
|-----------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
|                                                                                                           | Dicer animation<br> | Displays the dicer components: chuck, workpiece, Height device, Dress block (if configured), Spindle and vision system. |
|                                                                                                           |                     | Real-time image of workpiece                                                                                            |
| Dynamic Parameters<br> | Cutting Speed                                                                                        | For editing in process pause/stop mode                                                                                  |
|                                                                                                           | Depth                                                                                                |                                                                                                                         |
| Log of operations<br>  | Displays the various performed operations                                                            | Used mainly for debugging                                                                                               |
| Animated buttons<br>   | Vision Control button                                                                                | See Table 2-4 above                                                                                                     |
|                                                                                                           | Spindle buttons                                                                                      | See Table 2-4 above                                                                                                     |
|                                                                                                           | Workpiece button                                                                                     | See Table 2-4 above                                                                                                     |

## 2.4 Menu Navigation Chart

The following chart shows the Main Menu buttons and the screens that can be accessed from each of those buttons.



## 2.5 Modes of Operation

Model 71XX series operates in the following working modes:

- **Auto Mode:** enables processing workpieces without user intervention.
- **Manual Mode:** enables performing procedures separately, manually etc.
- **Unprotected Mode:** reserved for advanced users only and employed when performing diagnostic and debugging maintenance operations. Note that unauthorized users can cause damage to the system and the workpiece when working in the unprotected mode.

### 2.5.1 Protected/Unprotected Mode

The Model 71XX application, by default, operates in the protected mode whereby the application prevents performing diagnostic operations.



**Danger:**

1. When working in the unprotected mode, you take complete responsibility for any tasks that you perform. Therefore, it is essential that you are familiar with Model 71XX operation before working in the unprotected mode.
2. Only users with access-level authorization can switch to the unprotected mode.
3. The following windows can be switched to the unprotect mode to enable changes:
  - Operation Setup
  - Axis Setup
  - Calibration
  - Functions and I/O
  - Permissions
  - System

**To switch to the unprotected mode (example):**

1. In the Main Menu window, press **Maintenance > Function and I/O**; the Function and I/O window appears.
2. Select a **Function and I/O** folder and press **Unlock**; the top title banner changes to green, indicating the unprotected mode is activated.

**To switch back to the protected mode:**

Select a new window; the Function and I/O window is relocked and the top title banner changes to light blue, indicating protected mode is active.

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## Chapter 3. Installing the System

Installing the System lists and describes the system specifications and prerequisites, as well as the procedures that must be performed to install the system.

Installing the System includes:

- ❖ System Specifications (section 3.1).
- ❖ Site Preparation (section 3.2).
- ❖ System Installation (section 3.3).

## 3.1 System Specifications

Specifying System Elements describes various elements in the system and includes the accepted specifications of each element:

- Workpieces.
- Blades.
- Broken Blade Detector (Optional).
- Spindle.
- X Axis (Feed).
- Y Axes (Indexing).
- Z Axis (Cut Depth).
- Theta Axis (Rotational).
- Vision System.
- Display.

### 3.1.1 Workpieces

The Model 71XX is capable of dicing round and rectangular Workpieces with a thickness up to 12 mm and the following maximum dimensions:

**Table 3-1. Workpiece Maximum Measurements**

| 71XX Model                           | Workpiece Type | Maximum Dimensions            |
|--------------------------------------|----------------|-------------------------------|
| Standard<br>(Models 7122 and 7124)   | Circular       | 200 mm (8 inch) diameter      |
|                                      | Rectangular    | 200 x 200 mm (8 x 8 inches)   |
| Large Area<br>(Models 7132 and 7134) | Circular       | 300 mm (12 inch) diameter     |
|                                      | Rectangular    | 300 x 300 mm (12 x 12 inches) |

### 3.1.2 Blades

The system can work with the following blades:

**Table 3-2. Blade Settings**

| 71XX Model                           | Size     | Type           |
|--------------------------------------|----------|----------------|
| Models 7124 and 7134<br>(4" Spindle) | 4" to 5" | Hubless        |
| Models 7122 and 7132<br>(2" Spindle) | 2" to 3" | Hub or Hubless |
| Model 71TS<br>(Tilted 2" Spindle)    | 2" to 3" | Hub or Hubless |
| Model 71MD<br>(Medical 2" Spindle)   | 2" to 3" | Hub or Hubless |



### 3.1.3 Broken Blade Detector (Optional)

The BBD is available on the 71XX Models 7122 and 7132 (2" Spindle).

The BBD uses a light transmitter and receiver that are mounted on the Cooling Block, and is used for detecting any breakage along the Blade perimeter having a minimum size of 1 mm in length and/or 0.5 mm in depth; regardless of the Spindle speed (up to 60,000 rpm).

### 3.1.4 Spindle

The 71XX Series machines are equipped with a front-mount, air bearing, water cooled 2" or 4" Spindle.

The Spindle specifications are as follows:

#### 3.1.4.1 2" Spindle

Table 3-3. 2" Spindle Specification

| Property                       | Details                |
|--------------------------------|------------------------|
| Spindle Type                   | DC, Brushless          |
| Maximum Rotation Speed         | 60,000 rpm             |
| Speed Stability During Cutting | ± 1%                   |
| Output power                   | 1.2 kW (2.4 kW option) |
| Programmable Speed Increment   | <100 rpm               |

#### 3.1.4.2 4" Spindle

Table 3-4. 4" Spindle Specification

| Property                       | Details                   |
|--------------------------------|---------------------------|
| Spindle Type                   | DC Brushless/AC Brushless |
| Maximum Rotation Speed         | 30,000 rpm                |
| Speed Stability During Cutting | ± 1% (DC); ± 3% (AC)      |
| Output power                   | 2.5 kW                    |
| Programmable Speed Increment   | <100 rpm                  |

### 3.1.5 X-Axis (Feed)

The X Axis is an air-bearing axis.

**Table 3-5. X-Axis Specification**

| Property            | Details                           |
|---------------------|-----------------------------------|
| Cut Stroke          | 416 mm (Standard)                 |
|                     | 445 mm (Large Area Configuration) |
| Feed Rate Range     | up to 600 mm/sec.                 |
| Feed Rate Stability | ±1%                               |

### 3.1.6 Y-Axis (Indexing)

The Y Axis includes ball bearing slides and a stepper motor.

**Table 3-6. Y-Axis Specification**

| Property                             | Details                           |
|--------------------------------------|-----------------------------------|
| Stroke                               | 210 mm (Standard)                 |
|                                      | 345 mm (Large Area Configuration) |
| Cumulative Accuracy (bi-directional) | ±1.5 µm/ 200 mm                   |
| Indexing Accuracy                    | ±1.0 µm                           |
| Minimum Programmable Step            | 0.2 µm                            |

### 3.1.7 Z-Axis (Cut Depth)

The Z Axis includes ball bearing slides and a stepper motor.

**Table 3-7. Z-Axis Specification**

| Property                  | Details              |
|---------------------------|----------------------|
| Stroke                    | 45 mm for 2" Spindle |
| Minimal Step (resolution) | 0.2 µm               |
| Accuracy (unidirectional) | 2 µm                 |
| Repeatability             | 1 µm                 |

### 3.1.8 Theta-Axis (Rotational)

The standard Theta (T) Axis includes cable drive with a stepper motor.

**Table 3-8. Theta-Axis Specification**

| Model                  | 712X and 713X                         | 71TS                                  | 71MD                                  |
|------------------------|---------------------------------------|---------------------------------------|---------------------------------------|
| Theta Axis Type        | Closed Loop Direct Drive DC Brushless | Closed Loop Direct Drive DC Brushless | Closed Loop Direct Drive DC Brushless |
| Stroke                 | 350°                                  | 350°                                  | 350°                                  |
| Resolution of Rotation | 0.125 arc-sec.                        | 0.125 arc-sec.                        | 0.125 arc-sec.                        |
| Repeatability          | 4 arc-sec.                            | 4 arc-sec.                            | 4 arc-sec.                            |
| Accuracy               | 4 arc-sec.                            | 4 arc-sec.                            | 4 arc-sec.                            |

### 3.1.9 Vision System

The Vision System includes the Digital Camera (Fire-Wire technology), Vertical and Oblique LED illumination.

**Table 3-9. Vision Specifications**

| Digital Magnification | x30    | x60    | x120   | x240    | Model 71XX Type |
|-----------------------|--------|--------|--------|---------|-----------------|
| Optical Magnification | 1      | 2      | 4      | 8       | All Models      |
| Resolution            | 10 µm  | 5.0 µm | 2.5 µm | 1.25 µm | All Models      |
| Field of View         | 4.8 mm | 2.4 mm | 1.2 mm | 0.6 mm  | All Models      |

### 3.1.10 Display

The Model 71XX employs a 17" touch screen and user interface.

## 3.2 Site Preparation

Before installing the system, verify that all of the following elements are present at the site and their specifications have been met:

- Environmental Requirements.
- Electricity Supply.
- Water.
- Pneumatics.
- Site Dimensions.



**Note:** If the system require fastening to the floor, contact ADT for the securing kit.

### 3.2.1 Environmental Requirements

To ensure proper system performance, verify that the following room specifications are met.

**Table 3-10. Environmental Requirements**

| Property              | Specifications                                   |
|-----------------------|--------------------------------------------------|
| Temperature           | 20°C – 25°C (68°F – 77°F)                        |
| Temperature Variation | ±1°C (±1.8°F)                                    |
| Humidity              | Less than 70% relative humidity (non-condensing) |
| Lighting              | Minimum 1000 lux.                                |

### 3.2.2 Electricity Supply

The system's internal main power connection has been preset by ADT to comply with the voltage requirements at your site, according to information that you provided at the time of ordering.

#### 3.2.2.1 Electric Supply Specifications

**Table 3-11. Electric Supply Specifications**

| Property              | Specifications                |
|-----------------------|-------------------------------|
| Voltage options (RMS) | 200 to 240 ±10% VAC, 1 phase  |
| Frequency             | 50/60 Hz ±2%                  |
| Power consumption     | 2.5 / 3.5 kVA (2"/4" Spindle) |

### 3.2.2.2 External Circuit Breaker

Table 3-12. External Circuit Breaker

| Element                        | Specifications                                |
|--------------------------------|-----------------------------------------------|
| Circuit breaker rate           | Minimum 25 A                                  |
| Circuit breaker inrush current | Minimum 100 A                                 |
| Location/Distance              | Maximum 2 meters from the rear of the system. |

### 3.2.3 Water

Several types of water, and for varying purposes, are used by the system. Verify that the water that is used meets the system's requirements.



**Caution:** Do not use high acid or alkaline water, or water with impurities such as iron, calcium, copper, chlorine, sodium or magnesium, as these can clog or corrode the piping.

#### 3.2.3.1 General Water Composition Requirements (Maximum Values)

Table 3-13. General Water Composition Requirements

| Property                     | Specifications                   |
|------------------------------|----------------------------------|
| Acidity (pH)                 | 6.8 – 7.8                        |
| Conductivity                 | 1000 $\mu$ S/cm                  |
| Total dissolved solids (TDS) | 750 mg/l                         |
| Total Hardness               | 270 mg/l as (CaCO <sub>3</sub> ) |
| Calcium (Ca)                 | 80 mg/l                          |
| Alkalinity                   | 150 mg/l as (CaCO <sub>3</sub> ) |
| Silica (SiO <sub>2</sub> )   | 30 mg/l                          |
| Sulfate (SO <sub>4</sub> )   | 100 mg/l                         |

#### 3.2.3.2 Cutting Water Supply

Table 3-14. Cutting Water Supply

| Property         | Specifications                                |
|------------------|-----------------------------------------------|
| Conductivity     | 0.05 – 1.0 $\mu$ S/cm                         |
| Pressure         | 4–5 atm. (0.4–0.5 MPa) or 58–73 Psi $\pm$ 10% |
| Temperature      | 20–25 °C $\pm$ 1 °C (68–77 °F $\pm$ 1.8 °F)   |
| Consumption      | Up to 9 Lpm                                   |
| Filtration level | 25 $\mu$ m or better                          |
| Inlet hose       | Ø 12 mm O.D. min.                             |

### 3.2.3.3 Spindle Cooling Water Supply



**Caution:** Do not use DI water for spindle cooling water, as it can aggressively attack the cooling surfaces inside the spindle and erode seal areas.

**Table 3-15. Spindle Cooling Water Supply**

| Property         | Specifications                              |
|------------------|---------------------------------------------|
| Pressure         | 2.8–6.2 atm. (40–90 psi) (0.28–0.62 MPa)    |
| Consumption      | 2.2 Lpm (0.29/0.21 gpm)                     |
| Temperature      | Ambient 20–25 °C ± 1 °C (68–77 °F ± 1.8 °F) |
| Filtration level | 25 µm or better                             |
| Inlet Hose       | Ø 8 mm I.D. min.                            |



**Note:** Significant temperature changes may affect the process and cut accuracy. Use a buffer tank or constant-temperature equipment to maintain a stable water temperature of: 20–25 °C (68–77 °F) ± 1 °C (± 1.8 °F) from initial water temperature during System operation.

### 3.2.3.4 Water Drain



**Caution:** Ensure that the drain pipe on the wall side is installed no higher than 400 mm from the floor, and allows free water flow.

**Table 3-16. Water Drain**

| Property             | Specifications            |
|----------------------|---------------------------|
| Drain extension hose | Ø 38 mm O.D.              |
| Outlet height        | 460 mm (18.1")            |
| Outlet diameter      | Ø 38 mm (1.5")            |
| Facility side        | Lower than 400 mm (15.7") |
| Maximum length       | 2 meters (40")            |

### 3.2.3.5 Mist Exhaust

**Table 3-17. Mist Exhaust**

| Property                            | Specifications      |
|-------------------------------------|---------------------|
| Mist extension hose                 | Ø 100 mm O.D.       |
| Outlet center height from the floor | 1175 mm (46.3")     |
| Minimum suction                     | 300 cm/h (177 cf/m) |

## 3.2.4 Pneumatics

Pneumatics describes the compressed air specifications.

**Table 3-18. Pneumatics**

| Property         | Specifications                                 |
|------------------|------------------------------------------------|
| Pressure         | 5.5–7 atm. (80–102 psi) 0.55–0.7 Mpa $\pm$ 10% |
| Consumption      | 350 Lpm (12.4 cfm)                             |
| Filtration level | 0.1 $\mu$ m or less                            |
| Oil residue      | Less than 0.1 ppm wt/wt                        |
| Dew Point        | Below -15°C (5 °F)                             |
| Air Temperature  | 20–25 °C $\pm$ 1 °C (68–77 °F $\pm$ 2 °F)      |
| Inlet Hose       | Ø 12 mm I.D. min                               |

## 3.2.5 Site Dimensions

The site dimensions are determined according to the following factors:

- System Dimensions.
- Center of Gravity.

### 3.2.5.1 System Dimensions

System dimensions refer to the physical size of the system.

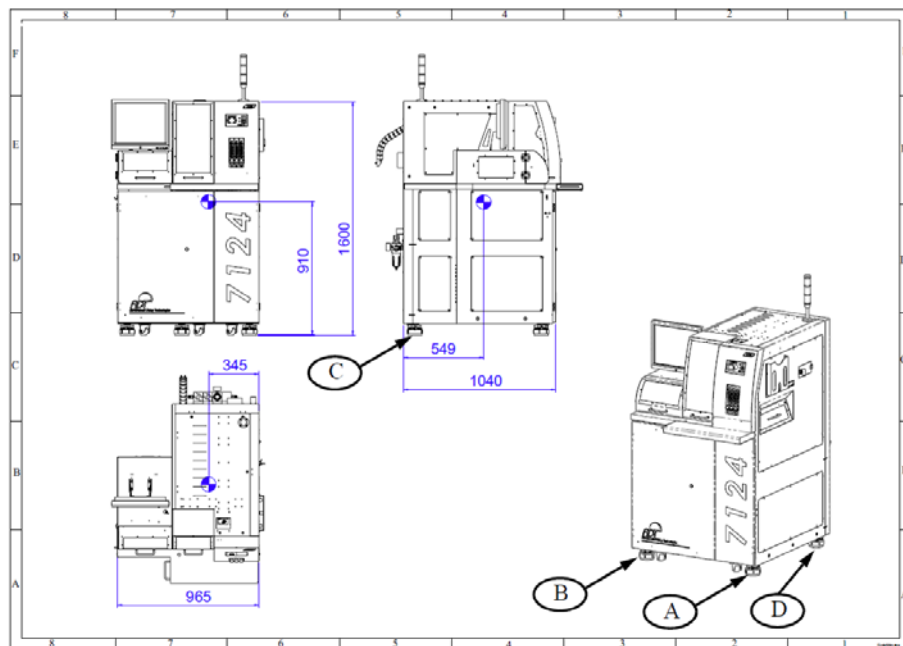


**Note:** The customer must supply all utility connectors and drain hoses; the main power cable (excluding the connector) is provided by ADT.

**Table 3-19. System Dimensions**

| Dimension                    | With Packing Crate    | Without Packing Crate |
|------------------------------|-----------------------|-----------------------|
| Width                        | 1250 mm (4'1.2")      | 1000 mm (3'3.4")      |
| Depth                        | 1580 mm (5'2.2")      | 1300 mm (4'3.2")      |
| Height (without Light Tower) | 1705 mm (5'7.1")      | 1600 mm (5'3")        |
| Overall Weight               | 1100 Kg (2425.08 lb.) | 900 Kg                |

### 3.2.5.2 Center of Gravity



**Figure 3-1. Figure 2-9: Model 71XX Weight Distribution**

The total weight of the system is distributed according to the following table.

**Table 3-20. Model 7900 Weight Distribution**

| Leg Position  | Item | Weight |
|---------------|------|--------|
| Front - Right | A    | 220 kg |
| Front - Left  | B    | 175 kg |
| Rear - Left   | C    | 245 kg |
| Rear - Right  | D    | 260 kg |
| Total Weight  |      | 900 kg |



**Note:** The leg positions are considered left or right according to the left and right sides of the user when facing the machine.

### 3.2.5.3 Installation Site

Installation site refers to the free space that is required around the system.

**Table 3-21. Installation Site**

| Location          | Space          |
|-------------------|----------------|
| Front System Path | 1.3 m (4'3.2") |
| Rear System Path  | 0.8 m (2'7.5") |
| Right System Path | 0.8 m (2'7.5") |
| Left System Path  | 0.8 m (2'7.5") |
| Height Path       | 1.6 m (5'3")   |



## 3.3 System Installation

System Installation describes the procedures for receiving, unpacking and installing the system, and the tests and adjustments that must be performed after installing the system.



**Notes:**

- ◆ The test procedures are also referred to in the Field Acceptance Test Supplement (document P/N 971XX-9007-000-10).
- ◆ If fastening the system to the floor is required, contact ADT and request the securing kit.

Read the instructions carefully before beginning the installation, and allow enough time to complete all of the tests and adjustments after installation.



**Caution:** These procedures should be performed by trained and experienced personnel.

Performing System Installation includes the following:

- Receiving the System.
- Initial Inspection.
- Unpacking the System.
- Parts and Accessories Shipped with the System.
- Positioning the System on your Production Floor.
- Securing the System to the Production Floor
- Connecting Utilities to the System.
- Connecting the Main Power Cord to the Power Supply.
- Setting Date and Time Properties.
- Installing the Chuck.
- Installing the Blade.

Follow these installation instructions in the order in which they appear.

### 3.3.1 Receiving the System

The entire system is shipped in a single packing crate. Each system is shipped separately, even when more than one unit has been ordered.

Before proceeding with the installation procedure, examine the shipping documents and packing crate label against your purchase order. In case of discrepancy, contact your local ADT Customer Support representative.

### 3.3.2 Initial Inspection

Upon receipt, inspect the exterior of the packing crate for any damage which might have occurred during transportation or delivery.

There are three indicators (of two different kinds) located on the exterior of the packing crate:

- TipWatch - indicator turns red if the crate has been tilted to more than the predetermined permissible angle.
- ShockWatch - indicator turns red if the crate has been subject to an impact or vibration beyond the predetermined permissible threshold.



**Figure 3-2. Indicators**

Inspect the color of the three indicators. If any have turned from clear to red, do not proceed with the installation. Leave the packing crate and the indicators as they are, and immediately contact the local ADT Customer Support representative.



**Note:** Failure to contact the ADT Customer Support representative immediately, in these cases, may result in cancellation of the equipment warranty.

If all indicators are clear, proceed to unpack the system.

In addition, inspect the shock and tilt indicators located on the front and rear of the casing.

### 3.3.3 Unpacking the System

The packing crate consists of a lower pallet, an attached upper pallet, and plywood panels, secured together with staples and screws.



**Caution:**

- Follow the unpacking procedure exactly as described below. Failure to do so may result in inadvertent damage to the system and the items shipped within.
- The system is bolted to the base of the crate; there are no other restraints

There are two phases to the unpacking procedure:

- Uncrating the System.
- Releasing All Moveable Parts and Unsealing the System.

### 3.3.3.1 Uncrating the System

The system arrives completely covered by two layers of plastic and enclosed in a steel strapped, wooden shipping crate.

**To uncrate the system:**



**Caution:**

1. Examine all of the Shock and Tilt indicators located on the crate panels. If any one of the indicators indicates that the machine had been tilted or jolted, do not proceed. Immediately contact your ADT representative.
2. Only when the Shock and Tilt indicators have not been disturbed, begin to uncrate.



**Danger:** Protect yourself when removing the steel banding strips that secure the crate. Steel banding strips under tension can spring open with force when cut.

1. Cut and remove the steel banding strips that secure the crate.



**Figure 3-3. Crate and Steel Bands**

2. Remove the screws from around the outer edges of the top of the crate and remove the top of the crate.



**Danger:** Take extra care during the next step, as the two remaining sidewalls of the crate will come apart at the same time when the screws are removed.

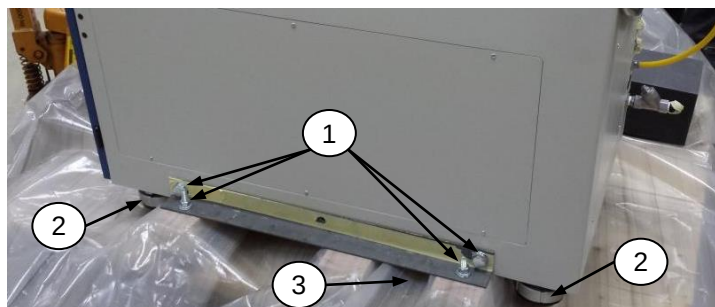
3. Remove the screws along the edges of the sidewall panels of the crate and individually remove the side panels.
4. Open up the heavy plastic sheet and carefully pull it off of the machine and cartons.



1. Model 71XX
2. Monitor, and Accessories
3. Pallet

**Figure 3-4. Packed System with Crate Walls Removed**

5. Remove the strap holding the Monitor and Other Accessories, and remove them (two boxes). Keep the boxes nearby but out of the way.
6. Remove all packing and wrapping materials.
7. Remove the Pallet Securing Bolts that secure the two Pallet Securing Jigs on the Pallet.



1. Pallet securing jigs and bolts
2. System levelers
3. Forklift entry point

**Figure 3-5. Pallet securing bolts (right side)**



**Danger:** Personnel must stay out of the way when the System is being lifted.



**Caution:** The System must only be lifted by inserting the fork into the securing jigs through the entry points (see Figure 3-5).

8. Use a forklift to lift the System off of the pallet to a height that enables access to the bolts that secure the jigs to the system.

9. Remove the four Jig Securing Bolts which secure the Pallet Securing Jigs to the System (two bolts per jig).
10. Open the four adjustable levelers (see Figure 3-5) to the maximum possible travel.
11. Lower the system to the floor so it stands firmly on the levelers, and drive the forklift backwards until it is out of the pallet securing jigs.
12. Pull the jigs from under the system.

### 3.3.3.2 Parts and Accessories Shipped with the System

Tables Table 3-22 through Table 3-26 list and describe the parts and accessories that are shipped with the system. Verify that all parts and accessories listed in the tables were received.

**Table 3-22. Standard System Accessories**

| In | Part Number       | Description and Use                                                                                                            | Quantity   |
|----|-------------------|--------------------------------------------------------------------------------------------------------------------------------|------------|
|    | 00990-0522-000-00 | Tubes Kit:<br>Air (12mm-Yellow, 3m + Fitting)<br>Spindle Water (8mm-Blue, 3m + Fitting)<br>DI Water (12mm-Black, 3m + Fitting) | Kit        |
|    | 83200-0800-038    | Drain Pipe (Clear - 40 mm, 2 m)                                                                                                | 1          |
|    | 11200-1000-033    | Band for Drain Pipe                                                                                                            | 1          |
|    | 23255-0002-06     | Plug Fitting for 6 mm tube (for spindle water circulation; on machine)                                                         | 1          |
|    | 07100-0903-000-00 | Button and NCHD Adjustment Bar Assembly                                                                                        | 1          |
|    | 07100-0959-000-01 | Main water filter                                                                                                              | 1          |
|    | 11950-1719-000    | Monitor 17" Flat (on machine)                                                                                                  | 1          |
|    | 07100-0050-027-00 | Monitor Securing Belt                                                                                                          | 1          |
|    | Various           | Keys to Main Circuit Breaker cover + Bypass Front Door + Cutting Cover                                                         | 1+1<br>2+2 |
|    | Drivers & License | Matrox Driver, Connector, Mother Board Driver                                                                                  | 1          |
|    | 07100-3802-002-33 | USB Flash Drive 4GB (for periodical backup)– on m/c                                                                            | 1          |
|    | N/A               | Image of saw                                                                                                                   | 1          |
|    | 07910-8__-__-__   | Install Shield for S/W Ver. _____                                                                                              | 1          |
|    | 07100-0928-000-01 | Air Purification (Option - on machine)                                                                                         | 1          |
|    | 07100-0800-000-02 | 6" Chuck                                                                                                                       | 1          |
|    | 07100-0801-000-02 | 8" Chuck                                                                                                                       | 1          |
|    | 07100-3041-000-00 | Light Tower                                                                                                                    | 1          |
|    | 07100-3339-000-00 | Light Tower with Buzzer (option)                                                                                               |            |

**Table 3-23. 4" System Accessories**

| In | Part Number       | Description and Use                        | Quantity |
|----|-------------------|--------------------------------------------|----------|
|    | 00775-8900-008    | 4" Spindle Nut                             | 1        |
|    | 00775-8922-000    | 4" Flange Wrench Tool                      | 1        |
|    | 00775-8921-000-01 | 4" Spindle Wrench                          | 1        |
|    | 06980-0705-028    | Tray Targets – (DIE for pixel Calibration) | 1        |
|    | 00990-0440-038-00 | 4" Cooling Nozzle (d=1.5mm)                | 1        |
|    | 00990-0440-039-00 | 4" Cooling Nozzle (d=2.5mm)                | 1        |
|    | 00990-0440-040-00 | 4" Cooling Nozzle (d=3.5mm)                | 1        |
|    | 4B785-3001-000    | 4" Torque Meter (Option)                   |          |

**Table 3-24. 2" System Accessories**

| In | Part Number       | Description and Use                                   | Quantity |
|----|-------------------|-------------------------------------------------------|----------|
|    | 13600-0059-003    | 2" Spindle Blade Nut                                  | 1        |
|    | 07700-0908-002-00 | Handle (B)                                            | 1        |
|    | 07700-0916-000-00 | Hex Key                                               | 1        |
|    | 4B785-4000-000    | 2" Flange Separator                                   | 1        |
|    | 06980-0705-028    | Tray Targets – In Package (DIE For pixel Calibration) | 1        |
|    | 0757-8711-000     | 2" Handling Tool (option)                             | 1        |
|    | 07200-0448-003-00 | 2" Cooling Nozzle (d=1.5mm)                           | 1        |
|    | 07200-0448-004-00 | 2" Cooling Nozzle (d=2.5mm) (on machine)              | 1        |
|    | 07200-0448-005-00 | 2" Cooling Nozzle (d=3.0mm) 1 07100-0905              | 1        |
|    | 00990-0441-055-00 | 2" Cooling Nozzle (d=2mm)                             | 1        |
|    | 00990-0441-056-00 | 2" Cooling Nozzle (d=1.5mm)                           | 1        |
|    | 07100-0905-000-00 | Torque and Extractor (option)                         |          |
|    | 27795-0012-000    | Torque meter (adjusted to 3.2 for Tilted, option)     |          |
|    | 27795-0015-002    | Torque Hex cup (for Tilted, option)                   |          |
|    | 27795-0015-003    | Torque Extension (for Tilted, option) 1               |          |

**Table 3-25. Documentation Items Shipped with Model 71XX**

| In | Part Number       | Description and Use                     | Quantity |
|----|-------------------|-----------------------------------------|----------|
|    | 971XX-9000-001-XX | Installation Instructions               | 1        |
|    | 971XX-9002-000-XX | Operation Manual (on USB flash drive)   | 1        |
|    | 971XX-9003-000-XX | Illustrated Parts Book (IPB)            | 1        |
|    | 971XX-9005-000-XX | Maintenance Manual (on USB flash drive) | 1        |
|    | 97100-3805-000-03 | MSC Configuration Document              | 1        |
|    | 971XX-8XXX-000-XX | Release Notes for S/W ver. X.X.X        | 1        |
|    | 971XX-9007-000-XX | Field Acceptance Test                   | 1        |

**Table 3-26. Additional Accessories not Listed Above**

| In | Part Number       | Description and Use               | Quantity |
|----|-------------------|-----------------------------------|----------|
|    | 071XX-0901-000-00 | Standard Spare Parts Kit          | 1        |
|    | 07100-0902-000-XX | Extended Spare Parts Kit (option) |          |
|    |                   | Chuck                             |          |
|    |                   | Chuck                             |          |
|    |                   | Blade                             |          |
|    |                   | Blade                             |          |
|    |                   | Flange                            |          |
|    |                   | Flange                            |          |

### 3.3.4 Positioning the Model 71XX

After uncrating the system and verifying that all parts and accessories have been received, the system can be moved to the production floor.



**Note:** The wheels are designed to allow movement over smooth surfaces. If the travel distance is long or the surface is not smooth, it is recommended using a forklift to protect sensitive components.

**To Position the Model 71XX:**

1. Move the System to the prepared location.
2. Carefully maneuver the Model 71XX into position.
3. Lower the four Leveling Mounts.
4. Place a Level on the Theta-Table in the X-Axis direction and level the System by adjusting the Leveling Mounts (clockwise to lower, counterclockwise to ×).
5. Continue leveling the System with the Level placed on the Theta Table in the Y-Axis direction.
6. Repeat steps 4 and 5 until the System is absolutely level.
7. [Optional] Anchor the levels to the floor.



8. Remove all Masking Tape according to the list that follows. Put a check mark in the “Removed” column in each table, after removing each item.



**Caution:** Failure to remove all Masking Tape before operating the System might result in severe damage to the System. ADT is not responsible for damage caused by Customers not removing all of the Masking Tape.



1. Plexiglas Covers
2. Shelf

Figure 3-6. Model 71XX before removing masking tape

Table 3-27. Masking Tape

| Number | Location                                                            | Removed |
|--------|---------------------------------------------------------------------|---------|
| 1      | All Plexiglas Covers                                                |         |
| 2      | Shelf                                                               |         |
| 3      | Keyboard (underneath Shelf)                                         |         |
| 4      | X Carriage Cover (securing two X Carriage Cover screws and one Key) |         |
| 5      | External Utility Inlets (Rear)                                      |         |
| 6      | Drain                                                               |         |



### 3.3.5 Checking the Internal Shock Indication

Inspect the two Shock Indicators, as shown in the following two figures.



Figure 3-7. Shock indicator at the rear of the system



Figure 3-8. Shock indicator near the Main System Controller (MSC)

Inspect the color of these Indicators. If any have turned from clear to the color described in Section 3.3.2, do not proceed with the installation. **Leave the packing crate and the Indicators as they are, and immediately contact the local ADT Customer Support Representative.**



**Note:** Failure to contact the ADT Customer Support Representative immediately in these cases may result in cancellation of the equipment warranty.

### 3.3.6 Removing the Desiccant Packages

The Model 71XX is shipped with Desiccant Packages (see following figures). Remove these packages before proceeding to the next unpacking step.



Figure 3-9. Desiccant package on the Theta Table



Figure 3-10. Desiccant package at the back of the system

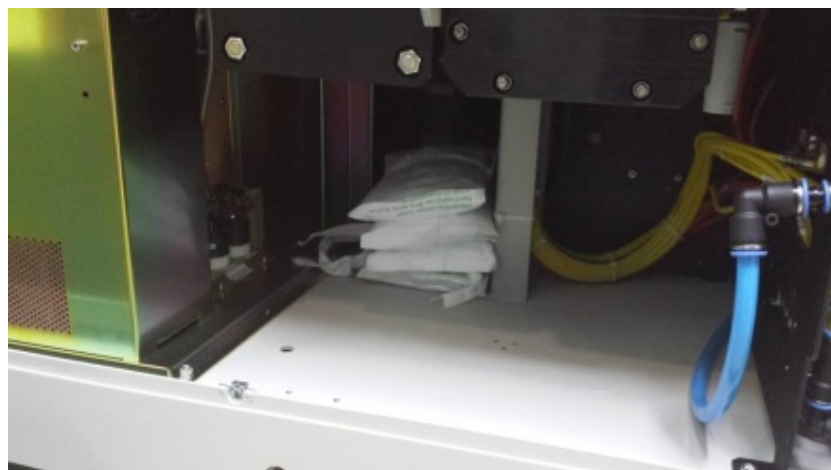


Figure 3-11. Desiccant package near the Electronic Bench

### 3.3.7 Removing the Securing Jigs

The System is shipped with bright yellow Securing Jigs that prevent the Z-axis, Y-axis and X-Carriage from moving during shipment.

The X-axis securing jig is attached to the X-Carriage left side bracket and locks the X-Leadscrew.

The Y-Z-axis securing jig is attached to the Spindle Case and locks it to the Y-Bridge.



**Caution:** When removing the X-Axis Securing Jig, the X-Axis assembly should be supplied with air. Failure to do so might result in damage to the Air Bearings.

**To remove the X-axis/Spindle securing jig:**

1. Remove the left side access door of the system to access the left side of the X-Carriage.

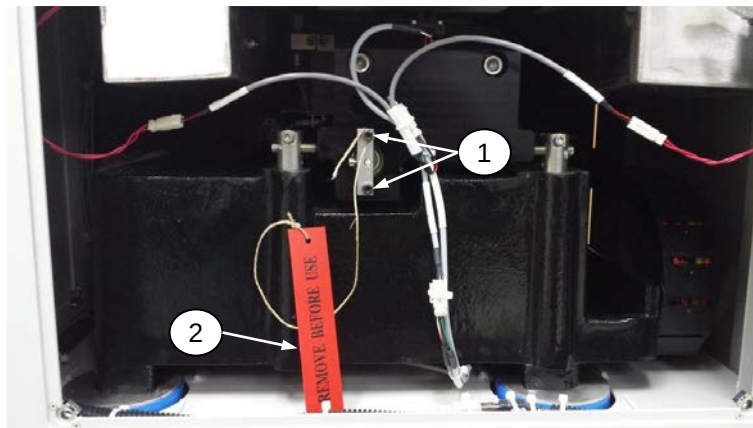


**Figure 3-12. Removing the left side access door**

2. Remove the two screws securing the X-Axis Securing Jig.



**Note:** To allow easy accesses, disconnect the air and water gun pipes.



1. X-Axis securing jig screws
2. X-Axis securing tag

**Figure 3-13. Removing the X-axis securing jig**

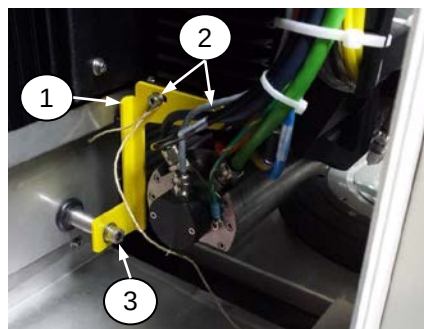
3. Remove the Securing Jig.
4. Put back the left side access door.

#### To remove the Y-Z-Axis securing jig

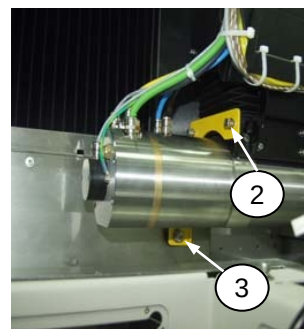
1. Remove the upper-rear panel to reveal the Y-Z-Axes manifold.
2. Remove the metal cover to gain access to the Y-Z-Axes Securing Jigs.
3. Remove the upper left side cover.
4. Loosen the Y-Axis Jig screws (item 2).
5. Loosen the Z-Axis Jig screw (item 3).
6. Remove the Y-Z-Axes Securing Jig (item 1).
7. Re-attach both back panels and upper left side panels.



**Note:** The 2" and 4" systems have different locking systems for the Y-Z axes.



2" Spindle



4" Spindle

1. Y-Z Axis Securing Jig
2. Y-Axis Jig Screws (x2)
3. Z-Axis Jig Screw

**Figure 3-14. Removing the Y-Z axes securing jig**



**Note:** The Theta table is released after connecting the system utilities, as described in section 3.3.9.

### 3.3.8 Securing the System to the Production Floor

Upon request, ADT can supply a special kit for securing the system to the floor. For more information, contact your ADT Customer Support representative.

### 3.3.9 Connecting Utilities to the System



**Caution:** The utilities must be connected by the ADT Customer Support representative only. Do not allow anyone, other than the representative, to connect the utilities to the system. Any damage caused will not be covered by the system warranty.

Contact the local ADT Customer Support Representative to complete the installation of the Model 71XX.

**To connect utilities to the system (ADT Customer Support rep):**

1. Flush the water and air supply pipes, for at-least 10 minutes, before connecting to the system.



**Caution:** Failure to do so could result in foreign matter entering crucial parts of the pneumatic system.

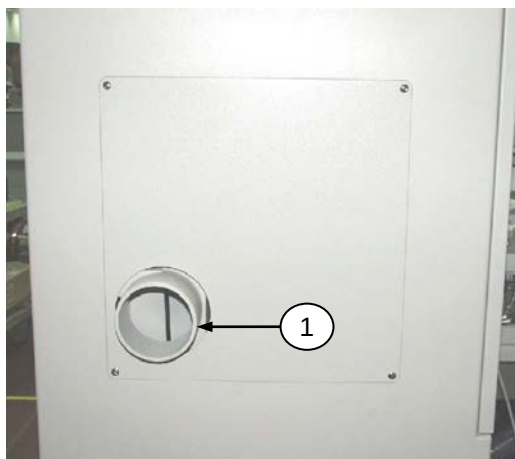
2. Connect all utilities to the system according to the requirements listed in the following table. The points of connection are shown in Figure 3-15 and Figure 3-16.



**Caution:** Make sure that the drain pipe on the wall side is installed no higher than 400 mm from the floor, and allows a free water flow.

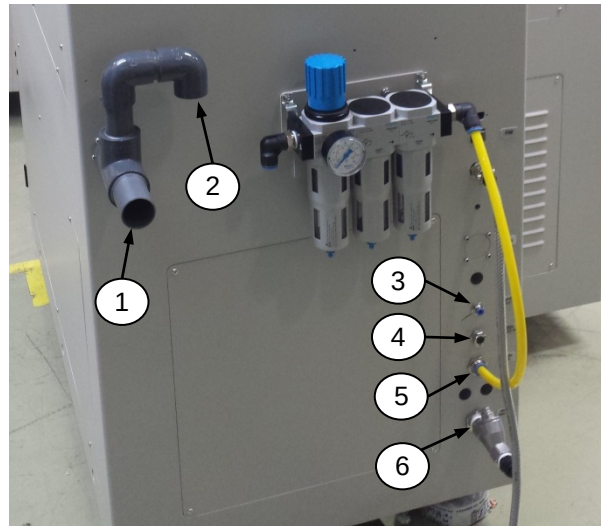
**Table 3-28. System Utility Connections**

| Utility                      | Diameter | Connecting Point |
|------------------------------|----------|------------------|
| Drain                        | 38 mm    | Rear of system.  |
| Spindle Cooling Water Supply | 8 mm     | Rear of system.  |
| Air Supply                   | 12 mm    | Rear of system.  |
| DI Water Supply              | 12 mm    | Rear of system.  |
| Exhaust                      | 100 mm   | Rear of system.  |



1. Exhaust Outlet

**Figure 3-15. Exhaust connection**



1. Drain Outlet
2. Overflow Drain Outlet
3. Connection for optional Spindle Cooling Water Recycler
4. Spindle Water Inlet
5. Air Inlet
6. DI Water Inlet

**Figure 3-16. Utility connection points**

**If a spindle cooling water chiller is to be use, do the following:**

1. Open the machine lower right side cover and locate the spindle water out hose inside the machine.
2. Transfer the spindle water out hose from the drain line to the outlet connection (balloon 3 Figure 3-16).
3. Transfer the plug connected to the spindle water out to the drain connection of the spindle water out, to prevent water from leaking into the system.



**Caution:** Before running any test procedures, check the water pipeline of the System for leakages.



### 3.3.10 Installing the Light Tower

The Light Tower is packed ready for installation, with the required screws and rubber washer.

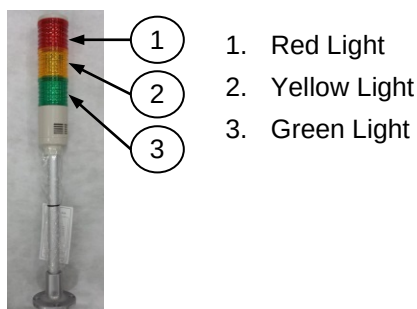


Figure 3-17. Light tower

#### To Install the Light Tower:

1. Remove the Rear Side Upper Left Panel of the System.
2. Feed the Light Tower cable through the rubber washer and hold the rubber washer on the underside of the Light Tower.
3. Feed the Light Tower cable through the opening in the Top Cover, and attach the Light Tower to the Top Cover using the provided four screws.



**Note:** Ensure that the rubber washer fits properly under the base of the Light Tower and that the screw holes line up correctly.

4. Attach the Light Tower connector to the connector inside the System.
5. Replace the Rear Side Upper Left Panel.

### 3.3.11 Connecting the Main Power Cord to the Power Supply



**Caution:** Before connecting the power cord of the machine, check and verify the voltage of the outlet has the correct voltage rating.

- For CE Marked systems: connect the main power cord to the main circuit breaker on the wall. No plug is used.
- For non-CE Marked systems: connect the main power cord plug to the main power utility outlet.

### 3.3.12 Powering On the System



**Note:** At this stage, the external water and air utilities supplied to the System are connected, as well as the air exhaust and the water drain.



**Caution:** When powering on the System, do not mark the System Init checkbox before verifying all securing jigs were removed and that the air supply is turned on.

**To power on the system:**

1. Ensure that the Emergency Stop button is in pulled out position.
2. Set the Main Circuit Breaker to the On position.
3. Press the **ON** button on the front panel.

### 3.3.13 Setting Date and Time Properties



**Note:** The system can work, even if the date and time are incorrect. However, the automatic real-time back-up is enabled only when date and time are set appropriately.

**To reset the time properties:**

1. Using the keyboard, press **Control** and **Escape**.
2. Press the hour displayed at the right bottom area of the screen.
3. In the Date and Time Properties, choose **Time Zone**.
4. From the drop-down box, select the relevant time zone and press **Apply**.
5. Select the **Date & Time** view and set the correct date and time.
6. Press **Apply** and then **OK**.

### 3.3.14 Installing the Chuck

Installing the Chuck includes the following sections. Read the information described in General Precautions before installing the chuck.

- General Precautions.
- Securing the Cutting Chuck to the Theta Table.
- Testing Chuck Flatness.



### 3.3.14.1 General Precautions

In addition to the safety precautions described at the beginning of this manual, follow the General Precautions to ensure accuracy of measurement.

- Before taking any measurements on the chuck, ensure that the range of motion required for testing will not cause the spindles or any axis, to collide with the measurement tool.
- Keep the speed of movement for each axis to a minimum during measurements.
- Verify that the measurement tool has a valid calibration tag.
- Before using precision measuring tools and instruments remove oil and fingerprint residues with isopropyl alcohol. Gently wipe the equipment dry with white, clean, paper towels.

It is recommended handling such equipment with gloves.

- Return precision measurement equipment to their protective containers immediately after use.

### 3.3.14.2 Securing the Cutting Chuck to the Theta Table

Carefully unpack the cutting chuck.

**To secure the cutting chuck to the table:**

1. Make sure that the theta table is clean. Clean it with alcohol and a clean cloth or with cleaning paper if necessary.
2. Place the cutting chuck on the theta table, ensuring that the tongue on the underside of the cutting chuck fits into the groove on the top of the theta table.
3. Activate the vacuum that secures the cutting chuck to the theta table:
  - a. In the application, from the Main Menu, press **Maintenance > Functions and I/O > Cutting Table**; the Functions and I/O window opens in the Cutting Table view.
  - b. Press **Unlock** to enable modification.
  - c. Select the Chuck Holder **Vacuum** check box and then select the Chuck Holder **Accept** button; the vacuum that secures the cutting chuck to the theta table is activated.

### 3.3.14.3 Testing Chuck Flatness

After placing the cutting chuck on the theta table, perform the chuck flatness test to verify that the chuck meets the following flatness tolerance specifications:

**Table 3-29. Chuck Flatness Tolerance**

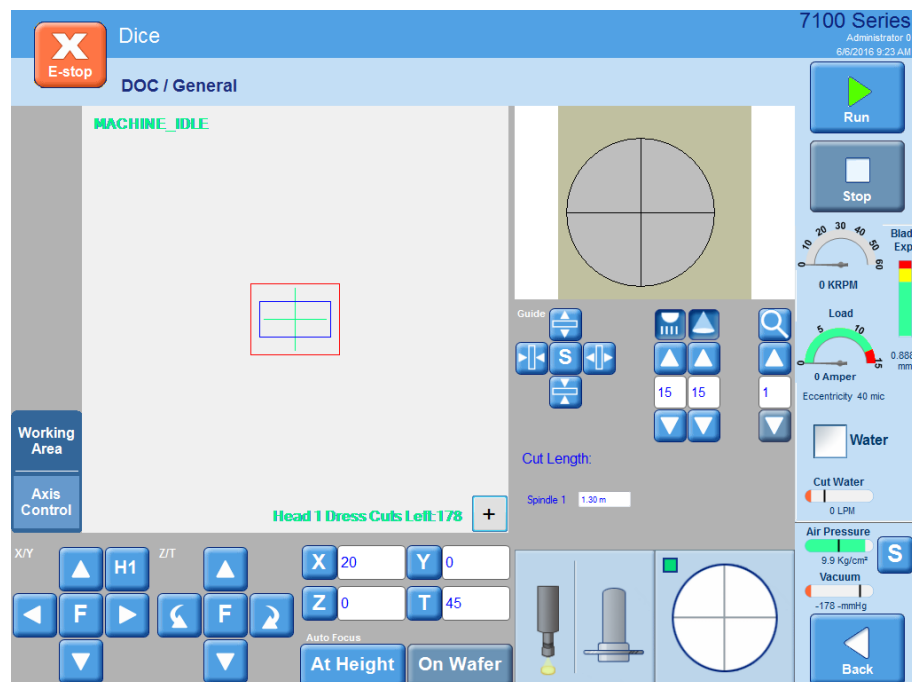
| Chuck Size | Flatness Tolerance |
|------------|--------------------|
| 6 Inch     | 7 $\mu\text{m}$    |
| 8 Inch     | 10 $\mu\text{m}$   |
| 12 inch    | 15 $\mu\text{m}$   |

#### To test chuck flatness:

1. Mount the measuring probe onto a spindle so that the measuring probe touches the cutting chuck surface.

Recommendation: Use a special jig holder to mount the micrometer jig holder on the upper side of the spindle holder, and position the measuring probe near the chuck table surface.

2. In the application, from the Main Menu, press **To Dice**; the Dice window appears in the Axis Control view.



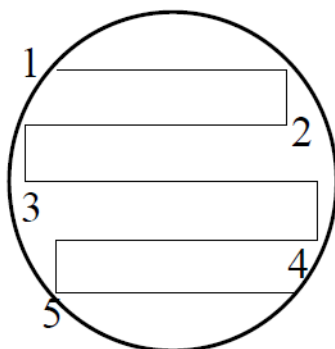
**Figure 3-18. Dicer Axis Control Window**

3. Verify that the theta table is in the **Angle 0** position.
4. Use the Z Up/Down keys to lower the Z-axis until the probe touches the chuck table surface.
5. Set the measuring probe to zero (probe reference position).



**Note:** From this point, do not touch the measuring probe or the measuring probe Zeroing button, and do not change the position of the Z Axis.

6. Use the X/Y Axis controls to move the measuring probe to position 1, as shown in following Figure 3-19.



**Figure 3-19. Chuck Flatness Position Diagram**

7. Use the X/Y Axis controls to move the spindles with the attached measuring probe along the path outlined Figure 3-19 to positions 2, 3, 4 and 5, while observing the changes on the measuring probe scale.



**Note:** Do not let the measuring probe fall off the end of the cutting chuck. Use the outer ring of the cutting chuck as the outer limit to which the measuring probe can reach.

8. Verify that the measured deviation is within the corresponding specifications.



**Note:** If the deviation is greater than the specification, contact the local ADT representative.

9. Use the X/Y Axis controls to move the measuring probe to the center of the cutting chuck and rotate the theta table to the **Angle 90** position. Ensure that the measuring probe is still touching the surface of the cutting chuck.
10. Repeat steps 7 and 8.
11. Remove the Long Probe Extension Holder from the spindle.

### 3.3.15 Additional Tests

Perform the tests described in the following sections:

- "Correcting Spindle Parallelism to the X Axis" (section 7.4.3.1).
- "Testing the Spindle Runout" (section 7.4.3.2).
- "Glass Perpendicularity Block Release Procedure" (section 4.4.3).

### 3.3.16 Installing the Blade

After performing the Chuck Flatness test, the blade should be mounted onto the spindle, as described in the following procedure.

#### To install the blade:

1. In the application, from the Main Menu, press **To Dice**; the Dice window appears.
2. Press the **Spindle** button and select **Change Blade**; the spindle stops rotating and moves automatically to the Blade Change position, enabling easy access to the blade. In the application, the Change Blade wizard appears in the Step 1 of 2: Define Blade window.

| Change Blade Step 1 of 2: Define Blade                                                   |                |               |                |
|------------------------------------------------------------------------------------------|----------------|---------------|----------------|
| Lot number                                                                               | a23            |               |                |
| Blade                                                                                    | Nickel Hub     |               |                |
| Flange Diameter                                                                          | 53.7972        | mm            |                |
| Flange                                                                                   | f1 - 54.174 mm |               |                |
| <input type="checkbox"/> Leave current <input checked="" type="checkbox"/> Change flange |                |               |                |
| Type:                                                                                    | Regular        | Maximum RPM   | 60000          |
| Name:                                                                                    | Nickel Hub     | Hub Type:     | Hub Blade      |
| Diameter (mm):                                                                           | 55.5752        | Matrix:       | Matrix Nickel  |
| Exposure (mm):                                                                           | 0.889          | Density:      | Density S      |
| Expansion Tolerance                                                                      | 0.17           | Size Diamond: | Size Diamond S |
| Thickness (mm):                                                                          | 0.035          | Bond Q:       | Bond Q         |
| Dress Wafer Recipe:                                                                      | Dressing Wafer | Company:      | Semitec        |
|                                                                                          |                | Next          | Cancel         |

Figure 3-20. Change Blade



#### Notes:

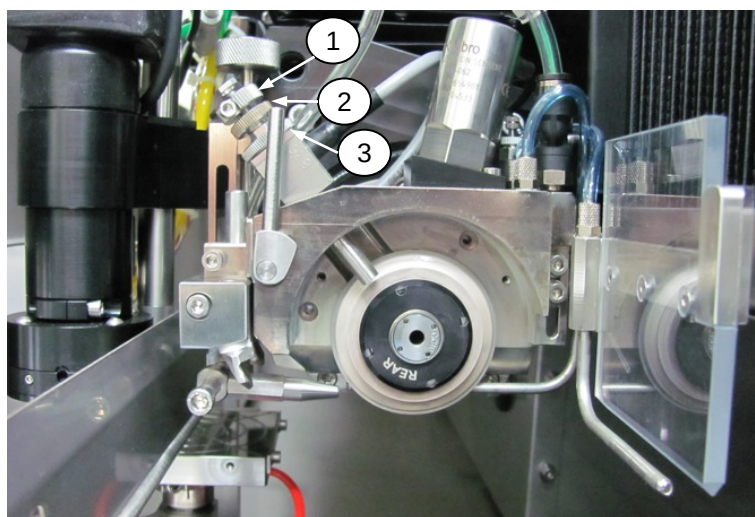
- ◆ Change Blade is a two-step procedure. Press **Next** to move to the next step and then press **Finish** to save the changes.
  - ◆ When the blade type is defined as Hubless, the Flange diameter field is activated.
  - ◆ The new exposure value appears after the process height is completed.
3. Depending on the blade type:
    - **For Hub Blade:** insert the new Blade into the Blade Holder (if supplied) by pulling up the Release Ring, placing the Blade Holder over the new Blade (the Blade must be face up) and releasing the Release Ring.
    - **For Hubless Blade:** mount the Blade on the flange, using the supplied toolkit.

4. Depending on the blade type:
  - **For Hub Blade:** mount the new Blade on the Spindle by placing the Blade Holder (if supplied) over the end of the Spindle and pulling back the Release Ring.
  - **For Hubless Blade:** manually mount the assembly on the Spindle; use Blade Holder, if supplied.
5. Install the holding nut and tighten it using the torque wrench (until a click is heard).



**Note:** At this stage, all the system utilities must be connected. Compressed air and spindle cooling water must be supplied.

6. In the Define Blade window, press **Next**; the Step 2 of 2: Tune BBD window appears.
7. If the system is equipped with a BBD, continue with following step 8, if not— go to step 9.
8. Tune the BBD:
  - a. Slightly release the BBD securing screw.



1. BBD Travel Securing Nut
2. Middle Securing Nut
3. Adjustment Nut

**Figure 3-21. Cooling Block with BBD**

- b. Slowly shift the BBD until the BBD%, of the target BBD reading, registers approximately 15%.



**Note:** The BBD threshold bar is pre-set with a 50% default value. Setting the threshold to a lower value causes the BBD to be more sensitive; setting it to a higher value causes it to be less sensitive.

- c. Tighten the BBD securing screw.
9. Close the cooling block and tighten the screw.
10. Press **Finish**; the Dice window regains focus and the system performs initialization. The spindle begins to revolve and the system performs a Height procedure.

### 3.3.16.1 Performing Chuck to Non-Chuck Delta Measurement

After mounting the blade, use the software to perform the Chuck to Non-Chuck Delta Measurement procedure. This teaches the system the difference in height between the height device and the cutting chuck.



**Note:** Use a dressed Blade for the Chuck to Non-Chuck delta Measurement procedure.

**To perform chuck to non-chuck delta measurement:**

1. In the application, from the Main Menu, press **Maintenance > Functions and I/O > Height**; the Functions and I/O window appears in the Height view.
2. Press the **Measure** button; the system performs Chuck Height, followed by NCH measurement, and then calibrates the two values. The values appear in the Process Height window.

### 3.3.17 Performing a Cut Test

Perform cut testing after all installation procedures have been completed.



**Note:** The blade should be dressed before performing a cut test (see the 71XX Operation Manual for details).

**To perform a cut test:**

1. Create a test recipe (see the Recipe Builder chapter in the Operations Manual).
2. Cut a “test” workpiece.
3. Examine the workpiece after cutting to verify that the required specifications have been achieved.

## Chapter 4. Periodic Maintenance

Regular and timely maintenance is essential for keeping the system in good operating condition. To achieve reliable operation, perform the maintenance procedures listed in this chapter, at the specified intervals.



**Note:** The Periodic Maintenance schedule and emphasis depends on system **usage** and type of **application**. In some cases, some of the examinations may need to be performed at a higher frequency.

The Periodic Maintenance chapter includes the maintenance procedures that must be performed.

The Maintaining the System chapter includes:

- ❖ Maintaining the System on a Daily Basis (section 4.1)
- ❖ Maintaining the System on a Monthly Basis (section 4.2)
- ❖ Maintaining the System on a Quarterly Basis (section 4.3)
- ❖ Maintaining the System on a Yearly Basis (section 4.4)

## 4.1 Maintaining the System on a Daily Basis

The Daily Maintenance Procedures section lists and describes the procedures to be performed daily, and the specifications that must be met.



**Caution:**

- All system operations must be stopped and the workpiece and blades removed, before performing these procedures.
- When a cleaning procedure requires a solvent, use the solvent documented.  
DO NOT USE ACETONE

### 4.1.1 Daily Maintenance Procedures

Maintenance Technician Name: \_\_\_\_\_

Date: \_\_\_\_\_

| Table 4-1. Daily Maintenance Procedures |                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                          |        |
|-----------------------------------------|-----------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
|                                         | Procedure                                                                               | Description & Specifications                                                                                                                                                                                                                                                                                                                                                                                                                             | Passed |
| 1                                       | Examine the level of air pressure for the spindle and X Axis, and adjust when necessary | Read the air pressure gauge on the Air Regulator panel located at the bottom rear side of the system (see Figure 11-18).<br>If the air pressure is below 72 psi (4–5 atm), adjust the regulator located above the gauge:<br>1) Pull the dial up.<br>2) Rotate the dial to correct the level.<br>3) Push the dial down back into place.                                                                                                                   |        |
| 2                                       | Check the spindle cooling-water supply                                                  | Stop the flow of cooling water to the Spindle. Verify that the appropriate Error Message is received.                                                                                                                                                                                                                                                                                                                                                    |        |
| 3                                       | Check and adjust the Water flow from the Cooling Block                                  | Open the cooling water passing through the Cooling Block jets and nozzles. <ul style="list-style-type: none"><li>• Verify that the flow of water to the Blade is constant and positioned correctly.</li><li>• Verify that the reading in the Sensors screen meets process specifications.</li></ul> <b>To Dice &gt; Sensors button (  ) &gt; Accessories pane).</b> |        |
| 4                                       | Verify Presence of water leaks outside the System                                       | Inspect the floor in the area surrounding the System for traces of water. Inspect all external Tube and hose connections.                                                                                                                                                                                                                                                                                                                                |        |
| 5                                       | Inspect for water leaks inside the system                                               | 1) Open the front, rear, and side panels and inspect the electronic bench and the X Casting gutter for signs of leakage.<br>2) Inspect the X Casting gutter for signs of blockage or for a kink in the drainage tube.<br>3) Inspect the area of the pneumatic panel for signs of leakage.                                                                                                                                                                |        |



**Table 4-1. Daily Maintenance Procedures**

|    | Procedure                                  | Description & Specifications                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Passed |
|----|--------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| 6  | Clean the Dicer                            | <ol style="list-style-type: none"> <li>1) Remove all waste material that has collected on the X Bellows.</li> <li>2) Clean the collecting waste filter pan.</li> <li>3) Use a sponge and clean water (DI water recommended) to remove sediment from the: <ul style="list-style-type: none"> <li>▪ X Protector</li> <li>▪ Cooling Block (metal parts)</li> <li>▪ Internal Covers</li> <li>▪ Wheel Mount</li> </ul> </li> </ol>                                                                                                                                                                                                                                                                              |        |
| 7  | Clean the Microscope Lens                  | Clean the Microscope lens with a cotton swab and isopropyl alcohol                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |        |
| 8  | Clean the Height device                    | <ul style="list-style-type: none"> <li>• For Systems with an optical, Non-Contact Height Device:<br/>Clean the internal vertical prism surfaces with a cotton swab and isopropyl alcohol, and water (clean only the internal vertical surfaces that face each other. Do not clean the two external angled surfaces).<br/>After cleaning, check the condition and positions of the water and air nozzles. Then, close the water and check if water still flows from the water nozzles; if yes, check the water solenoid for a clog or dirt causing the leak.</li> <li>• For Systems with a Mechanical Button:<br/>Clean the Button top surface with a cotton swab, isopropyl alcohol, and water.</li> </ul> |        |
| 9  | Clean and adjust the Broken Blade Detector | Clean the LED and Sensor with a cotton swab and isopropyl alcohol or water. Tune the BBD, as described in section 9.1.1.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |        |
| 10 | Clean the Transparent Covers               | Clean the inside of the transparent covers with a clean cloth and water.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |        |

## 4.2 Maintaining the System on a Monthly Basis

The Monthly Maintenance Procedures lists and describes the procedures to be performed monthly, and the specifications that must be met.



**Caution:**

- All system operations must be stopped and the workpiece and blades removed, before performing these procedures.
- When a cleaning procedure requires a solvent, use the solvent documented.  
DO NOT USE ACETONE

### 4.2.1 Monthly Maintenance Procedures

Maintenance Technician Name: \_\_\_\_\_

Date: \_\_\_\_\_

| Table 4-2. Monthly Maintenance Procedures |                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |        |
|-------------------------------------------|------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
|                                           | Procedure                                | Description & Specifications                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Passed |
| 1                                         | Perform the Daily Maintenance Procedures | See section 4.1.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |        |
| 2                                         | Clean the Dicer                          | Remove all waste materials collected on the X-axis bellows and check the condition of the X bellows for damage and leaks.                                                                                                                                                                                                                                                                                                                                                                                |        |
| 3                                         | Clean the System Internal Parts          | Open the front door and remove the covers to inspect and clean.<br>Clean the PC unit, X-casting, Electric Bench, Pneumatic panel, Y Axis and the Spindle unit.                                                                                                                                                                                                                                                                                                                                           |        |
| 4                                         | Clean the X Axis                         | 1) Power down the system (without turning off air).<br>2) Detach the X Bellows from the X Carriage.<br>3) Clean the surface of the X Axis with a lint-free cloth (or paper towels) and isopropyl alcohol.                                                                                                                                                                                                                                                                                                |        |
| 5                                         | Chuck and Turntable                      | Release the cutting chuck vacuum and remove the chuck: <ul style="list-style-type: none"><li>• Check for presence of water.</li><li>• Clean the Theta Table surface (gently go over it with an Arkansas stone).</li><li>• Inspect the chuck O-rings for possible cracks or damage, and grease with vacuum grease before re-installing. Replace if necessary.</li><li>• Install the cutting chuck and turn on the vacuum. Verify the vacuum level reaches its maximum reading (above 600 mmHg).</li></ul> |        |
| 6                                         | Spindle Filter                           | Remove the spindle brass filter; then, clean, dry and re-install.                                                                                                                                                                                                                                                                                                                                                                                                                                        |        |

**Table 4-2. Monthly Maintenance Procedures**

|    | Procedure                                             | Description & Specifications                                                                                                                                                                                                                                                                                                                                                                                                             | Passed |
|----|-------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| 7  | Clean the Cooling Block Assembly                      | Do not turn off the air supply to the Spindle.<br>1) Remove the Blade and Flange Assembly and remove the Cooling Block.<br>2) Inspect all tubes.<br>3) Clean the Spindle shaft front side and the wheel-mount backside from saw dust.<br>4) Re-Install the Cooling Block and ensure correct position and alignment of the jets nozzles.<br>5) Mount the flange and blade assembly (verify the flange is not touching the cooling block). |        |
| 8  | Check and adjust the wheel-mount front surface runout | The wheel-mount front surface runout should not be greater than 2.5 microns (see section 7.3.1)                                                                                                                                                                                                                                                                                                                                          |        |
| 9  | Gauge Values                                          | Verify all values are according to the system specifications. Refer to the tables in Chapter 3.                                                                                                                                                                                                                                                                                                                                          |        |
| 10 | Inspect and replace the Spindle Brushes               | 1) Open the spindle carbon brush housing, as described in section 7.2.3.2, and remove the carbon brushes.<br>2) Blow air inside and clean the internal part with a cotton swab and alcohol; then dry it.<br>It is recommended to replace the carbon brushes.<br>3) Check the amount of graphite left in the spindle brush. Replace if the brush length is 3–4 mm or less.                                                                |        |
| 11 | Verify/adjust the level of system air pressure.       | Read the air pressure gauge on the Air Regulator Unit (see Figure 11-18).<br>If the air pressure is below 72 psi (4.8 atm), adjust the regulator located behind the front lower panel:<br>a) Pull the dial up.<br>b) Rotate the dial.<br>c) Push the dial back into place.                                                                                                                                                               |        |
| 12 | Perform Periodical Backup                             | Enter the backup Flash drive and verify that the file has been saved.                                                                                                                                                                                                                                                                                                                                                                    |        |

## 4.3 Maintaining the System on a Quarterly Basis

The Quarterly Maintenance Procedures lists and describes the procedures to be performed quarterly, and the specifications that must be met.



**Caution:**

- All system operations must be stopped and the workpiece and blades removed, before performing these procedures.
- When a cleaning procedure requires a solvent, use the solvent documented.  
DO NOT USE ACETONE

### 4.3.1 Quarterly Maintenance Procedures

Maintenance Technician Name: \_\_\_\_\_

Date: \_\_\_\_\_

| Table 4-3. Quarterly Maintenance Procedures |                                                                       |                                                                                                                                                                                                                                                                                                        |        |
|---------------------------------------------|-----------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
|                                             | Procedure                                                             | Description & Specifications                                                                                                                                                                                                                                                                           | Passed |
| 1                                           | Perform Daily Maintenance Procedures                                  | See section 4.1                                                                                                                                                                                                                                                                                        |        |
| 2                                           | Perform Monthly Maintenance Procedures                                | See section 4.2                                                                                                                                                                                                                                                                                        |        |
| 3                                           | Test the main circuit breaker switch and the Emergency Stop button    | 1) While the system is powered on:<br>a) Turn off the main circuit breaker (located at the rear of the system).<br>b) Verify that system operation stops immediately.<br>2) While the system is operating:<br>a) Push the Emergency Stop button.<br>b) Verify that system operation stops immediately. |        |
| 4                                           | Remove old grease from the X, Y and Z Axes, and then grease the axes. | See instructions in section 4.4.2.                                                                                                                                                                                                                                                                     |        |
| 5                                           | Verify the accuracy of Y Axis positioning                             | Place a workpiece with a known index on the cutting chuck. Perform auto-alignment, and then measure the distance between the first and last streets to calculate the Index.<br>See more details in section 4.4.3.                                                                                      |        |
| 6                                           | Clean the Y Axis linear encoder                                       | If a problem occurs with the Y Axis, open the front cover, remove the Y-bellows, and clean the Linear Encoder with a cotton swab and methanol (not isopropyl alcohol).                                                                                                                                 |        |
| 7                                           | Verify Z Axis repeatability                                           | See section 8.7.1.<br>Specification: 1.0 micron.                                                                                                                                                                                                                                                       |        |

**Table 4-3. Quarterly Maintenance Procedures**

|    | Procedure                                              | Description & Specifications                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Passed |
|----|--------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| 8  | NCH Calibration                                        | See section 9.2.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |        |
| 9  | Check the condition of the cutting chuck               | <p>Inspect the condition of the cutting chuck:</p> <ul style="list-style-type: none"> <li>• When dirt is apparent on the chuck or the underside of the Tape.</li> <li>• When error messages associated with the cutting chuck appear, and other potential causes have been ruled out.</li> </ul> <p>Use ultrasonic cleaning equipment and purified water for cleaning. Use an Arkansas Stone to finely abrade the surface and remove small protrusions. Clean the underside of the chuck by blowing an alcohol-air mixture, then blow with dry air until the chuck is completely dry.</p> |        |
| 10 | Measure the cutting chuck flatness                     | <p>The specifications are:</p> <p>6 inch (150 mm) Chuck - 7 microns<br/>8 inch (200 mm) Chuck - 10 microns<br/>12 inch (300 mm) Chuck - 15 microns</p>                                                                                                                                                                                                                                                                                                                                                                                                                                    |        |
| 11 | Inspect for correct operation of both overflow sensors | <ol style="list-style-type: none"> <li>1) Turn on the cutting water.</li> <li>2) Carefully move the sensor float upwards. Expected results: <ol style="list-style-type: none"> <li>a) An error message should appear.</li> <li>b) All water in the system should stop.</li> <li>c) The Overflow signal box should turn green (to view the Overflow signal box, go to <b>To Dice &gt; Sensors button  &gt; Accessories pane</b>).</li> </ol> </li> </ol>                                              |        |
| 12 | Water Separator                                        | Check operation of Water Separator; refer section 11.6.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |        |
| 13 | Main Water Filter                                      | Clean and replace if required.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |        |
| 14 | Check Stations                                         | Check, and adjust if required, all dicer Stations.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |        |

## 4.4 Maintaining the System on a Yearly Basis

The Yearly Maintenance Procedures section lists and describes the procedures to be performed yearly, and the specifications that must be met.



**Caution:**

- All system operations must be stopped and the workpiece and blades removed, before performing these procedures.
- When a cleaning procedure requires a solvent, use the solvent documented.  
DO NOT USE ACETONE

### 4.4.1 Yearly Maintenance Procedures

Maintenance Technician Name: \_\_\_\_\_

Date: \_\_\_\_\_

| Table 4-4. Yearly Maintenance Procedures (1 of 2) |                                          |                                                                                                                                                                                                                                                                                                                 |        |
|---------------------------------------------------|------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
|                                                   | Procedure                                | Definition & Specifications                                                                                                                                                                                                                                                                                     | Passed |
| 1                                                 | Weekly Maintenance Procedures            | See section 4.1.                                                                                                                                                                                                                                                                                                |        |
| 2                                                 | Monthly Maintenance Procedures           | See section 4.2.                                                                                                                                                                                                                                                                                                |        |
| 3                                                 | Quarterly Maintenance Procedures         | See section 4.3.                                                                                                                                                                                                                                                                                                |        |
| 4                                                 | Verify the spindle to X Axis parallelism | See section 7.4.3.1.<br>Specification: 5 microns.                                                                                                                                                                                                                                                               |        |
| 5                                                 | Verify Spindle Runout (radial & axial)   | See section 7.3.1<br>Specification: 1 micron.                                                                                                                                                                                                                                                                   |        |
| 6                                                 |                                          |                                                                                                                                                                                                                                                                                                                 |        |
| 7                                                 |                                          |                                                                                                                                                                                                                                                                                                                 |        |
| 8                                                 | Axes Condition                           | Check and clean from dirt and possible corrosion (use contact cleaner if necessary): <ul style="list-style-type: none"><li>• The X, Y, Z, Theta and main Controller cable connections.</li><li>• The X Axis Driver.</li><li>• The Spindle Driver and Regenerative Unit.</li><li>• The Main Controller</li></ul> |        |
| 9                                                 | Verify Cutting Chuck Flatness            | See details in section 3.3.14.3.                                                                                                                                                                                                                                                                                |        |

| Table 4-4. Yearly Maintenance Procedures (1 of 2) |                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |        |
|---------------------------------------------------|-------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
|                                                   | Procedure                                                                     | Definition & Specifications                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Passed |
| 10                                                | Inspect the condition of the cutting chuck and clean the chuck when necessary | <ol style="list-style-type: none"> <li>1) Inspect the condition of the cutting chuck and clean the chuck when dirt appears on the chuck or on the underside of the tape.</li> <li>2) Pour alcohol on top of the chuck (porous side) and blow air underneath the hole to remove any clog in the porous.</li> <li>3) Finely abrade the chuck surface and remove small protrusions with an Arkansas Stone.</li> <li>4) Clean the underside of the chuck by blowing an alcohol-air mixture, then blowing with dry air until the chuck is completely dry.</li> </ol> |        |
| 11                                                | Replace Filters                                                               | <p>Replaced the air filter to the Spindle.</p> <p>Check, and replace if required, the vacuum exhaust filters.</p> <p>If equipped with the Air Purification Kit option, replace all filters.</p>                                                                                                                                                                                                                                                                                                                                                                 |        |

## 4.4.2 Greasing Procedures

The Greasing Procedures describes how to grease the system axes and leadscrews to ensure reliable system operation and prevent wear. It is recommended performing the procedures once a year, after (and not in place of) completing the Weekly, Monthly, and Yearly Maintenance Procedures.



### Caution:

- All system operations must be stopped and the workpiece and blades removed, before performing these procedures.
- When a cleaning procedure requires a solvent, use the solvent documented. DO NOT USE ACETONE.
- Failure to follow the greasing procedures will result in unnecessary wear to the system axes and leadscrews, which may cause inaccuracies in system operation.



**Note:** Greasing requires basic knowledge of system operation. Refer to the ADT Model 71XX Operations Manual for the relevant operation procedures, as well as the safety precautions in the Getting Started chapter, before starting the greasing procedures.

The following components should be greased once a year, or more often, depending on usage and conditions:

- X Axis.
- Y Axis.
- Z Axis.

### 4.4.2.1 Greasing Equipment

#### 4.4.2.1.1 Grease Gun

ADT highly recommends the Model KH-120 Grease Gun (P/N 27795-0501-000). The grease gun is manufactured to meet the JISB9808-Grease Gun-L100 standard.



**Note:** Any Grease Gun that meets the JISB9808 standard may be used. The system grease nipples meet the JISB1575 standard.

The grease gun can have a press or a pull-type lever for applying grease.

#### 4.4.2.1.2 Grease Type

Use ISOFLEX NBU15 grease (P/N 39087-0016-000). Contact ADT Customer Support for purchasing information.



**Note:** If the Model KH-120 Grease Gun is ordered from ADT, it is supplied with ISOFLEX NBU15 grease.



**Caution:** The leadscrews and linear guides are greased before the system is shipped. ADT uses long-life grease that exhibits minimal liquid-oil separation. If a different type of grease is used, there may be changes in the greasing interval requirements, as well as chemical reactions between the new grease and the remaining ISOFLEX NBU15 grease. It is therefore essential to use the correct grease.



#### 4.4.2.1.3 Cleaning Rags

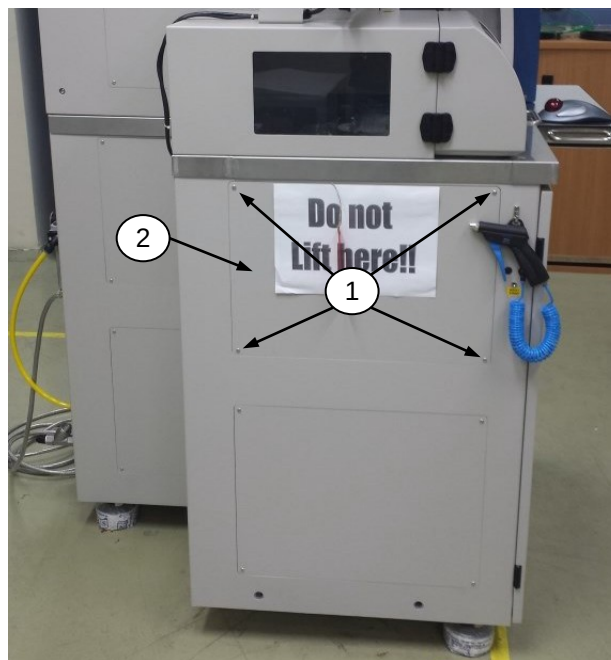
Make sure that there is a sufficient supply of clean rags or paper toweling available during greasing. A clean rag or paper towel is used for wiping off old grease or excess grease that is ejected during greasing. If greasing is done in clean-room conditions, use a clean, dust-free rag or paper towel.

#### 4.4.2.2 Greasing the X-axis

Grease the X Axis leadscrew on a yearly basis. Remove the old grease from the X Axis before applying new grease.

**To remove old grease from the X axis:**

1. In the application, from the Main Menu, press **Maintenance > Axis Setup > X**; the Maintenance-Axis Setup window appears.
2. From the Control Panel-X, set the X Axis to **0**; the X Axis moves left, so that the grease nipple is easily accessible to the grease gun.
3. Open the left top panel to access the X Axis leadscrew, by removing its four securing screws.



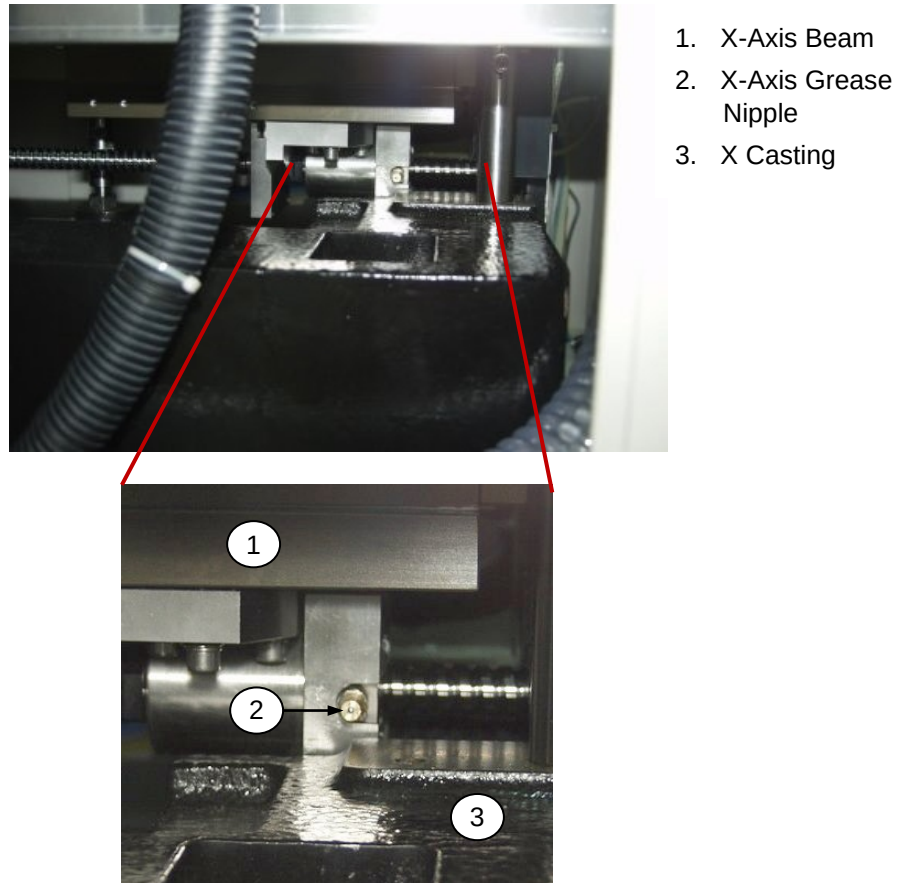
1. Securing  
Screws
2. Left Side  
Top Cover

**Figure 4-1. Left Side Panel**

4. Open the backside lower left panel to access the grease nipples of the bearing house.



**Note:** You may want to disconnect the pipes that supply water and air to the water and air gun. The water and air supply to the System must be closed.



**Figure 4-2. X Axis Grease Nipple**

5. Remove the old grease from as much of the leadscrew that can be accessed, using a clean rag or paper towels.

**To grease the X axis:**



**Note:** Remove the old grease before applying the new grease. See “Removing the Old Grease from the X Axis” in previous section.

1. Move the X Axis to the **0** position.
2. Apply grease to the X Axis grease nipple using the grease gun and apply grease to the leadscrews and linear guides using your finger.



**Note:** Press down on the grease gun lever and pump the grease through the grease nipple until you see that the old grease is coming out of the bearing house.

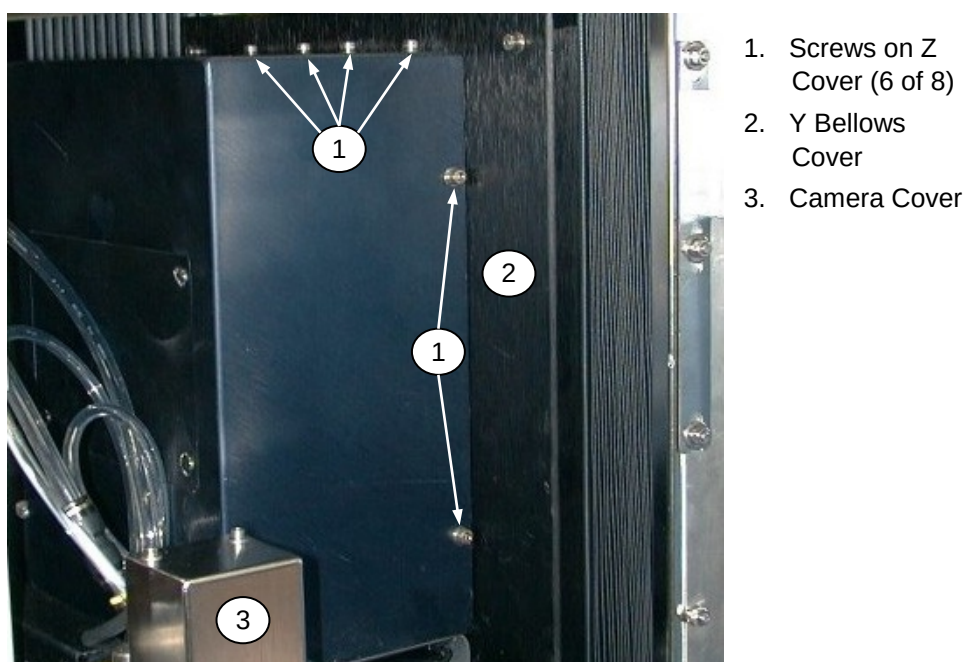
3. Use a clean rag or paper towel to wipe off any excess grease.
4. Initialize and operate the newly greased X Axis:
  - a. From the Main Menu, press **Maintenance > Axis Setup > X**.
  - b. Press **Init** to initialize the axis.
  - c. Press **Start Self-Test**; the X Axis begins moving automatically between its two limits. Allow Self-Test for one minute.
  - d. Press the **Stop Self-Test**.
5. Check under the X table and surroundings for any excess grease that may have flown-off or scattered, especially on the X beam, and clean it.

#### 4.4.2.2.1 Greasing the Y and Z Axes

Grease the Y and Z Axes simultaneously. Remove the old grease from the Y and Z Axes before applying new grease.

##### To remove old grease from the Y and Z axes

1. In the application, from the Main Menu, press **Maintenance > Axis Setup**; the Maintenance-Axis Setup window appears.
2. Press **Z** and in Control Panel-Z, set Z to **0** (the Z-axis should be at its lowest position).
3. Open the Front Clear Cover.
4. Remove the eight screws (2 on front, 4 on top, 2 on back) that secure the Z Casting Cover and remove the Cover.

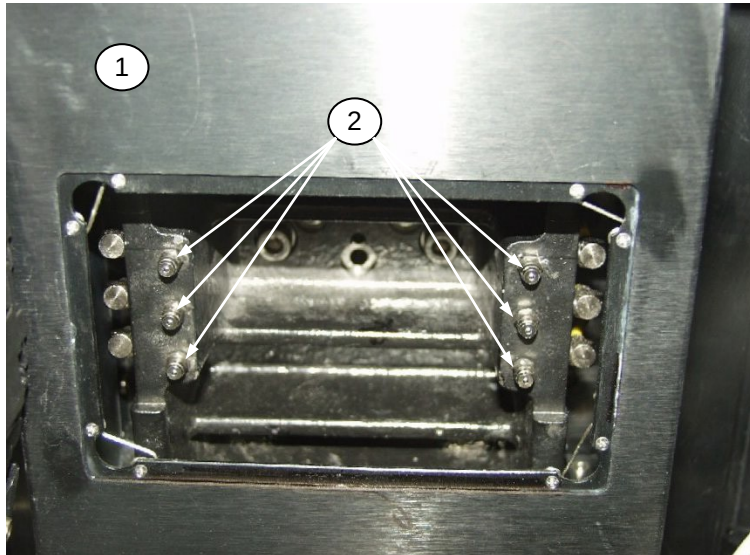


**Figure 4-3. Z Casting Cover**

5. Remove the four Y Bellows Cover screws and move the Bellows to one side.
6. Remove the old grease from the Leadscrew using waste paper.
7. Press **Y** and in Control Panel-Y, set Y to **0** (the furthest back position).
8. Remove the old grease from the Y leadscrews and linear guides using paper toweling.

**To grease the Y and Z axes:**

1. Remove the Maintenance Panel.
2. Remove the four screws securing the Z Casting Access Panel to the Z Casting Cover, and remove the Z Casting Access Panel.



1. Z Casting
2. Grease Nipples

**Figure 4-4. Z Casting Grease Nipples**



**Note:** Press the lever of the Grease Gun three times to feed sufficient grease into the nipples.

3. Apply grease to the six grease nipples, three on each side of the Z Casting. Use waste cloth to wipe off any excess grease.
4. Apply grease to the Y nipple using the grease gun, and to the linear guides using your finger.
5. Re-attach the Y Bellows to the Y Bellows Cover, using the screws previously removed.
6. Re-attach the Z Casting Cover, using the eight screws previously removed.
7. Close the hinged Spindle Cover.
8. Initialize and operate the newly greased Y and Z axes:
  - a. In the application, from the Main Menu, press **Maintenance > Axis Setup > Y**.
  - b. Press **Init** to initialize the axis.
  - c. Press **Start Self-Test**; the Y Axis begins moving automatically between its two limits.
  - d. Allow the Y Axis to continue moving for one minute.
  - e. Press the **Stop Self-Test**.
9. Repeat for the Z-axis.
10. Close all open panels.

### 4.4.3 Perpendicularity Test

Use a Perpendicular Block for geometrical adjustments and for accuracy and repeatability tests.

**To perform the Perpendicularity test:**

1. Mount tape on a designated frame and place it on the cutting chuck.
2. Mount the Perpendicularity Block onto the chuck, on top of the tape.
3. Using the camera and main video screen:
  - a. Select two points (left and right sides) on the upper edge and align the block in the X Axis. Use the T Axis rotation for precise alignment.
  - b. After getting a proper alignment, do not change the T Axis position.
4. Move the camera to the intersection in the upper edge and set the X and Y coordinates to **0**.
5. Move the Y Axis forward 150 mm and check if there is a deviation. If a deviation is found, call and consult with ADT customer support.



**Note:** When using a Measurement Probe (Dial Indicator) for Parallelism and Axes test, use any magnetic plate and mount the plate on the X Table Protector.

[Optionally, use the Bracket for Measurement Probe (Dial Indicator), which can be ordered from ADT.

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## Chapter 5. Application Definitions

Application Definitions describes how to define the basic functionality of Maintenance and Operation Setup windows and includes the following sections:

- ❖ Axis Setup Window (section 5.1).
- ❖ Calibration (section 5.2).
- ❖ Functions and I/O (section 5.3).



**Note:** For a detailed explanation of the setup parameters and parameter values, refer to the Model 71XX Operations Manual.

## 5.1 Axis Setup Window

From the Axis Setup window, the parameters of the following two groups can be defined:

- Axes.
- Stations.



**Note:** The system must be initialized before modifying the parameters. The system can be initialized when logging in, from the Main Menu window, or by pressing the Init button for each axis, spindle, or station.

### To open the Axis Setup window:

From the Main Menu, press **Maintenance > Axis Setup**; the Axis Setup window appears.

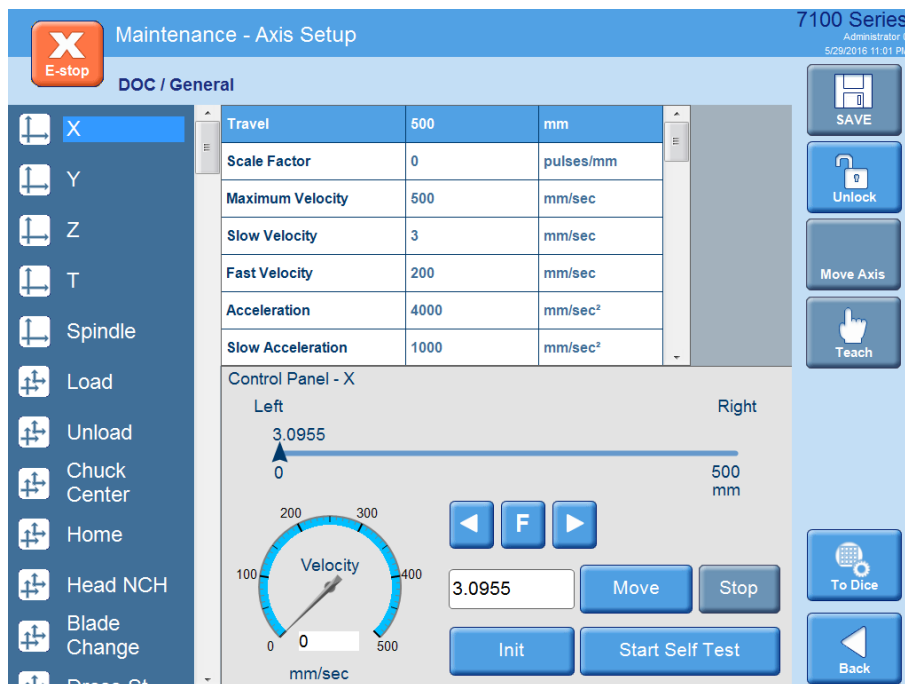


Figure 5-1. Axis Setup Window

### 5.1.1 Axes

In the Axes group, the following can be performed:

- Initialize an Axis.
- Manually Define an Axis Position.
- Define the Axis Parameter.
- Perform a Self-Test.

#### 5.1.1.1 Initializing an Axis

Each axis can be initialized independently.



**Note:** To protect the system, initialize the Z Axis before initializing the other axes.



**To initialize an axis:**

Select the target axis and press **Init**; the axis is initialized.

### 5.1.1.2 Manually Moving an Axis to a Selected Position

The axis position can manually be defined for the purpose of maintenance, evaluation, etc.

**To manually define the position of an axis:**

1. From the Main Menu, press **Maintenance > Axis Setup**; the Axis Setup window appears.
2. Select an axis; the axis view appears.
3. Perform one of the following to move the axis to a required position:
  - Press the Fast (**F**) or Slow (**S**) control button, to select if to move the slider in a fast or slow motion; then, press and hold the **Movement Control** button (e.g.,  or ) to move the axis to the desired location.

Or

- Enter a value into the Control Panel field box and press **Move**; to move to the set position.

### 5.1.1.3 Manually Rotating the Spindle

The spindle can be manually rotated at a set velocity.

**To manually rotate the spindle at a set velocity:**

1. From the Main Menu, press **Maintenance > Axis Setup**; the Axis Setup window appears.
2. Select the Spindle; the spindle view appears.
3. Perform one of the following to rotate the spindle at a desired velocity:
  - Press and hold the **Movement Control** button (e.g.,  or ) until the desired velocity is reached.

Or

- Enter a value into the Control Panel field box and press **Set Speed**.

### 5.1.1.4 Performing a Self-Test

The Self-Test operation moves the axis back and forth to evaluate its performance.

**To perform a Self-Test:**

Select the target axis and press **Start Self-Test**; the axis automatically toggles between its home position and its limit until the **Stop Self-Test** button is pressed.



**Note:** There is no Self-Test for the spindle.

### 5.1.1.5 Defining the Axis Parameters

The parameters of each axis can be defined.



**Caution:** Only authorized personnel should perform changes to the axis parameters.

**To define axis parameters:**

1. Select the target axis; the axis view appears.
2. Press **Unlock** to enable changes.
3. Press the target parameter; the Keypad appears.
4. Enter a new parameter (which does not exceed the Min and Max values) and press **Close**; the new parameter appears in the Parameters list.
5. Press **SAVE**.



**Note:** For a complete list of parameters, see following Table 5-1.

### 5.1.2 Axis Setup Parameters

Table 5-1 lists and describes the various axis and spindle parameters.

| Table 5-1. Axis Setup Parameters |                   |                       |                                                                                                                                                                                                                                                                                                                              |
|----------------------------------|-------------------|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| No.                              | Parameter         | Axis/Spindle          | Description                                                                                                                                                                                                                                                                                                                  |
| 1                                | Travel            | All axes and spindle  | The distance that the axes can travel.<br>X, Y, and T Axes take into account limits imposed by mechanical bumper guards and optical controls.<br>Z Axis takes into account an upper bumper guard and the lowest physical travel point minus the Z Safety Position.<br>The Spindle can travel 360 degrees and without limits. |
| 2                                | Scale Factor      | All axes and spindles | Factor of pulsing; necessary for calculating accurate distances.                                                                                                                                                                                                                                                             |
| 3                                | Maximum Velocity  | All axes and spindles | Maximum velocity that the specific axis/spindle can travel.                                                                                                                                                                                                                                                                  |
| 4                                | Slow Velocity     | All axes and spindles | The slow axis/spindle speed for manually moving the axes in the Dice window.                                                                                                                                                                                                                                                 |
| 5                                | Fast Velocity     | All axes and spindles | The fast axis/spindle speed for manually moving the axes in the Dice window.                                                                                                                                                                                                                                                 |
| 6                                | Acceleration      | All axes and spindles | Accelerating speed; from the start of axis/spindle movement to the defined working speed.                                                                                                                                                                                                                                    |
| 7                                | Slow Acceleration | All axes and spindles | A slow accelerating speed; until the axis/spindle reaches the working speed (for manual axis operation).                                                                                                                                                                                                                     |
| 8                                | Deceleration      | All axes and spindles | Decelerating speed; from the defined axis/spindle working speed to a complete stop of movement.                                                                                                                                                                                                                              |

**Table 5-1. Axis Setup Parameters**

| No. | Parameter                   | Axis/Spindle          | Description                                                                                                                                                                               |
|-----|-----------------------------|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 9   | Kill Deceleration           | All axes and spindles | The deceleration value when pressing the E-Stop button.                                                                                                                                   |
| 10  | Self-Test Delay             | All axes              | Time between the self-test cycles.                                                                                                                                                        |
| 11  | Camera Calibration Point    | All axes              | Station for calibrating the camera.                                                                                                                                                       |
| 12  | Axis X Safety Position      | X axis                | Position on the axis that provides a safety margin.                                                                                                                                       |
| 13  | Axis X Safety Left Position | X axis                | Position on the axis that provides a safety margin on the left side (LA option).                                                                                                          |
| 14  | Dress Station               | X, Y and Z axes       | Position of the axes above the front-left corner of the dress station, where the dress station procedure is performed. (Note: Relevant only when there is a dress station in the system). |
| 15  | Warmup Start Position       | All axes              | N/A                                                                                                                                                                                       |
| 16  | Warmup Finish Position      | All axes              | N/A                                                                                                                                                                                       |
| 17  | Warmup Idle Position        | All axes              | The idling position on the axes for the warmup procedure.                                                                                                                                 |
| 18  | Home                        | All axes and spindle  | Position of axes following initialization.                                                                                                                                                |
| 19  | Load                        | All axes              | Position of the axes when the workpiece is loaded.                                                                                                                                        |
| 20  | Unload                      | All axes              | Position of the axes when the workpiece is unloaded.                                                                                                                                      |
| 21  | Chuck Center                | X, Y and Z axes       | Position of camera above center of chuck rotation.                                                                                                                                        |
| 22  | Blade Change Position       | X, Y and Z axes       | Position of the axes when a blade is changed                                                                                                                                              |
| 23  | Axis Pos 1                  | X, Y and Z axes       | Axis start position                                                                                                                                                                       |
| 24  | Axis Pos 2                  | X, Y and Z axes       | Axis end position                                                                                                                                                                         |
| 25  | Dress Block Change Position | All axes              | Position of the axes during the dress-block change procedure (option).                                                                                                                    |
| 26  | Pre Non-contact Height      | All axes              | Station before beginning NCH (takes into account the blade with the largest diameter.                                                                                                     |
| 27  | Calibration Start Position  | All axes              | Position on all of the axes that provides a safety margin during the calibration procedure.                                                                                               |
| 28  | Start Align Position        | T axis                | The theta axis position when beginning alignment.                                                                                                                                         |
| 29  | Chuck Height Start Position | Z axis                | The Z axes position when beginning the chuck height procedure.                                                                                                                            |
| 30  | Chuck Height Stop Position  | Z axis                | The Z axes position when ending the chuck height procedure.                                                                                                                               |

| Table 5-1. Axis Setup Parameters |                                |              |                                                                    |
|----------------------------------|--------------------------------|--------------|--------------------------------------------------------------------|
| No.                              | Parameter                      | Axis/Spindle | Description                                                        |
| 31                               | Auto Focus on Dicer Chuck Pos. | Z axis       | Z Axis position.                                                   |
| 32                               | Z Height Movement              | Z axis       | Calculated coordinator - the lowest point that Z can travel.       |
| 33                               | Axis Z Safety Position         | Z axis       | Lowest non-cutting position on the Z Axis during manual operation. |
| 34                               | Chuck Position                 | Z axis       | Height, whereby the blade contacts the Chuck table.                |
| 35                               | Light Align Angle              | T axis       | Align angle                                                        |

### 5.1.3 Stations

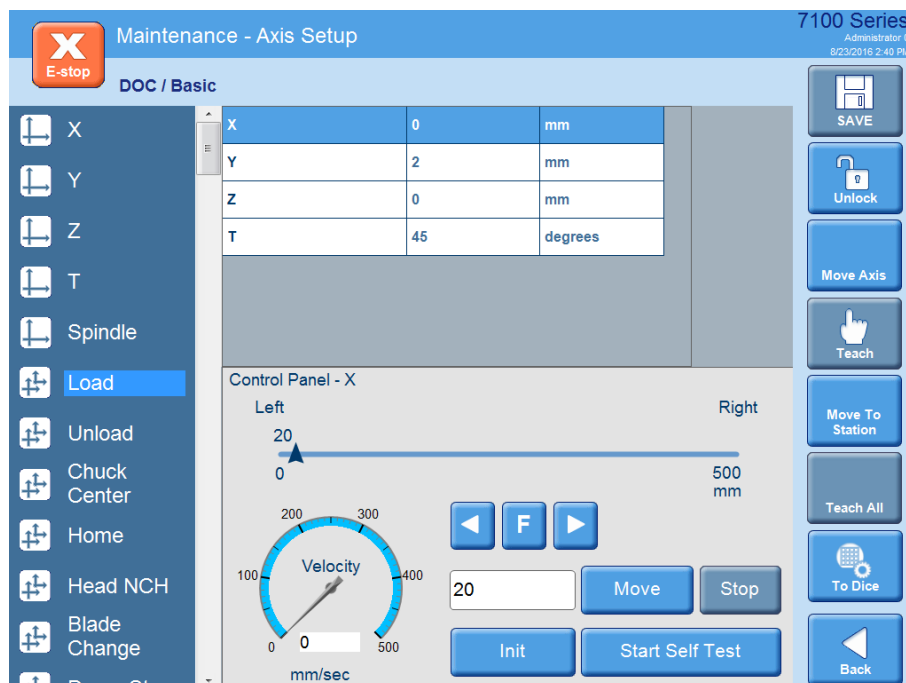


Figure 5-2. Axis Setup Stations

Stations are defined positions on the axes at which the following actions occur:

- **Load/Unload:** The position of the X, Y, Z and T axes when workpieces are loaded onto and unloaded from the cutting chuck.
- **Chuck Center:** The position of the X, Y, Z and T axes when the microscope is over the center of the cutting chuck.
- **Home:** The home position of the X, Y, Z and T axes.
- **Head NCH:** The position of the X, Y, Z and T axes when the spindle is above the Non-Contact Height device.

- **Blade Change:** The position of the X, Y, Z and T axes during the Blade Change procedure.
- **Warmup Station:** The position of the X, Y, Z and T axes during the warmup procedure.
- **Dress Station:** If configured.
- **HMT Stations:** If configured.

The position of each axis can be modified and then taught to the system.



**Note:** It is recommended writing down the station coordinates before changing them.

After making positions changes to the Station axes, the following procedures can be performed:

- **Move Axis:** the system moves the selected station axis to the defined axis coordinate. This movement is reflected in the Control Panel slider.
- **Teach:** Requires first pressing **Unlock**; the system is taught the selected station axis coordinate according to the current axis position.
- **Move To Station:** the system moves the axes to the axis coordinates of the selected station. This movement is reflected in the Control Panel slider.
- **Teach All:** Requires first pressing **Unlock**; the selected station is taught according to the axis coordinates at the current axes positions.

## 5.2 Calibration

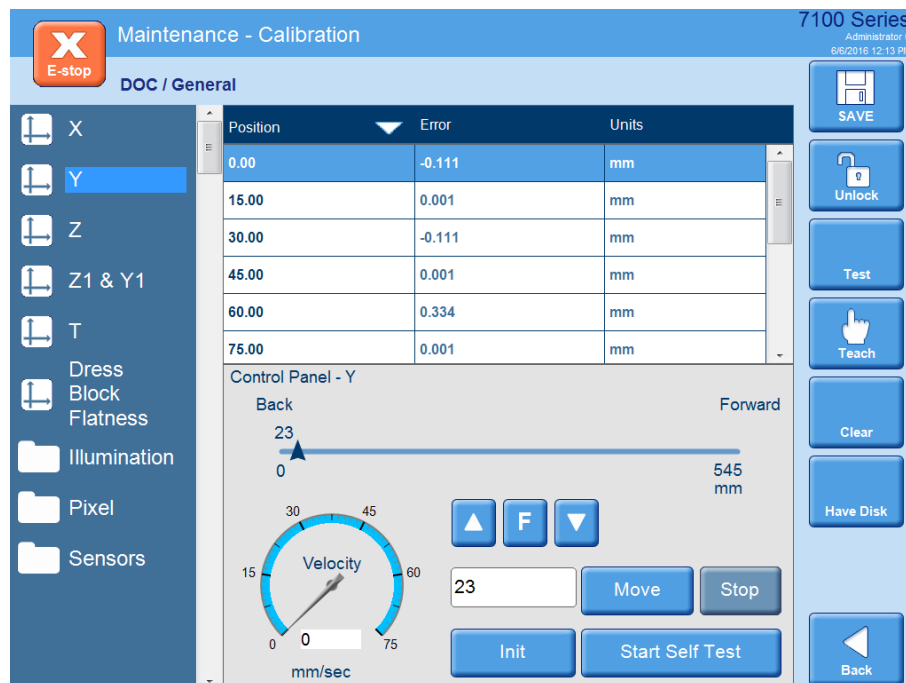
The Calibration window enables teaching the system absolute axis, illumination, pixel, and sensor accuracy.



**Caution:** Calibration procedures should be performed by ADT personnel only.

### 5.2.1 Axes

Axis calibration is mapped out in the factory, before the system is shipped. All axis positions and associated error margins are recorded into the system.



**Figure 5-3. Calibration Axis Window**

## 5.2.2 Illumination

Illumination can be calibrated so that the illuminations set on various systems are similar, allowing transferring recipes and dicing accordingly, using the same illumination values on other dicing systems.



**Note:** The Illumination graph is predefined at ADT. Presently, calibration change cannot be performed in this version. In the event of problems or difficulties in connection to illumination, refer to your ADT representative.

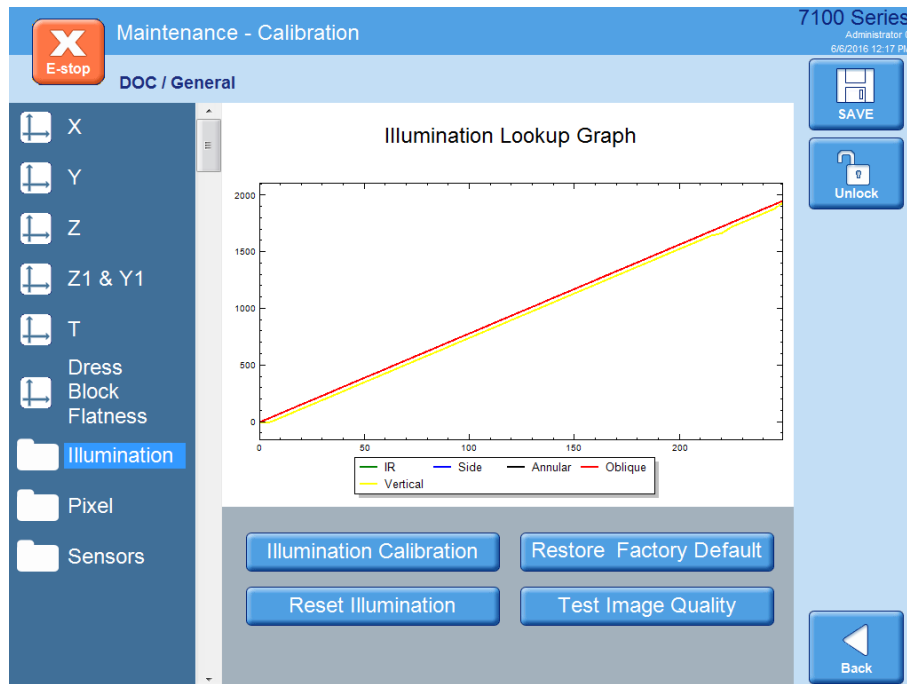


Figure 5-4. Calibration Illumination Window

## 5.2.3 Pixel

The size of the pixel that appears through the microscope and camera can be calibrated.

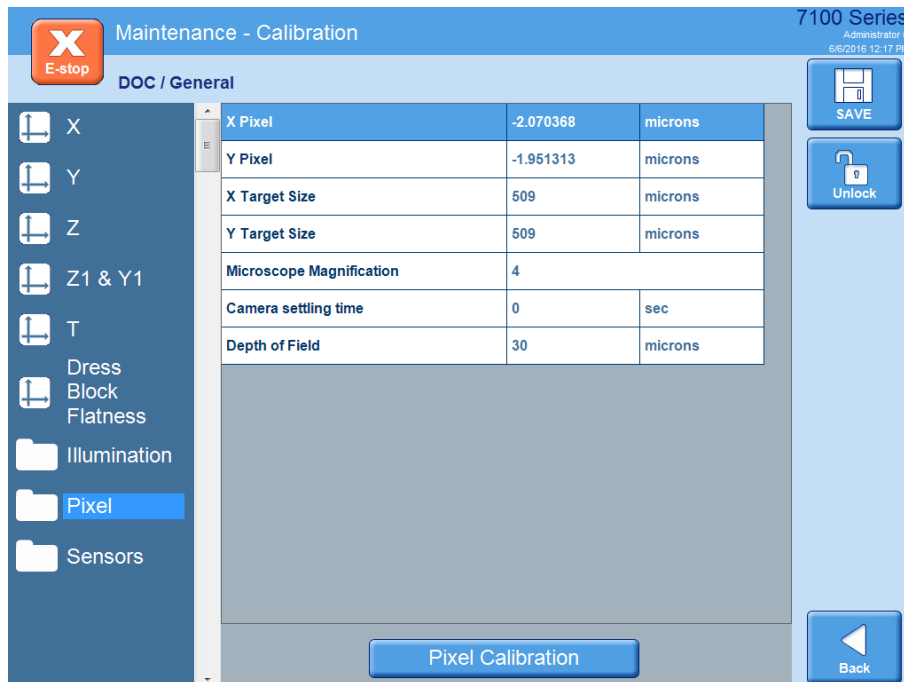


Figure 5-5. Calibration Pixel Window

### To calibrate the pixel size:

1. Load a special Target workpiece (for information on Target workpieces, contact your local ADT representative).
2. In the pixel view, press **Pixel Calibration**; the Pixel Calibration wizard appears in the Dice window.
3. Follow the wizard steps:
  - a. Move the vision system to the center of the workpiece.
  - b. In the Vision Control view, press the high-magnification **Zoom** button  and then use the **At Height** and **On Wafer** Auto Focus buttons to adjust the focus.
  - c. In the Axis Control view, use the **Guide** controls to arrange the Search and Teach areas around the reticle:
    - ♦ Use the **R** button controls to manage the reticle (green cross lines).
    - ♦ Use the **S** button to position the Search area (red box).
    - ♦ Use the **T** button to position the Teach area (blue box).
  - d. Press **Teach** to teach the position to the system.
  - e. Press **Wizard Finish**.



## 5.2.4 Sensors

To improve the precision of the system sensors, their coefficient and constant parameters can be modified.



**Note:** Modifying requires logged in to the application as an ADT Administrator.

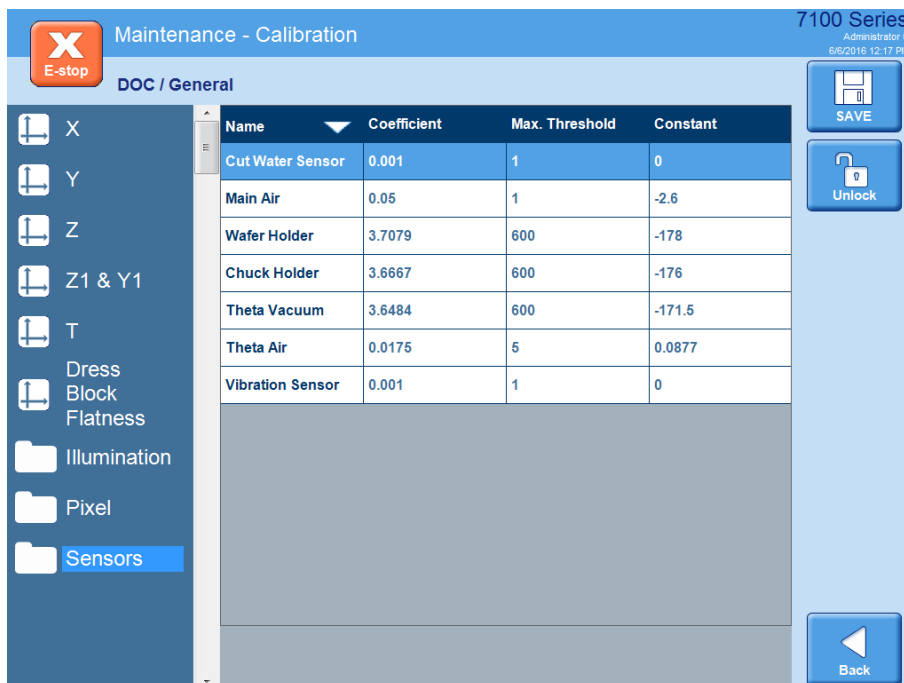


Figure 5-6. Calibration Sensors Window

**To modify sensor coefficient and constant parameters:**

1. Press **Unlock** to enable changes.
2. Select a value to change; the Keypad appears.
3. Enter the modified value and press **Close**.
4. Repeat steps 2–3 for each value that should be changed.
5. Press **SAVE**.

## 5.3 Functions and I/O

From the Functions and I/O window, you can perform various height procedures, control the vacuum systems, perform a chuck change, and examine and control the head system elements.



**Note:** If the system is configured with additional options, they may also appear in this window.

Functions and I/O includes:

- Performing Dicer Functions and I/O.
- Performing Height Functions and I/O.
- Performing Cutting Table Functions and I/O.
- Performing Head System Functions and I/O.

### 5.3.1 Dicer Functions and I/O

The spindle and moving axes are enclosed in a transparent door to prevent injury to the operator and/or damage to the system. When the spindle is rotating, the door is closed and locked by an interlock (mechanical locking device with a sensor). The interlock can be manually enabled and disabled.

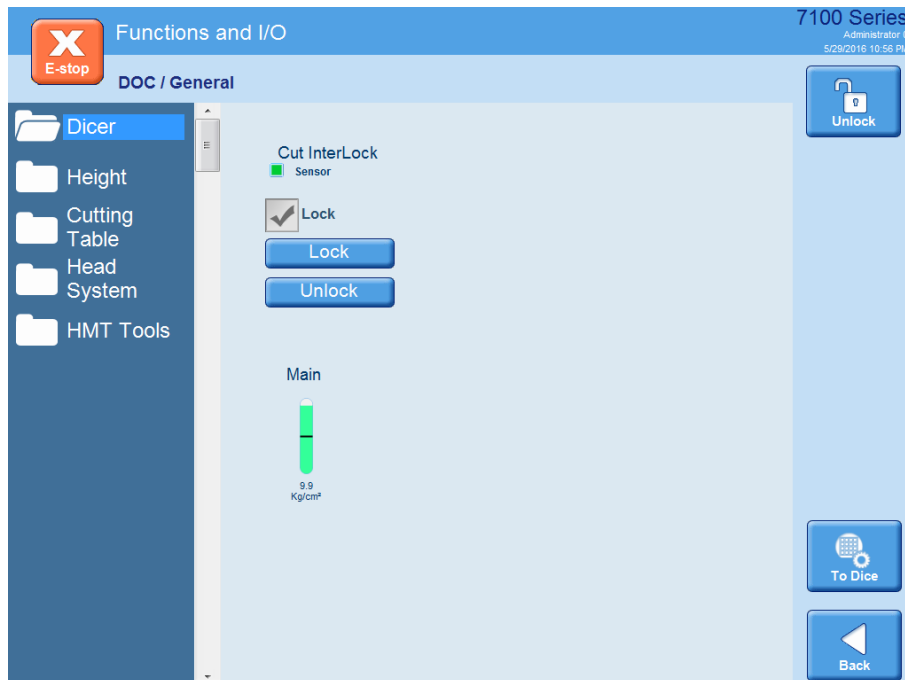


Figure 5-7. Functions and I/O Dicer Window



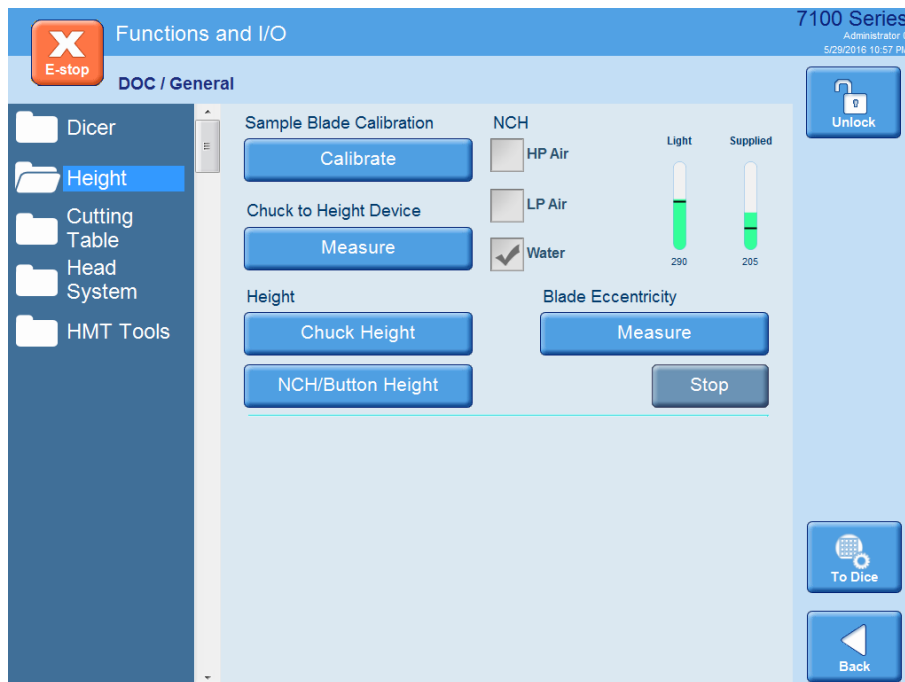
**Notes:**

- ◆ When the interlock device is enabled, the door locks when the spindle is active.
- ◆ When the interlock-controlled door is open, all axes are disabled.

**To manually enable/disable the interlock device:**

1. From the Main Menu, press **Maintenance > Functions and I/O > Dicer**; the Functions and I/O window appears in the Dicer view.
2. Press the **Unlock** button  to enable changes.
3. Press the **Cut InterLock Lock** or **Unlock** button to either manually enable or disable the interlock device.

## 5.3.2 Height Functions and I/O



**Figure 5-8. Functions and I/O Height Window**

From the Height windows, the following Height procedures can be initiated:

- **Sample Blade Calibration:** defines the spindle position value (when using a new blade of known diameter). This value is used to determine the accurate exposure of the current blade when performing Process Height.
- **Chuck to Height Device Measure:** determines the difference between the chuck height and the height device. The system calibrates the two measurements and uses the height device as a reference point. Performed after changing the position of the cutting chuck relative to the height device, or vice versa.
- **Height**
  - **Chuck Height** measurement: determines the cutting chuck position along the Z Axis. The blade automatically descends to a pre-specified position above the chuck and continues descending until the blade touches the cutting chuck metallic surface.
  - **NCH/Button Height** measurement: determines the NCH/Button position along the Z Axis. The blade automatically descends to a pre-specified position above the NCH/Button and continues descending until the blade edge breaks the light beam emitted by the NCH device, or touches the mechanical button top surface.
- **Blade Eccentricity Measure:** measures the eccentricity of the mounted blade to avoid too deep or too shallow cuts (due to blade eccentricity).
- **Stop:** enables stopping the procedure.
- **NCH**

- **HP Air** activation: the solenoid that controls the airflow through the NCH prism nozzles is activated. This flow of air clears water and debris from between the NCH prisms, or from the surface of the mechanical button. The actual flow can be re-adjusted from the air regulator unit. See details in section 11.4.
- **LP Air / Water:** activates and deactivates the solenoids that control the low-pressure airflow and water flow on the NCH device, operating during the height procedure.



**Note:** For more on the Height procedures, see Preparing the System for Dicing in the Operations manual.

### 5.3.3 Cutting Table Functions and I/O



**Figure 5-9. Functions and I/O Cutting Table Window**

From the Cutting Table window the following can be performed:

- Release and engage vacuum for the following elements:
  - The workpiece.
  - The chuck.
  - The theta table.
- Verify the Chuck 8 VDC for Chuck Height performance.
- Initiate the Chuck Change procedure.

**To engage and release the wafer and chuck holder vacuums:**

1. Press **Unlock** to enable changes.
2. Press **Release** to release the vacuum and **Accept** to engage the vacuum holding the workpiece (wafer holder area) or chuck (chuck holder area); the gauge in each area indicates the level of the vacuum when engaged.

**To engage and release the T Table vacuum:**

1. Press **Release** to release the vacuum and **Brake** to engage the vacuum holding the Theta Table in place; the Theta Table gauge indicates the level of the vacuum when the Theta Table has been braked.



**Note:** In Unlocked mode, the vacuum can be enabled by selecting the related Vacuum check boxes and enable the air release by selecting the Air check box.

**To verify Chuck 8 VDC for Chuck Height performance:**

1. Press **Unlock** to enable changes.
2. Select the **Voltage** box; the check box turns green.
3. Select the **Gnd** box; the check box should change back to gray.

**To change the chuck:**

1. Press **Chuck Change** to initiate the Chuck Change procedure.



**Notes:** For a complete description of the chuck change procedure, see section 9.3.1.

2. [Recommended] Update the Auto-Focus on the chuck. See section 10.2.2.1.

## 5.3.4 Head System Functions and I/O

From Head System, the following systems can be tested and/or activated:

- Broken Blade Detector (BBD).
- Spindle.
- Water Flow Control.
- Camera

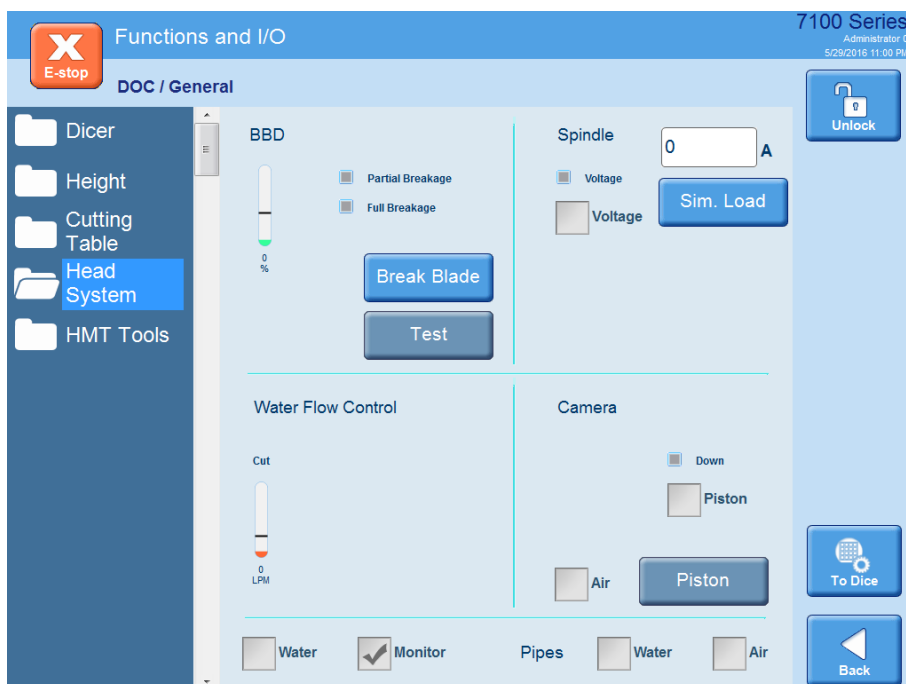


Figure 5-10. Functions and I/O Head System Window

### 5.3.4.1 Broken Blade Detector (BBD)

The BBD panel displays a visual report of the present status of the blade:

- The BBD sensor shows if the blade is broken or not:
  - Green indicates the blade is not broken and is usable.
  - Orange indicates the blade is broken and not usable, and it should be replaced.
  - The black line is the threshold; when the orange color passes this mark, the blade is detected as broken and should be replaced.
- Breakage Indicators:
  - Partial Breakage: turns from green to yellow when there is slight damage to the blade.
  - Full Breakage: turns from green to red when there is major damage to the blade.

#### To test the blade for breakage:



**Note:** To test the blade for breakage, the spindle must rotate.

1. Press **Unlock**; the Test button is activated.
2. Press the **Test** button:
  - If the blade is in good condition, the Breakage Indicators remain green-colored.
  - If the blade is damaged, an Error message appears and the Breakage Indicators change color.

### 5.3.4.2 Spindle

From the Spindle pane the spindle measured load (current [A]) can be monitored. This pane also allows checking the voltage signal sent to the spindle brushes, then to the blade.



**Danger:** When working with or around the spindle, turn off the spindle and verify it is not rotating.

#### To test spindle voltage:

1. Press **Unlock**; the Voltage check box is activated.
2. Select the **Voltage** check box:
  - If there is a voltage connection from the brushes to the blade, the voltage indicator remains green-colored.
  - If there is no voltage connection from the brushes to the blade, the voltage indicator turns gray.

### 5.3.4.3 Water Flow Control

The Water Flow Control panel displays a visual report of the present status of the water that is applied during the cutting procedure.



**Note:** To mechanically adjust the water flow to the cooling block, see section 9.4.2.

From the Water Flow Control pane, the following can be performed:

- Monitor the water flow to verify that water is in fact flowing from the flow exits.
- Open and close the blade cut-water flow (main jet).

#### To control the water flow to the blade:

1. From the main menu, go to **Maintenance > Functions and I/O > Head System**.
2. Press **Unlock**; the Water check box is activated:
3. Select the **Water** check box to open the water to the blade.
4. Adjust the water flow, with the flowmeter control valve located on the system front panel, to the desired flowrate.
5. Clear the **Water** check box to close the water to the blade.

#### To monitor the water flow:

1. Press **Unlock**; the Monitor check box is activated:
2. Select the **Monitor** check box to verify that water flows from the flow exits.
3. Clear the **Monitor** check box to cancel water-flow verification.



**Caution:** The Monitor check box must be selected during the dicing process.

### 5.3.4.4 Camera

From the Camera panel, the following can be performed:

- Open and close the high-pressure blow air that clears the debris from the workpiece directly beneath the vision system (microscope and camera).

#### To control the air to the workpiece:

1. Press **Unlock**; the Air check box is activated:
2. Select the **Air** check box to open the air to the workpiece.
3. Open the system lower back panel and adjust the needle valve for the camera air blow (see Figure 11-1 in section 11.4).
4. Clear the **Air** check box to close the air to the workpiece.
5. Close the system opened panel.

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## Chapter 6. Electrical and Electronic Systems

This chapter discusses the Power Supply System of the Model 71XX, and includes the following sections:

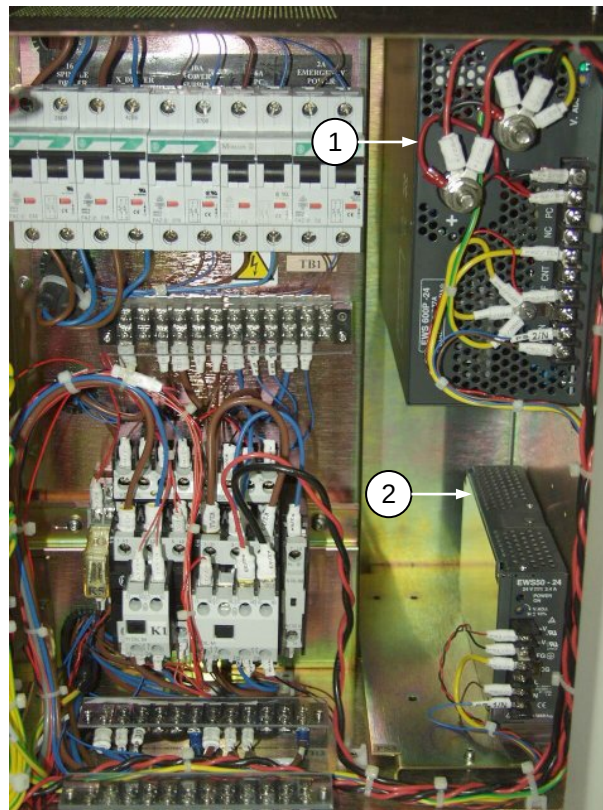
- ❖ Power Supply System (section 6.1)
- ❖ Mains Power Input (section 6.2)
- ❖ When the System is Powered Down (section 6.3)
- ❖ When the System is Powered Up (section 6.4)

## 6.1 Power Supply System

The Model 71XX Power Supply System consists of the following three Power Supply units:

- 24 VDC, 600 W, Main Power Supply
- 24 VDC, 50 W, Emergency Power Supply
- 24 VDC Illumination Power Supply (optional, for the Halogen Illumination type)

The Power Supply System, located on the right side of the Electronic Bench at the front of the System, extracts the various voltages required for all the operating subsystems from the input voltage.



1. Power Supply (24 VDC)
2. Emergency Power Supply (24 VDC)

**Figure 6-1. Model 71XX Power Supply System**

In accordance with the logic built into the System software, power is supplied to subsystems in sequence, which protects the User, the System and Workpieces from injury and damage. Voltage for the Controller and the Drivers, for example, is not supplied before the Main System Controller (MSC) has turned on and launched the Model 71XX software.

## 6.2 Mains Power Input

The single-phase Mains Power Input to the Model 71XX enters the System via the Main Circuit Breaker, located at the right side of the System, from where it passes through an RFI filter and connects to the Circuit Breakers inside the Power Supply. From the Circuit Breakers, Mains Power Input is delivered to each of the two Power Supply units.

The Mains Input can range from 200 to 240 VAC.

## 6.3 When the System is Powered Down

When the System is powered down and the Main Circuit Breaker is turned off, no voltage reaches the System. After the Main Circuit Breaker has been turned on, the power is supplied to the Emergency power Supply unit. On the Control Panel the OFF button is lit, as well as the AC lamp, which indicates the presence of power in the System.

## 6.4 When the System is Powered Up

The Model 71XX is powered up by pressing the ON button on the Control Panel. When the ON button is pressed, the following sequence occurs:

1. K1 is operated, supplying 230V to the MSC and Monitor. 230V is supplied to the Main power supply. +24 VDC is supplied to Controller, SSB0, SSB1, Bench Fan, Z and Theta Drivers, Lamp 'ON', and Lamp '+24V'
2. The Controller checks that the System is functioning normally, and signals the K2 circuit breaker for the Y-axis motor driver. The amber LEDs in the K1 and K2 circuit breakers indicate normal function.
3. Following the run command from the Controller:
  - a. The Y Driver receives +24 VDC from the additional contacts of the K2 contactor.
  - b. The Spindle Driver receives 230 VAC through the K2 contactor and F7 circuit breaker.
  - c. The X Driver receives 230 VAC through the K2 contactor and F6 circuit breaker.

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## Chapter 7. Maintaining the Spindle

Maintaining the Spindle describes the procedures that can be performed when the system spindle malfunctions.

Maintaining the Spindle includes:

- ❖ Introduction (section 7.1).
- ❖ The Spindle Electronic (section 7.2).
- ❖ The Spindle Wheel Mounts (section 7.3).
- ❖ Replacing the Spindle (section 7.4).

## 7.1 Introduction

Maintaining the Spindle describes the procedures for replacing the following system elements:

- Spindle Driver.
- Regenerative Unit.
- Spindle.

Maintaining the Spindles also describes the tests and adjustments that must be performed when installing a new spindle.



**Notes:**

- ◆ Read all the instructions carefully before beginning the procedures described in this chapter and allow enough time to complete all the tests and adjustments after spindle installation.
- ◆ Keep the spindle and surrounding areas free of dust and dirt.



**Caution:**

- These procedures should not be attempted by untrained or inexperienced personnel.
- The spindle is equipped with air bearings to enable free rotation. If, for any reason, there is no air or water flow, spindle initialization is not possible. The flow of air to the spindle is adjusted in the Air Regulator Unit, as described in section 11.8.  
Do not attempt to rotate the spindle, even manually, while the air supply is turned off, as this can cause damage to the spindle internal parts.

## 7.2 The Spindle Electronics

When a spindle malfunctions, first make sure that correct air pressure is supplied to the spindle. Following, examine the mechanical condition of the spindle to verify it is not stuck and it can be manually rotated freely.

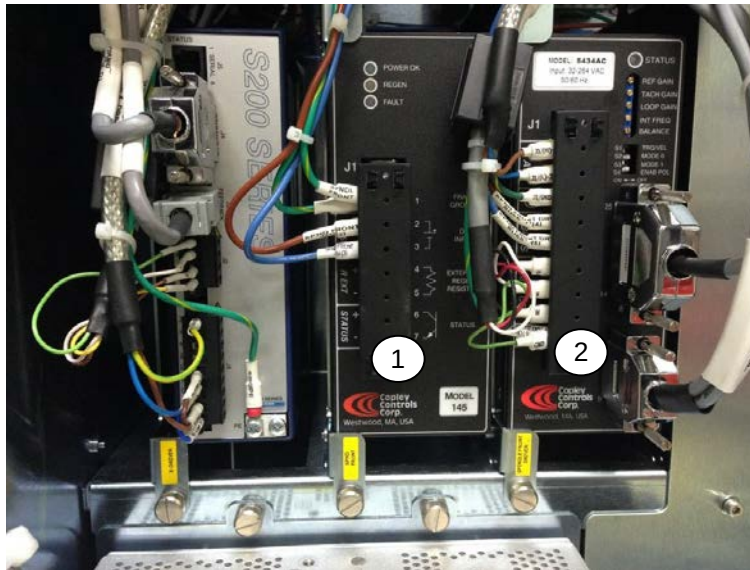
After verifying the spindle is not stuck, examine the spindle's electronic systems before replacing the spindle itself

The Spindle Electronics section includes the following elements:

- Spindle Driver
- Regenerative Unit.
- Spindle Brush Unit.

## 7.2.1 Spindle Driver

The spindle driver is located on the middle portion of the electronic bench. The spindle driver is a DC motor driver.



1. Spindle Driver
2. Regenerative Unit

**Figure 7-1. Spindle Driver and Regenerative Unit**

When a spindle problem occurs, replace the spindle driver before replacing either the spindle brushes or the spindle itself, regardless of the error message displayed in the application. The reason for this is that the spindle driver is responsible for sending error messages to the system software. If the spindle driver is defective, it could send an inaccurate message.



**Caution:** Consult authorized Advanced Dicing Technologies personnel before replacing any spindle component.

Spindle Drivers operations includes:

- Interpreting the Spindle Driver LED Codes.
- Replacing the Spindle Driver.

### 7.2.1.1 Interpreting the Spindle Driver LED Codes

The spindle driver has three LED indicators on its front panel, indicating the driver status:

**Table 7-1. Spindle Driver LEDs**

| Color            | Indication                              |
|------------------|-----------------------------------------|
| Green (blinking) | Spindle driver is not initiated.        |
| Green (steady)   | Spindle driver is functioning properly. |
| Red              | Spindle driver is malfunctioning.       |

### 7.2.1.2 Replacing the Spindle Driver

Replace the spindle driver as the first step in solving problems with the spindle.

**To replace the spindle driver:**

1. Power down the system.
2. Open the front door to gain access to the electronic bench.
3. Locate the faulty spindle driver (see Figure 7-1), and disconnect all wires and cables. Note that the wires and cables are labeled to correspond with the proper pins. Each pin also has a position number printed to its right.
4. Remove the screws securing the spindle driver and pull the driver towards you.
5. Install the new spindle driver in the system, using the screws that were previously removed.
6. Reconnect all cables and wires; check the labels to ensure proper placement.
7. Power up the system, and check the LED indicator (see Table 7-1).
8. Initialize the system and verify that the spindle functions properly.
9. Close the front door.

## 7.2.2 Regenerative Unit

The Regenerative Unit is located in the middle portion of the electronic bench, to the right of the spindle driver. Its purpose is to prevent excessive voltage from reaching the spindle driver. See Figure 7-1.

Regenerative Unit operations include:

- Interpreting the Regenerative Unit LED Codes.
- Replacing the Regenerative Unit.

### 7.2.2.1 Interpreting the Regenerative Unit LED Codes

The Regenerative Unit has one LED indicator on its front panel, indicating the status of the unit:

**Table 7-2. Regenerative Unit LEDs**

| Label | Color  | Indication                                                         |
|-------|--------|--------------------------------------------------------------------|
| Power | Green  | The Regenerative Unit is functioning normally.                     |
| Regen | Yellow | Power flow of over 360V. The Regenerative Unit begins functioning. |
| Fault | Red    | The Regenerative Unit is malfunctioning and should be replaced.    |



### 7.2.2.2 Replacing the Regenerative Units

The red Fault LED indicates that the Regenerative Unit is malfunctioning, and should be replaced.

**To replace the regenerative unit:**

1. Power down the system.
2. Open the front door to gain access to the electronic bench.
3. Locate the faulty Regenerative Unit (see Figure 7-1), and disconnect all wires and cables.



**Note:** All wires and cables are labeled to correspond with the proper pins, and each pin has a position number printed to its right.

4. Remove the screws securing the Regenerative Unit to the electronic bench and remove the Regenerative Unit by pulling it towards you.
5. Install the new Regenerative Unit, replacing the screws that were previously removed.
6. Reconnect all cables and wires; check the labels to ensure proper placement.
7. Power up the system, and check the LED indicator (see Table 7-2).
8. Initialize the system and verify that the spindle functions properly.
9. Close the front door.

### 7.2.3 Spindle Brush Unit

At the spindle rear side is a housing with two carbon brushes. The brushes are used to conduct a low-voltage current to the blade, thereby enabling performing a Contact Height on the cutting chuck.

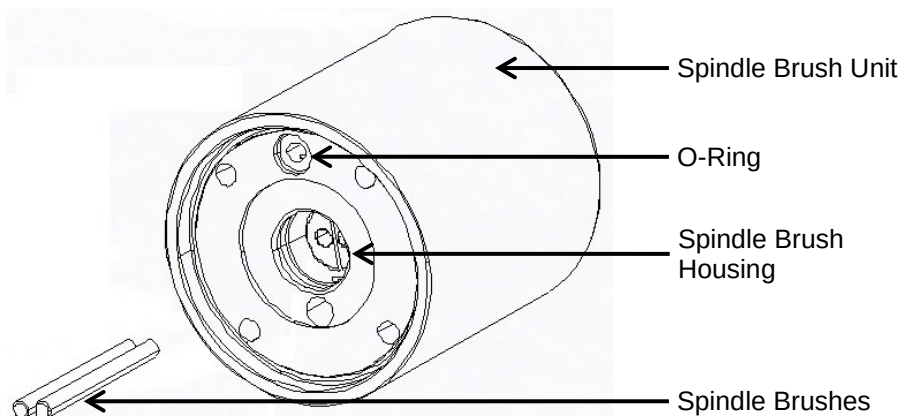


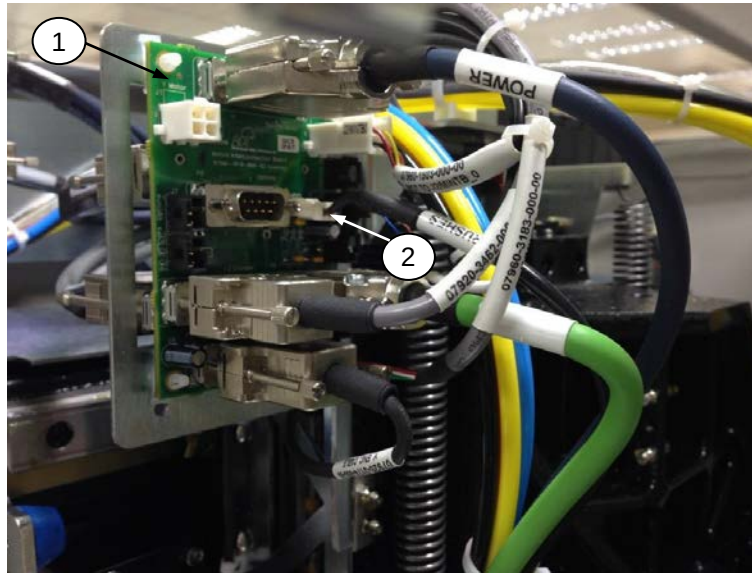
Figure 7-2. Spindle Brush Unit

#### 7.2.3.1 Inspecting the Spindle Brush Wiring Connection

If there is an indication of a spindle brush problem, examine the spindle brush wiring connection without removing the unit from the spindle.

**To inspect the spindle brush wiring connection:**

1. Open the backside service cover.
2. Locate and disconnect the connector that connects the spindle brushes to the relay board.



1. Relay Board
2. Spindle Brush Connector

**Figure 7-3. Spindle Relay Board**

3. Examine the resistance between the two brushes, using an ohmmeter. The value should be between 0 and 5 Ohms. If the value is not within this range, there is probably a problem with one of the brushes.
4. To determine which brush is causing a problem, examine the resistance between:
  - a. Pin 1 and the spindle rotor.
  - b. Pin 2 and the spindle rotor.

### **7.2.3.2 Replacing the Spindle Brushes**

When an error message appears, indicating spindle brush failure, the brushes should be replaced (not cleaned). Both brushes should be replaced even if it appears that one is still functioning properly.

**To replace the spindle brush unit:**

1. Turn off the power and air supply to the spindle.
2. Open the upper left-side panel and the back panel (to gain access to the spindle rear side).
3. Unscrew the two screws that secure the sensor assembly at the rear of the spindle and remove the sensor assembly from the spindle.
4. Remove the used brushes. See Figure 7-2.
5. Clean off the carbon powder from the brush holders located in the sensor assembly and from the copper contact at the end of the spindle shaft.



**Note:** Verify that the brush tip is directed towards the copper contact plate at the end of the spindle shaft.

6. Place the new brushes inside the brush holders.
7. Verify that the O-Ring seal is in place at the time of installation.
8. Place the sensor assembly to the rear of the spindle and gently tighten the two securing screws.
9. Close the upper left and rear panels.
10. Power up the system.
11. Initialize the system and verify that the spindle functions properly.

## 7.3 The Spindle Wheel Mounts

The blade is mounted on the spindle wheel mount, which also balances the spindle rotation. Do not remove the wheel mount, as removal can cause problems with spindle balance and runouts.

If a problem with a wheel mount is recognized, contact your ADT representative.

### 7.3.1 Performing Wheel-Mount Run-Out Measurements

Perform the Wheel Mount Front Surface Runout test to determine whether the wheel mount rotates according to system specifications.

**To perform wheel mount (front surface) run-out measurements:**

1. Remove the blade from the spindle.



**Note:** You may want to remove the Cooling Block.

2. Power down the system and then power up again.
3. In the Login window, do not select the Init system check box.
4. Initialize each system component, except for the spindle:
  - a. From the Main Menu, press **Maintenance > Axis Setup**; the Axis Setup window appears.
  - b. Select each of the following axes and press the **Init** button to initialize each specific axis:



**Note:** Do not initialize the Spindle axis.

- ♦ Z (the Z axes must be initialized first).
- ♦ X
- ♦ Y
- ♦ T

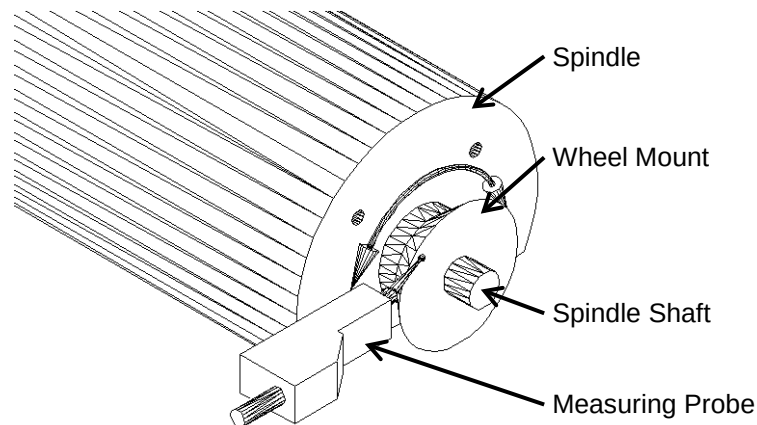
5. Remove the cutting chuck from the theta table:

- a. From the Main Menu, press **Maintenance > Functions and I/O > Cutting Table**; the Functions and I/O window appears in the Cutting Table view.
- b. Press the Chuck Holder **Release** button; the vacuum that secures the cutting chuck to the theta table is released.
6. Clean the spindle shaft and wheel mount face thoroughly with isopropyl alcohol.
7. Move the theta table to the left side limit on the X Axis:
  - a. From the Main Menu, press **Maintenance > Axis Setup**; the Maintenance window appears in the Axis Setup view.
  - b. Press **X** and in the Control Panel-X, press  to move the X Axis to the Left 0 position.
8. Attach a test indicator, connected to a magnetic holder, to a magnetic table jig.



**Note:** Place a piece of tape on the underside of the magnetic table jig, so that it does not come into direct contact with the theta table.

9. Position the table jig on the theta table, to the left of the spindle shaft.



**Figure 7-4. Top View - Axial Runout**

10. Position the measuring probe so that it lightly touches the wheel mount immediately inside the outer raised area and slightly above the spindle shaft center.
11. Set the measuring probe to zero.
12. Rotate the spindle shaft manually to verify whether the deviation is within the specified 2.5 microns.



**Note:** If the deviation is greater than the specified 2.5 microns, contact your local ADT representative.

## 7.4 Replacing the Spindle

Replacing the Spindle includes the following procedures:

- Replacing the Spindle (2" or 4").
- Testing the Spindle Position.



**Notes:**

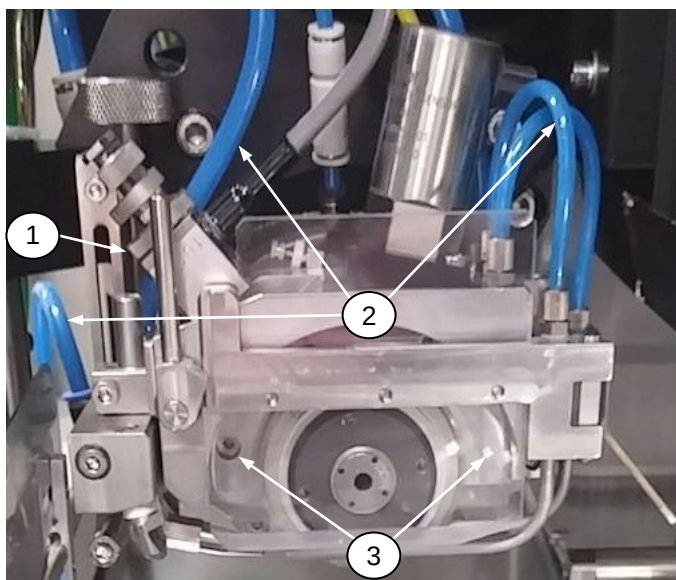
- ◆ The Model 71XX may be equipped with a 2" or 4" spindle.
- ◆ Spindle replacement is followed by a positioning test.

### 7.4.1 Replacing a 2" Spindle

Replacing a spindle involves the following procedure:

**To remove the spindle:**

1. Remove the blade from the spindle and remove the Cooling Block.
2. Power down the system; the X and Y Axes move to the Home positions, allowing easy access to the spindle.
3. Turn off the water and air supplies.
4. Remove the upper-rear, lower, and upper left-side panels.
5. Open the spindle and workpiece covers.
6. Open the spindle "snake-type" cable carrier.
  - a. Disconnect the spindle cables from the spindle extension cable.
  - b. Disconnect the spindle carbon-brush cable from the Contact board.
7. Disconnect all plugs, and the air and water tubes from the rear of the spindle.
8. Disconnect the three transparent water tubes from the cooling block.



1. BBD with Adjusting Nuts
2. Water Tubes
3. Cooling Block Connecting Screws (X2)

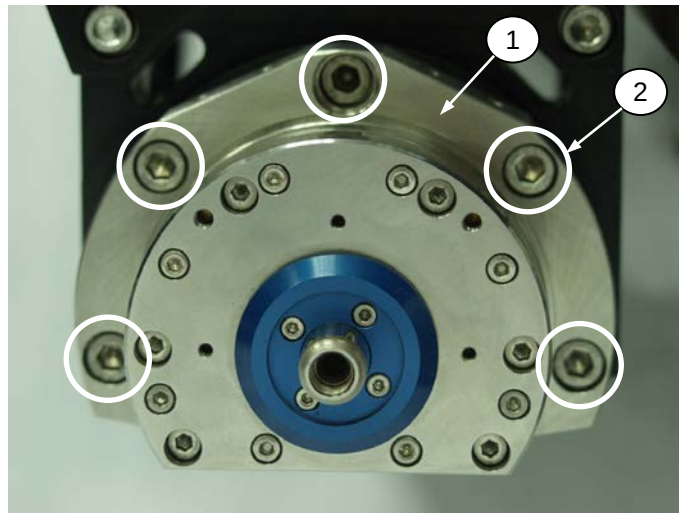
**Figure 7-5. Cooling Block**

9. Remove the Broken Blade Detector (BBD) from the cooling block (if installed) and place it in a safe location without disconnecting it from the BBD cable.
10. Remove the two (2) screws securing the cooling block to the spindle and remove the cooling block.



**Caution:** Place a protective covering beneath the spindle before removal.

11. Remove the five screws securing the spindle to the spindle holder.



1. Spindle
2. Spindle Holding Screws (X5)

**Figure 7-6. Spindle Bracket**

12. Half turn the spindle and carefully remove it from the holder.

### To install a new 2" spindle:



**Caution:** During installation, do not allow the spindle to come into contact with the height device, as this can damage the height device.

1. Position and secure the new spindle onto the spindle bracket, using the five spindle screws.
2. Reconnect the spindle cables and carbon brush cable; then lay it in the "snake-type" cable carrier and close it.
3. Reconnect the plugs and air and water tubes to the rear of the spindle.
4. Close the cable carrier bars.
5. Examine all tube connections and replace all panels.
6. Turn on the air and water supplies to the system and verify that there are no leaks in the tube connections.
7. Power up the system and verify that the new spindle works, before performing any further tests or adjustments.

### To test the new spindle:

1. Manually rotate the spindle wheel mount to verify smooth movement.
2. Operate the spindle at different speeds, ranging from the minimum to the maximum speeds:
  - a. From the Main Menu, press **Maintenance > Axis Setup > Spindle**.
  - b. In the Control Panel-Spindle field box, enter 1,000 and press **Set Speed**; the new spindle begins rotating at 1,000 rpm.
  - c. Repeat step b, entering values of 20,000, 40,000 and 60,000 into the field box.
3. Perform the Spindle Positioning Test (see following section 7.4.3).
4. Attach the cooling block to the spindle with its two securing screws.
5. Reconnect the water tubes to the cooling block.

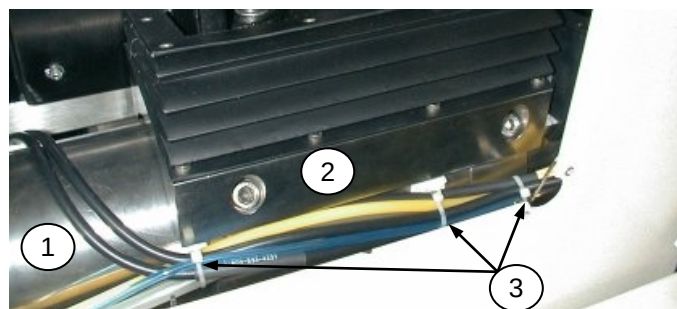


## 7.4.2 Replacing a 4" Spindle

Replacing a 4" Spindle involves removing the faulty Spindle with the Microscope and Bracket still attached, as these components are attached to the Spindle housing.

### To prepare the spindle for removal:

1. Remove the blade from the spindle and remove the Cooling Block.
2. Power down the system; the X and Y Axes move to the Home positions, allowing easy access to the spindle.
3. Turn off the water and air supplies.
4. Remove the upper-rear, lower left, and upper-left panels.
5. Open the spindle and workpiece covers.
6. Open the spindle "snake-type" cable carrier.
  - a. Disconnect the spindle cables from the spindle extension cable.
  - b. Disconnect the spindle carbon-brush cable from the Contact board.
7. Disconnect all plugs, and the air and water tubes from the rear of the spindle.
8. Remove the Maintenance Panel on the left side of the Spindle Cover.
9. Remove the eight screws securing the Z Casting Cover to the Z Casting.
10. Manually move the Y Axis all the way to the front of the System.
11. Remove the cables and tubes from the vision system:
  - a. Cut the plastic ties securing the Vision System Cables to the Spindle.

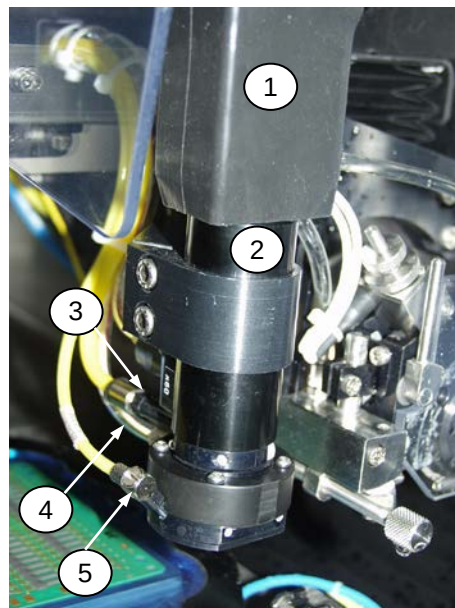


1. Spindle
2. Spindle Bracket
3. Plastic Ties

**Figure 7-7. Vision System Cables**

- b. Disconnect the Cable on top of the Camera





1. Camera
2. Microscope
3. Vertical Illumination Cable
4. Oblique Illumination Cable
5. Microscope Air Tube

**Figure 7-8. Microscope Illumination Cable**

- c. Disconnect the Illumination Cables from the Microscope (the cabling may vary according to the illumination type used).
- d. Disconnect the Cleaning Air Tube from the Microscope.

12. Disconnect the transparent water tubes from the cooling block.



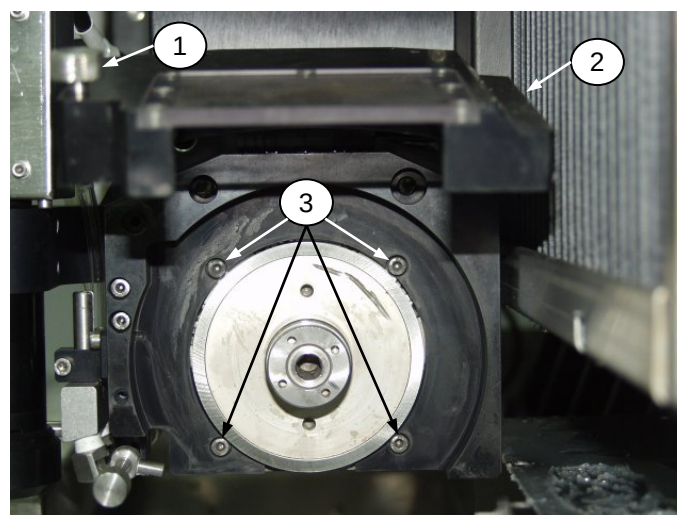
**Caution:** The 4" Spindle is heavy. Have someone hold the Spindle from underneath before performing the next step to prevent the Spindle, when released, from falling down and damaging the System.



**Note:** Cover the Chuck to protect it from possible mechanical contact.

### To remove the spindle

1. Remove the four screws securing the Cooling Block to the Spindle.

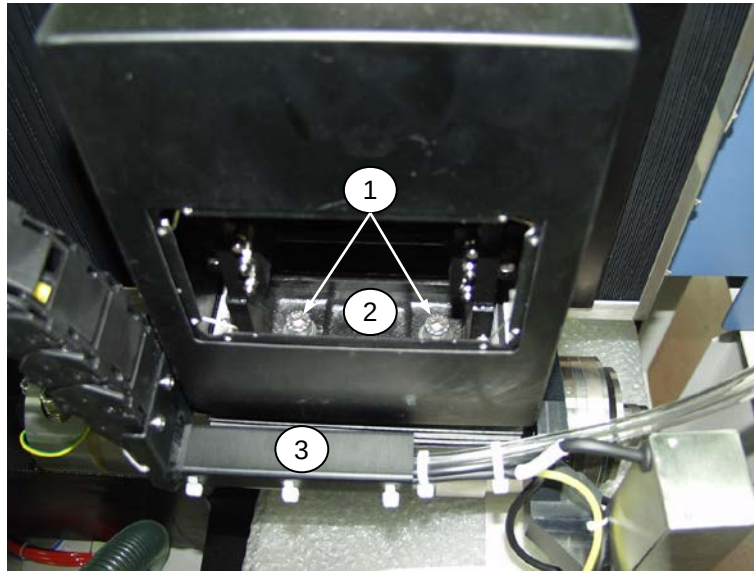


1. Cooling Block Thumbscrew
2. Cooling Block Door
3. Cooling Block Screws

**Figure 7-9. Cooling Block for 4" Spindle**

2. Remove the Cooling Block from the spindle.

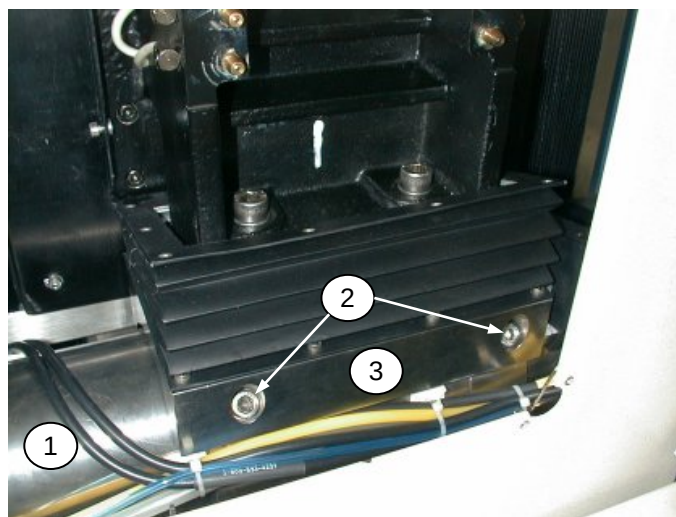
3. Remove the chain front-mount bracket holding the cables and tubes from the spindle housing.



1. Spindle Bracket Securing Screws
2. Z Casting
3. Chain Front Mount

**Figure 7-10. Spindle Bracket**

4. Remove the eight screws that secure the Z casting cover to the Z casting (see Figure 4-3 for reference).
5. Remove the two screws securing the spindle bracket to the Z casting. The spindle, together with the spindle bracket, Z cover, Z bellows and microscope are released.
6. Remove the four screws securing the Spindle to the spindle bracket (two, as shown in following Figure 7-11, and two on the opposite side of the Spindle), and remove the Spindle.



1. Spindle
2. Spindle Bracket Screws
3. Spindle Bracket

**Figure 7-11. Spindle and Spindle Bracket**

### To install a 4" new spindle:



**Caution:** During installation, do not allow the spindle to come into contact with the height device, as this can damage the height device.

1. Secure the spindle to the spindle bracket using the four screws that were removed earlier.
2. Secure the spindle bracket to the Z casting using the two screws removed earlier.
3. Connect the Z casting cover with the bellows.
4. Reconnect the chain front mount.
5. Reconnect the yellow air tube to the left side of the microscope.
6. Reconnect the illumination cables (the cabling may vary according to the illumination type used) to the microscope.
7. Reconnect the cable to the top of the camera.
8. Secure the vision system cables to the spindle using plastic ties.
9. Manually move the Y-axis all the way to the back of the System (the Home position).
10. Reconnect all plugs and air and water tubes to the rear of the spindle.
11. Turn on the air and water supplies to the system and verify that there are no leaks in the tube connections.
12. Power up the system and verify that the new spindle works, before performing any further tests or adjustments.

### To test the new spindle:

1. Manually rotate the spindle wheel mount to verify smooth movement.
2. Operate the spindle at different speeds, ranging from the minimum to the maximum speeds:
  - a. From the Main Menu, press **Maintenance > Axis Setup > Spindle**.
  - b. In the Control Panel-Spindle field box, enter 1,000 and press **Set Speed**; the new spindle begins rotating at 1,000 rpm.
  - c. Repeat step b, entering values of 20,000 and 30,000 into the field box.
3. Perform the Spindle Positioning Test (see following section 7.4.3).
4. Secure the cooling block to the spindle using the four screws that were removed earlier.
5. Reconnect the water tube to the cooling block.

### 7.4.3 Testing the Spindle Position

The purpose of the test is to determine if the spindle is exactly parallel to the X Axis along the horizontal plane.

Testing the Spindle Position includes:

- Correcting Spindle Parallelism to the X Axis.
- Testing the Spindle Runout.

Use the following tools:

- Measuring Probe.
- Perpendicularity Block.
- Magnetic Stand.



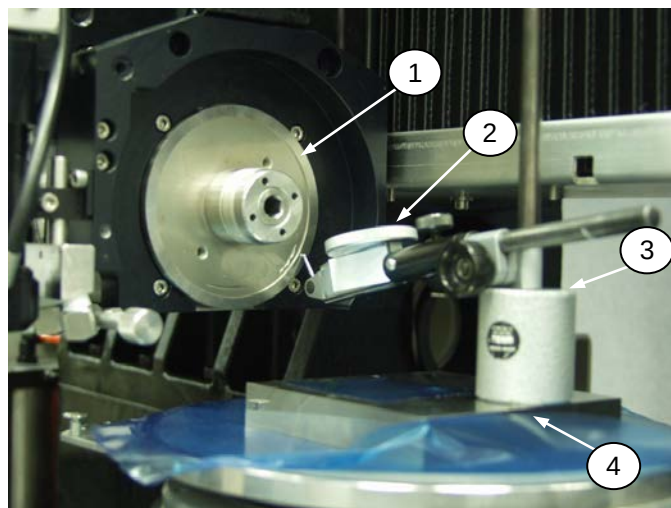
**Note:** Wear cotton gloves when handling the Perpendicularity Block to prevent leaving residue that can interfere with the measurements.

**To verify the new spindle is horizontally parallel to the X Axis:**

1. Clean the wheel mount with isopropyl alcohol.
2. Mount tape on a designated frame and place it on the cutting chuck.
3. Mount the Perpendicularity Block onto the chuck, on top of the tape.



**Note:** The images below are for the purpose of reference only and do not reflect the spindles used in the 7910 system.

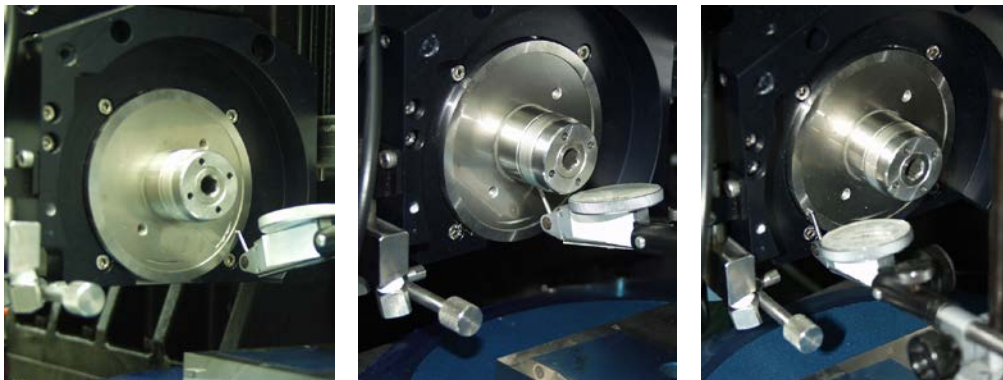


1. Wheel Mount Rim
2. Measuring Probe
3. Magnetic Stand
4. Perpendicularity Block

**Figure 7-12. Wheel Mount Rim and Measuring Probe**

4. Attach the Measuring Probe to the Magnetic Stand and place them on the Perpendicularity Block such that the Measuring Probe lightly touches the outer rim of the wheel mount at a point on the rim that is just below the line of the spindle axis. The Measuring Probe should be situated such that when the Measuring Probe moves, it passes just below the spindle shaft.
5. Set the Measuring Probe to 0.
6. Position the X Axis:

- a. From the Main Menu, press **Dice > Axis Control**; the Dice window appears in the Axis Control view.
- b. Using the X/Y Axis controls, move the X Axis from a point just under the line of the spindle axis on one side of the wheel mount rim to a point just under the line on the other side of the wheel mount rim.



**Figure 7-13. Testing Horizontal Parallelism of the Spindle to the X-Axis**

7. Verify that the gap between the two points does not exceed 2.5 microns. If the test result does exceed the specification, correct the spindle parallelism to the X Axis as described in following section 7.4.3.1.
8. Disconnect the Measuring Probe and Magnetic Stand from the system.
9. Remove the Perpendicularity Block and the tape from the system.



### 7.4.3.1 Correcting Spindle Parallelism to the X Axis

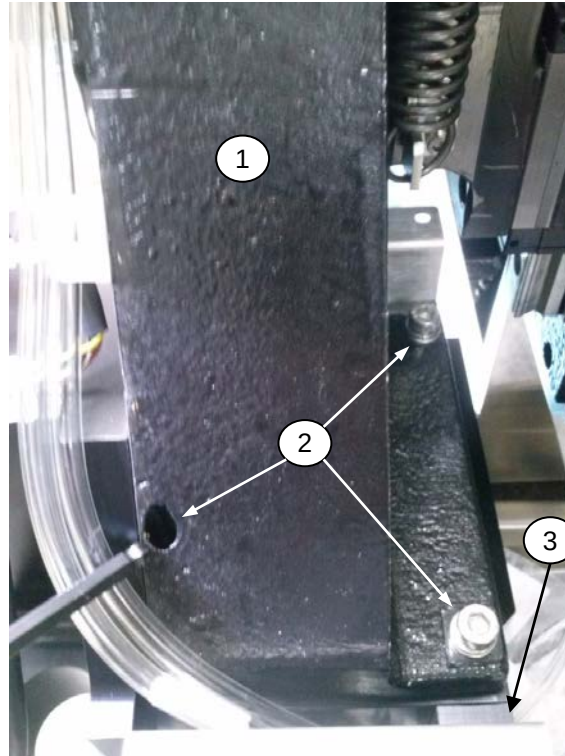
When a deviation measured while testing the spindle position is greater than the 5 microns specified, correct the parallelism.

**To correct parallelism of the spindle to the X Axis:**



**Note:** Before performing the steps in this procedure, perform steps 2 to 7 in section 7.4.3 above.

1. Slightly loosen the two front screws and one rear screw, connecting the spindle bracket to the Z-axis casting.



1. Z-axis Casting
2. Spindle Holder Mounting Screws (4<sup>th</sup> not visible)
3. Spindle Bracket

**Figure 7-14. Spindle Holder Mounting Screws**

2. Adjust the spindle parallelism using the rear screw as a pivot.
3. Retighten all four screws.
4. Retest the parallelism (see section 7.4.3 above).



**Note:** If the results of the second test are still unsatisfactory, contact your local ADT representative.

### 7.4.3.2 Testing the Spindle Runout

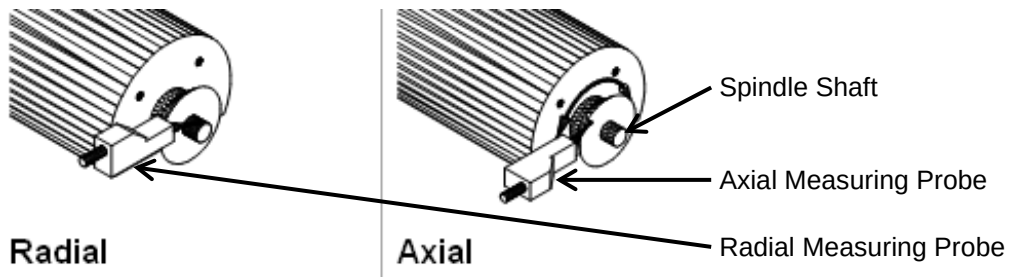
The Radial Runout test determines whether the new spindle rotates within the specifications.



**Caution:** The Micrometer (Measuring Probe) must be set to zero when performing these tests. Be sure to wait until the gauge has settled before beginning each test.

**To perform the radial runout accuracy test:**

1. Clean the spindle shaft thoroughly with isopropyl alcohol (this is important, as a particle of dirt on the spindle shaft may be larger than the 1-micron variation allowed by the test).
2. Attach the Measuring Probe Cable Connector to the receptacle on the Measuring Probe.
3. Place a piece of tape on the underside of the table jig so that it does not come into direct contact with the theta table.
4. Attach the Measuring Probe Arm to the Magnetic Table Jig.
5. Position the Magnetic Table Jig on the theta table so that the Measuring Probe is perpendicular to the spindle shaft.
6. Position the Measuring Probe so that it lightly touches the wheel mount hub at a point just behind the tightening nut.



**Figure 7-15. Measuring Probe on Spindle Shaft**

7. Set the Measuring Probe to zero.
8. Rotate the spindle shaft manually to determine whether the deviation meets the 1-micron specification.



**Note:** If the deviation is greater than the specified 1 micron, contact your local ADT service representative.

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## Chapter 8. Maintaining the Axes

Maintaining the Axes describes the procedures for replacing axes motors, bellow assemblies, drivers and controllers, and the tests and adjustments to perform after installing these new parts.

Read all of the instructions carefully before beginning, and allow sufficient time to complete all the tests and adjustments after installation.



**Caution:** These procedures should not be attempted by untrained or inexperienced personnel.

Maintaining the Axes includes:

- ❖ Describing the Axis Assemblies (section 8.1).
- ❖ Troubleshooting the Axes (section 8.2).
- ❖ Maintaining the Axes Drivers (section 8.3).
- ❖ Replacing the Controller (section 8.4).
- ❖ Removing and Installing the X Axis Motor (section 8.5).
- ❖ Removing and Installing the Y Axis Motor (section 8.6).
- ❖ Removing and Installing the Z Axis Motor (section 8.7).
- ❖ Maintaining the Theta Axis (section 8.8).
- ❖ Calibrating the Sine & Cosine Parameters (section 8.9).



**Caution:**

- Be sure to keep the axes and surrounding area free of cutting materials dust and dirt.
- After powering down the system, always wait at least 10 minutes before commencing any maintenance procedures, to enable any active current to dissipate from the system.

## 8.1 Describing the Axis Assemblies

The system is composed of the following axis assemblies:

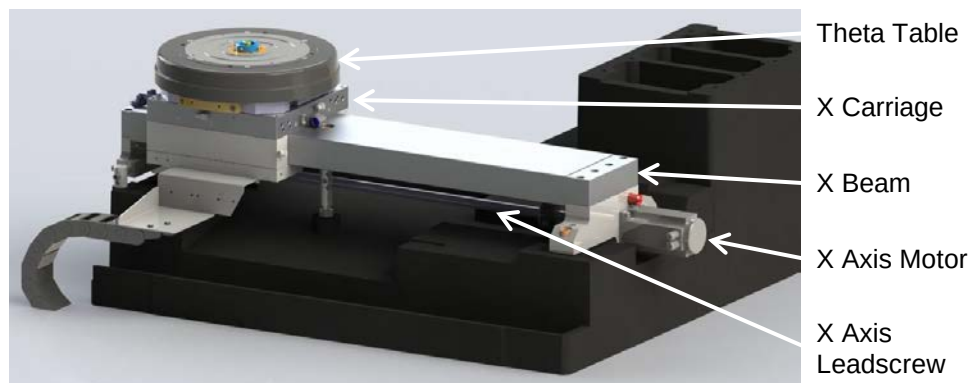
- X Axis Assembly.
- Y Axis Assembly.
- Z Axis Assembly
- Theta Axis Assembly

### 8.1.1 X Axis Assembly

The X Axis assembly moves the cutting chuck left and right. It is composed of the following main subassemblies:

- X Casting.
- X Beam.
- X Carriage.
- Brushless DC Motor.
- Bellows Assembly.

The X Carriage moves along the X Beam. The Theta Axis, which enables the cutting chuck's rotation movement, is installed on the X Carriage. The bellows assemblies prevent water from reaching the inner parts of the system. Any excess water that collects on the bellows runs off into the drainage pan.



**Figure 8-1. X Axis Assembly**

When a problem occurs, first check the X Axis motor driver and controller, and replace them if necessary before considering replacing the X Axis motor; regardless of the error message displayed in the application.

The reason for this is that the X Axis motor driver is responsible for sending error messages to the application. If the X Axis motor driver is defective, it can send an inaccurate message.



**Note:** The X Axis is supplied with airflow to enable smooth movement. To adjust the X Axis airflow, see section 11.8.



**Caution:** Never move the X Axis without having verified the correct air supply. An incorrect air supply can cause permanent damage to the system.

## 8.1.2 Y Axis Assembly

The Y Axis assembly moves the spindle horizontally toward and away from the user. It is composed of the following main subassemblies:

- Stepper Motor.
- Y Base.
- Y Casting.
- Y Axis Leadscrew.
- Y Axis Bellows.
- Y Axis Guides (upper, lower).
- Linear Encoder.

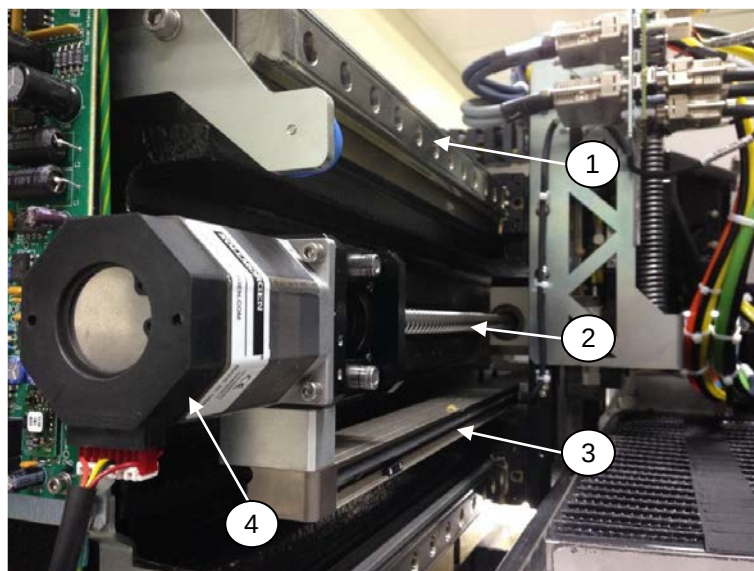


**Note:** The Linear-Encoder scale is connected to the Y central plate. The encoder head is attached to the Y-Casting, enabling recognizing the location of Y Axis.

The Y Axis moves independently along the Y Axis guides, which are fitted to the Y Casting. The stepper motor is mounted between the two guides on the Y Casting.

When a problem occurs, first check the Y Axis motor driver and controller, and replace them if necessary before considering replacing the Y Axis motor, regardless of the error message displayed in the application.

The reason for this is that the Y Axis motor driver is responsible for sending error messages to the application. If the Y Axis motor driver is defective, it can send inaccurate messages.



1. Y Axes Upper Guide
2. Y Axes Leadscrew
3. Y Axes Linear Encoder
4. Stepper Motor

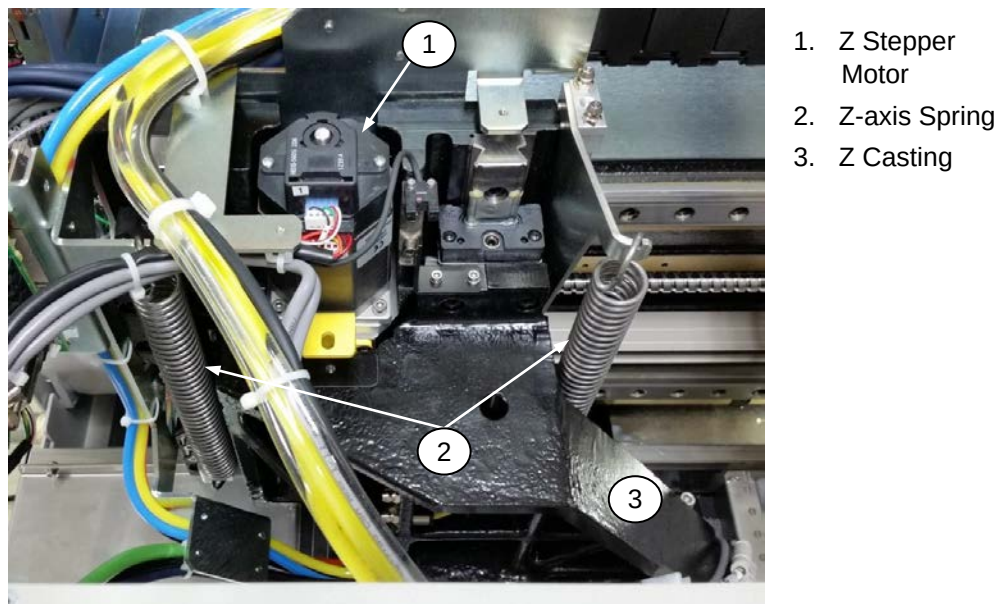
Figure 8-2. Y Axis

### 8.1.3 Z Axis Assembly

The Z Axis assembly moves the spindles vertically towards and away from the cutting chuck. The Z Axis assembly is comprised of the following subassemblies:

- Stepper Motor
- Z Axis Springs
- Z Axis Guides
- Z Axis Leadscrew
- Z Axis Casting

The Z Axes move independently along the Z Axis guides, which are fitted to the Y Axis. The stepper motor is mounted between the two guides on the Z Casting.



**Figure 8-3. Z Axis**

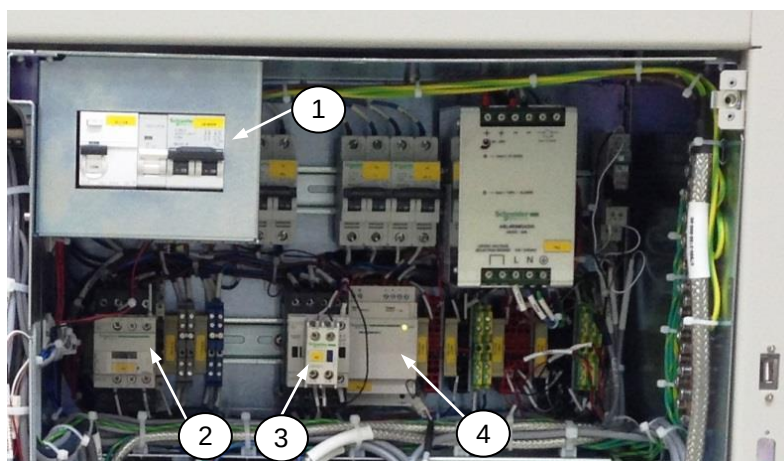
When a problem occurs, check the Z Axis motor driver and controller, and replace them if necessary before considering replacing the Z Axis motor, regardless of the error message displayed in the application. The reason for this is that the Z Axis motor driver is responsible for sending error messages to the application. If the Z Axis motor driver is defective, it can send inaccurate messages.

## 8.2 Troubleshooting the Axes

When an axis does not function properly, there are several steps to perform to determine the source of the malfunction. By performing these steps, it can be determined if the problem is in the power supply, the controller, the axis motor driver, or the axis motor.

### To troubleshoot the axes:

1. Open the bottom-front panel to access the power supply and the electronic bench.
  - a. Examine the main circuit breakers.
  - b. Examine the circuit breakers.
  - c. Turn on (move to the ON position) any circuit breakers that have turned off.



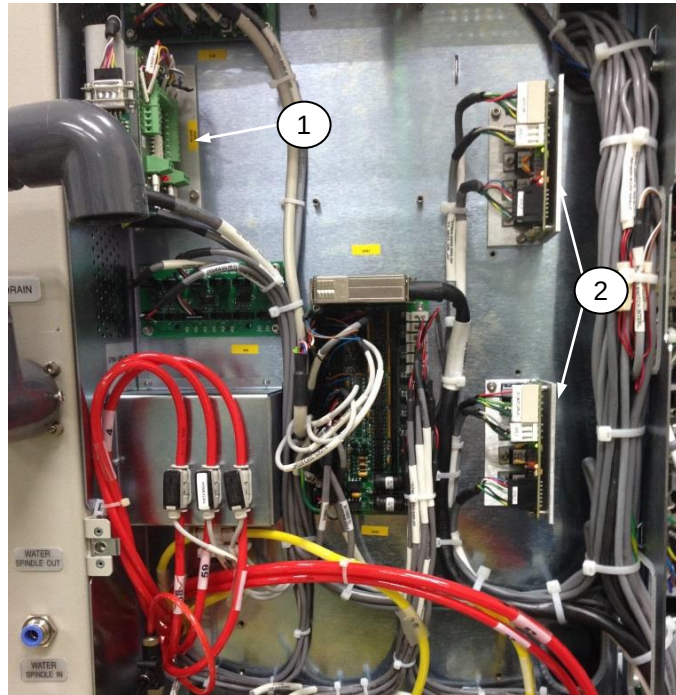
1. Main Power Supply (PS1)
2. Contactor (K1)
3. Contactor (K2)
4. Emergency Stop Power Supply (PS2)

**Figure 8-4. Power Supply**

1. Power down the system, then power up the system.
2. Examine the power supply to the following units:
  - a. Power Supply 1 (PS1): Green LED indicates proper function.
  - b. Emergency Stop Power Supply 2 (PS2): Green LED indicates proper function.
  - c. K1 and K2 Contactors: A red marker indicates that the specific contactor is not conveying power.
3. Verify the status of the axis motor drivers and controllers, as follows:
  - a. Controllers: Green LEDs (two per controller) indicate proper function.
  - b. Regenerative Unit: Red LED indicates a malfunction.
  - c. X Motor Driver: Red LED indicates a malfunction.
  - d. Spindle Driver: Red LED indicates a malfunction.



- e. Y, Z, and Theta Drivers: Each driver has two LEDs. On the left, a red LED and on the right, a green LED. The red LED, appearing alone, indicates a malfunction. After the software is initialized, the red LED should turn to amber, indicating proper functioning.
4. Examine the Y, Z, and/or Theta Motor driver fuse with a digital voltmeter. Replace the fuse if necessary.



- 1. Theta Motor Driver
- 2. Y and Z Axes Motor Drivers

**Figure 8-5. Y and Z Motor Driver Fuse**

5. Examine the Y and Z motor drivers by swapping them. The Y and Z motor drivers are interchangeable, provided the resistors are in the proper place. For information about removing and replacing these drivers, see section 8.3.4.
6. After determining that all drivers are functioning normally, try replacing the controller as described in section 8.4.
7. Replace the axis motor only after eliminating the possibility of problems with the power supply, driver, and controller.



**Note:** For information on replacing the X Axis motor, see section 8.5. For information on replacing the Y Axis Motor, see section 8.6. For information on replacing the Z Axis Motor, refer to section 8.7.

## 8.3 Maintaining the Axes Drivers

The axes drivers are located in the electronic bench at the rear of the system. Each axis driver controls the movement of a single axis motor.

### 8.3.1 Describing the X Axis Motor Driver

The X Axis motor driver is a DC motor driver located on the central-left side of the electronic bench.



1. X Axis  
Motor Driver

Figure 8-6. X Axis Motor Driver

The X Axis motor driver contains a single LED that indicates the status of the driver, as follows:

Table 8-1. X Axis Motor Driver LED

| Color            | Indication                                   |
|------------------|----------------------------------------------|
| Green (blinking) | X Axis motor driver is receiving power.      |
| Green (steady)   | X Axis motor driver is functioning properly. |
| Red              | X Axis motor driver is malfunctioning.       |

## 8.3.2 Replacing the X Axis Motor Driver

When the red LED appears, the X Axis motor driver should be replaced. If the red LED does not appear, but the X Axis motor does not function properly, replace the X Axis motor driver (or the controller) before replacing the X Axis motor itself.

### To replace the X Axis motor driver:

1. Power down the system.
2. Open the rear panel to access the electronic bench.
3. Locate the faulty X Axis motor driver (see Figure 8-6), and disconnect all wires and cables.



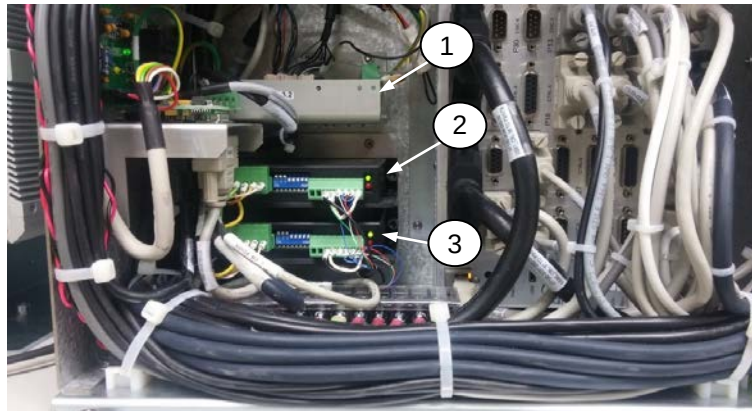
**Note:** Remove the complete plug unit; not the separate wires.

4. Remove the screws securing the X Axis motor driver frame, and pull the driver frame towards you.
5. Install the new X Axis motor driver into the frame and push the driver frame back into place.
6. Replace the screws that secure the driver frame to the electronic bench.
7. Reconnect all cables and wires (examine the labels to ensure proper placement).
8. Close the rear panel.
9. Power up the system.
10. Perform X Sin Offset (see section 8.9).
11. Test the axis.



### 8.3.3 Maintaining the Axes Motor Drivers

The Y-axis and Z-axis drivers are stepper motor drivers, and are located in the left section of the electronic bench. The drivers are interchangeable.



1. Theta Motor Driver
2. Y Axis Motor Drivers
3. Z Axis Motor Drivers

**Figure 8-7. Y, Z, and Theta Axis Motor Drivers**

The Closed Loop Theta axis has a separate driver.

The Y and Z motor drivers each contain two LEDs that indicate the status of the drivers, as follows:

**Table 8-2. Y and Z Axes Motor Driver LED**

| Left  | Right | Indication                                                                          |
|-------|-------|-------------------------------------------------------------------------------------|
| Red   | Green | Axis motor driver is receiving power (24V).                                         |
| Amber | Green | Application has been initialized and the Axis motor driver is functioning normally. |
| Red   | Off   | Motor driver is functioning normally.                                               |

### 8.3.4 Replacing the Y and Z Axes Motor Drivers

When a motor driver red LED appears alone, the specific axis motor driver should be replaced. If the red LED does not appear alone, but the axis motor still does not function properly, replace the specific axis motor driver, or the controller, before considering replacing the axis motor itself.

**To replace the Y and Z Axes motor drivers:**



**Note:** Ensure that the resistors on the driver are moved to the correct position for the specific axis.

1. Power down the system.
2. Open the rear panel to access the electronic bench.
3. Locate the faulty axis motor driver (see Figure 8-7), and disconnect the three plugs.
4. Remove the two screws securing the axis motor driver and heat sink, and remove the driver (including the heat sink) from the system.



**Note:** If the new axis motor driver does not include a heat sink, remove the heat sink from the old axis motor driver by removing the four screws on top of the driver, as well as the screws that connect each transistor and its insulation pad to the back of the heat sink. Secure the new axis motor driver to the heat sink, making sure to place an insulation pad behind each transistor.

5. Install the new axis motor driver and heat sink into the system, using the screws that were removed earlier.
6. Reconnect the three plugs to the axis motor driver.
7. Close the rear panel.
8. Power up the system.
9. Perform the Y Sine Offset procedure (see section 8.9).
10. Test the axes.

## 8.4 Replacing the Controller

The controller is located in the lower-right corner of the electronic bench, directly below the spindle driver. The controller links between the Main System Controller (MSC) and the system component; as controlling the movement of the axes, operating the solenoids, reading the sensors, etc.

The controller has two Status LEDs and a Reset button. When the controller is functioning normally, the LEDs are green.

If there is a communication problem between the controller and the system, press the **Reset** button to reboot the controller.

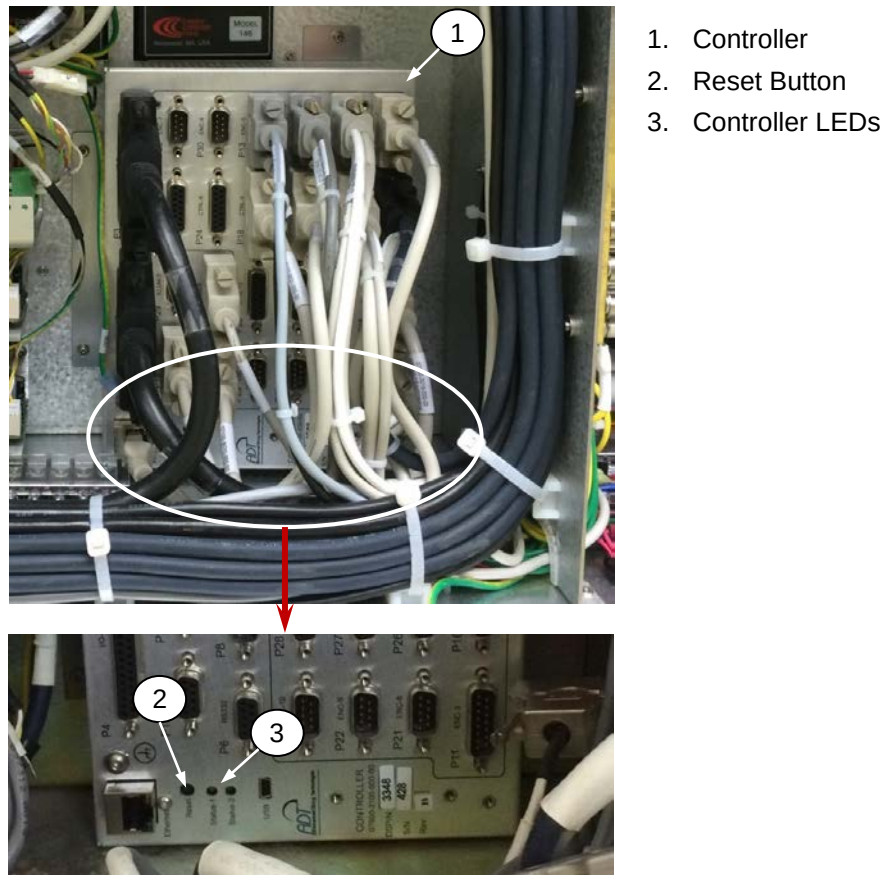


Figure 8-8. Controller

The controller is connected to the MSC, with an Ethernet cable, from the controller Ethernet socket to the bottom communication socket of the MSC.

If there is a problem with an axis, and the driver is not the source of the problem, replace the controller before replacing the axis motor.



**Note:** When replacing an old type of controller with the new type, the NCH board also needs to be replaced and additional software setups are required. For details, see the "Old to New Controller Replacing Instructions", P/N 97120-0987-000-00.

**To replace a controller:**

1. Power down the system.
2. Open the front door to gain access to the electronic bench.
3. Locate the controller and disconnect all of the plugs connected to the controller (note that the plugs are labeled), including the ground cables.
4. Remove the three screws that connect the rear plate of the controller to the electronic bench, and remove the controller.
5. Remove the four screws that secure the rear plate to the back of the controller and remove the rear plate.
6. Secure the rear plate to the back of the new controller, using the four screws used in the previous step.
7. Install the new controller into the electronic bench by replacing the three screws that were previously removed.
8. Reconnect all the removed plugs to the controller. Examine the labels and ensure proper placement.
9. Close the electronic bench door.
10. Power up the system.
11. In the application, initialize the controller by confirming software upgrade.

## 8.5 Removing and Installing the X Axis Motor

The X Axis (servo-) motor is connected by a coupling to the X Axis leadscrew. The motor is located on the right side of the system.

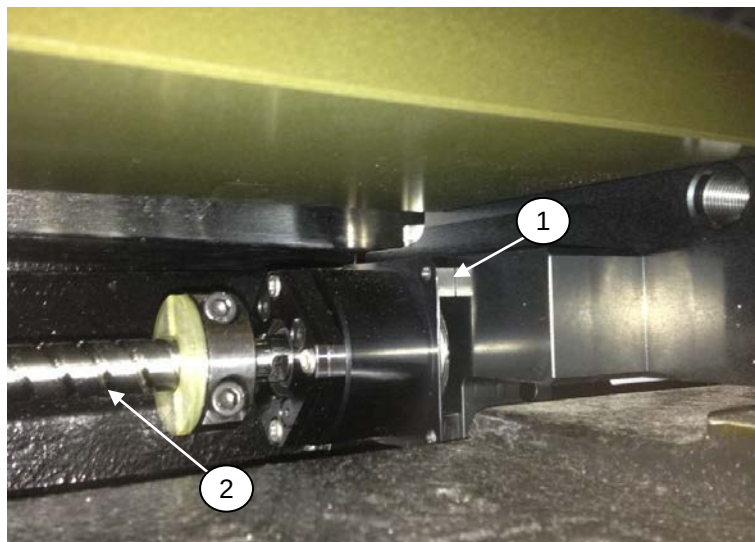
Replace the X Axis motor only after verifying that the driver and controller are not the source of the malfunction.

### To remove the X Axis motor:

1. Power down the system (make sure that the air supply to the system remains on).
2. Remove the upper-right and front panels.
3. Release the X Bellows by removing the four screws that secure the bellows to the X Carriage.
4. Disconnect the X motor plug, located behind the front-right corner of the bench frame.
5. Manually slide the X Carriage to the left side of the system, until the first setscrew is accessible.
6. Using a 2.5 mm hex wrench (Allen key #4 for M3 screw), loosen the screw that holds the motor coupling (access is through the hole on the motor bracket).
7. Move the X Carriage until the second screw is accessible.



**Note:** Move the X Carriage to manually rotate the leadscrew until the each screw aligns with the coupling hole.



1. Access Hole to screw (Securing Motor Coupling)
2. X Leadscrew

**Figure 8-9. X Leadscrew**

8. Remove the four screws securing the X Axis motor in place.
9. Remove the motor.

**To install the X Axis motor:**

1. Insert the replacement X Axis motor into the system.
2. Insert and turn the four screws that secure the motor into place but do not tighten the screws.
3. Tighten the two setscrews in the motor coupling.
4. Tighten the four screws that secure the motor.
5. Connect the P10 and X motor plugs behind the front-right corner of the bench frame.
6. Power up the system.
7. Test the new X Axis motor from the application:
  - a. In the Main Menu, press **Dice > Axis Control**.
  - b. Use the X/Y Axis Controls to move the X Axis.
8. Perform the Sine and Cosine Offset procedure (see section 8.9).
9. Test the axis.

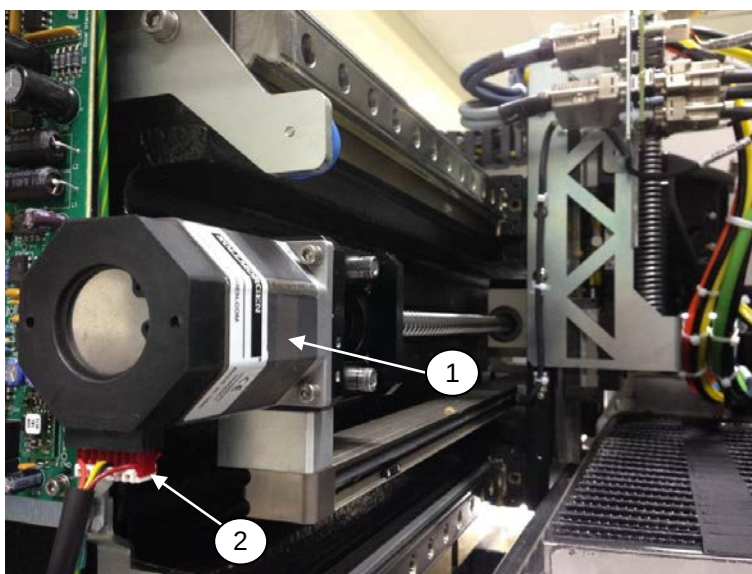
## 8.6 Removing and Installing the Y Axis Motor

The Y Axis motor is a stepper motor connected by couplings to the Y Axis leadscrew. The Y Axis motor is located at the back of the system.

The Y Axis motor should be replaced only after verifying that the driver and controller is not the source of the malfunction.

### To remove the Y Axis motor:

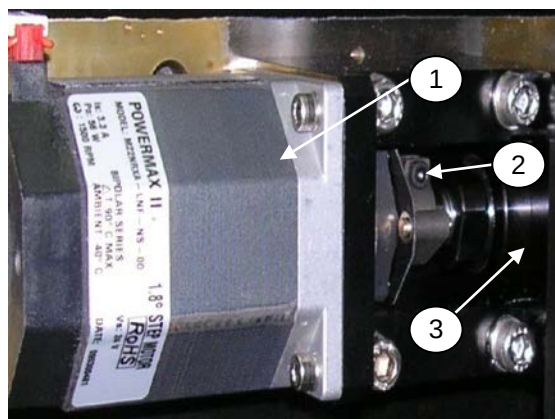
1. Power down the system.
2. Remove the rear-upper panel cover.
3. Disconnect the Y motor plug.



1. Y Axis Motor
2. Y Motor Plug

Figure 8-10. Y Axis Motor

4. Using a 2.5 mm hex wrench, loosen the screw on the motor coupling of the motor (if necessary, move the Y Axis until the screw is lined up with the hole in the coupling and is accessible).



1. Y Axis Motor
2. Setscrew
3. Motor Coupling

Figure 8-11. Y Axis Motor Coupling

5. Remove the four screws holding the Y Axis motor to the coupling bracket.
6. Remove the Y Axis motor by sliding it towards you.

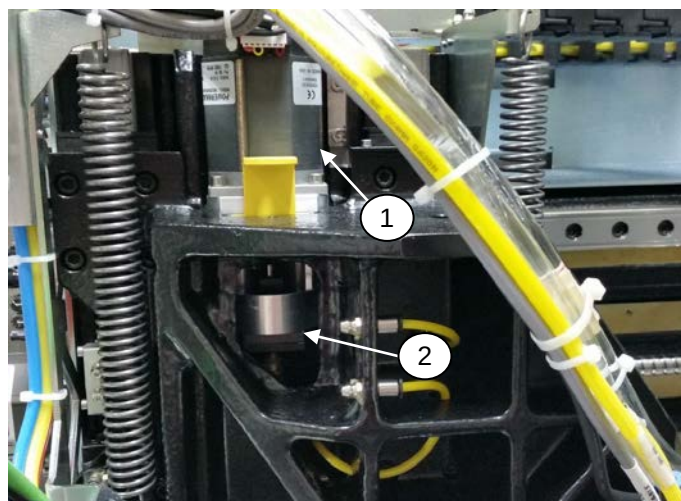
**To install the Y Axis motor:**

1. Slide the replacement Y Axis motor into place.
2. Replace the four screws that secure the motor to the coupling bracket but do not tighten them.
3. Tighten the screw that secures the coupling to the motor (if necessary, manually move the Y Axis until the screw is accessible).
4. Tighten the four screws that secure the motor to the coupling bracket.
5. Connect the Y motor plug above the motor coupling.
6. Power up the system.
7. Test the new Y Axis motor from the application:
  - a. In the Main Menu, press **Dice > Axis Control**.
  - b. Use the X/Y axis controls to move the Y Axis.
8. Perform Sine and Cosine Offset procedures (see section 8.9).
9. Test the axis.



## 8.7 Removing and Installing the Z Axis Motor

The Z Axis motor is a stepper motors connected by a coupling to the Z Axis leadscrew. The Z Axis motor is located on top of the Z Axis.



1. Z Axis Motor
2. Z Axis Coupling

Figure 8-12. Z Axis Motor]

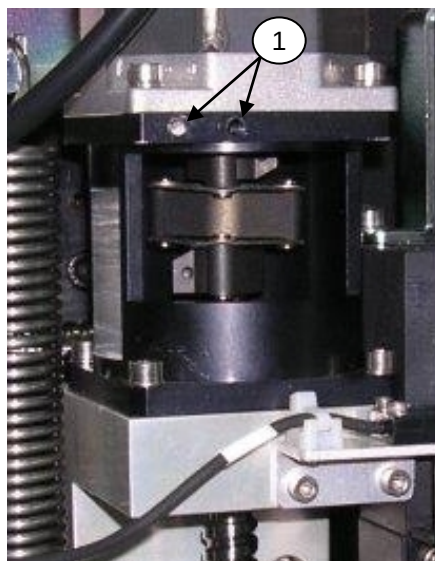
The Z Axis motor should be replaced only after verifying that the driver and controller is not the source of the malfunction.

### To remove a Z-axis motor:

1. Power down the system; the Z Axis motor automatically moves to its highest position and then moves slightly down by gravity.
2. Remove the top panel.
3. Position a supporting device beneath the spindles and lift the Z Axis motor until the screw in the coupling is visible.



**Caution:** When loosening the coupling screw, gravity tends to slowly pull the spindle downwards. Verify that the spindle is securely supported before loosening the screws.



1. Screws

Figure 8-13. Z Motor Coupling Screws

4. Loosen the coupling screw (on the motor side) and disconnect the Z Axis motor plug.
5. Remove the four screws which secure the Z Axis motor to the Z casting and remove the motor.

**To install the new Z Axis motor:**

1. Put the replacement Z Axis motor into place on top of the Z casting.
2. Tighten the coupling screw.
3. Replace and tighten the four screws which secure the Z-axis motor to the Z casting.
4. Connect the Z Axis motor plug.
5. Remove the spindle supports.
6. Power up the system.
7. Test the new Z Axis motor from the application:
  - a. In the Main Menu, press **Dice > Axis Control**.
  - b. Use the Z/T Axis Controls to move the Z Axis.
8. Perform Sine and Cos Offset (see section 8.9).
9. Test the axis.

### 8.7.1 Testing for Z Axis Repeatability

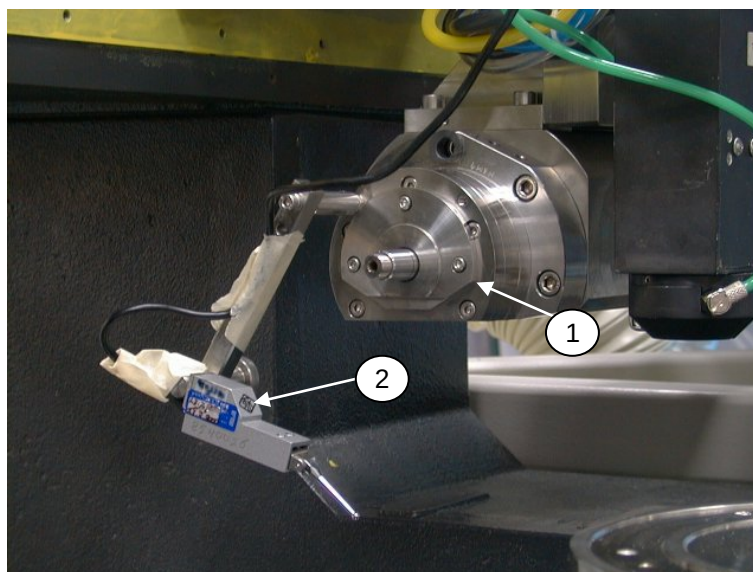
After installing a new Z Axis motor, confirm Z Axis repeatability accuracy.

**To test Z Axis repeatability:**

1. Remove the blade from the spindle (for removing a blade, refer to the Operations manual).
2. Disconnect the tubes from the cooling block.
3. Open the screws attaching the cooling block to the spindle, and remove the cooling block.
4. Attach the measuring probe to the jig.
5. Attach the jig to the left screw hole on the front of the spindle (see Figure 8-14).



**Note:** Figure 8-14 is not an entirely accurate depiction of the 71XX Y Spindle Assembly and is for reference only.



- 1. Spindle
- 2. Jig

**Figure 8-14. Jig Fitted to Spindle**

6. In the application, from the Main Menu, press **Dice > Axis Control**.
  - a. Use the Z/T axis controls to position the Z Axis 7 mm above the chuck.
  - b. Position the measuring probe so that it lightly touches the top surface of the cutting chuck and set the measuring system to **0**.
7. Use the application to raise and lower the spindle fifty times by a distance of 1.0 mm:
  - a. From the Main Menu, press **Maintenance > Calibration**; the Maintenance-Calibration window appears.
  - b. Select **Z**; the Z view, including Control Panel-Z, appears.
  - c. Press **Teach**; the Teach window appears.
    - i. In the **Start** field, enter the start position (7 mm from the chuck).
    - ii. In the **Stop** field, enter the stop position (16 mm from the chuck).
    - iii. In the **Step** field, enter the steps (1).



**Note:** To calculate the precise Start and Stop positions, use the Z Axes parameters as reference.

- d. Press **OK**.
  - e. Press the **Test** button (located above the Teach button); the Test window appears.
  - f. Enter the following values in the fields provided:
    - ♦ Cycle Number: 100
    - ♦ Delay (sec): 2
  - g. Press **Start**.
8. Verify that, for 100 repetitions, the position error (minimum to maximum) does not exceed 1.5 microns.



**Note:** If the deviation is greater than 1.5 microns, call your local ADT representative for assistance.

9. Remove the long, horizontal jig from the spindle.
10. Reattach the cooling block to the spindle with the two screws on the cooling block.
11. Reattach the blade.

## 8.8 Maintaining the Theta Axis

The system utilizes the Closed Loop (Linear Encoder) Theta Axis turntable. Maintaining the Theta Axis includes the following:

- Maintaining the Closed Loop Theta Linear Encoder/Scale Surface
- Cleaning the Closed Loop Linear Encoder Optical Aperture and Scale

### 8.8.1 Maintaining the Closed Loop Theta Linear Encoder/Scale Surface

It is important to keep the Closed Loop Linear Encoder/Scale surface clean.

On a quarterly basis:

- Inspect the Closed Loop Linear Encoder Optical Aperture and Scale.
- Clean parts when debris has collected on the Closed Loop Linear Encoder and Scale.

#### 8.8.1.1 Cleaning the Closed Loop Linear Encoder Optical Aperture and Scale

The Closed Loop Linear Encoder comprises:

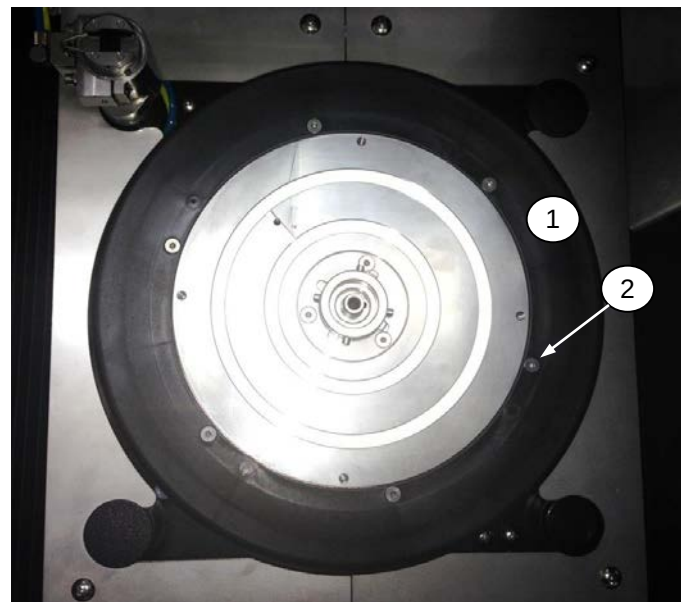
- Reading Head.
- Scale (glued to the table surface).

When cleaning:

- Use an airline brush to remove coarse and abrasive particles.
- Complete the cleaning process by gently wiping the surfaces with a clean cloth and isopropyl alcohol.

#### To clean the scale:

1. Stop system operation.
2. Slide-open the transparent loading cover.
3. Remove the chuck from the theta table (see section 9.3.1).
4. Remove the six screws that secure the plastic skirt covering the theta table.
5. Remove the plastic skirt cover.



1. Turntable Plastic Skirt
2. Connecting Screws (×6)

**Figure 8-15. Theta Table Cover**

6. Clean the scale as described above.

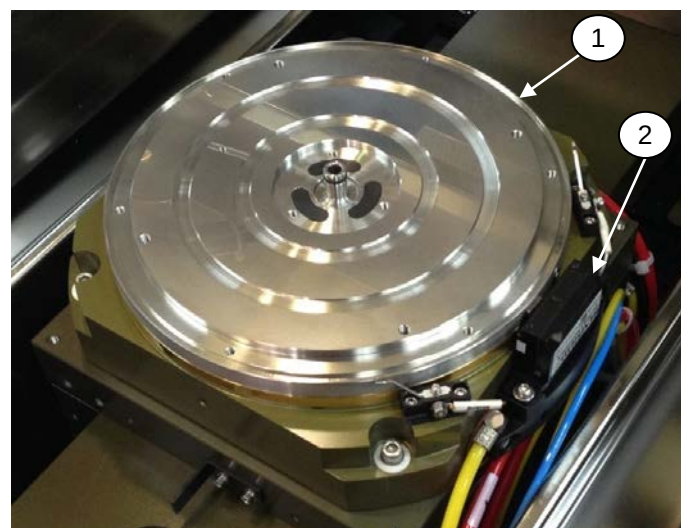


**Note:** Make sure that the rubber O-ring (gasket) is in place. Add a small amount of theta table grease to ensure the seal. For the correct grease, contact your ADT representative.

7. Replace the plastic skirt cover and secure it with the screws.
8. Initialize the theta table.
9. Ensure that theta table full travel is smooth.

**To adjust the Renishaw Reading Head:**

1. Remove the plastic skirt cover.



1. Theta Table
2. Renishaw Reader Head

**Figure 8-16. Theta Table with Renishaw Reader Head**

2. Check the sensor color on the Renishaw Reading Head:
  - Green: properly adjusted.
  - Red: must be readjusted.



**Note:** From the application, rotate the Theta table to verify correct adjustment throughout the entire axis travel. Make sure that the green light stays on throughout the duration of the Theta rotation

3. If there is the need for adjustment, open (half turn) both screws holding the Renishaw Reading Head.
4. Adjust the Reading Head position until the sensor turns green.
5. From the application, rotate the Theta table again.



**Note:** The green light can blink when reading the tape position.

6. Tighten the screws.
7. Replace the cover and secure it with the screws.
8. Initialize the Theta Table.
9. Ensure that full Theta Table travel is smooth.

## 8.9 Calibrating the Sine & Cosine Parameters for Y and Z Axes

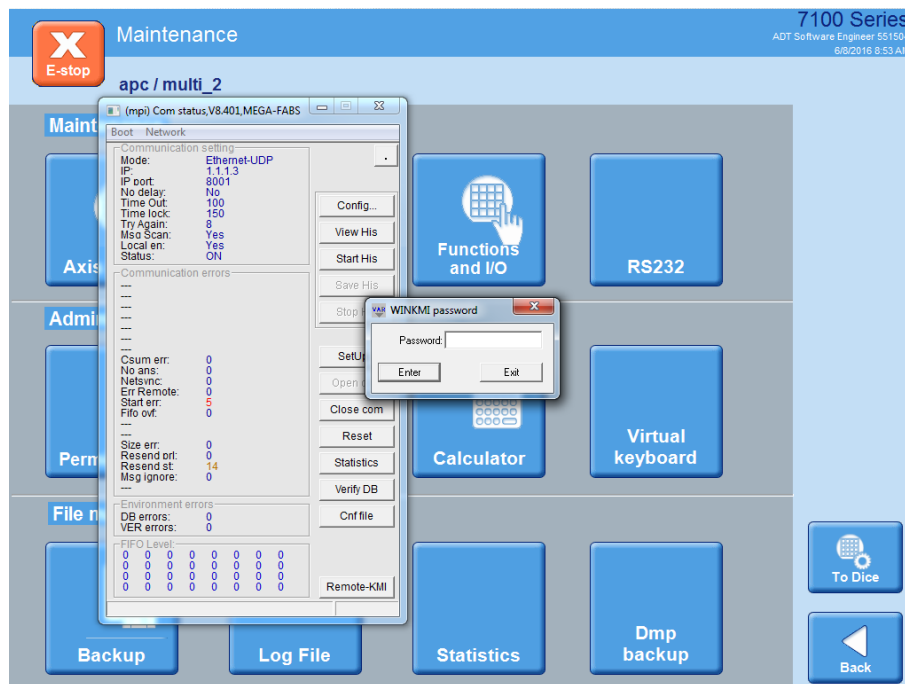
The Axes motor drivers should be calibrated when the driver is replaced. This calibration helps to make axis movement smooth and void of noises.



**Note:** Calibrating the Sine and Cosine parameters requires entering the KMI, which requires a password. Obtain the password before performing.

**To Open the KMI axes window:**

1. In the application, from the Main Menu, press **Maintenance > RS232**; the KMI window appears.



**Figure 8-17. KMI Window**

2. Press **Remote-KMI**, type the password and press **Enter**.
3. Select the “**KMI Ver 1361**” window.
4. Press **F3** to open the new “**Sys/User Variables**” window.



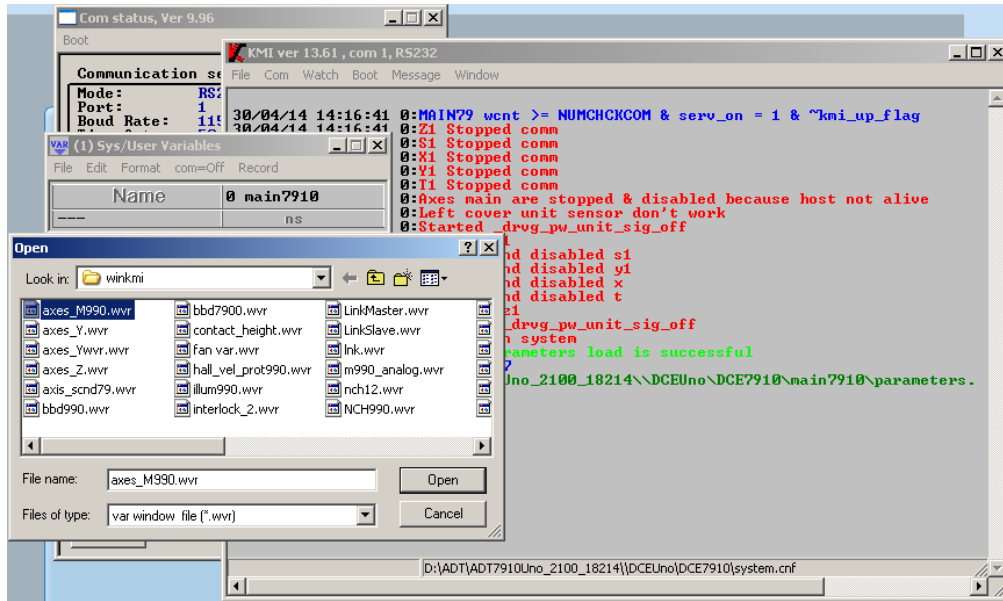


Figure 8-18. KMI browsing window

5. In the menu, press **Load File**, select **axes\_M990.wvr** and press **Open**; the axes M990 window appears.

| Name      | 0 main7910 | y1          | 0 main7910 | s1          | 0 main7910 | z1          | 0 main7910 | x           | 0 main7910 | t           |
|-----------|------------|-------------|------------|-------------|------------|-------------|------------|-------------|------------|-------------|
| en        | ld         | 0           | ld         | 0           | ld         | 0           | ld         | 0           | ld         | 0           |
| command   | f          | 0.000000    | f          | 0.000000    | f          | 0.000000    | f          | 0.000000    | f          | 0.000000    |
| ref_pos   | ld         | 0           | ld         | 0           | ld         | 0           | ld         | 0           | ld         | 0           |
| rel_pos   | ld         | 0           | ld         | 0           | ld         | 0           | ld         | 0           | ld         | 0           |
| enc_pos   | ld         | 2           | ld         | 1           | ld         | 0           | ld         | -3          | ld         | 1           |
| pos_err   | ld         | 0           | ld         | 0           | ld         | 0           | ld         | 0           | ld         | 0           |
| max_err   | ld         | 1000        | ld         | 0           | ld         | 1000        | ld         | 2000        | ld         | 0           |
| vel_err   | f          | 0.000000    | f          | 0.000000    | f          | 0.000000    | f          | 0.000000    | f          | 0.000000    |
| target    | ld         | 0           | ld         | 0           | ld         | 0           | ld         | 0           | ld         | 0           |
| trg       | ld         | 0           | ld         | 0           | ld         | 0           | ld         | 0           | ld         | 0           |
| prf_st    | ld         | 6           | ld         | 6           | ld         | 6           | ld         | 6           | ld         | 6           |
| stop_m    | ld         | 1           | ld         | 1           | ld         | 1           | ld         | 1           | ld         | 1           |
| pe_fl     | ld         | 0           | ld         | 0           | ld         | 0           | ld         | 0           | ld         | 0           |
| vel_max   | ld         | 300000      | ld         | 15000       | ld         | 40000       | ld         | 600000      | ld         | 800000      |
| jv1       | ld         | 0           | ld         | 0           | ld         | 0           | ld         | 0           | ld         | 0           |
| acc       | f          | 5.00000e+06 | f          | 200000.     | f          | 1.00000e+06 | f          | 200000.     | f          | 2.50000e+06 |
| dcc       | f          | 5.00000e+06 | f          | 200000.     | f          | 1.00000e+06 | f          | 200000.     | f          | 2.50000e+06 |
| dcc_kill  | f          | 1.00000e+07 | f          | 2.00000e+06 | f          | 1.00000e+07 | f          | 2.00000e+06 | f          | 2.50000e+06 |
| jog_vel   | ld         | 0           | ld         | 1           | ld         | 0           | ld         | 0           | ld         | 0           |
| sw1_pos   | ld         | 0           | ld         | 0           | ld         | 0           | ld         | 0           | ld         | 0           |
| sw2_pos   | ld         | 570000      | ld         | 0           | ld         | 250000      | ld         | 40000       | ld         | 1000000     |
| prcs_t    | ld         | 333         | ld         | 333         | ld         | 333         | ld         | 333         | ld         | 333         |
| prot_t    | ld         | 333         | ld         | 333         | ld         | 333         | ld         | 333         | ld         | 333         |
| kill_t    | ld         | 333         | ld         | 333         | ld         | 333         | ld         | 333         | ld         | 333         |
| t_t       | ld         | 333         | ld         | 0           | ld         | 333         | ld         | 333         | ld         | 333         |
| run_curr  | f          | 600.000     | f          | 0.000000    | f          | 600.000     | f          | 0.000000    | f          | 400.000     |
| idle_curr | f          | 200.000     | f          | 0.000000    | f          | 200.000     | f          | 0.000000    | f          | 200.000     |
| sm_ampl   | f          | 0.000000    | f          | 0.000000    | f          | 200.000     | f          | 300.000     | f          | 0.000000    |
| p_p_g     | f          | 0.100000    | f          | 0.000000    | f          | 0.000000    | f          | 0.000000    | f          | 0.000000    |
| v_p_g     | f          | 1.00000     | f          | 0.000000    | f          | 0.000000    | f          | 400.000     | f          | 0.000000    |
| vel_p_g   | f          | 1.00000     | ns         |             | f          | 1.00000     | ns         |             | f          | 2240.00     |
| pos_p_g   | f          | 0.100000    | ns         |             | f          | 0.100000    | ns         |             | f          | 0.0075000   |
| p_i_out   | f          | 0.000000    | f          | 0.000000    | f          | 0.000000    | f          | 0.000000    | f          | 0.000000    |
| v_i_out   | f          | 0.000000    | f          | 0.000000    | f          | 0.000000    | f          | 0.000000    | f          | 0.000000    |
| pg_vel    | ld         | 0           | ld         | 0           | ld         | 0           | ld         | 0           | ld         | 0           |
| enc_sgn   | f          | 2.50000     | f          | 1.00000     | f          | 9.76563     | f          | -3.00000    | f          | 1.00000     |
| sin_offs  | f          | 31.0000     | f          | 0.000000    | f          | 25.0000     | f          | 0.000000    | f          | 0.000000    |
| cos_offs  | f          | 12.0000     | f          | 0.000000    | f          | 14.0000     | f          | 0.000000    | f          | 0.000000    |

Figure 8-19. Axis M990 Window

To calibrate the Sine and Cosine parameters for Y and Z Axes:

1. Change the value of the **sm\_ampl** parameter in the Y1 and X tables from **200** to **0** (null).
2. Perform Sinus Offset calibration:
  - a. Set the multimeter in the DC voltage position.

- b. Open the rear panel of the system and connect the multimeter to the wire out of pin No. 1 or No. 2 on the driver being calibrated (Y-axis or Z-axis).
  - c. Perform the measurement.
  - d. Change the value of **sin\_offs** parameter in the axes M990 window until the multimeter readout is **0**.
  - e. Write down the value (the values of the **sin\_offs** and **cos\_offs** parameter should be between -20 and 20).
3. Perform Cosine Offset calibration:
  - a. Set the multimeter in the DC voltage position.
  - b. Connect the multimeter to the wire out of pin No. 3 or No. 4 on the driver being calibrated (Y-axis or Z-axis).
  - c. Perform measurement.
  - d. Change the value of **cos\_offs** parameter in the axes M990 window until the multimeter readout is **0**.
  - e. Write down the value (the values of the **sin\_offs** and **cos\_offs** parameters should be between -20 and 20).
4. Close the **KMI** and **Communication** windows.
5. Return to the Main Menu and go to: **Operation Setup > DCE Parameters**.
6. Enter the new **Sin\_offs** and **Cos\_offs** values into the parameters table.
7. Save the changes.
8. Initiate the axes to verify smooth movement.

## Chapter 9. Maintaining System Components

Maintaining System Components describes the procedures for replacing damaged or faulty system components, and includes the tests and adjustments that must be performed after installing such new components.

Maintaining System Components includes:

- ❖ Maintaining the Broken Blade Detector (BBD) (section 9.1).
- ❖ Maintaining the Height Device (section 9.2).
- ❖ Maintaining the Cutting Chuck (section 9.2.2).
- ❖ Maintaining the Cooling Block (section 9.4).

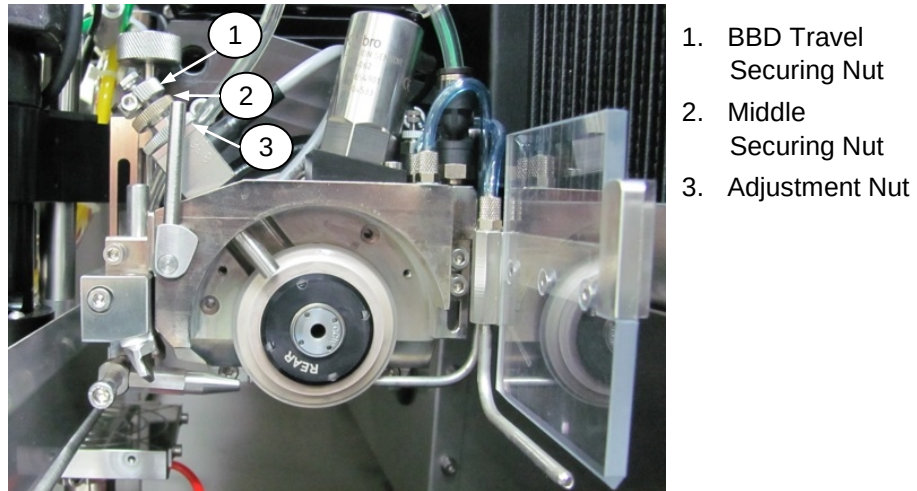
Read all of the instructions carefully before replacing a component, and allow yourself enough time to complete all of the tests and adjustments after installing the new component.



**Caution:** These procedures should not be performed by untrained or inexperienced personnel.

## 9.1 Maintaining the Broken Blade Detector (BBD)

The optional Broken Blade Detector (BBD) for the Model 71XX is for use on Systems equipped with a 2" Spindle. The BBD uses an Optical Sensor to detect partial or total Blade breakage. Both the BBD Transmitter, which provides the source of light, and the BBD Receiver are located above the Cooling Block. The Blade Checker Board converts the signal from the Optical Sensor for use by the System.



**Figure 9-1. Cooling Block with the Broken Blade Detector**

Maintaining the Broken Blade Detector includes the following sections:

- Tuning an existing BBD.
- Replacing the BBD.
- Tuning a New BBD.
- Testing the BBD.
- Replacing the Blade Checker Board.
- Adjusting the Blade Checker Board.



**Note:** For BBD periodic maintenance, see section 4.1.

### 9.1.1 Tuning an Existing BBD

The BBD must be tuned from time to time so that the BBD transmitter constantly maintains the correct setting.

#### To tune an existing BBD:

1. Go to the **Maintenance > Functions and I/O > Head System** screen.

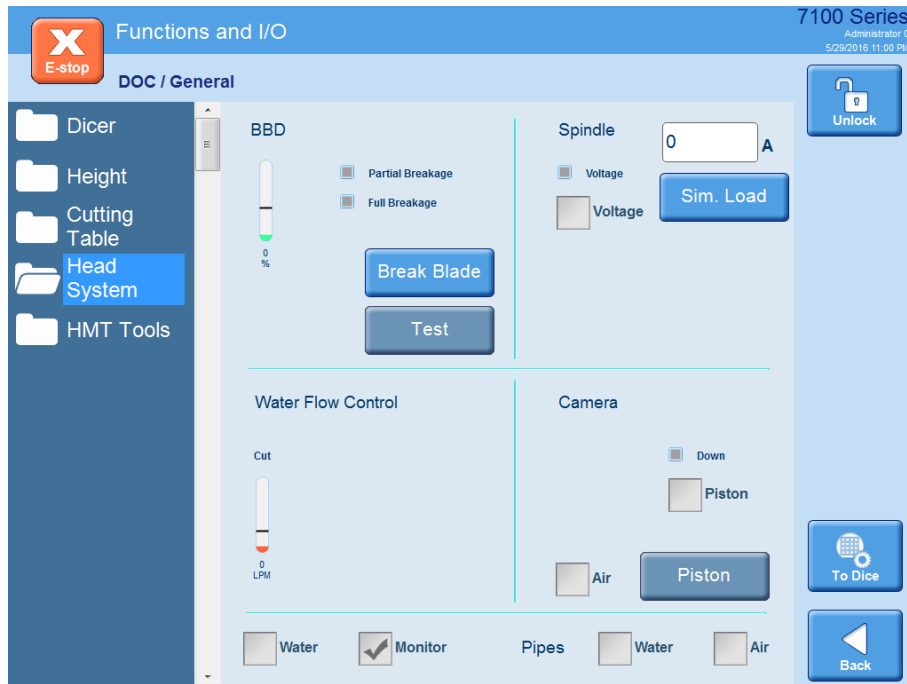


Figure 9-2. Head System Screen

2. Press **Unlock**.
3. Press the **Test** button.
4. Slightly release the BBD securing screw.
5. Slowly shift the BBD until the BBD reading (%) registers approximately 15%.
6. Tighten the BBD securing screw.

## 9.1.2 Replacing the BBD

When the LED output from the BBD transmitter is too low, first clean the transmitter and the receiver. If cleaning does not improve LED output, replace the BBD.

### To replace the BBD:

1. In the application, from the Main Menu, press **To Dice**; the Dice window appears.
2. Press **Stop** to stop spindle rotation.
3. Perform the Change Blade procedure:
  - a. Press the **Spindle** button > **Change Blade**; the Change Blade wizard appears.

| Change Blade Step 1 of 2: Define Blade                                                   |                |               |                |
|------------------------------------------------------------------------------------------|----------------|---------------|----------------|
| Lot number                                                                               | a23            |               |                |
| Blade                                                                                    | Nickel Hub     |               |                |
| Flange Diameter                                                                          | 53.5752        | mm            |                |
| Flange                                                                                   | f1 - 54.174 mm |               |                |
| <input checked="" type="checkbox"/> Leave current <input type="checkbox"/> Change flange |                |               |                |
| Type:                                                                                    | Regular        | Maximum RPM   | 60000          |
| Name:                                                                                    | Nickel Hub     | Hub Type:     | Hub Blade      |
| Diameter (mm):                                                                           | 55.5752        | Matrix:       | Matrix Nickel  |
| Exposure (mm):                                                                           | 1              | Density:      | Density S      |
| Expansion Tolerance                                                                      | 0.17           | Size Diamond: | Size Diamond S |
| Thickness (mm):                                                                          | 0.035          | Bond Q:       | Bond Q         |
| Dress Wafer Recipe:                                                                      | None           | Company:      | Semitec        |
|                                                                                          |                | Next          | Cancel         |

**Figure 9-3. Change Blade Window (1 of 2)**

- b. Clear the **Leave current blade** check box.
  - c. Press **Next**; Step 2 appears.
4. Remove the blade.
    - Open the system rear panel.



**Caution:** When completing any task in the area beneath the top panel, make sure that the cables and tubes do not block axis movement and/or cannot be damaged by any axis travel movement.

5. Disconnect the Sensor Cable from the Blade Checker Board.
6. Release and remove the BBD Travel Securing Nut.

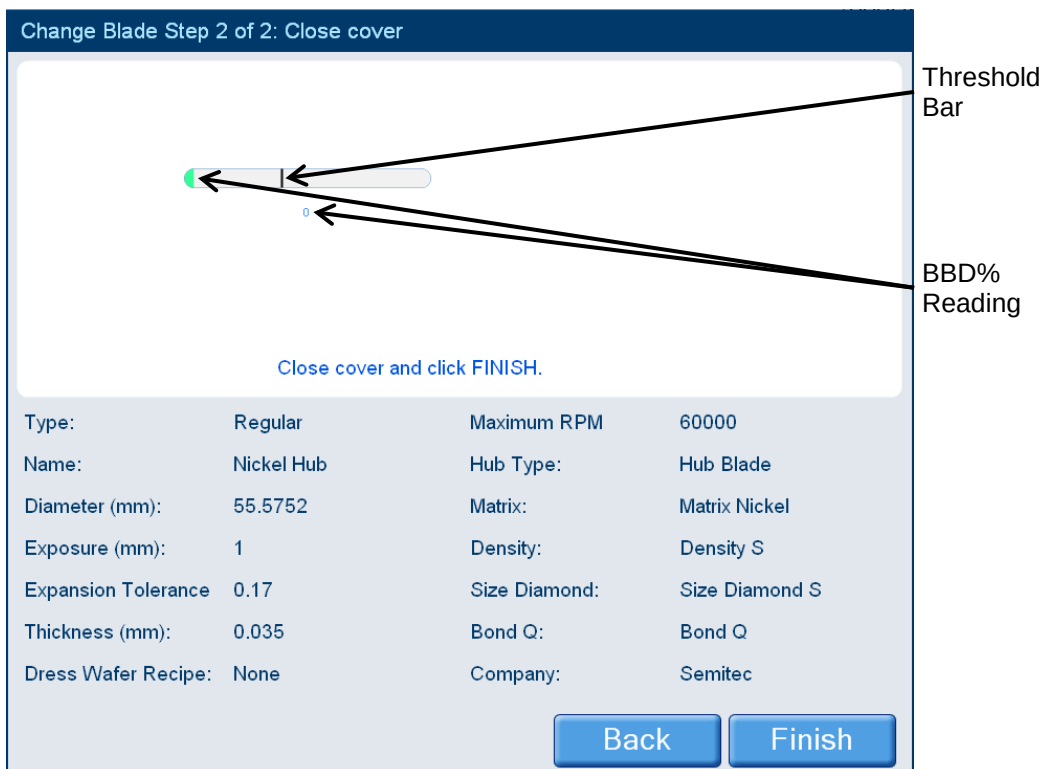
7. Lift out the old BBD Assembly.
8. Place the new BBD Assembly in position on top of the Cooling Block.
9. Tighten the BBD Travel Securing Nut.
10. Attach the Sensor Cable
11. Replace the blade.
12. Tune the BBD assembly as described in following section 9.1.3.

### 9.1.3 Tuning a New BBD

After the new BBD assembly has been installed, it must be tuned so that the BBD transmitter is at the correct setting.

#### To tune the BBD:

1. Continue with Step 2 of the Change Blade procedure:



**Figure 9-4. Blade Change Window (2 of 2)**

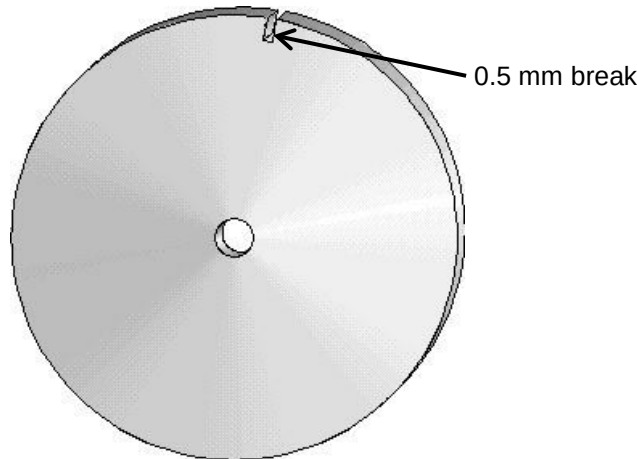
- a. Slightly release the BBD securing screw.
- b. Slowly shift the BBD until the BBD% reading registers approximately 15%.
- c. Tighten the BBD securing screw.
- d. Press **Finish**.



**Note:** The BBD threshold bar is pre-set with a 50% default value. Setting the threshold to a lower value causes the BBD to be more sensitive; setting it to a higher value causes it to be less sensitive.

### 9.1.4 Testing the BBD

After replacing and adjusting the new BBD, install the Checking Blade to test the abilities of the new BBD. The Checking Blade contains a 0.5 mm break along its edge. When the new BBD has been correctly installed and adjusted, the break should be detected by the optical sensor.



**Figure 9-5. Checking Blade**

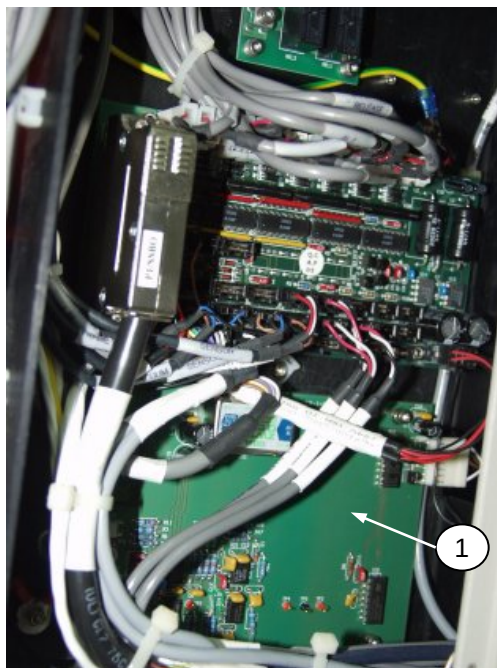
**To test the BBD:**

1. Mount the Checking Blade on the spindle (for instruction on mounting blades, see the 7120/7130 Dicing Series Operations Manual).
2. Activate the spindle:
  - a. In the application, from the Main Menu, press **Dice > Spindle** button; the spindle menu appears.
  - b. Select **Operate Spindle**; the spindle is activated.
3. Perform Height:
  - a. Press (again) the **Spindle** button; the spindle menu appears.
  - b. Press **Height**; the system begins to perform Height and should stop and display an error message indicating that the BBD has detected a broken blade.
    - i. If an error message is not received, then the new BBD has either not been installed correctly, or has not been correctly adjusted.
    - ii. Repeat the installation and adjustment procedures, and then repeat the test.



### 9.1.5 Replacing the Blade Checker Board

The system contains a Blade Checker board, which converts the signal from the BBD transmitter for use by the system. The Blade Checker Board is located on the Board Panel to the right of the Pneumatic Panel at the rear of the System.



1. Blade Checker Board

**Figure 9-6. Blade Checker Board Location**

#### **To replace the Blade Checker board:**

1. Power down the system.
2. Turn off the Main Circuit Breaker at the right side of the System, and disconnect the System from Mains power
3. Remove the rear center panel to reveal the Blade Checker Board.
4. Remove the J1, J3, J4 connectors and the two ground lugs from the Blade Checker board.
5. Remove the screws securing the Blade Checker board to the electrical panel, and remove the board.
6. Insert the new board onto the mounting spacers, and secure with the screws that were previously removed.
7. Attach the J1, J3, J4 connectors and the two ground lugs to their appropriate positions on the board.
8. Reconnect the System to Mains power, turn on the Main Circuit Breaker and power up the System.
9. Adjust the Blade Checker board (see following section 9.1.6).

## 9.1.6 Adjusting the Blade Checker Board

After replacing the Blade Checker board, it must be adjusted.

### To adjust the Blade Checker board:

1. Remove the JP1 jumper from the Blade Checker board to turn off the BBD Transmitter LED.
2. Connect a voltmeter to the Blade Checker board, as follows:
  - Common probe to TPG.
  - Positive probe to TP1.
3. Adjust the VR2 potentiometer screw so that 0.0 VDC appears on the voltmeter display.
4. Remove the voltmeter probes.
5. Replace the JP1 jumper.
6. In the application, from the Main Menu, press **Maintenance > Axis Setup > Spindle**.
7. In the **Set Speed** field enter 45,000 and press **Set Speed**; the spindle begins to rotate at 45,000 RPM.
8. Test the BBD, as described in section 9.1.4 above.
9. Place back the rear center panel.

## 9.2 Maintaining the Height Device

The Model 71XX is equipped with one of the following Height Devices:

- Non-Contact Height Device (NCHD)
- Mechanical Button (option)

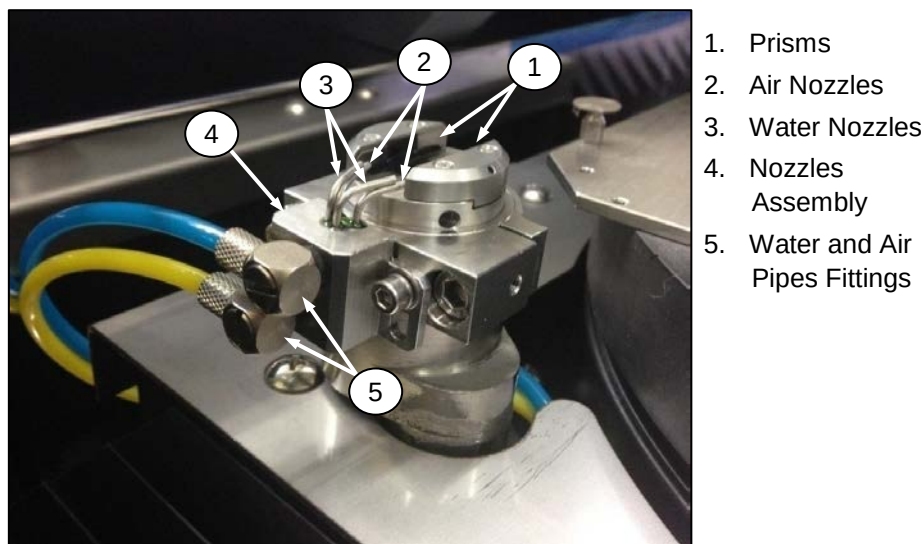
The Height Device is located on the X Carriage Cover, to the left rear of the Theta Table. It is used to perform the height procedure, which enables determining chuck position along the Z Axis. The height procedure must be performed regularly to enable Z position compensation due to blade wear.

Both types of Height Devices must be properly adjusted and kept clean from dust and debris. They can be replaced if they malfunction.

### 9.2.1 Non-Contact Height Device (NCHD)

The NCH device (Non-Contact Height device) is an optical device that uses an LED transmitter and receiver between two prisms. The height procedure is performed by lowering the blade over the NCH device until the edge of the blade breaks the beam of light emitted by the LED transmitter.

The NCH is supplied with a constant stream of water in order to keep it free of debris. Immediately before height is performed, the stream of water stops and air is applied to the NCH device to dry the prisms and ensure their proper function. The stream of water resumes immediately after the Non-Contact Height procedure has been completed.



**Figure 9-7. Non-Contact Height Device**



**Note:** The NCH device must be regularly cleaned and adjusted to ensure proper height measurements.

Maintaining the Height Device includes:

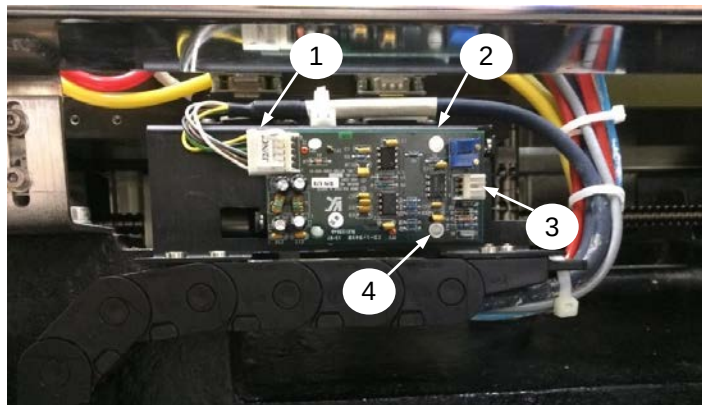
- Replacing the Non-Contact Height Device.
- Setting up the Non-Contact Height Device.

### 9.2.1.1 Replacing the Non-Contact Height Device

The NCH device must be replaced if the LED or detector inside the NCH device fails, or if the blade accidentally makes contact with one of the NCH device prisms.

#### To remove the Non-Contact Height device:

1. Clean the area around the NCH device so that it is free of dust and water.
2. Power down the system.
3. Turn off the water and air supplies to the system.
4. Open the front cover.
5. Disconnect the NCHD Plug from the cable that connects to the NCHD Board at the front of the system.



1. J2 connector
2. NCH Card Board
3. JP1 connector
4. Plastic fasteners (×4)

Figure 9-8. NCH Device Card

6. Loosen the screws securing the holding band to the NCH device and remove the band from the NCH device.
7. Lift the NCH device up to remove it from the system.

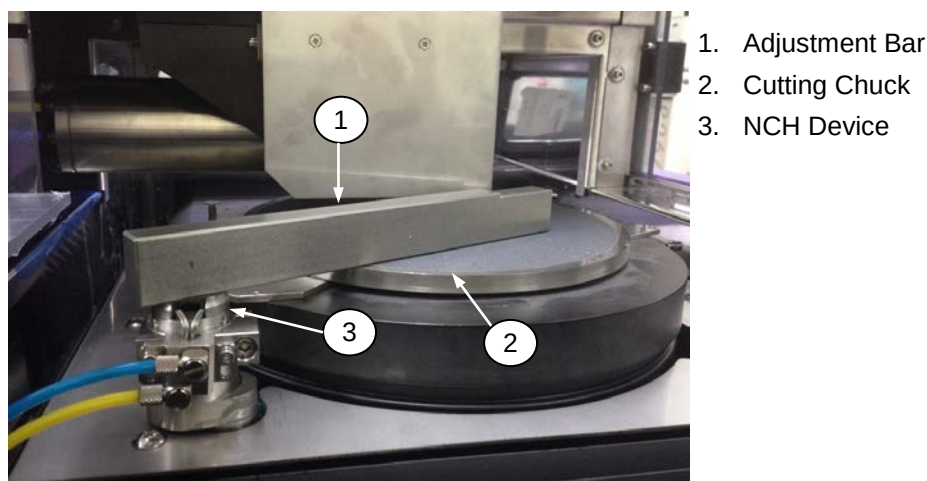
#### To connect the Non-Contact Height device:

1. Thread the new NCH device plug through the opening in the X Carriage and insert the new NCH device.
2. Connect the NCHD Plug to the cable that connects to the NCHD Board.
3. Align the prisms on the NCH device so that they are parallel to the X-axis. The top surface of the NCH device should be at about the same height as the chuck's top surface.
4. Replace the holding band using the screws previously removed (do not tighten the screws).
5. Ensure that the air nozzle tips are about 1 mm above the bottom of the NCH device groove.



**Caution:** Ensure that nothing can touch, hit, or bend the nozzle tips during the installation.

6. Adjust the nozzles so that the spray of water rinses the prisms.
7. Connect the NCH device plug to the cable that connects to the NCH device board.
8. Use the supplied adjustment bar (see Figure 9-9) to perform Z parallelism adjustment:
  - a. Place the adjustment bar on the cutting chuck so that one end of the bar is above the NCH device.



**Figure 9-9. NCH parallelism**

- ➡ **Note:** Ensure that the bar is snug on the cutting chuck.
- b. Gently move the NCH device assembly upwards until it just touches the bar.
- ➡ **Note:** It is important to keep the NCH device groove parallel to the X Axis while adjusting Z parallelism.
- c. Tighten the band securing screw.
9. Open the air and cutting water supplies and power up the system.

### 9.2.1.2 Setting up the Non-Contact Height Device

The NCH device setup comprises the following adjustments and settings:

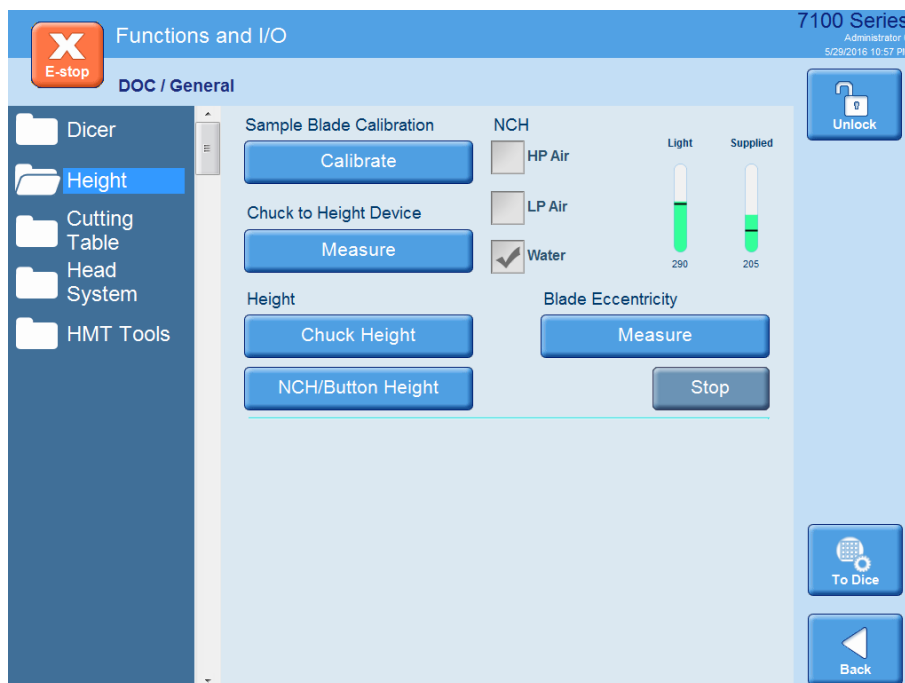
- Adjusting and Verifying Air and Water Flow.
- Preparing the System for the Non-contact Height Procedure.
- Adjusting the Signal Intensity.
- Measuring the Chuck to Non-Chuck Delta.



**Note:** The adjustments are performed through the application.

**To verify and adjust the air and water flow:**

1. In the application, from the Main Menu, press **Maintenance > Functions and I/O**; the Functions and I/O window appears.
2. Select **Height**; the Height view appears.



**Figure 9-10. Air Deactivated and Water Activated**

3. Press the **Unlock** button to enable modification.
4. Clear the **HP Air** and **LP Air** check boxes and then select the **Water** check box to deactivate the air pressure and activate the water pressure.
5. Adjust the valve on the cleaning water solenoid, so that the water is not over-pressurized but under enough pressure to keep the NCH device prisms clean.
6. In the Functions and I/O window, clear the **Water** check box and select the **LP Air** (low-pressure) check box.



**Note:** The water solenoid is normally open, while the air solenoid is normally closed.

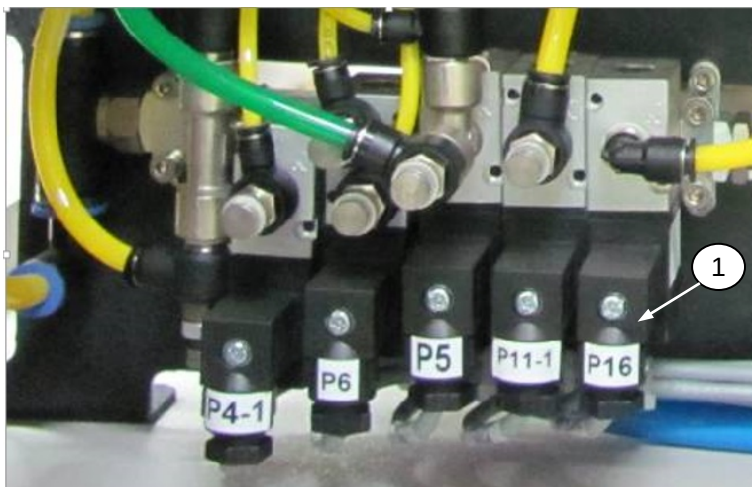
7. Select the **HP Air** (high-pressure) check box, to activate the air pressure.



8. Adjust the HP air supply so that the HP air is under enough pressure to keep the prisms dry.



**Note:** The NCH device HP air supply is found on the Air and Vacuum unit located at the rear of the system.



1. High air pressure adjusting Valve (P16)

**Figure 9-11. Pressure Adjustment Controls on the Pneumatic Panel**

**To prepare the system for the NCH procedure:**

1. Install a blade on the spindle.
2. Teach Pre Non-contact Height:
  - a. In the application, from the Main Menu, press **Dice > Axis Control**; the Axis Control view appears.
  - b. Use X/Y and Z/T axis controls to bring the blade above the NCH device so that the lowest point of the blade is approximately 3 mm above the prisms and centered along the NCH groove.
  - c. From the Main Menu, press **Maintenance > Axis Setup > Z**; the Z view appears.
  - d. Select **Pre Non-contact Height**; the Teach button is activated.
  - e. Press **Teach**; the Pre Non-contact Height value for the Z Axis is taught to the system.
  - f. Press **SAVE**.
3. Teach NCHD Station:
  - a. From the Main Menu, press **Dice > Axis Control**; the Axis Control view appears.
  - b. Use X/Y and Z/T axis controls to bring the blade above the NCH device so that the lowest point of the blade is approximately 3 mm above the prisms and centered along the NCH groove.
  - c. From the Main Menu, press **Maintenance > Axis Setup**; the Axis Setup view appears.
  - d. Select **Head NCH**.

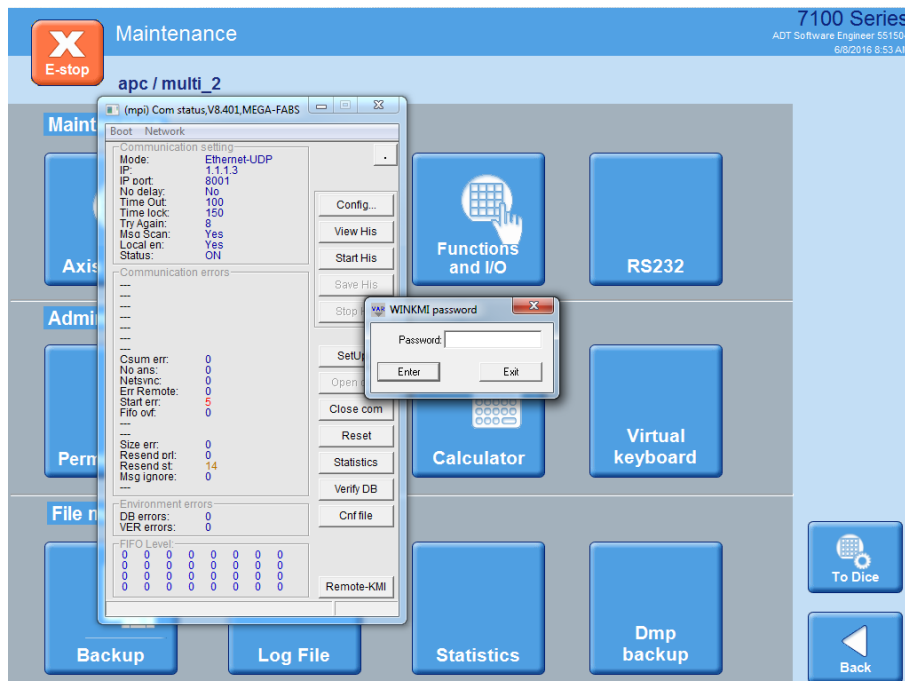
- e. Press **Unlock** to enable changes.
- f. Press **Teach All**; the Front Head NCH value for the Z Axis is taught to the system.
- g. Press **SAVE**.

**To adjust the signal intensity:**



**Note:** Have the KMI daily password ready.

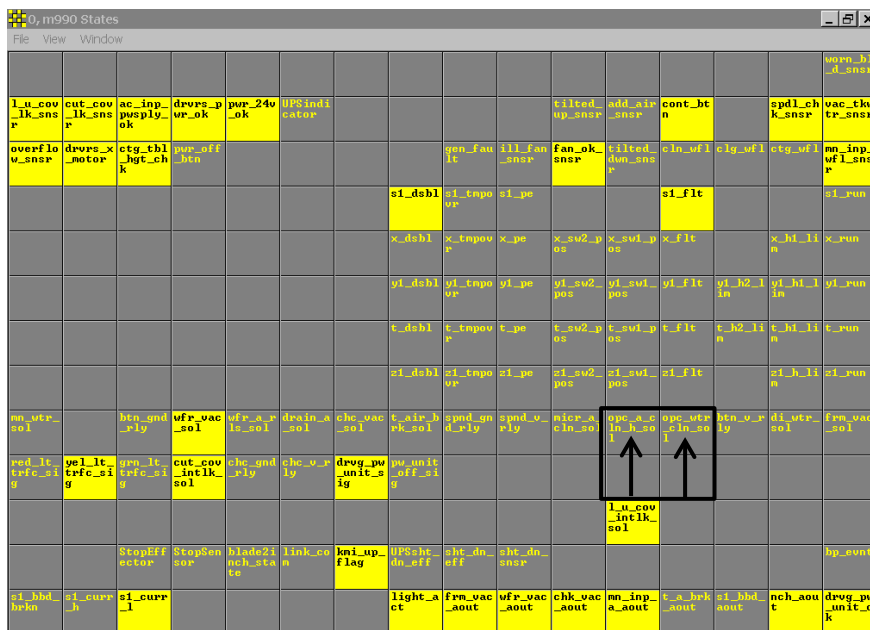
1. In the application, from the Main Menu, press **Maintenance > RS232**; the Communication Status window appears



**Figure 9-12. Com Status Window**

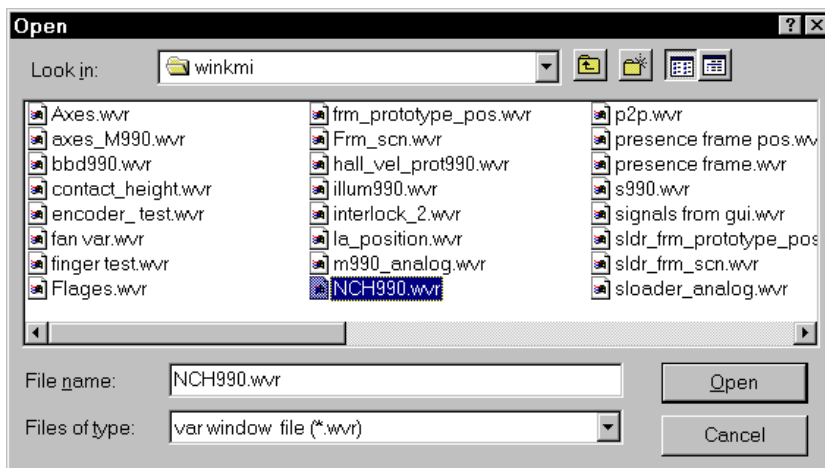
2. Press **Run KMI**; the KMI Password dialog box appears.
3. Enter the password and press **OK**; the KMI Ver window appears.
4. Press **Watch > States > 0 m990**; the States window appears.





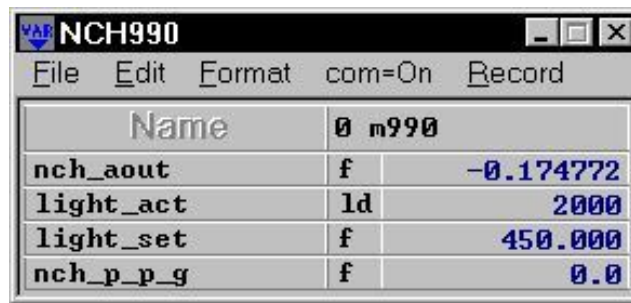
### Figure 9-13. States Window

- a. Close the NCH cleaning water solenoid, by holding down the **Ctrl** key and pressing the **opc\_a\_** square.
- b. Close the NCH high-pressure cleaning air, by holding down the **Ctrl** key and pressing the **opc\_wt\_** square.
5. Clean the NCH prisms with a lint-free cloth or swab.
6. In the KMI ver window, press **Watch > Open Vars window** or press **F3**; the System User Variables window appears.
7. Press **File > Load from file**; the Load from File window appears.



**Figure 9-14. Load from File Window**

8. Select **NCH 990.wvr** and press **Open**; the NCH990 window appears.



| NCH990                         |    |           |
|--------------------------------|----|-----------|
| File Edit Format com=On Record |    |           |
| Name                           |    | 0 m990    |
| nch_aout                       | f  | -0.174772 |
| light_act                      | ld | 2000      |
| light_set                      | f  | 450.000   |
| nch_p_p_g                      | f  | 0.0       |

Figure 9-15. NCH 990 window

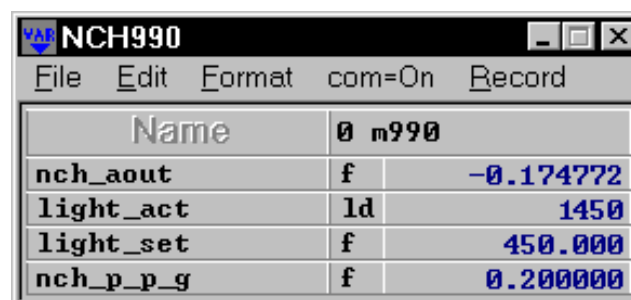
- a. Press the **nch\_p\_p\_g** value and enter **0.2**.
  - b. Press the **light\_set** value, enter **0**.
9. On the NCH board, adjust the VR2 potentiometer until the **nach\_aout** value comes as close to **0** as possible.



1. VR1 potentiometer adjusting screw

Figure 9-16. Adjusting the Signal's Intensity

10. In the NCH990 window, press the **light\_set** value and enter **450**.
11. Adjust the signal intensity by turning the VR1 potentiometer screw until the **light\_act** value reaches 1450.



| NCH990                         |    |           |
|--------------------------------|----|-----------|
| File Edit Format com=On Record |    |           |
| Name                           |    | 0 m990    |
| nch_aout                       | f  | -0.174772 |
| light_act                      | ld | 1450      |
| light_set                      | f  | 450.000   |
| nch_p_p_g                      | f  | 0.200000  |

Figure 9-17. NCH light\_act value

12. In the NCH990 window, press the **nch\_p\_p\_g** value and enter **0**.
13. Close the KMI window after completing the signal adjustment.



**Caution:** Do not replace the NCHD card cover while the system is powered on.

### 9.2.1.3 Measuring the Chuck to Non-Chuck Delta

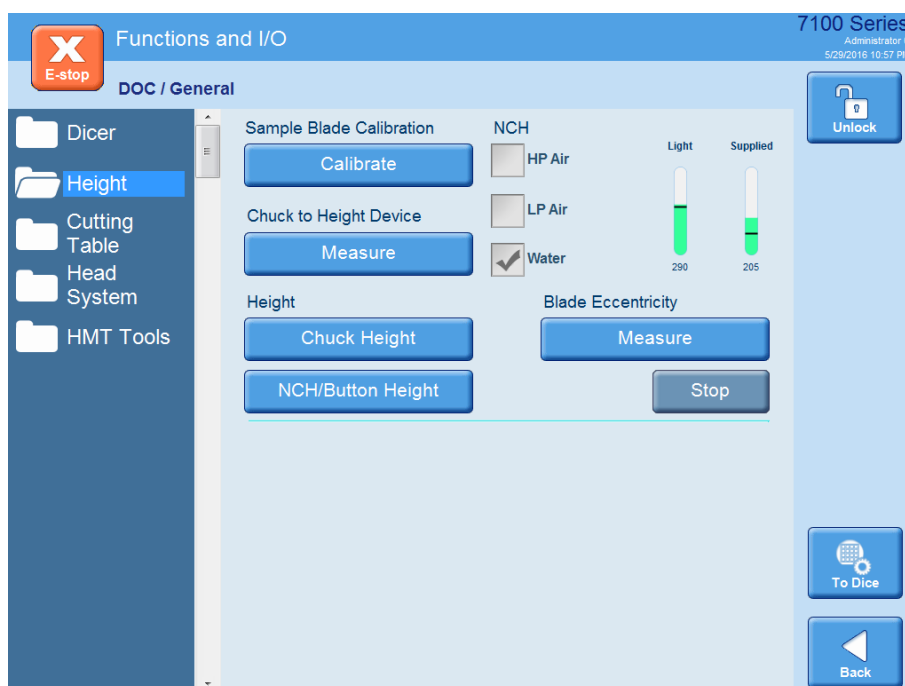
After replacing the NCH device, the chuck to non-chuck delta must be taught to the system.



**Note:** The Delta Z measurements should be performed with a dressed blade and best after running the saw with water.

**To perform chuck to non-chuck delta measurement:**

1. In the application, from the Main Menu, press **Maintenance > Functions and I/O**; the Functions and I/O window appears.
2. Press **Height**; the Height screen appears.



**Figure 9-18. Functions and I/O Window**

3. Press the **Chuck to Height Device/Measure** button; the system performs Chuck Height, followed by NCH measurement, and then calibrates the two values.
4. Wait for the system to finish the calibration.
5. Perform a NCH procedure to verify that the system works properly.

## 9.2.2 Mechanical Button (option)

When the Mechanical Button is the installed Height device, the Height procedure is performed by lowering the Blade onto the mechanical Button to close a circuit. In order to ensure accurate results, the system uses mapped coordinates based on blade type and size to make contact with the Button surface at a different location, each time Button Height is performed. Once a new contact area is no longer available, the Button must be replaced.

The System keeps track of the number of Height procedures that can still be performed on the mechanical Button, and displays this information in the Activity Log at the lower left of the screen. A message is displayed when the Button needs to be replaced.

During operation, the Button is supplied with a constant stream of air in order to keep it free of debris. The mechanical Button should be cleaned daily.

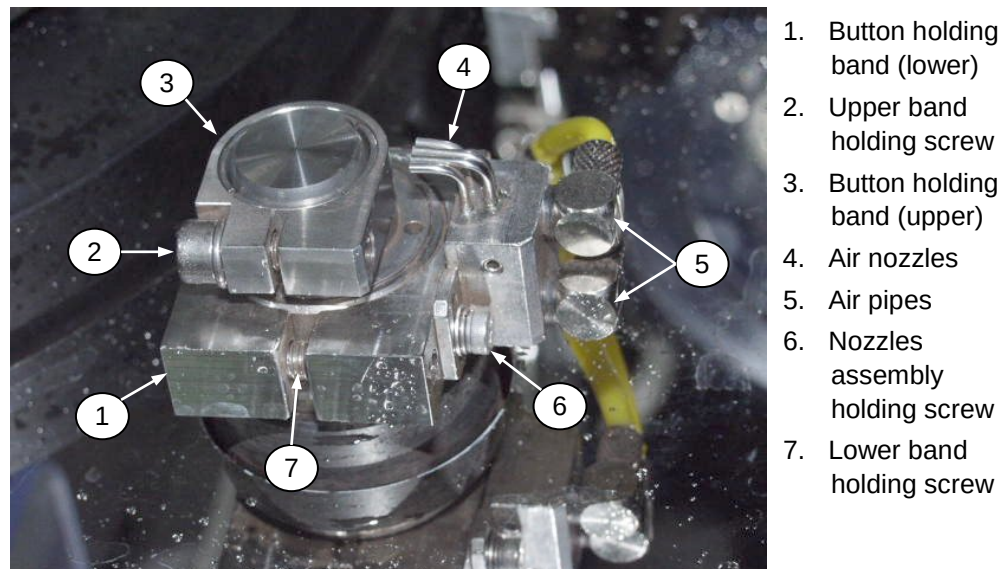


Figure 9-19. Height Button

### 9.2.2.1 Replacing the Mechanical Button

When using a mechanical Button as the Height device, the system automatically tracks the number of Height procedures that can be performed before the Button must be replaced. This number is displayed after each Height procedure in the Activity Log at the lower left pane of the screen. When the limit is reached, the System displays a message instructing the user to perform the Change Button procedure to replace the worn Button with a new one.



**Note:** It is recommended to replace the Mechanical Button, while a used (dressed) Blade is mounted.

**To Replace the Mechanical Button:**

1. Clean the area around the mechanical Button so that it is free of dust and water.
2. In the application, from the Main Menu, press **Maintenance > Functions and I/O**; the Functions and I/O window appears.
3. Press **Height**; the Height screen appears.
4. Press **Change Button**. The software switches to the Dice (Axis Control) screen and displays the Wizard with instructions that guide the User step by step through the procedure.



**Note:** The instructions given in the wizard do not replace the instructions given in steps 5 through 12 of the present section.

5. On the Mechanical Button housing, loosen the screw securing the upper C-clamp and remove the C-clamp.
6. Remove the Mechanical Button by inserting a small screwdriver into any of the four slots and prying upwards to release the Button.
7. Insert the new Mechanical Button into its housing, leaving it slightly higher than it needs to be.
8. Replace and partially tighten the upper C-clamp.
9. Place a Straightness Block across the Chuck and Mechanical Button and use the Block to press firmly on the Mechanical Button to ensure that it is flat and parallel to the top surface of the Cutting Chuck.
10. Firmly tighten the upper C-clamp on the Mechanical Button housing.
11. In the Wizard in the Video Workspace, click Next and then Finish (according to the wizard). The System automatically performs a flatness check on the Mechanical Button and then performs the Chuck to Non-Chuck Delta Measurement procedure to calibrate the difference in height between the Mechanical Button and the Cutting Chuck.
12. Adjust the height and flatness of the Mechanical Button as described in following section 9.2.2.2.

### 9.2.2.2 Adjusting the Mechanical Button

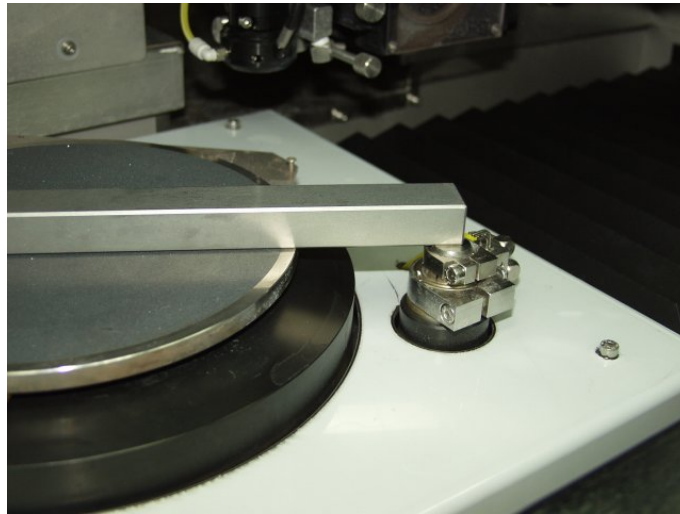
The surface of the Mechanical Button must be flat and parallel to the Chuck in order to function properly. Adjust the Mechanical Button by performing the following procedure.



**Note:** When working with thick workpieces, the height of the Mechanical Button should be adjusted accordingly in order to ensure accurate results.

#### To Adjust the Mechanical Button:

1. Place a straightness block across the chuck and mechanical Button and use the block to press firmly on the mechanical Button to ensure it is flat.



**Figure 9-20. Adjusting Button Height**

2. Firmly tighten the upper C-clamp on the Mechanical Button housing.
3. In the Wizard in the Video Workspace, click Finish. The System automatically performs a flatness check on the Mechanical Button and then performs the Chuck to Non Chuck Delta Measurement procedure to calibrate the difference in height between the Mechanical Button and the Cutting Chuck.
4. Adjust the airflow and the nozzles position.

### 9.2.2.3 Cleaning the Mechanical Button

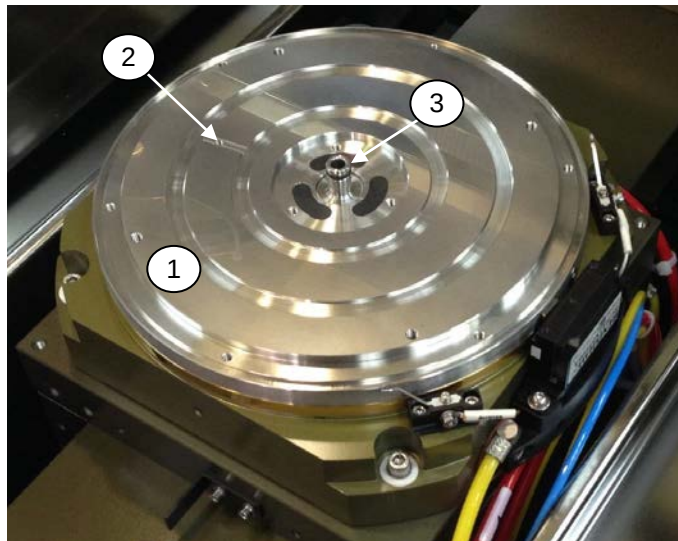
The surface of the Mechanical Button must be kept clean and free of dust at all times. Clean the surface of the Mechanical Button daily with a cotton swab, isopropyl alcohol and water.



## 9.3 Maintaining the Cutting Chuck

The system is equipped with a 'quick change' cutting chuck, which is held in place on the theta table by vacuum. The theta table has two vacuum holes:

- The single workpiece hole, at the center of the theta table, provides vacuum to the chuck's upper surface to hold the workpiece in place.
- The single chuck hole provides vacuum to the two theta table grooves for holding the chuck in place.



1. Theta Table
2. Chuck Holding Vacuum Holes
3. Vacuum Hole

**Figure 9-21. Theta Table**



**Note:** The system also uses the vacuum hole at the center of the Theta to supply an air puff when releasing a workpiece from the cutting chuck. Adjusting this release air is performed, if required, on the Pneumatic Panel.

Maintaining the Cutting Chuck includes the following sections:

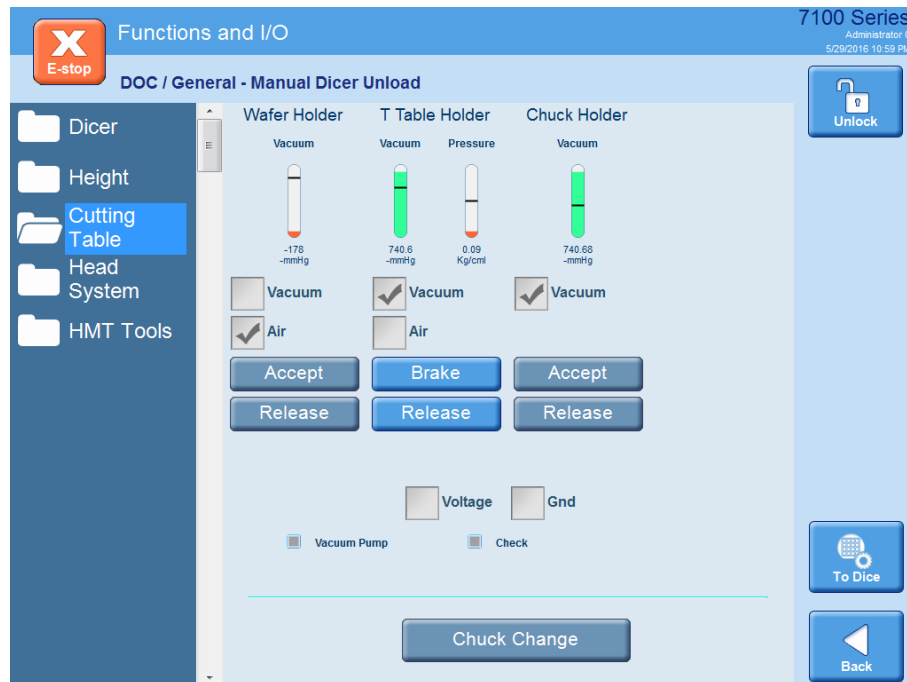
- Replacing the Cutting Chuck.
- Performing Chuck Height.

### 9.3.1 Replacing the Cutting Chuck

During the course of working with the system, the chuck can require replacement because of change in work requirements, due damage or malfunctioning.

**To replace the cutting chuck:**

1. In the application, from the Main Menu, press **Maintenance > Functions and I/O**; the Functions and I/O window appears.
2. Press **Cutting Table**; the Cutting Table view appears.



**Figure 9-22. Changing the Chuck**

3. Press **Chuck Change**; the Dice window in the Axis Control view appears.
4. Remove the cutting chuck from the theta table.
5. Clean the theta table and the underside of the new cutting chuck with a lint-free cloth and isopropyl alcohol.
6. Place the new cutting chuck on the theta table. Make sure that the placement pins are properly aligned and the O-ring at the bottom of the chuck is present.



**Caution:** If the new chuck is higher (thicker than the old one), verify that the Z Calibration Start Position value (**Maintenance > Axis Setup > Z**) is set so that the lowest point of the largest blade used on the system is about 3 mm above the chuck upper surface.

7. Follow the wizard steps:
  - When replacing the chuck with an identical chuck:
    - i. Press **Finish**.



- ii. Close the cover and press **Finish** for a second time; the system automatically performs the Chuck to Non-Chuck Delta Measurement.
- When the new chuck dimensions are different than the dimensions of the original chuck:
  - i. Press **Define**; the Operation Setup window appears.
  - ii. Press **Unlock**, define the new chuck values, and press **SAVE**.
  - iii. Press **Back** to return to the Dice window.
  - iv. Press **Finish**.
  - v. Close the cover and press **Finish** for a second time; the system automatically performs the Chuck to Non-Chuck Delta Measurement.



**Note:** If the system is equipped with a NCH device, the Delta Z procedure should be performed with a dressed blade.

8. Perform the Chuck Flatness Test as described in section 3.3.14.3.

### 9.3.2 Performing Chuck Height

Chuck Height teaches the system the chuck position by lowering the blade onto the cutting chuck. When chuck height is performed, the cutting chuck and the spindle shaft are disconnected from the grounding. The blade is lowered onto the metal ring at the edge of the cutting chuck. (The size of the ring is defined by the Contact Height Band Width parameter in **Operation Setup > Cutting Table**) The contact between the blade and the metal ring closes an electrical circuit sending a signal to the spindle driver that enables the location of the blade edge along the Z Axis to be measured.

#### To perform chuck height:

1. In the application, from the Main Menu, press **Maintenance > Functions and I/O**; the Functions and I/O window appears.
2. Press **Height**; the Height view appears.
3. Press **Chuck Height**; the system lowers the blade until it contacts the ring around the outer edge of the cutting chuck (the height results are reported in the Activity Log).



#### Notes:

- ◆ If the chuck is ceramic, the system automatically performs Height.
- ◆ If the chuck is metallic, the system requests to locate the point of contact by the visual system.

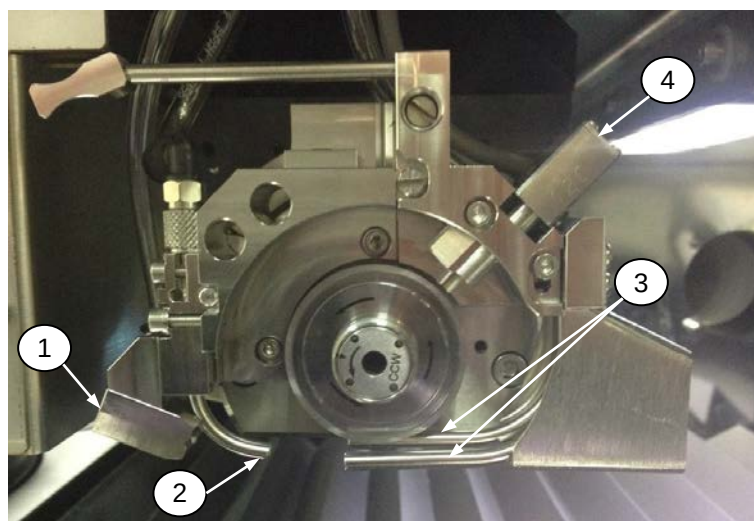
## 9.4 Maintaining the Cooling Block

The Cooling Block nozzles supply water to cool the Blade and to clean the surface of the Workpiece. The Model 71XX is equipped with one of three Cooling Blocks, which can be changed to fit different needs:

- •2"/3" Cooling Block for the 2" Spindle
- •5" Cooling Block for the 4" Spindle

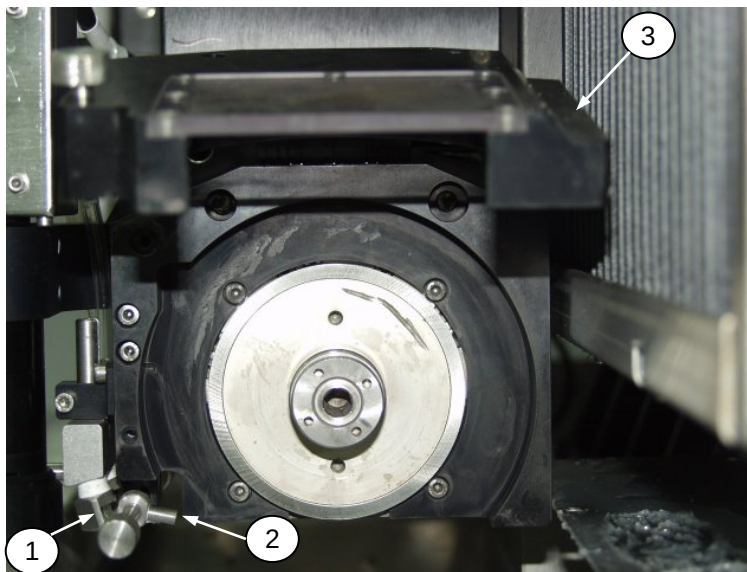


**Cautions:** It is imperative that the cooling block nozzles are correctly positioned so that they deliver maximum water to the blade and surface of the workpiece. They must also not be positioned too close to the surface of the workpiece where they could damage the surface of the workpiece.



- 1. Cleaning Jets
- 1. Main Jet
- 2. Spray Bars
- 1. BBD (option)

Figure 9-23. Cooling Block (2")



- 1. Cleaning Jets
- 2. Main Jet
- 3. Cooling Block Door

Figure 9-24. Cooling Block (4")

The Cooling Block contains the following cooling and cleaning elements:

- **Main Jet:** High-pressure water directed at the center of the blade cutting edge, to cool the blade and keep the cutting edge clean of debris.
- **Spray Bars (2"):** High-pressure water directed onto the two sides of the blade cutting edge, to cool the blade and keep the sides of the cutting edge clean of debris.
- **Cleaning Jets:** High-pressure water directed onto the workpiece, to keep the workpiece clean of debris.

### 9.4.1 Replacing a Cooling Block

During the course of working with the system, a cooling block can require replacement due to change in work requirements, damage or malfunction.

**To replace a cooling block:**

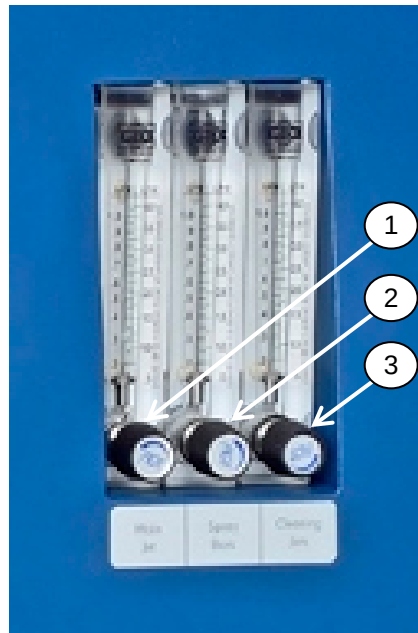
1. Turn off the air and water supplies to the system.
2. Remove the blade (see section 3.3.16 for reference).
3. (2" spindle option) Open the BBD locking screw and remove the BBD, if one is installed.
4. Disconnect the water tubes connected to the cooling block.
5. Remove the screws that secure the cooling block to the spindle.
6. Attach the new cooling block to the spindle with the screws
7. Attach the water tubes to the new cooling block.
8. (2" spindle option) Secure and adjust the BBD to the new cooling block, and tune the BBD (see section 9.1 for reference).
9. Turn on the water and air connections to the system.
10. From the application, turn on the water to the cooling block.
  - From the Main Menu, press **To Dice > Spindle** button > **Open Water**.
11. Use the nozzle adjusting screws to adjust the position of the nozzles:
  - Adjust the position of the Cutting Nozzle using the adjusting screw, so that the water flow is equally divided by the blade.
  - Adjust the height of the Cooling Nozzles with the screws on the back of each nozzle, so that they are approximately 1 mm above the surface of the workpiece.
  - Adjust the height of the Cleaning Nozzles as required, using the adjusting screw.



**Caution:** Make sure to position the cleaning nozzles so that they wash away the cutting debris. The debris must not be allowed to dry on the surface of the workpiece where it can harden and be difficult to remove.

## 9.4.2 Adjusting the Water Flow to the Cooling Block

The water flow to the cooling block can be controlled from the Cooling Block Water Flow Adjustment Unit located at the front of the system. The flow of water (liters per minute) is set according to the planned work procedure.



1. Main Jet
2. Spray Bars
3. Cleaning Jets

**Figure 9-25. Cooling Block Water Flow Adjustment Unit**

The Main Jet is connected to the application through a flow sensor. Therefore, adjustments to the Main Jet water flow can be seen from the application (see section 5.3.4.3 for reference). All other water adjustments to the cooling block are strictly mechanical.



**Note:** Flow sensors also enable stopping the cutting process when the detected Main Jet water pressure is below the set threshold.

## Chapter 10. Maintaining the Vision System

The system incorporates a high performance vision system that enables precise workpiece positioning for the purpose of alignment and kerf checking.

Maintaining the Vision System includes:

- ❖ Maintaining the Camera (section 10.1).
- ❖ Maintaining the Microscope (section 10.2).
- ❖ Calibrating the Vision System (section 10.3)

## 10.1 Maintaining the Camera

Maintaining the Camera includes:

- Replacing the Camera (with the same type of camera).
- Verifying the Driver
- Aligning the Camera.

### 10.1.1 Replacing the Camera

There are three types of camera replacements: analog to analog, analog to digital and digital to digital.

When replacing the camera, perform the following procedure is required.

**To replace the camera:**

1. Power down the system.
2. Disconnect the System from Mains power.
3. Open the spindle cover and remove the camera cover.



1. Camera (with cover)
2. Microscope Bracket Clamp Screws
3. Microscope

**Figure 10-1. Vision System**

4. Detach the camera cable from the top of the camera.
5. Remove the camera from the system by rotating the camera counterclockwise until it detaches from the microscope.
6. Install the new camera, rotating the new camera clockwise into the microscope's camera fitting until the camera is attached firmly.
7. Reattach the camera cable, and replace the camera cover.

8. Close the spindle cover.
9. Reconnect the system to the main power source and power up the system.
10. Calibrate the Vision if necessary (see section 10.3).

## 10.1.2 Verifying the Driver

After replacing the camera, verify that the driver is installed and recognized by the system. First, read following section 10.1.2.1.

### 10.1.2.1 MSC Image and Users Review

The Main System Controller (MSC) can be configured in one of the following two Image types:

- **Closed Image**

The MSC closed image type includes two defined user profiles:

- **Operator**

This is the system default user, with limited access rights, and is used for all standard system operations.

- **Administrator**

The Administrator user is required for performing special operations, as reverting to previous software versions or for system troubleshooting, and requires a log-in password.

- **Open Image**

The MSC or PC open image type includes only one user profile, with full access rights (no user configuration limitations).



**Note:** Verifying the driver is performed when logged-in as an Administrator.

**To identify the image profile:**

1. Exit to Windows.
2. Click **Start** or press **Ctrl+Esc** on the keypad. The Start menu content identifies the image type and profile.

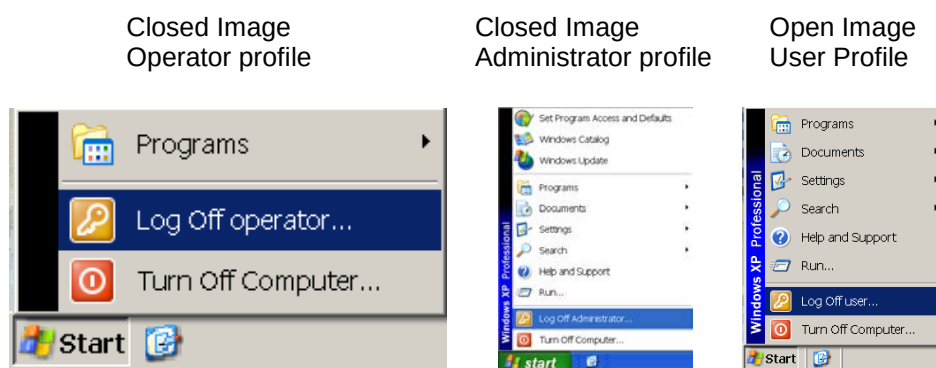


Figure 10-2. Image and Profiles



**To switch to the Administrator profile:**

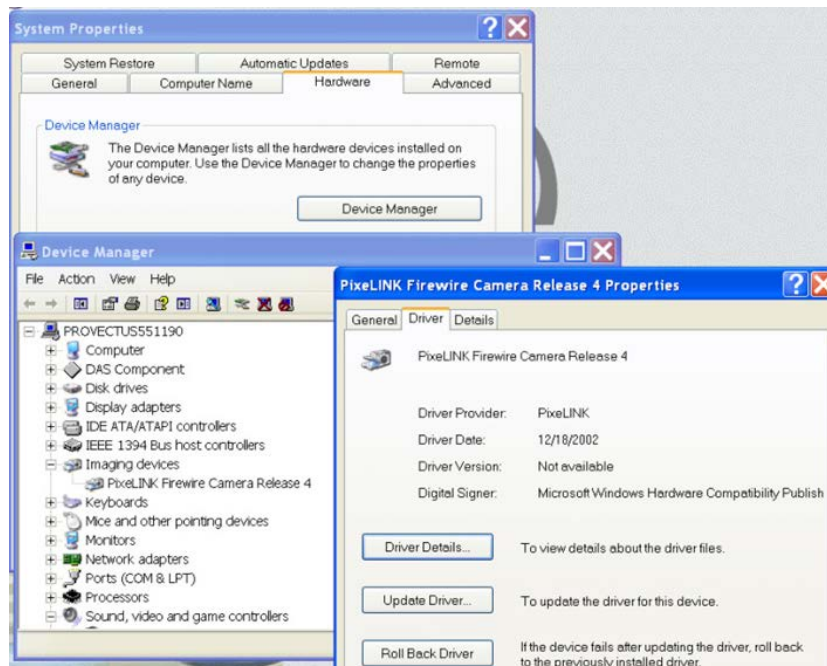
1. From the desktop press **Ctrl+Esc**.
2. Click **Start-> Log Off operator...**
3. Press the keyboard keys **Ctrl+Alt+Del** twice.
4. The Log On to Windows window appears.
5. Select **Administrator** as the User name.
6. Type the Password: **adminadt**.
7. Click **OK**; the profile changes to Administrator.

### 10.1.2.2 PixeLINK Camera Driver Verification

After installing the PixeLINK camera, verify the driver software is installed and recognized by the system.

**To verify the PixeLINK camera driver is installed and recognized:**

1. Switch from Operator to Administrator user.
2. Click **Start > Settings > Control Panel > System**; the System Properties window appears.
3. Click **Hardware > Device Manager**; the Device Manager window appears.
4. From the Device Manager list, double click **Imaging devices**.
5. Under Imaging devices, select **PixeLINK Firewire Camera Release 4**.
6. Double click **PixeLINK Firewire Camera Release 4**, the PixeLINK Firewire Camera Release 4 Properties window appears.



**Figure 10-3. Device Manager window**

If the driver is not PixeLINK, the driver provider should be updated. For updating the driver, see following section 10.1.2.3.



### 10.1.2.3 Updating the Driver Provider for the PixelINK Camera

1. From the **PixelINK Firewire Camera Release 4 Properties** window Driver tab, click **Update Driver**; the Hardware Update Wizard appears.

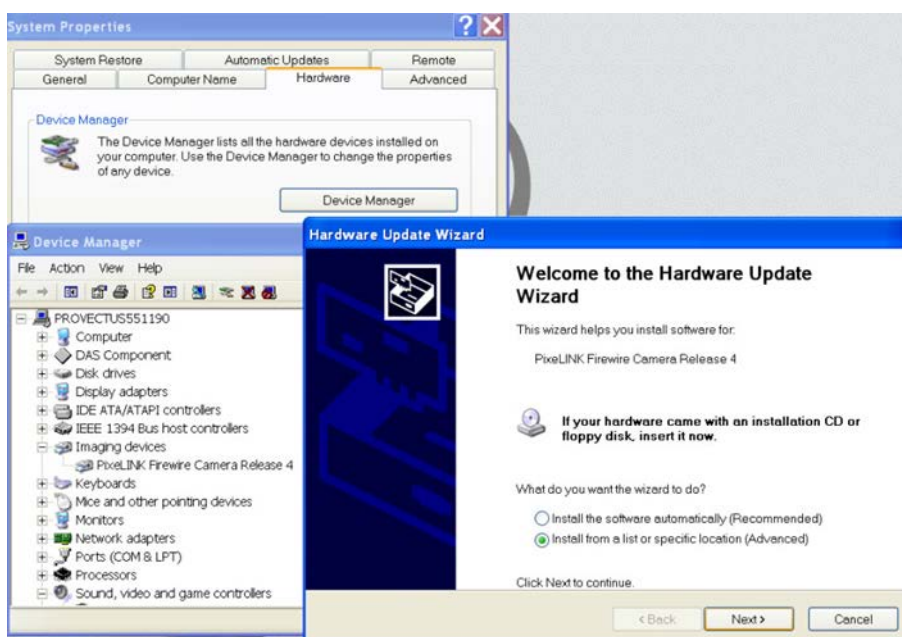


Figure 10-4. Hardware Update Wizard window

2. Select **Install from a list or specific location (Advanced)** and click **Next**.

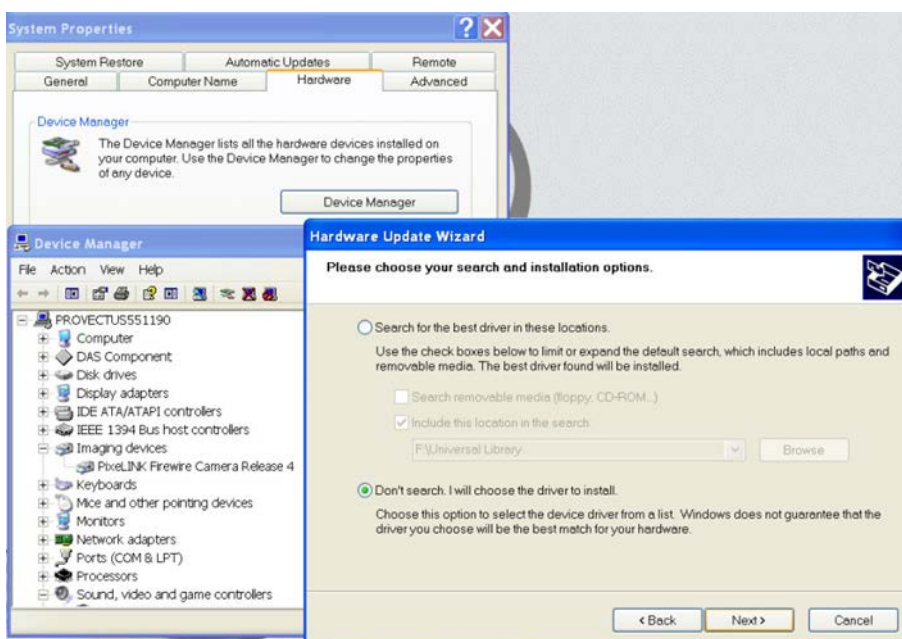


Figure 10-5. Choose search and Installation options window

3. Select **Don't Search. I will choose the driver to install**, and click **Next**; the Select the device driver window appears.

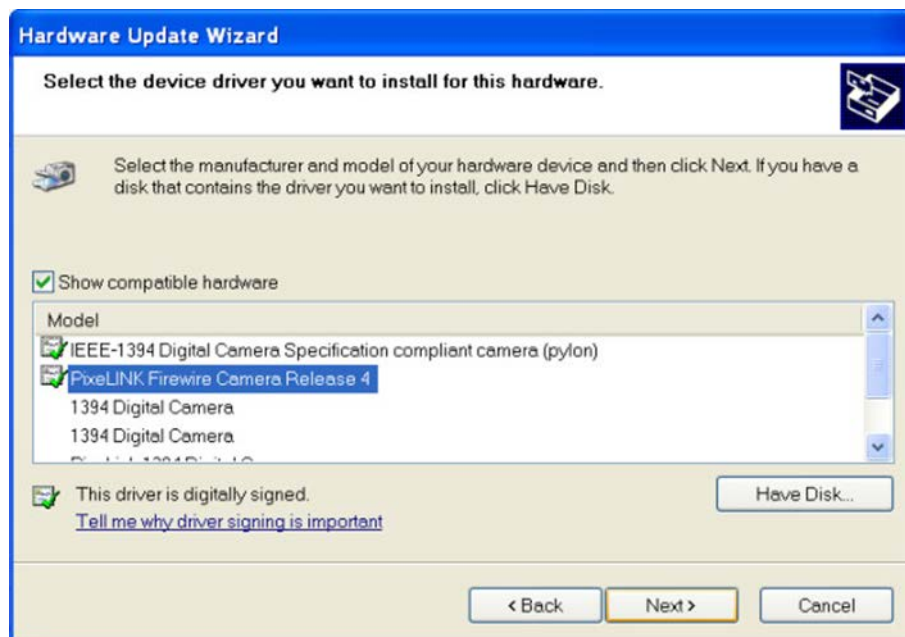


Figure 10-6. Select the device driver window

4. Select **PixeLINK Firewire Camera Release 4** and click **Next**.

The required files will be copied; the Completing the Hardware Update Wizard window appears.

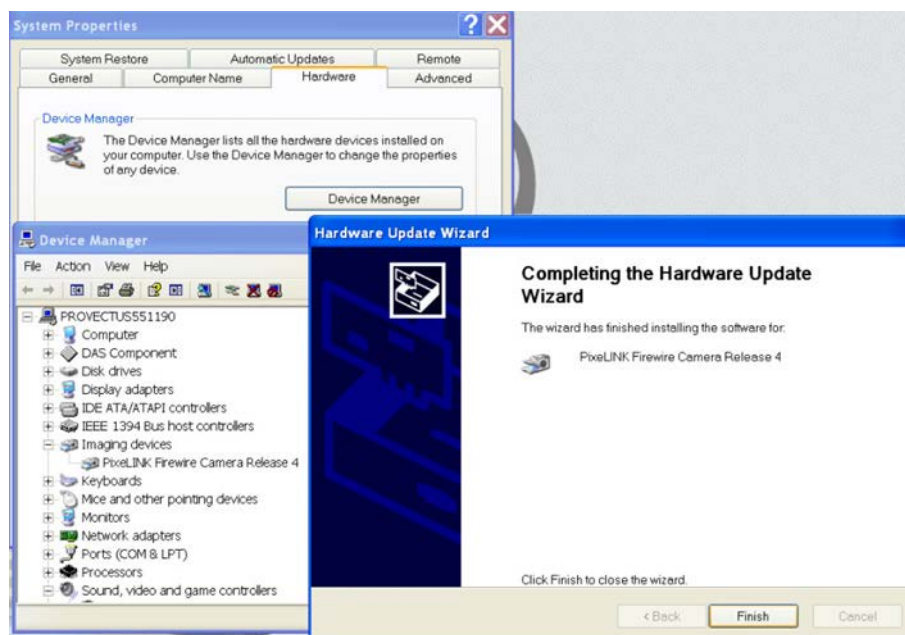


Figure 10-7. Completing the Hardware Update Wizard window

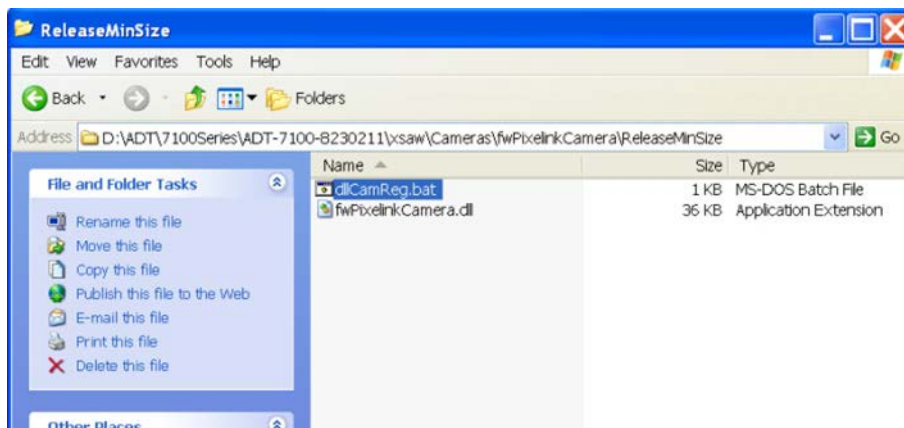
5. Click **Finish** to completing the hardware wizard.

### 10.1.2.4 Configuring the PixeLINK Camera Driver

After verifying that the camera driver is installed and recognized by the system, configure the PixeLINK camera driver in the application.

**To configure the PixeLINK camera driver:**

1. Go to **D:\ADT\7100Series\ADT-7100-XXXXXXX\xsaw\Cameras\fwPixelinkCamera\ReleaseMinSize**.



**Figure 10-8. PixeLINK batch File**

2. Double-click the **dlCamReg.bat** file; the application appears.
3. Click **ADTsystem** shortcut to run the ADT dicing application.
4. Login as Administrator, and from the Main Menu go to **Maintenance > Administration > System > Hardware Configuration** and verify the Camera Type is **PixeLink**.



**Figure 10-9. Hardware Configuration**

### To Adjust the Camera Parameters:

1. Open the ADT dicing application and from the **Axis Control** window bring the camera into focus.
2. Press the **Working Area** button, and from the **Illumination** icon select **Camera Parameters**; the Camera Parameters window appears.

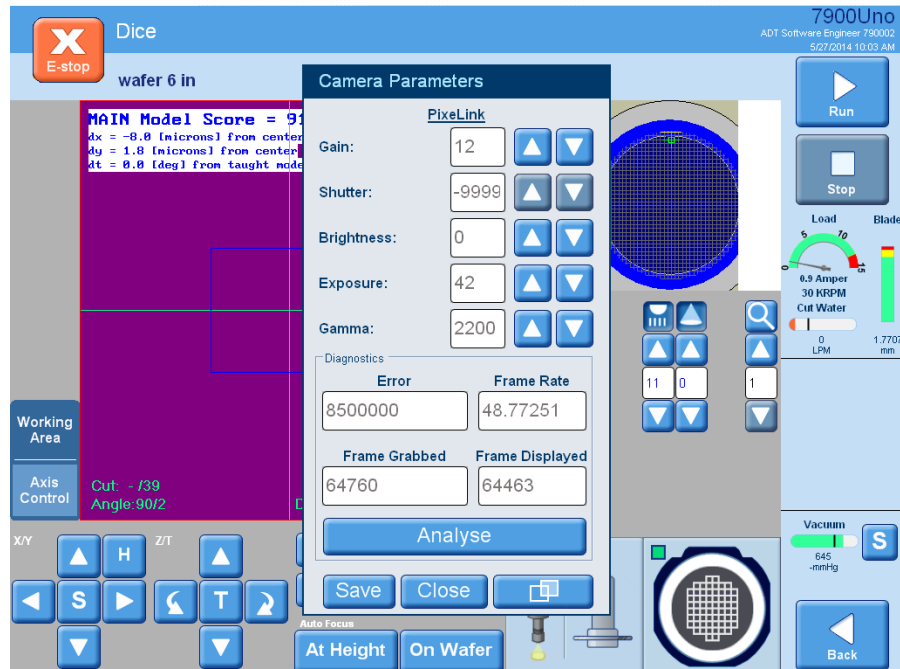


Figure 10-10. PixeLINK Camera Parameters

3. Adjust the settings according to visual reception on the workpiece.
4. Verify that the visual system is aligned to the X and Y Axes (see section 10.1.3).
5. Calibrate the vision system; if necessary (see section 10.3).
6. Reassemble the vision system covers, and close the spindle cover.
7. Shut down the system and then power on to verify that the system is functioning correctly.

## 10.1.3 Aligning the Camera

After installing a new camera, align the camera along the X Axis before resuming dicing.

### To align the camera:

1. Load a workpiece onto the cutting chuck.
2. Manually align the workpiece (for Manual Alignment, see the 7120/7130 Operation Manual).
  - a. Use the Dice window **X/Y** axis controls to move the camera from one side of the workpiece to the other.
  - b. Examine the display in the video workspace for alignment.

- ♦ If the display appears to be misaligned, continue with step 3.
  - ♦ If the display appears to be aligned, resume dicing.
3. Align the camera along the X Axis:
    - a. Open the spindle cover to gain access to the camera.
    - b. If very slight adjustment is needed, move the entire visual system by hand, in either a clockwise or a counterclockwise direction. Make sure that the camera remains firmly screwed into the microscope.
    - c. If a greater adjustment is needed, continue with step 4.
  4. To perform a fine alignment adjustment:
    - a. Loosen the two microscope bracket clamp screws (see Figure 10-1 above) and adjust the camera/microscope unit (keep the screw sufficiently tight so as prevent the vision system from falling out of the bracket but loose enough to afford controlled movement by hand).
    - b. If the camera is properly aligned but the microscope requires alignment, gently pull the vision system down until the setscrew is exposed.
    - c. Open the setscrew and rotate only the microscope until the microscope is aligned to the X Axis.
    - d. Tighten the setscrew and gently push the vision system back up into the bracket clamp.
    - e. Tighten the bracket clamp screws.
    - f. Close the spindle cover.



**Note:** After moving the vision system, Auto Focus on Dice Chuck value needs to be updated. For details, see section 10.2.2.1.

## 10.2 Maintaining the Microscope

The microscope is located to the left of the spindle, and is attached by a bracket to the spindle. The camera is positioned above the microscope.



**Note:** When the onscreen display shows signs that the microscope lens is dirty, remove the lens protector and clean the lens with a cotton swab and isopropyl alcohol.

Maintaining the Microscope includes:

- Replacing the Microscope.
- Adjusting the Microscope.

### 10.2.1 Replacing the Microscope

When a microscope with a different magnification is required, or if the microscope is faulty, the microscope should be replaced.

#### **To remove a microscope:**

1. Power down the system; the spindles automatically move to the Home position and provide you access to the vision.
2. Open the spindle cover and detach the cable from the top of the camera.
3. Measure and record the distance between the bracket bottom edge and the microscope bottom edge (this distance is required for installation).
4. Remove the camera from the system by rotating the camera counterclockwise until it detaches from the microscope.
5. Disconnect the illumination assembly with the air tube from the microscope.
6. Carefully loosen the screw that secures the microscope to the microscope bracket clamp.
7. Remove the microscope from the system by sliding it out from underneath the microscope bracket clamp.

#### **To install the new microscope:**

1. Insert the new microscope into the microscope bracket clamp from underneath.
  - a. Gently push the microscope up until the setscrew is hidden by the microscope bracket.
  - b. Verify that the distance between the bracket bottom edge and the microscope bottom edge is the same as measured in step 3 above, and adjust accordingly.
  - c. Tighten the bracket securing screw.
2. Connect the illumination assembly with the air tube.
3. Attach the camera to the top of the microscope by inserting it into the fitting and rotating clockwise until tight.

4. Attach the camera cable to the top of the camera.
5. Close the spindle cover.
6. Power up the system.
7. Align the camera and microscope to the X Axis (see section 10.1.3 above).
8. Calibrate the vision system, as described in following section 10.3.

## 10.2.2 Adjusting the Microscope

After a new microscope is installed, aligning it along the X, Y, and Z Axes and the following procedures must be performed, before resuming dicing:

- Teaching the Chuck Center to the System.
- Teaching Focus on Chuck to the System.
- Performing Manual Y Offset.

### 10.2.2.1 Teaching Focus on Chuck

After the chuck center has been taught, reteach the system the chuck focus.



**Note:** If continuing from "Teaching the Chuck Center", skip to step 6.

#### To reteach focus on chuck:

1. In the application, from the Main Menu, press **Dice > Axis Control**.
2. Using the **Z/T** axis controls, manually focus the new microscope on the cutting chuck.
3. Press **Back > Maintenance > Axis Setup**; the Axis Setup window appears.
4. Press **Unlock** to enable modifications.
5. Press **Z**; the Z view appears.
6. Scroll to **Auto focus on dicer chuck position**; the Teach button becomes active.
7. Press **Teach** to teach the system the focus on the chuck.
8. Press **SAVE**.
9. Perform the camera alignment procedure as described in section 10.1.3.

### 10.2.2.2 Teaching the Chuck Center

Teach the system the location of the center of the cutting chuck.

#### To teach the chuck center to the system:

1. In the application, from the Main Menu, press **Maintenance > Axis Setup**; the Axis Setup window appears.
2. In the left panel, scroll to and press **Chuck Center**; the Chuck Center view appears.
3. Press **Unlock** to enable changes.

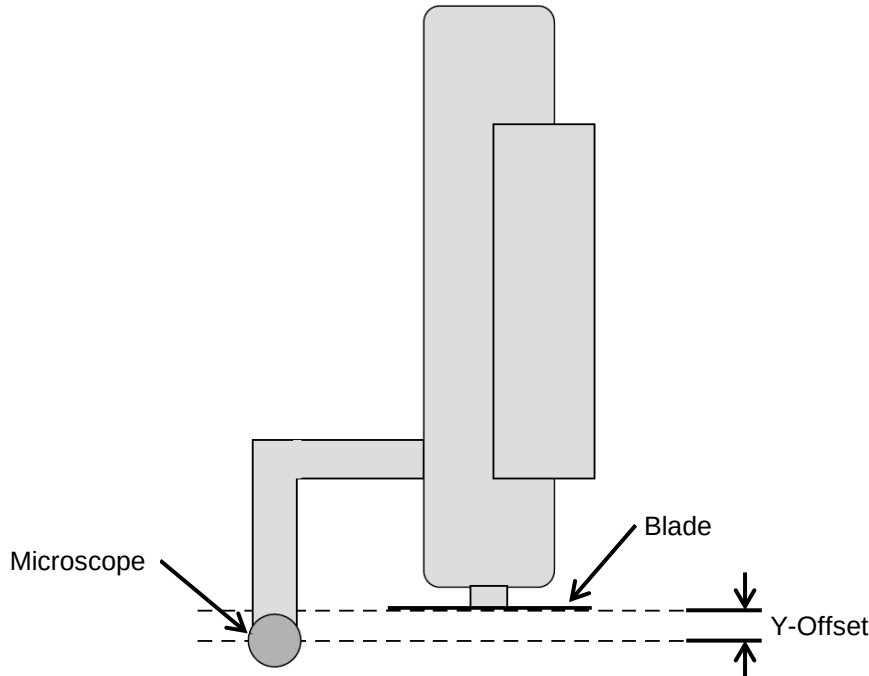


4. Press **Move To Station**; the microscope moves to the approximate center of the cutting chuck.
5. Press **Back > Back > To Dice > Axis Control**; the Dice window appears in the Vision Control view.
6. Using the **Light** controls, manually adjust the illumination until a clear image of the chuck appears.
7. Press the **At Height** button (for **Auto Focus**); the system automatically sets the microscope's Z position above the chuck.
8. Use the **Z/T** axis controls to rotate the cutting chuck on the theta axis.
9. Verify in the video display that the rotation occurs at the center of the display.
  - When a deviation can be seen, proceed with step 10.
  - When there is no deviation, continue with step 12.
10. Use the **X/Y** axis controls to move the microscope to the chuck center.
11. Use the **Z/T** axis controls to rotate the chuck and verify that the reticle is at, or close to, the absolute center of the cutting chuck.
12. Press **Back > Maintenance > Axis Setup > Chuck Center**; the Axis Setup window reappears in the Chuck Center view.
13. Press **Teach All**; the new chuck center position is taught to the system.
14. Press **SAVE**.



### 10.2.2.3 Performing Manual Y Offset

The final adjustment performed after installing a new microscope is Manual Y Offset. Manual Y Offset is a procedure that teaches the system the difference in position between the blade and the microscope along the Y Axis.



**Figure 10-11. Y Offset Between Microscope and Blades**

Teaching the Y-offset involves performing a single cut and positioning the microscope exactly over the cut. The single cut can be performed within the Cut Map or outside of the Cut Map (for example, on the tape).

#### 10.2.2.3.1 Defining Manual Y Offset Recipe Parameters

Before the Manual Y Offset adjustment can be performed, the system must be taught where to place the single cut that will be used in the procedure. To teach this to the system, the Manual Y Offset parameters first need to be defined in the application.

**To define the Manual Y Offset recipe parameters:**

1. From the Main Menu, press **Assign Job**; the Assign Job window appears.
2. Select the target recipe and press **Assign**.
3. Return to the Main Menu and press **Recipe Builder**; the Recipe Builder window appears.
4. Press **Edit**; the Recipe Editor window opens.
5. Scroll to the **Y Offset** category, and press the **In Cut Map Only** parameter:
  - **Yes**: to perform a single cut within the Cut Map.
  - **No**: to perform a single cut outside of the Cut Map. When set to No, the following two parameters need to be defined:

- ♦ Define End Location from Edge: the distance from the upper and lower edge of the workpiece towards the center of the workpiece where the single cut can be performed.
- ♦ Define Start Location from Edge: the distance from the upper and lower edge of the workpiece towards the edge of the frame where the single cut can be performed.



**Note:** For information on adding parameters to a recipe, refer to the Performing Dicing Procedures chapter in the ADT 7120/7130 Operations Manual.

**To perform Manual Y Offset:**

1. From the Main Menu, press **To Dice**; the Dice window appears.
2. Press the **Workpiece** button and in the menu, select **Load Workpiece**; the workpiece is loaded onto the chuck.



**Note:** To perform Y-offset in a Cut Map, generate a Cut Map (see “Manually Aligning the Workpiece” section in the ADT 7120/7130 Operations Manual).

3. Press the **Spindle** button and select **Y Offset**; the Manual Y Offset dialog box appears.
4. Press **Yes** to perform a manual single cut; the wizard appears and directs you to find the Y Cut Center.



**Note:** When a single cut was performed before selecting Manual Y Offset, the microscope automatically moves to the center of that cut.

5. Use the Axis controls to move the reticle to the center of the cut:
  - S (X/Y controls): moves the microscope slowly.
  - P (X/Y controls): moves the microscope one pixel at a time.
  - T (Guide controls): sizes the blue Teach Window around the cut. The borders should be aligned over the edges of the kerf.
6. In the wizard, press **Finish**; the system is automatically updated.

**To verify Y Offset parameters:**

1. From the Main Menu, press **Operation Setup > Head System**; the Head System view appears.
2. View the Y Offset Correction.

## 10.3 Calibrating the Vision System

Vision Calibration is critical for all procedures that make use of the vision system, such as Alignment and Kerf Checking. The vision algorithms that control the vision procedures must have a precise pixel-to-micron conversion rate in order to achieve accurate results.

There are two calibration types that can be performed on the vision system:

- Pixel Size Calibration.
- Illumination Calibration.

### 10.3.1 Performing Pixel Size Calibration

Pixel Size Calibration measures the exact size of a pixel, in microns, as seen in the FOV. Calibration should be performed on the Special Calibration workpiece (ADT P/N 06980-0174-002), included with the system.



**Note:** Pixel size calibration can be performed on any other target with exact dimensions. When using the Special Calibration workpiece, change the X Target Size and Y Target Size parameter values accordingly.

Pixel Size Calibration should be performed whenever the camera or microscope is replaced. If after performing a vision component replacement the light intensity is satisfactory, there is no need to perform the Illumination Calibration. However, Pixel Calibration must be performed.

**To perform Pixel Size Calibration:**

1. From the Main Menu, press **To Dice**; the Dice window appears.
2. Press the **Workpiece** button and select **Load Workpiece**; the Load Workpiece dialog box appears.
3. Place the Special Calibration workpiece onto the chuck and press **OK**.
4. Press **Back** and from the Main Menu press **Maintenance > Calibration > Pixel**; the Maintenance-Calibration Pixel view appears.
5. Verify that both X Target Size and Y Target Size are 509 microns and that Microscope Magnification is 4.
6. If the Target Sizes are correct, press **Pixel Calibration**; the Pixel Calibration wizard appears in the Dice window.
  - a. Move the vision system to the center of the workpiece.
  - b. From the Vision Control view, set the magnification, use the light controls to set the illumination for optimum contrast and use the **Z/T Axis Controls** to correct the focus.
  - c. Use the **Z/T Axis Controls** to align the target.
  - d. Use the Guide Controls to size the blue Search box and red Teach box around the reticle in the center of the Field of View (FOV).

- e. Press the **Teach** button; the system measures the pixel size, using the target as a reference, and displays the results, in microns, in the FOV.
- f. Double-press any point in the FOV to verify that the calibration was successful; the reticle should be right on the point aimed at (in both X and Y direction). The results should be within the limits given in the following Table 10-1.

**Table 10-1. Camera Values**

| Camera Type | Magnification | Range for X Pixel<br>[μm] | Range for Y Pixel<br>[μm] |
|-------------|---------------|---------------------------|---------------------------|
| PixeLink    | × 4 (×120)    | 0.9 ± 0.1                 | 0.9 ± 0.1                 |

- g. Press **Finish** to complete the procedure.

**To verify the accuracy of the values:**

1. Double-press any location in the FOV; the selected point should move to the center of the reticle.
2. Measure a known distance and compare the results.

## 10.3.2 Performing Illumination Calibration

Illumination Calibration is performed to define a series of values that measure the sensitivity of the camera to coaxial and oblique illumination. The illumination parameters are defined as default and no additional adjustments are required.

However, each type of illumination can be set according to a scale ranging from 0 (completely dark) to 255 (completely white), which is useful in case of a special application.



**Note:** The system is set with a default linear graph for the illumination calibration. Any change to the illumination calibration graph will affect all vision tasks of the recipes (such as taught models for Auto Alignment, Cut Verify models, Kerf Checks).



**Caution:** After changing the calibration, the previous or default calibration graph is lost. The calibration should be performed with the Special Calibration workpiece, included with the system. A wizard leads through the calibration procedure.

### To perform illumination calibration:

1. From the Main Menu, press **To Dice**; the Dice window appears.
2. Press the **Workpiece** button and select **Load Workpiece**; the Load Workpiece dialog box appears.
3. Place the Special Calibration workpiece onto the chuck and press **OK**.
4. Press **Back** and from the Main Menu press **Maintenance > Calibration > Illumination**; the Illumination calibration-graph appears.

Illumination Lookup Graph

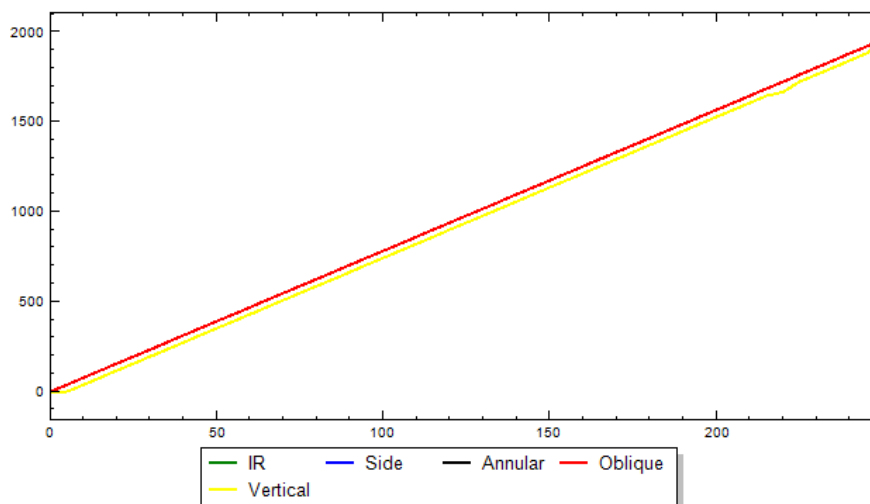


Figure 10-12. Illumination Calibration

In the Illumination Lookup Graph, the red line represents the oblique illumination values while the yellow line represents the coaxial illumination values.

1. Press **Illumination Calibration**; the Illumination wizard appears in the Vision Control view of the Dice window and the vision system moves to a position over the center of the Special Calibration workpiece.
  - a. In the Axis Control view, set to high magnification, and use the **Z/T** Axis controls to adjust the focus.
  - b. Press the **Next** button; the System automatically calibrates coaxial and oblique illumination values for the camera (the process can take several minutes to complete).
2. Press **Finish** to complete the procedure (or **Retry** to perform again).

When the procedure is complete, the results can be viewed in the Illumination Lookup Graph (see Figure 10-12 above).



**Note:** If the system is unable to perform illumination calibration, the system will stop and request to reset the illumination level and refocus on the target. After this is done, press Next to have the system attempt calibration again.

# Chapter 11. Maintaining Internal and External Utilities

The internal and external utilities are controlled and monitored by valve units located on the pneumatic panel. Maintaining Internal and External Utilities explains how to replace, adjust and test these valves, in case of malfunction.

The Maintaining Internal and External Utilities chapter includes the following sections:

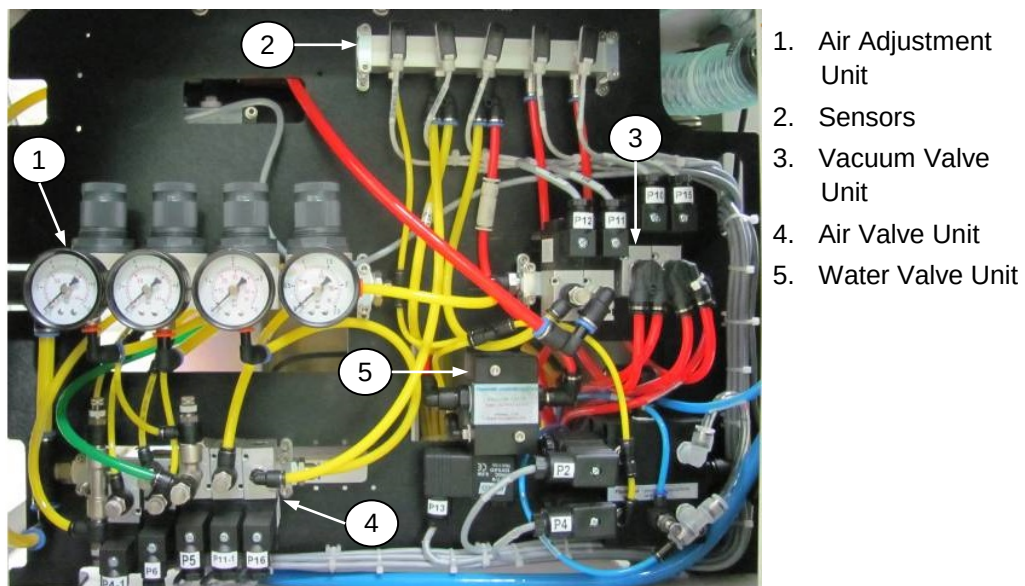
- ❖ Introduction (section 11.1)
- ❖ Maintaining the Vacuum Valve Unit (section 11.2)
- ❖ Maintaining the Water Valve Unit (section 11.3)
- ❖ Maintaining the Air Valve Unit (section 11.4)
- ❖ Maintaining the Air Tank (section 11.5)
- ❖ Maintaining the Water Separator (section 11.6)
- ❖ Calibrating the Pressure Sensors (section 11.7)
- ❖ Adjusting Air Pressure using the Air Regulator Unit (section 11.8)

## 11.1 Introduction

The system employs the following utilities:

- Main Air: for the X Axis and spindle air bearing.
- Vacuum: internally generated by a venturi vacuum pump located on the canister:
  - For securing workpieces during processing.
  - For holding the cutting chuck in place on the theta table.
  - For holding the theta table during dicing.
- Tap Water: for cooling the spindle.
- DI (Deionized) Water or Tap Water
  - For cooling the cutting blade.
  - For cleaning the workpiece and NCH device of cutting debris.

The valves and solenoids that control the air and water are mounted on the Pneumatic Panel.



**Figure 11-1. Pneumatic Panel**

This chapter explains how to replace, adjust and test a valve, should one become faulty.

In addition, this chapter describes how to replace the elements in the two Air Filters, which are mounted on the back of the Pneumatic Panel.

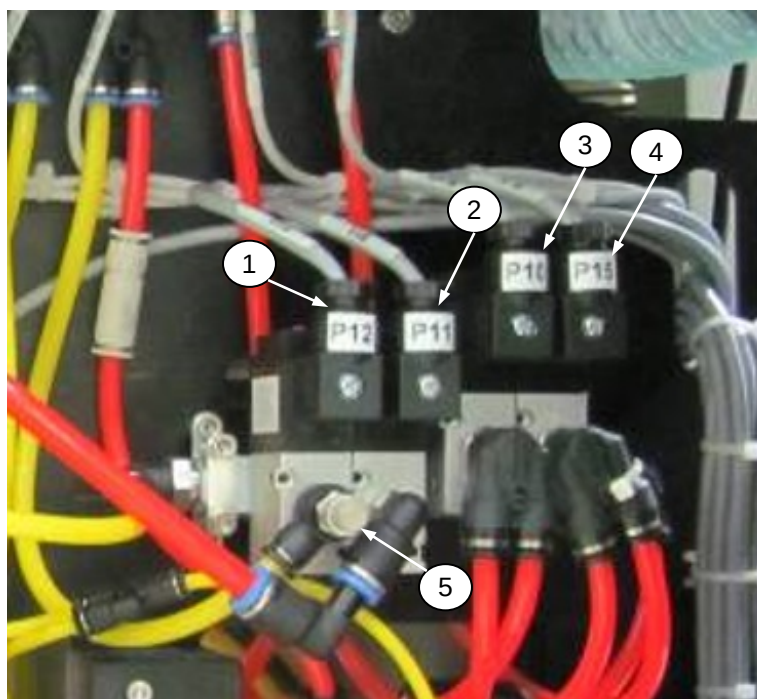


## 11.2 Maintaining the Vacuum Valve Unit

The vacuum valve unit contains solenoids that control the flow of air to the water separator, drain, and chuck vacuum.

The vacuum valve unit, located at the back bottom-left of the system, contains the following solenoids:

- **Solenoid P12:** controls the air supply that releases the workpiece from the chuck (air puff).
- **Solenoid P11:** controls the drain air-release vacuum, which removes accumulated water from the water separator.
- **Solenoid P10:** controls the chuck vacuum, which holds the cutting chuck to the theta table. The chuck vacuum is supplied by the vacuum pump connected to the solenoid.



1. Solenoid P12
2. Solenoid P11
3. Solenoid P10
4. N/A (P15)
5. Wafer Air Release Regulator

**Figure 11-2. Air and Vacuum Valve Unit**

Maintaining the Vacuum Valve Unit is composed of the following sections:

- Replacing a Vacuum Valve.
- Adjusting the Vacuum Valve Sensor Threshold.
- Testing the Vacuum Valve.

## 11.2.1 Replacing a Vacuum Valve

A faulty vacuum valve solenoid can be replaced without having to change the entire vacuum valve unit.

### To replace a faulty vacuum valve solenoid:

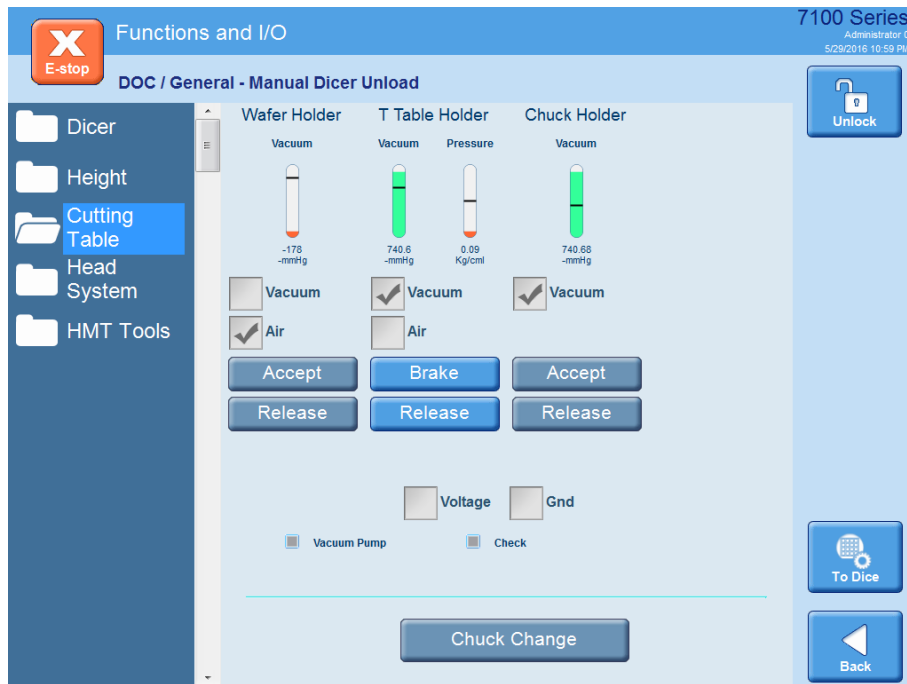
1. Power down the system.
2. Turn off the air and water supply to the system.
3. Remove the rear panel; the vacuum valve unit is on the bottom left.
4. Remove the vacuum air tube from the faulty valve.
5. Remove the electrical connector screw and disconnect the electrical connector (when removing the electrical connector, the solenoid securing bolt is revealed).
6. Remove the solenoid securing bolt and remove the solenoid along with its cover.
7. Secure the new solenoid and cover in place, using the solenoid securing bolt.
8. Attach the electrical connector and tighten the electrical connector screw.
9. Connect the tube removed in step 4.
10. Place back the rear panel.
11. Turn on the air and water supply to the system.
12. Power up the system.
13. Verify solenoid functionality from the interface (see section 11.2.3).

## 11.2.2 Adjusting the Vacuum Valve Sensor Threshold

After replacing a vacuum valve or solenoid, adjust the vacuum valve sensor threshold.

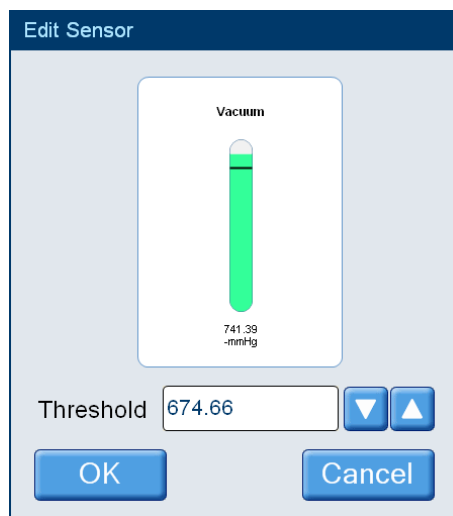
### To adjust a vacuum valve sensor threshold:

1. In the application, from the Main Menu, press **Maintenance > Functions and I/O > Cutting Table**; the Functions and I/O window appears in the Cutting Table view.



**Figure 11-3. Adjusting Vacuum Levels**

2. Press **Unlock** to enable modification.
3. Press the image of the vacuum sensor to adjust; the Edit Sensor window appears.



**Figure 11-4. Edit Sensor**

4. Press the **Threshold** number or the increase/decrease buttons to change the setting.
5. Press **OK** to save the settings.
6. Repeat steps 3 to 5 for each vacuum image.



**Note:** For the Sensor level default settings, see section 11.7.1 on page 11-205.

### 11.2.3 Testing the Vacuum Valve Solenoids

After replacing a vacuum valve solenoid, the solenoid must be tested to verify that it is operating properly and that there are no leaks. This is accomplished by activating the solenoid through the application.

#### To test Solenoid P12 for workpiece air release:

1. From the Main Menu, press **Maintenance > Functions and I/O > Cutting Table**; the Functions and I/O window appears in the Cutting Table view.

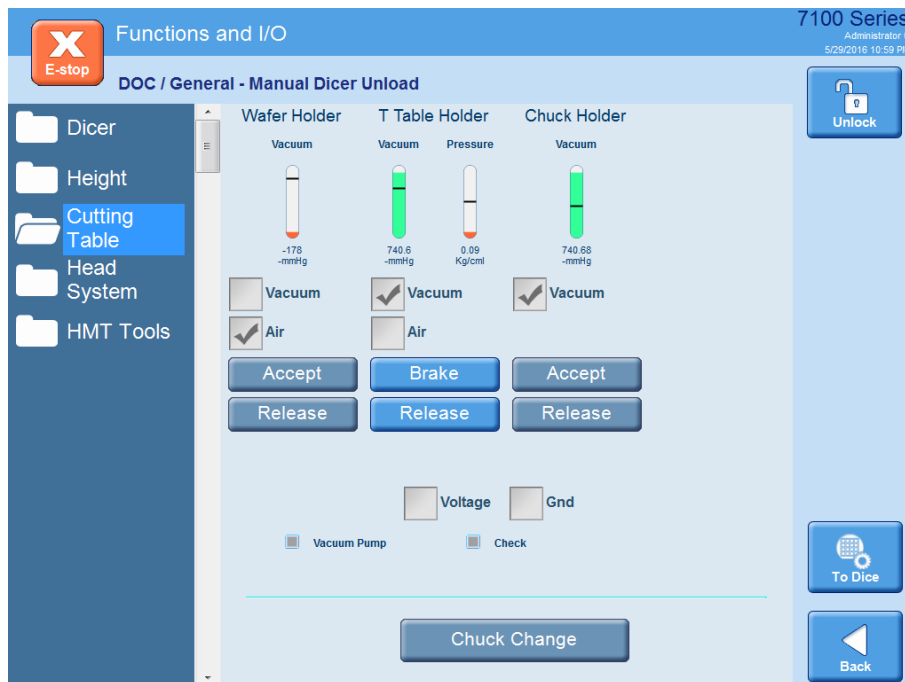


Figure 11-5. Adjusting Vacuum Levels

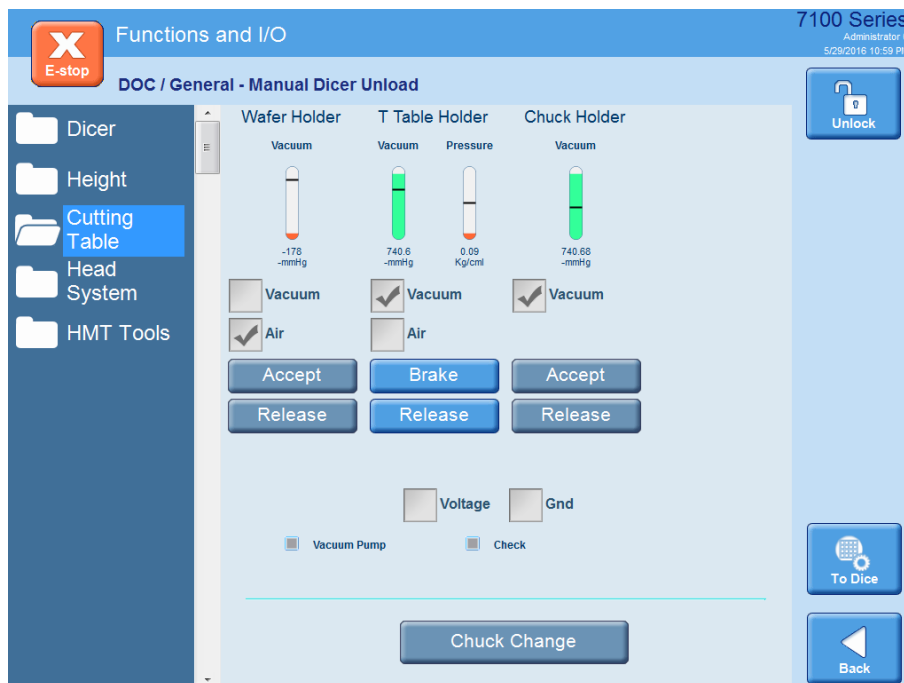
2. Press **Unlock** to enable modification.
3. Press the **Accept** button to activate the vacuum valve to test; the Vacuum check box is selected and the vacuum level appears beneath the Vacuum gauge (the black Threshold line should be in the green zone).
4. Examine the system and the vacuum unit to verify that the solenoid is functioning as required.
5. Press the **Release** button to clear the Vacuum check box and deactivate the vacuum; the test is completed.

#### To test Solenoid P11 for drain air release:

Fill the lower chamber of the water separator with water. The solenoid should become activated and draw the water out of the chamber.

**To test Solenoid P10 for chuck vacuum:**

1. Ensure that there is a cutting chuck on the theta table.
2. From the Main Menu, press **Maintenance > Functions and I/O > Cutting Table**; the Functions and I/O window appears in the Cutting Table view.



**Figure 11-6. Adjusting Vacuum Levels**

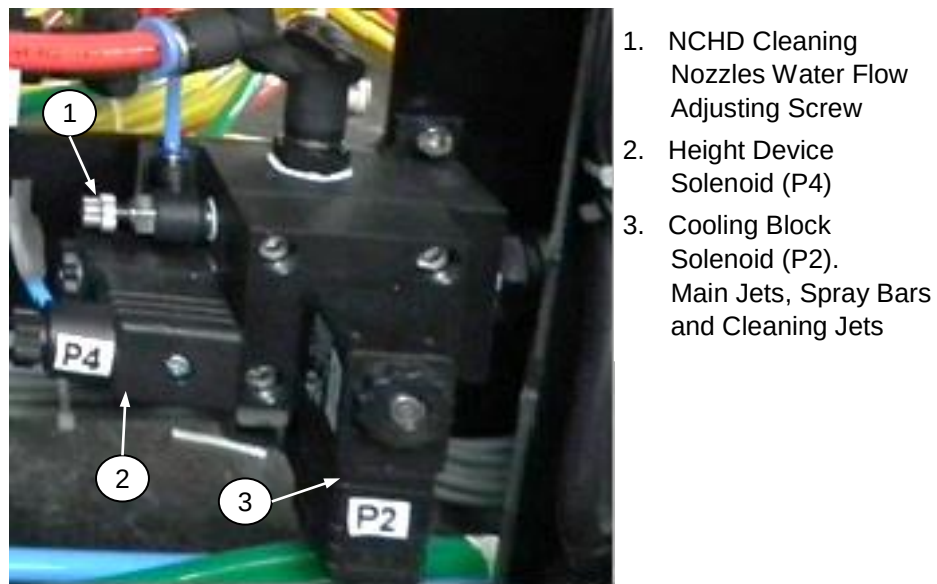
3. Press **Unlock** to enable modifications.
4. Press the Chuck Holder **Accept** button or select the **Vacuum** check box to activate the vacuum; the vacuum level appears beneath the Vacuum gauge (the black Threshold line should be in the green zone).
5. Examine the system and the vacuum unit to verify that the solenoid is functioning as required.
6. Press the **Release** button to clear the Vacuum check box and deactivate the vacuum; the test is completed.

## 11.3 Maintaining the Water Valve Unit

The water valve unit contains solenoids that control the water flow to the height device and the DI (Deionized) water supply to the cooling block. The solenoid, which controls the vacuum supply to the workpiece, is also located on this unit, as water is sometimes present in the air flowing through this vacuum. The water valve unit is located at the lower right of the pneumatic panel.

The water valve unit contains the following solenoids:

- **Solenoid P13:** controls the levels of workpiece vacuum, holding the workpiece to the cutting chuck.
- **Solenoid P4:** controls the flow of water to the height device and keeps the height device free of debris.
- **Solenoid P2:** controls the flow of DI water to the cooling block (main jets, cooling block spray bars, and cleaning jets).



**Figure 11-7. Water Valve Unit**

Maintaining the Water Valve Unit includes the following sections:

- Replacing a Water Valve Solenoid.
- Adjusting a Water Valve.
- Testing a Water Valve Solenoid.

### **11.3.1 Replacing a Water Valve Solenoid**

A faulty water valve solenoid can be replaced without having to change the entire water valve unit.

**To replace a water valve solenoid:**

1. Power down the system.
2. Turn off the air and water supply to the system.
3. Remove the rear panel to expose the water valve unit.
4. Remove the tube from the faulty valve.
5. Remove the electrical connector screw and disconnect the electrical connector.
6. Loosen the solenoid cover bolt and turn the solenoid cover to one side to expose the solenoid securing bolt.
7. Remove the solenoid securing bolt and remove the solenoid along with the solenoid cover.
8. Remove the needle valve and attach it to the new solenoid with solenoid cover.
9. Secure the new solenoid, with its cover, into place using the solenoid securing bolt. The electrical prongs should be facing downward.
10. Connect the electrical connector and tighten the electrical connector screw.
11. Connect the tube on the needle valve.
12. Place back the rear panel.
13. Turn on the air and water supply to the system.
14. Power up the system.

### **11.3.2 Adjusting a Water Valve**

After installing a new water valve unit or water valve solenoid, adjust the unit by using the needle valves connected to the solenoid.

### 11.3.3 Testing a Water Valve Solenoid

After replacing a water valve solenoid, the solenoid should be tested to verify it is operating properly and that there are no leaks from the connections.

**To test Solenoid P13 for workpiece vacuum:**

Follow the instruction provided in section 11.2.3 above.

**To test Solenoid P4 for the Height Device:**

1. From the Main Menu, press **Maintenance > Functions and I/O**; the Functions and I/O window appears.
2. Press **Height**; the Height view appears.
3. Press **Unlock** to enable modification.
4. Select the NCH **Water** check box to activate the flow of water to the height device.
5. Adjust the NCH device water valve to a point that the water coming out of the prism water nozzles covers the prisms with a spray of water.
6. Examine the gauges to verify the water flow, and examine all connections for leaks.
7. Clear the **Water** check box to complete the test.

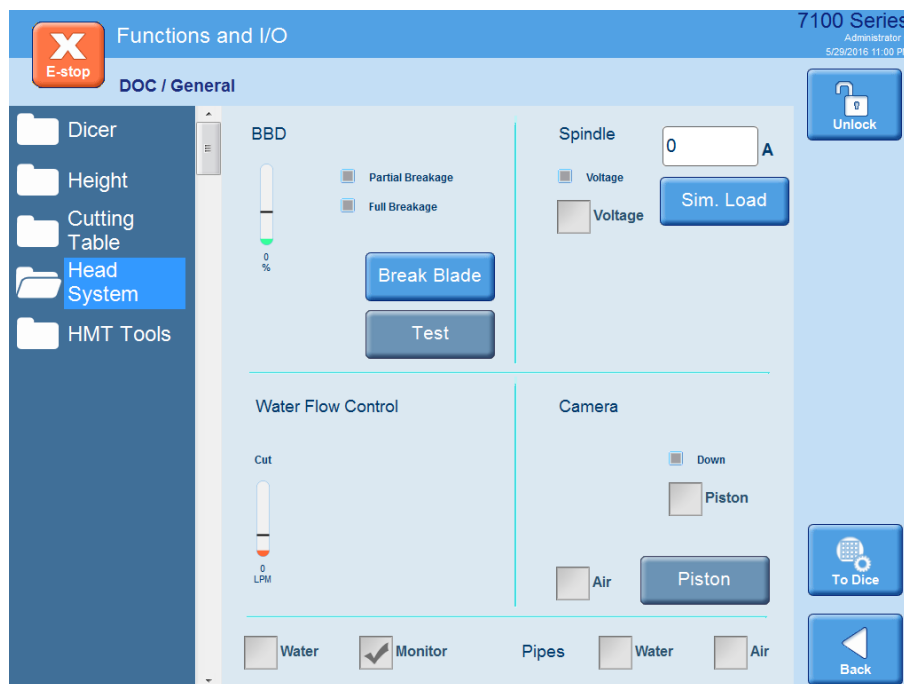


### To test Solenoid P2 for the cooling block:



**Note:** The solenoid P2 also controls the water for the main jets, spray bars, and cleaning jets.

1. From the Main Menu, press **Maintenance > Functions and I/O**; the Functions and I/O window appears.
2. Press **Head System**; the Head System view appears.



**Figure 11-8. Testing the Water Flow to the Cooling Block**

3. Press **Unlock** to enable modification.
4. In the Water Flow Control panel, select the **Water** check box; the water begins to flow.



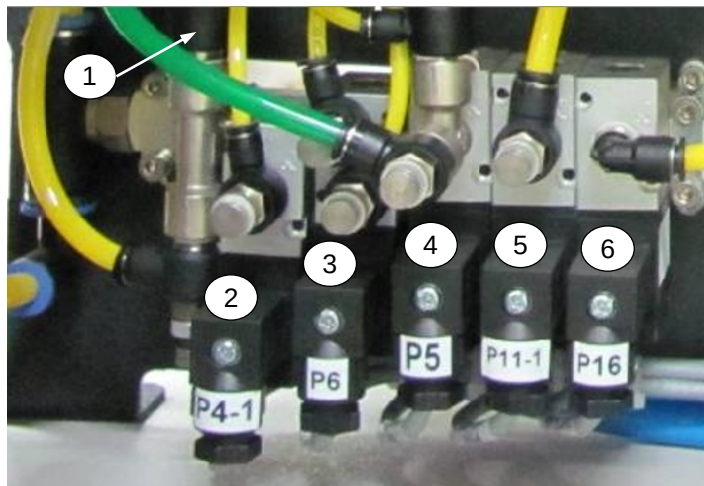
**Note:** The Monitor check box, which enables the cut water sensor, must also be selected. When selected, the dicing stops if water-flow is not detected.

5. Examine the gauges to verify the water flow, and check all connections for leaks.
6. Clear the **Water** check box to complete the test.

## 11.4 Maintaining the Air Valve Unit

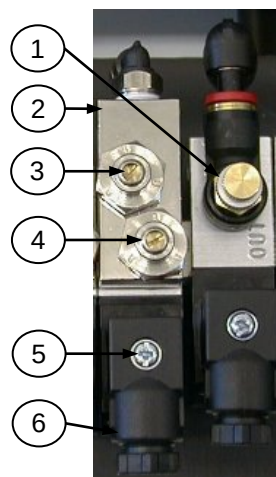
The air valve unit contains solenoids that control the airflow to the height device, microscope, water separator, and theta table. The air valve unit, located at the back, lower-left of the system, contains the following solenoids:

- **Solenoid P6:** controls the microscope cleaning air. The upper screw controls the low pressure (during dicing) while the lower screw controls the high pressure (during vision tasks).
- **Solenoid P5:** controls the high-pressure airflow that keeps the height device free of debris and water.
- **Solenoid P4-1:** controls the low-pressure airflow that keeps the height device free of debris and water.
- **Solenoid P11-1:** works in conjunction with Solenoid 11 in the vacuum unit to control the drain air release, which facilitates removal of accumulated water.
- **Solenoid P16:** controls the flow of air to the Closed Loop Theta Table.



1. Air Gun Needle Valve
2. Non-Contact Height (low pressure) Solenoid (P4-1)
3. Microscope Cleaning Solenoid (P6)
4. Non-Contact Height (high pressure) Solenoid (P5)
5. Drain Air Release Solenoid (P11-1)
6. Theta Table Solenoid (P16)

Figure 11-9. Air Valves Unit



1. Needle Valve
2. Solenoid with Cover
3. LP Needle Valve
4. HP Needle Valve
5. Electrical Connector Screw
6. Electrical Connector

Figure 11-10. Air Valves (close-up)

### 11.4.1 Replacing an Air Valve Solenoid

A faulty air valve solenoid can be replaced without having to change the entire air valve unit.

**To replace an air valve solenoid:**

1. Power down the system.
2. Turn off the air and water supply to the system.
3. Remove the rear panel; the air and vacuum unit is seen.
4. Remove the tube from the faulty valve.
5. Remove the electrical connector screw and disconnect the electrical connector.
6. Loosen the solenoid cover bolt and turn the solenoid cover to one side to expose the solenoid securing bolt.
7. Remove the solenoid securing bolt and remove the solenoid along with the solenoid cover.
8. Remove the needle valve and attach it to the new solenoid with solenoid cover.
9. Secure the new solenoid, with its cover, into place using the solenoid securing bolt. The electrical prongs should be facing downward.
10. Connect the electrical connector and tighten the electrical connector screw.
11. Replace the tube on the needle valve.
12. Replace the rear panel.
13. Turn on the air and water supply to the system.
14. Power up the system.

### 11.4.2 Adjusting an Air Valve

After replacing an air valve solenoid, the air valve should be adjusted, using the needle valve on top of each solenoid.



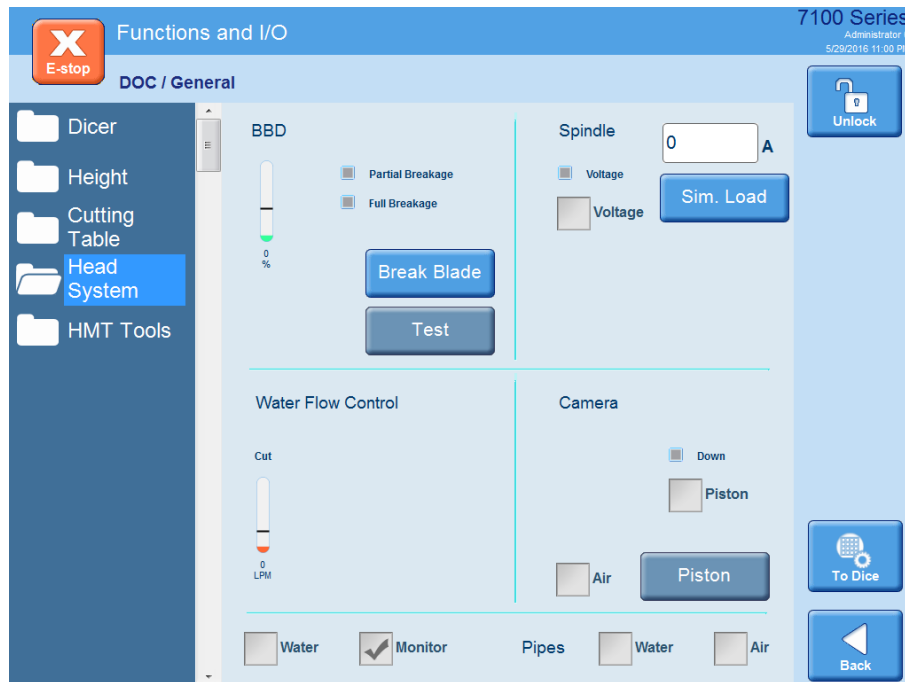
**Note:** Solenoid 16 (optional), which controls the flow of air to the Theta table, has no Needle Valve. If necessary, it can be adjusted using the Theta Air Adjustor on the Adjustment Unit as shown in Figure 11-1.

### 11.4.3 Testing an Air Valve Solenoid

After replacing an air valve solenoid, the solenoid should be tested to verify that it is operating properly and that there are no leaks in the connections.

**To test Solenoid 6 for cleaning the microscope:**

1. From the Main Menu, press **Maintenance > Functions and I/O**; the Functions and I/O window appears.
2. Press **Head System**; the Head System view appears.



**Figure 11-11. Testing for Microscope Cleaning**

3. Press **Unlock** to enable modification.
4. In the Camera area, select the **Air** check box to activate the flow of air to the camera lens and workpiece.
5. Examine the flow of air to the camera lens and workpiece and examine all connections for leaks.
6. Clear the **Air** check box to complete the test.

**To test Solenoids P4-1 and P5 for cleaning the height device:**

1. From the Main Menu, press **Maintenance > Functions and I/O**; the Functions and I/O window appears.
2. Press **Height**; the Height view appears.
3. Press **Unlock** to enable modification.
4. Clear the **Water** check box; the low air pressure is activated, as these two solenoids are connected in parallel (the water solenoid is normally open, while the air solenoid is normally closed). Do not reselect the Water check box before this procedure is completed.
5. Select the **LP Air** check box.
6. Use the P4-1 valve to adjust the low air pressure.
7. Select the **HP Air** check box to activate the high air pressure.
8. Use the P5 valve to adjust the high air pressure.

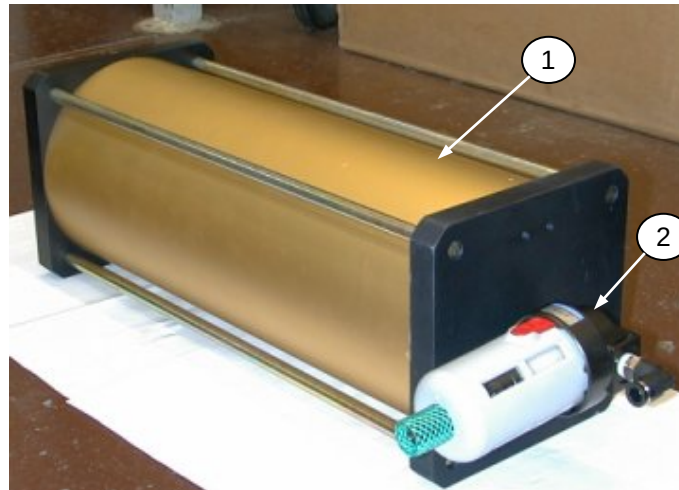
**To test Solenoid P11-1 for the drain air release:**



**Note:** Solenoid P11-1 works in conjunction with Solenoid P11 in the vacuum valve unit, and is tested together with it (for details, see section 11.2.3 “Testing the Vacuum Valve Solenoids” on page 11-190).

## 11.5 Maintaining the Air Tank

The system includes an air tank that continues to supply air to the spindle for a short period of time in case of an emergency shutdown. This prevents damage to the spindle by enabling spindle rotation to gradually stop. The air tank is equipped with a filter.



- 1. Air Tank
- 2. Air Filter

Figure 11-12. Air Tank with Filter

### 11.5.1 Maintaining the Air Tank Filter

During the course of operation, the air filter can become clogged with minute particles.

**To replace the air tank filter:**

1. Stop system operation.
2. Open the right panel; the air tank and filter can be seen.



**Note:** There is still air pressure inside the tank, so removing the air filter housing will be difficult or impossible.

3. Turn off the air supply to the machine, and wait for the air inside the tank to be released before removing the filter housing.
4. Remove the filter housing by pressing the red knob on the filter housing and rotating it clockwise.
5. Remove the old filter.

Note there are two O-rings; one mounted on the inner diameter of the filter housing and the other on the filter thread outer diameter. When a new filter is ordered from ADT, it comes along with a new O-ring. Use this new O-ring and put back the original O-ring of the filter housing.

6. Re-Install the filter housing with the new filter, and turn it counterclockwise until you hear the red knob snap and lock.
7. Open the spindle air and check for any leakage.
8. Start up the system.

## 11.6 Maintaining the Water Separator

The system includes a water separator which is responsible for collecting and draining water from the workpiece vacuum space. If, due to a malfunction, the water separator needs to be replaced, perform the following procedures:

- Replacing the Water Separator.
- Adjusting the Water Separator.
- Testing the Water Separator.

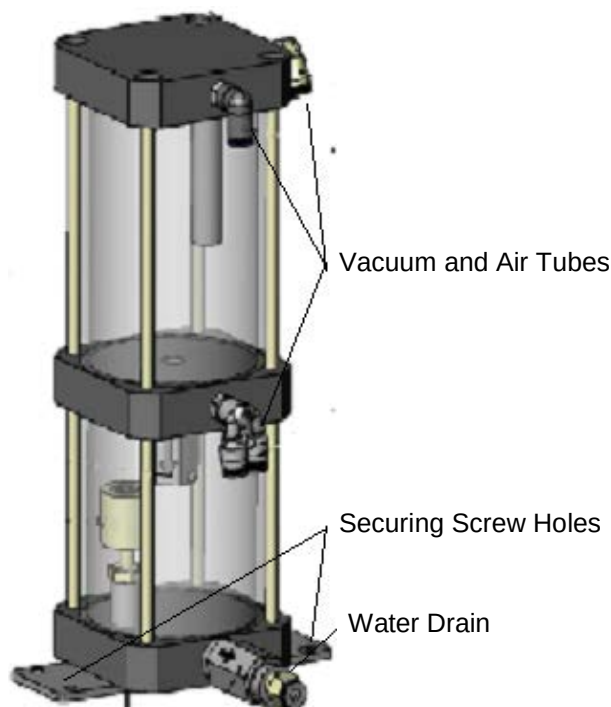


Figure 11-13. Water Separator.

### 11.6.1 Replacing the Water Separator

Replace the water separator when it malfunctions.

**To replace the water separator:**

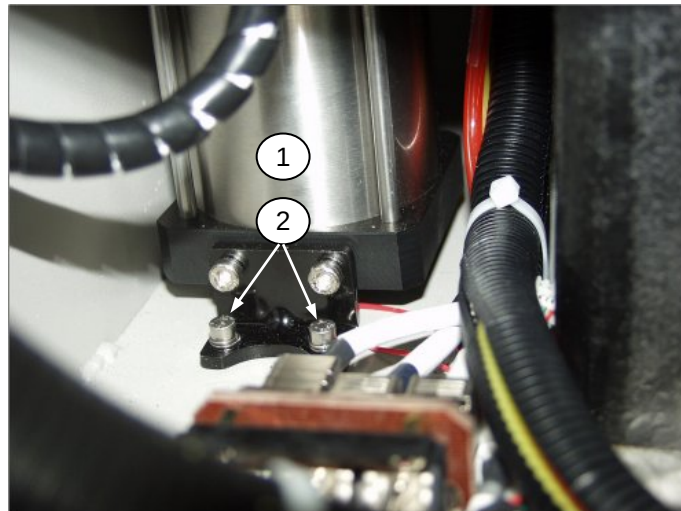
1. Close air and water supply.
2. Open the rear panel.
3. Disconnect all vacuum and air tubes connected to the water separator.
4. Disconnect the water drain tube.
5. Disconnect the electronic level sensor connector.





**Figure 11-14. Level Sensor Connector**

6. Disconnect the two screws securing the Water Separator to the Bench (from the other side, the Water Separator module is rested on two pins).



1. Water Separator
2. Securing Screws

**Figure 11-15. Water Separator Securing Screws**

7. Replace the water separator.
8. Reconnect the screws securing the water separator to the bench.
9. Reconnect the water drain tube.
10. Reconnect the vacuum and air tubes.
11. Reconnect the level sensor connector.



## 11.6.2 Adjusting the Water Separator

The water separator must be adjusted after replacement.

**To adjust the water separator:**

1. Open the rear panel; the air and vacuum unit is seen (see Figure 11-9).
2. Completely close the P11-1 valve and then open it 3 turns.

## 11.6.3 Testing the Water Separator

After replacing the water separator, it should be tested.

**To test the Water Separator:**

1. Open the right panel to enable a clear view of the Water Separator.
2. In the interface, from the Main Menu, press **Operation Setup > Dicer**; the Operation Setup window appears in the Dicer view.
3. Press **Unlock** to enable changes.
4. Press the **Drain on Unload** value and select **Yes**.
5. Press **SAVE**.
6. Return to the Main Menu and press **Dice**; the Dice window appears.
7. Press the **Workpiece** button and from the menu, select **Unload Workpiece** (if there is no loaded workpiece, first select **Load Workpiece**).
8. Verify that the Water Separator is draining water, signifying the Water Separator is functioning correctly.
9. Close the right panel and from the interface, press **Operation Setup > Dicer**; the Operation Setup window appears in the Dicer view.
10. Press **Unlock** to enable changes.
11. Press the **Drain on Unload** value and select the required value (Yes/No).
12. Press **SAVE**.

## 11.7 Calibrating the Pressure Sensors

The pressure sensors in the system can deviate from the actual pressure values. This variance can result from the sensor itself or from the Analog to Digital device located in the controller. It is recommended to use the procedure provided below in case of a difference of up to 20%. If the variance is greater than 20%, replace the sensor or the controller.

After replacing the sensors or the controller, the reading accuracy of the sensors should be verified again.

### To calibrate the sensor:

1. Connect a manometer in parallel to the sensor.
2. Write down the reading of the two sensors in two different states (pressures) as shown in the table below.

| State | GUI Sensor Readout | Manometer Readout |
|-------|--------------------|-------------------|
| 1     |                    |                   |
| 2     |                    |                   |

3. In the application, from the Main Menu, press **Maintenance > Calibration > Sensors**; the Maintenance - Calibration window appears in the Sensors view.

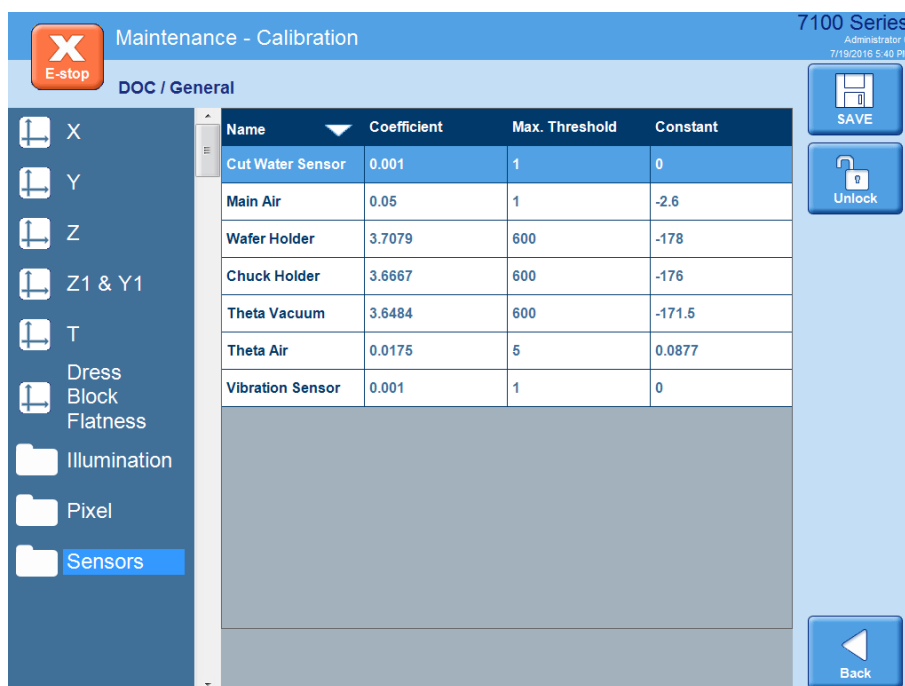


Figure 11-16. Sensor Calibration Parameters

4. Write down the current sensor reading values that appear in the Calibration window.
5. Calculate the new coefficient and constant using the following formula:

$$coefficient_{new} = \frac{Mano\_reading_2 - Mano\_reading_1}{GUI\_reading_2 - GUI\_reading_1} \bullet coefficient_{old}$$

$$Constant_{new} = Mano\_reading_1 - coefficient_{new} \bullet \frac{GUI\_reading_1 - Constant_{old}}{coefficient_{old}}$$

**Figure 11-17. Calibration Formula**

6. Press **Unlock** to enable modification.
7. Change the **Coefficient** and the **Constant** sensor values.
8. Press **SAVE**.

## 11.7.1 Reference Values for Calibration

**Table 11-1. Regulator Default Settings**

| Regulator Description                        | Spec Value                                 |
|----------------------------------------------|--------------------------------------------|
| General Air Supply                           | 80 ± 5 PSI (5.6 ± 0.4 kg/cm <sup>2</sup> ) |
| X-Axis Air Flow                              | 80 ± 5 PSI (5.6 ± 0.4 kg/cm <sup>2</sup> ) |
| Spindle Air Flow                             | 80 ± 5 PSI (5.6 ± 0.4 kg/cm <sup>2</sup> ) |
| Theta Air Flow                               | 40 ± 5 PSI (2.8 ± 0.4 kg/cm <sup>2</sup> ) |
| Wafer Air Release                            | 0.5-0.7 Bar (0.5-0.7 kg/cm <sup>2</sup> )  |
| Wall (facility) Air Flow Regulator           | 80-102 PSI (5.6–7.2 kg/cm <sup>2</sup> )   |
| Wall (facility) Spindle Water Flow Regulator | 40-90 PSI (2.8–6.3 kg/cm <sup>2</sup> )    |

**Table 11-2. Air and Vacuum Threshold Default Values**

| Threshold Description             | Spec Value              |
|-----------------------------------|-------------------------|
| Wafer Holder Vacuum GUI Threshold | 500±20 mm Hg            |
| Chuck Holder Vacuum GUI Threshold | 500±20 mm Hg            |
| T-table Air Pressure Threshold    | 1.17 kg/cm <sup>2</sup> |
| T-table Vacuum Threshold          | 500 mm Hg               |

## 11.8 Adjusting Air Pressure using the Air Regulator Unit

The Air Regulator Unit is used to adjust the air pressure for the X Axis, Spindle, Theta and the Workpiece Air Release.

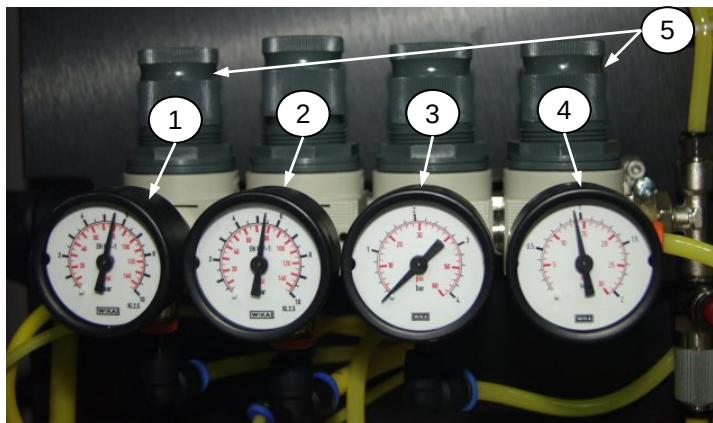
### To adjust air pressure:

1. For each regulator, pull out the air adjustment button to enable performing adjustments.
2. Turn the air adjustment button (clockwise to open or counterclockwise to close) until the pressure dial registers the suitable air pressure.
3. Push the air adjustment button back into place.



### Notes:

- ◆ To fine-tune the main regulator for workpiece air release, use the valve on Solenoid P12 (see Figure 11-9. Air Valves Unit).
- ◆ For default valve settings, see tables in “Reference Values for Calibration” on page 11-205.



1. X Axis Air Pressure Gauge
2. Spindle Air Pressure Gauge
3. Theta Air Pressure Gauge
4. Wafer Air Release Gauge
5. Air Regulator Adjusting Knobs

**Figure 11-18. Air Regulator Unit**



**Note:** The Adjustment Unit is used to adjust the air pressure for the X Axis, Spindle, Theta and the Workpiece Air Release, as shown in Figure 11-18. The air pressure to each can be adjusted, if necessary, by lifting the upper cap, twisting as required and pushing the cap back into place. In addition to the Main Regulator for the Workpiece Air Release, there is a Fine Tuning Regulator on the out of Valve P12 (see Figure 11-2). The Default Valve settings are provided in section 11.7.1.

## 11.9 Vacuum Canister

The Model 71XX dicing saws use air for operating various systems, for which the main air consumers, which constantly consume a large amount of air, are the venturi vacuum pumps.

To reduce the total amount of air required by the System, the internal venturi vacuum pumps can be replaced by a reduced air consumption system (vacuum pump and tank), which accumulates and provides the required vacuum.

In such a configuration, the vacuum to the machine vacuum consumers is supplied by the reduced air consumption system. When the level of vacuum in the canister reaches its lower level, the machine supplies air to the pump; increasing the pressure of the vacuum canister tank.

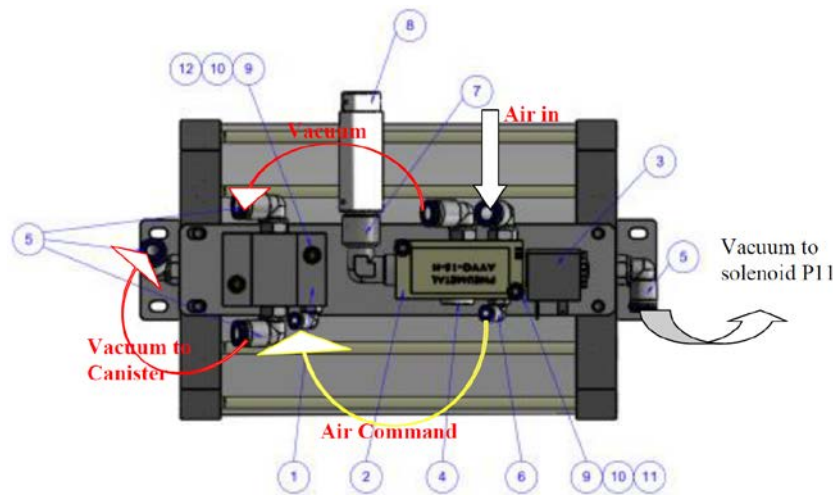


Figure 11-19. Vacuum Canister System

### 11.9.1 Vacuum System Operation

The following figures describe the vacuum system operation during various operations of the dicing saw.

- Vacuum on for wafer, chuck and Theta table (see in Figure 10-25).
- Vacuum on for chuck, Theta and wafer air puff (see Figure 10-26).
- Vacuum on during water-separator operation (see Figure 10-27).

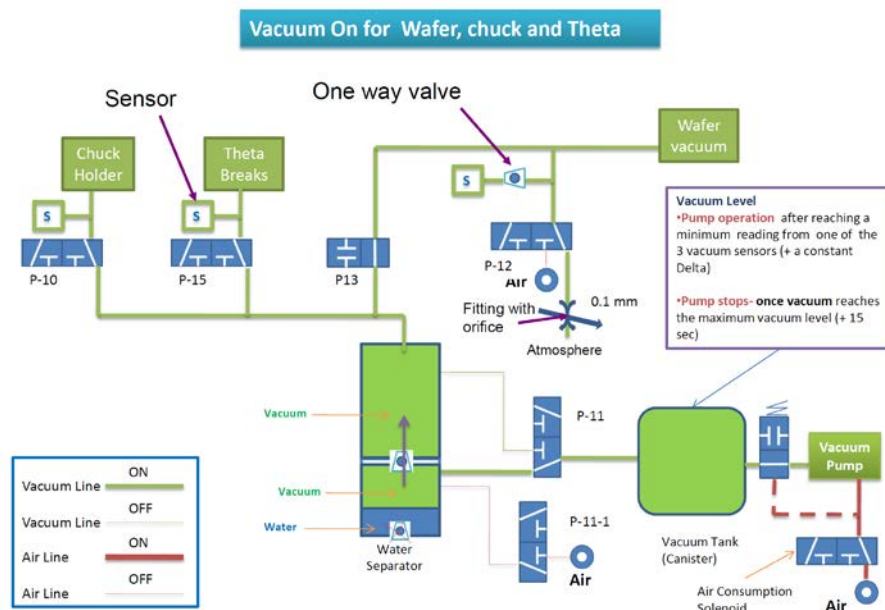


Figure 11-20. Vacuum on for wafer, chuck and Theta table

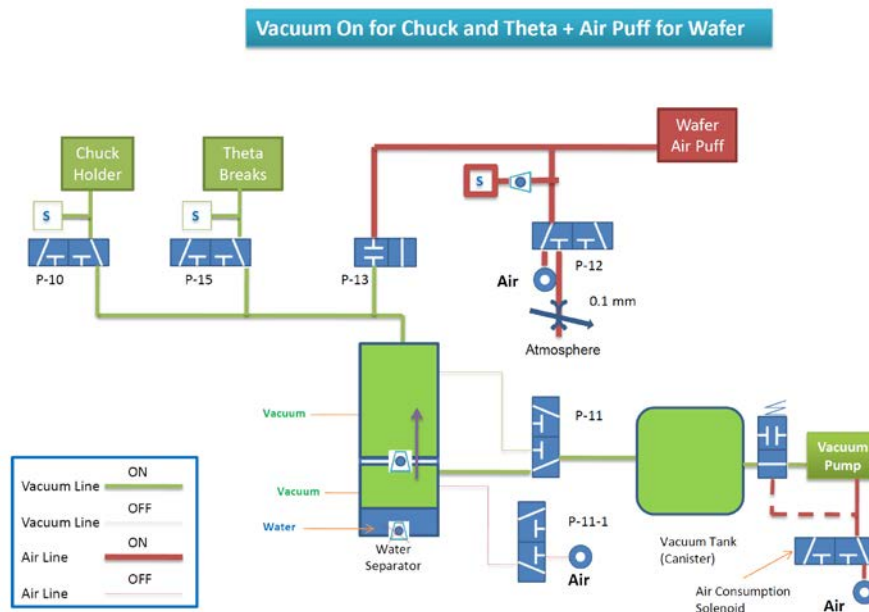


Figure 11-21. Vacuum on for chuck, Theta and wafer air puff



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## Chapter 12. Troubleshooting the System

Troubleshooting the System is a dynamic chapter that addresses issues that have been reported, and documents the solutions provided by Advanced Dicing Technologies Ltd.

Troubleshooting the System includes:

- ❖ Managing Reported Error Messages (section 12.1)
- ❖ Managing Recipe Problems (section 12.2)
- ❖ Solving Height Problems (section 12.3)
- ❖ Initializing the System (section 12.212.4)
- ❖ When Nothing Helps (section 12.5)

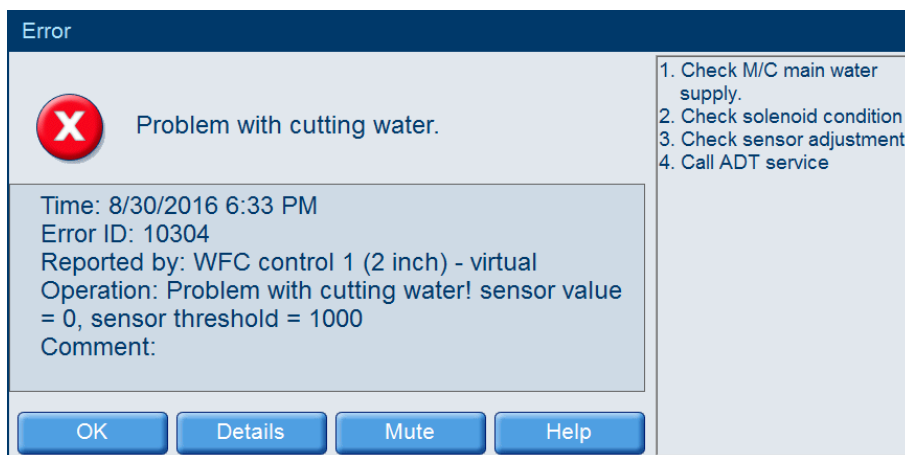
## 12.1 Managing Reported Error Messages

Error messages can be used to help diagnose and solve problems that occur while operating the system.

Error messages are divided into the following sections:

- Error panel: identifies the problem
- Details panel: describes the problem
- Troubleshooting panel: may provide possible solutions
- Error Messages Browser: when several errors occur simultaneously, enables displaying one error message at a time.

The user can identify the problem source using the data given in the Error panel and pressing the Details Screen. Selecting the Help button displays troubleshooting steps that may help solve the problem. The steps are numbered as separate tasks. The first step is the most likely to solve the problem.



**Figure 12-1. Sample Error Message (red)**

Errors are also indicated by the Light Tower. For more details about the Light Tower states, see section 2.3.1.3 on page 2-11.

On systems configured with a Buzzer, pressing Mute stops the buzzer.



**Note:** When a problem occurs that cannot be resolved using the details and/or troubleshooting information, write down the information displayed in the Error and Details panels and then contact your ADT service representative.

**To save error messages for reference:**

1. After an error message appears, select the **Detail** check box.
2. Connect a keyboard and press the **Print Screen** key on the keyboard.
3. Press **Ctrl+Esc** on the keyboard.
4. Go to **Programs > Accessories > Paint**.
5. Paste (**Ctrl+V**) the copied screen.
6. Save the file.

## 12.2 Managing Recipe Problems

When the system reports a recipe problem (e.g., Missing Model message), perform the first of the following steps. If this does not solve the problem, continue with the succeeding steps, until solving the problem:

1. Examine the recipe parameters.
2. Make sure that the active recipe is suited to the workpiece on the Dicer. If they are not suited, either re-assigns the job, or unload and reload the workpiece.
3. Reteach alignment.
4. Recreate the recipe (in some cases, because the file itself is corrupted, the recipe needs to be created from the beginning). It is important to use a working recipe or template, and not the corrupted file, for duplication.

## 12.3 Solving Height Problems

The height procedure determines the cutting chuck position along the Z Axis. If the dicing cut depth is not accurate, or if there is a problem with the Height device, perform the following solutions, in the order that they appear, until the problem is solved:

1. Change the blade (see details in the 7120/7130 Operations Manual).
2. Perform a height procedure, if still not solved, a Chuck to Height Device procedure, last— a Sample Blade Calibration procedure (see section 5.3.2).
3. Examine the height device for a malfunction (see details in section 9.2).

## 12.4 Initializing the System

When a problem occurs in the Model 71XX, the user can try solving the problem by re-initializing the System. This will often solve the problem and enable normal operation to resume.

Before re-initializing the whole system, it is recommended first initializing a specific system component (see section 12.4.1); only if the problem persists, initialize the whole system.



**Caution:** Before running an initialization procedure, it is recommended removing the workpiece from the cutting chuck.

### 12.4.1 Initializing Dicer Components

When a problem occurs in a component, that particular component can be reinitialized. This will often solve the problem and enable normal operation to resume.

**To initialize dicer components:**

1. Remove the workpiece, if installed, from the cutting chuck.
2. In the application, from the Main Menu, press **Maintenance > Axis Setup**; the Maintenance-Axis Setup window appears.
3. Press the component name in the dicer component list (e.g., **X**); the selected component view appears.
4. In the Control Panel, press the **Init** button; the specific component is initialized.
5. Repeat steps 3 and 4 for each dicer component that should be initialized.



**Notes:**

- ◆ When initializing more than one dicer component, initialize according to the following order: Z, X, Y, Theta.
- ◆ There is no need to wait for one initialization procedure to finish before performing the next one.

### 12.4.2 Initializing the Entire System

When initializing specific components does not solve the specific problem, re-initialize the entire system.

**To initialize the system:**

In the Main Menu, press **Sys Init**.

## **12.5 When Nothing Helps**

As in most computerized systems, not all the problems, which occur while running the system, are identifiable. If the steps, suggested in the Troubleshooting pane, do not solve the problem, powering the system down and then back up can be a solution.

### **12.5.1 KMI and Database**

The system has two fundamental components: the KMI software and the Model 71XX database. Higher level troubleshooting may require access to these components. Contact ADT for details.

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## Chapter 13. Appendix

The appendix includes:

- ❖ Axes Default Values (section 13.1)
- ❖ Spare Part Kits (section 13.2)

## 13.1 Axes Default Values

Table 13-1 represents the default values for the system's axes.



**Note:** The values provided in **Table 13-1** should not be changed without ADT approval or instructions.

**Table 13-1. Default Values for the System Axes**

| Parameter \ Axis                               | X    | Y   | Z    | Theta | Spindle<br>(2")                  | Spindle<br>(4")                  |
|------------------------------------------------|------|-----|------|-------|----------------------------------|----------------------------------|
| Max. Velocity<br>[mm/sec]                      | 600  | 75  | 15   | 60    | 60000                            | 30000                            |
| Slow Velocity<br>[mm/sec]                      | 15   | 15  | 1    | 3     | 1000                             | 1000                             |
| Fast Velocity<br>[mm/sec]                      | 200  | 50  | 5    | 30    | 60000                            | 30000                            |
| Acceleration<br>[mm/sec <sup>2</sup> ]         | 4000 | 250 | 3500 | 150   | 1<br>(krev/min <sup>2</sup> )    | 0.25<br>(krev/min <sup>2</sup> ) |
| Slow<br>Acceleration<br>[mm/sec <sup>2</sup> ] | 1000 | 100 | 3500 | 150   | 1<br>(krev/min <sup>2</sup> )    | 0.25<br>(krev/min <sup>2</sup> ) |
| Deceleration<br>[mm/sec <sup>2</sup> ]         | 4000 | 250 | 3500 | 150   | 1<br>(krev/min <sup>2</sup> )    | 0.25<br>(krev/min <sup>2</sup> ) |
| Kill Deceleration<br>[mm/sec <sup>2</sup> ]    | 4000 | 800 | 3500 | 800   | 1.25<br>(krev/min <sup>2</sup> ) | 0.5<br>(krev/min <sup>2</sup> )  |



## 13.2 Spare Part Kits

The following tables list the contents of the recommended and extended spare parts kits for the model 7120/7130.

### 13.2.1 Recommended Spare Part Kit

Table 13-2 lists the contents of the Recommended Spare Part Kit (ADT P/N 07100-0901-000-XX) for the Model 71XX.

**Table 13-2. Recommended Spare Parts Kit**

| ADT Part Number                  | Part Name                    | Qt. |
|----------------------------------|------------------------------|-----|
| <b>Utility &amp; Solenoid</b>    |                              |     |
| 28150-0053-000                   | Vacuum and Air solenoid      | 1   |
| 10400-0010-002                   | Water operator (P2&P4)       | 2   |
| 00990-0519-000-00                | Wafer vacuum solenoid        | 1   |
| 13600-0041-300                   | Brush for 2" or 4" spindle   | 2   |
| <b>Sensors &amp; Accessories</b> |                              |     |
| 19079-0006-001                   | Water sensor                 | 1   |
| 07100-3058-000-01                | Vacuum sensor                | 1   |
| 07100-3059-000-00 Z              | Limit sensor                 | 1   |
| 28562-0015-100                   | Gas spring                   | 1   |
| 23100-0012-001                   | Air filter element           | 1   |
| 60504-0016-024                   | O-ring for Q.C.C.            | 2   |
| 61400-0005-001                   | O-ring for Chuck             | 1   |
| <b>Electronics Boards</b>        |                              |     |
| 07700-4270-000-00                | Contact /Brush Check Bd.     | 1   |
| 07700-4250-000-01                | Sensor interface Bd.         | 1   |
| 07100-2100-000-00                | Axis driver board            | 1   |
| <b>Lamps</b>                     |                              |     |
| 28595-7500-100                   | Vertical/horizon lamp        | 1   |
| 13100-0021-000                   | On/Off Green lamp Indicators | 2   |
| 16200-5004-003                   | On/Off operation switch lamp | 2   |
| 13100-0020-008                   | Tower Light lamp             | 2   |
| <b>Fuses</b>                     |                              |     |
| 18543-2005-100                   | Fuse 5 Amp                   | 4   |

## 13.2.2 Extended Spare Part Kit

Table 13-3 lists the contents of the Extended Spare Part Kit (ADT P/N 07100-0902-000-XX) for Model 71XX.

**Table 13-3. Extended Spare Parts Kit**

| ADT Part Number                 | Part Name                    | Qt. |
|---------------------------------|------------------------------|-----|
| <b>Utility &amp;Sensors</b>     |                              |     |
| 13600-0041-300                  | Brush for 2" spindle         | 2   |
| 28150-0053-000                  | Vacuum and Air solenoid      | 1   |
| 28150-0052-000                  | Water solenoid               | 1   |
| 00990-0519-000-00               | Wafer Vacuum solenoid        | 1   |
| 00990-0125-000-01               | Non-Contact Height Ass'y     | 1   |
| 33850-0006-000 CCD              | Camera                       | 1   |
| <b>Sensors &amp;Accessories</b> |                              |     |
| 19079-0006-001                  | Water sensor                 | 1   |
| 07100-3049-000-01               | Air sensor                   | 1   |
| 07100-3058-000-01               | Vacuum sensor                | 1   |
| 28595-0511-000                  | X Bellow                     | 2   |
| 07100-3059-000-00               | Z Limit sensor               | 2   |
| 07100-3057-000-00               | Y Limit sensor               | 1   |
| 07100-3056-000-00               | X Limit sensor               | 2   |
| <b>Electronics Boards</b>       |                              |     |
| 07700-4280-000-01               | Non-Contact Height board     | 1   |
| 07700-4270-000-00               | Contact /Brush Check Bd      | 1   |
| 07700-4240-000-01               | Illumination driver Bd       | 1   |
| 07700-2100-000-01               | Slave Controller             | 1   |
| 07700-4250-000-01               | Sensor interface Bd          | 1   |
| 19065-6021-000                  | Matrix board                 | 1   |
| 13600-0001-504                  | Spindle driver               | 1   |
| 13600-0001-604                  | Copley, X Servo AMP          | 1   |
| 07100-2100-000-00               | Axis driver board            | 1   |
| <b>Lamps</b>                    |                              |     |
| 28595-7500-100                  | Vertical/horizon lamp        | 1   |
| 13100-0021-000                  | On/Off Green lamp Indicators | 2   |
| 16200-5004-003                  | On/Off operation switch lamp | 2   |
| 13100-0020-008                  | Tower Light lamp             | 2   |
| <b>Fuses</b>                    |                              |     |
| 18543-2005-100                  | Fuse 5 Amp                   | 2   |

## Chapter 14. Glossary

The Glossary includes terminology used in the 7900 Uno series Operations Manual.

### **Access Level**

Privilege categories that define the user's ability to view and/or perform specific activities, and view and/or change program-specific parameters.

### **Administrator**

The highest-ranked access level within your business. The administrator defines the various access level parameters. In addition, the administrator can add users to, and remove users from the system as well as define the access level of each user.

### **Align Kerf Model**

Orientation referencing model, based on a performed cut position and not on the eye-point.

### **Alignment**

Precise rotation of the workpiece in relation to the X-Axis (on the chuck) in preparation for cutting.

### **Alignment Center**

Position of the center of the street that was taught during Teach Alignment.

### **Alignment Types**

Classification of alignment conditions, by activity.

### **Angle**

A set of parallel streets on a workpiece.

### **Auto Mode**

A semi-automatic system operation. You are only required to manually place the workpiece on, and remove the workpiece, from the chuck.

### **Axis (pl. Axes)**

Coordinates along which the system components move; also used to describe the components themselves.

### **Blade**

The cutting tool used by the system to dice the workpiece. Blades are classified into two main categories: Hub Blade and Hubless Blade.

### **Blade Center**

The center of the cut line.

### **Blade Exposure**

The portion of the blade that is exposed and can be used to cut the workpiece.

- For Hubless blade: Blade Exposure = (Blade OD – Flange OD) / 2

- For Hub blade: Initial Blade Exposure is provided by the blade supplier.

**Blade Properties**

The characteristics of a particular blade.

**Button Height**

See Height.

**Bottom Chip Area**

Entire bottom chipped area of the kerf, from the  $W_{min}$  to the  $W_{max}$  (according to the vision workspace).

**Bottom Chipping**

Maximum chipped point measured from the  $W_{min}$  to the bottom of the chipped kerf area.

**Center to Max Chipping**

Distance from the aligned center of the kerf to the maximum chipped point.

**Chipping Area**

The area of chipping measured on either side of the kerf.

**Chuck**

The rotating table on which a workpiece is mounted and held in place by vacuum, during processing.

**Chuck Height**

Determines the chuck position by lowering the blade onto the cutting chuck.

**Chuck to Non-Chuck Delta Measurement**

Measure the difference between the chuck height and the NCH device in order to calculate the delta value.

**Cooling Block**

A device on the spindle, which uses water to cool the blade during cutting.

A mechanical component that is connected to the spindle front, providing cooling, cutting and cleaning water to the blade, and holds the BBD (option).

**Configuration**

Defines a particular arrangement, presence, or absence of system components (such as the Non-Contact Height Device).

**Cut Center**

Center of the actual kerf.

**Cut Depth**

The depth the blade is inserted, measured from the top of the workpiece surface.

**Cut Line**

The cutting path (kerf).

**Cut Map**

When alignment is complete, the system displays a graphic image of the workpiece with the defined alignment lines.

**Cutting Chuck**

See Chuck.

**Cutting Table**

See Chuck.

**Delta**

Denotes the difference between two points.

**Developer**

ADT application development personnel.

**DI Water**

De-ionized water.

**Dice (verb)**

To cut a workpiece into segments.

**Dice (noun-plural), Die (noun singular)**

The portions of a workpiece that have been cut into individual segments.

**Die Model**

Reference eye-points used in two-point alignment.

**Down-cut**

The blade enters the workpiece from the top and exits from the bottom.

**Dress**

To prepare a blade for use by cutting into a dedicated dressing block or workpiece.

**Dressing Block**

Used to prepare the blade. A dressing block can be used on a dressing station or on the chuck.

**Dress Station**

An optional dicer component designed to dress the blade as part of a processing program

**DY**

Delta Y, the deviation of the kerf center from the alignment center (also called Y-Offset).

**Engineer**

The second highest Access Level within your business.

**Exposure**

See Blade Exposure.

**Feed Rate**

The speed at which the workpiece is moved into the blade.

**Flange**

The rim that holds a hubless blade in place on the spindle shaft.

**Flow Rate**

The rate at which the water cools and cleans the blade and the workpiece cutting area.

**Frame**

The ring that holds the tape onto which the workpiece is mounted.

**Height**

The Height procedure is performed in order to determine the chuck position along the Z-Axis. Height is performed by lowering the spindle until the edge of the blade makes contact with either the chuck or a height device. Height is updated regularly by the system in order to compensate for differences in the Z-Axis position, due to blade wear. There are four different types of Height procedures:

- Chuck Height.
- Non-Contact Height Device.
- Chuck to Non-Chuck Delta Measurement.
- Sample Blade Calibration.

**Hub Blade**

A circular blade that has an integral center section and does not require a flange to hold it to the spindle.

**Hubless Blade**

A circular blade that consists only of a cutting surface, which must be held to the spindle shaft by an external rim (flange).

**Index**

The distance between each cut.

**Initialization**

The act of finding the specific points for different axes and components.

**Inspection**

The process of stopping the cutting procedure for the purpose of inspecting the quality of the cut.

**Kerf**

The cut made on the workpiece by the blade.

**Kerf Check**

Visual inspection of the kerf. Can be performed manually or automatically by the Pattern Recognition System (PRS).

**Load Monitoring**

Enables you to monitor or read the load on the spindle (measured in amps) caused by the resistance encountered by the blade during the cutting process.

**Login**

Enables the use of the system according to the designated Access Level.

**Loop Cut**

The repetition of a cut or a block.

**Low Model**

Used for correcting the Theta and locating a Main model. A low model can only be a Die Model.

**Main Model**

Required to perform auto alignment and used for fine/accurate alignment. The alignment process will always end with the Main Model alignment.

**Manual Mode**

Requires user intervention for specific procedures (e.g. Manual Alignment).

**Mixed**

The workpiece is cut in both directions; i.e. it is fed in the right-to-left direction at one index and left-to-right at the next index.

**Model**

A set of points, on the workpiece, which constitute a pattern. Models are used by the vision system for locational and orientation referencing.

**Model Scoring**

Rating an Alignment Model in order to determine reliability of the recognizable patterns.

**Nominal Distance**

When using a specific Shrinkage application, the Nominal Distance measures the distance between two points.

**Non-Contact Height Device**

The blade is lowered into a light beam emitted from the Non-Contact Height Device (NCH). The blade height is computed at the point the light beam is blocked. The NCH limits the need to calibrate the blade exposure and height directly on the chuck, causing blade and chuck wear.

**Operator**

Lowest ranking Access Level within your business; operators can only perform basic operations.

**Override**

Dressing a blade while cutting a regular workpiece. Initial cutting speed is reduced, gradually returning to the regular cutting speed according to the recipe steps.

**Pattern Recognition System (PRS)**

The aspect of the vision system that learns and identifies patterns used as locational reference points.

**Recipe**

Setting algorithms or parameters that control the operation of the system. A recipe can be assigned to an individual workpiece or an unlimited number of workpieces that share the same characteristics.

**Reticle**

The green cross hairs used in the Teach Box (located in the video workspace) to align the kerf.

**Sample Blade Calibration**

Performed after a spindle adjustment or replacement. Uses a known blade diameter to set the chuck position value. The chuck position value is used during operation to determine the accurate and exact exposure of the current blade by performing Chuck Height.

**Setup**

Preparing and adjusting the system or any system component in order to perform assigned tasks.

**Skew**

The maximum offset in the Y-Axis between the current model and the last model found.

**Spindle**

The DC brushless air bearing motor (dicer rotating component), onto which the blade is mounted.

**Street**

The cutting path. Streets are physically predefined during workpiece manufacturing.

**Sub-model**

Models used for Main model verification, when the Main model is unclear or not repeatable.

**Substrate**

See Workpiece.

**Teach**

The system is taught specific models. In Teach Alignment, the system is taught to find the center of the streets to be cut. In Teach Kerf Check, the system is taught which part of the kerf to inspect.

**Technician**

Middle rank Access Level. Technicians have received training and are experienced in maintaining the system.



**Theta (T) Axis**

Theta components rotate around the T Axis. Also referred to as the turntable on which the chuck is mounted. Also see Axis (pl. Axes).

**Top Chip Area**

Entire chipped area of the kerf, from the Wmin to the Wmax (according to the FOV workspace).

**Top Chipping**

Maximum chipped point measured from the Wmin to the top of the chipped area of the kerf (viewed in the Vision system).

**UPS (Uninterrupted Power Supply)**

The backup power source that allows completion of the currently running operation in case of a power failure.

**User**

A general term to describe any person who operates, adjusts or maintains the system, regardless of their access level.

**Verification Model**

During the dicing process, when a Main or Submodel becomes unusable (when the model has been diced or is concealed by debris) a Verification model is used. They are located towards the center of the workpiece, and are used to verify the cut position only.

**Warmup**

A procedure where the spindle rotates (without contacting the workpiece) at a predefined velocity and predefined time, in order to keep the spindle warm until the cutting process begins. During the Warmup, the water supply is activated.

**Wash Pipe**

A metal pipe mounted on the dicer. The wash pipe is used to clean and moisturize the workpiece before, during, and after dicing.

**WMax**

Maximum kerf width.

**WMin**

Minimum kerf width.

**Workpiece**

Also known as:

- Wafer.
- Substrate.
- Semiconductor material.

**X-Axis**

X-Axis components move left and right. Also see Axis (pl. Axes).

**Y-Axis**

Y-Axis components move from front to back. Also see Axis (pl. Axes).

**Y-Offset**

The deviation of the blade center from the microscope\ vision system center.

**Y-Offset Correction**

Automatic compensation and correction for the deviation of the blade center from the microscope\ vision system center.

**Z-Axis**

Z-Axis components move up and down. Also see Axis (pl. Axes).

